

38. Serial Communications Interface (SCI)

This is the SCI_B version of the SCI peripheral module.

SCI_B is referred to as SCI in this chapter.

38.1 Overview

The Serial Communications Interface (SCI) × 10 channels have asynchronous and synchronous serial interfaces:

- Asynchronous interfaces (UART and Asynchronous Communications Interface Adapter (ACIA))
- 8-bit clock synchronous interface
- Simple IIC (master-only)
- Simple SPI
- Simple LIN
- Smart card interface
- Manchester interface

The smart card interface complies with the ISO/IEC 7816-3 standard for electronic signals and transmission protocol. SCIn (n = 0 to 9) has FIFO buffers to enable continuous and full-duplex communication, and the data transfer speed can be configured independently using an on-chip baud rate generator.

In this section, PCLK refers to PCLKA.

Table 38.1 lists the SCI specifications, Table 38.2 shows the functions of each SCI channel, Figure 38.1 shows a block diagram of SCI, Figure 38.2 shows a clock source selector block diagram, and Table 38.3 lists the I/O pins.

Table 38.1 SCI specifications (1 of 4)

Parameter	Specifications
Number of modules	10 (SCIn (n = 0 to 9))
Serial communication modes	• Asynchronous • Clock synchronous • Simple IIC • Simple SPI • Simple LIN (SCIn) • Smart card interface • Manchester interface (SCIn)
Operation clock (TCLK)	Synchronized clock (PCLK) or independent clock (SCICLK) can be selected
Transfer speed	Bit rate specifiable with the on-chip baud rate generator
Full-duplex communications	• Transmitter: Continuous transmission possible using double-buffering • Receiver: Continuous reception possible using double-buffering
Half-duplex communications	Half-duplex communication is possible by using only TXDn pins
Data transfer	Selectable as LSB-first or MSB-first transfer
Inverter for communication pins (RXDn, TXDn)	Selectable inverter for each pin (RXDn, TXDn)
Interrupt sources	Transmit end, transmit data empty, receive data full, receive error, receive data ready, address match. Break Field detection/output, Bus collision detection, Active edge detection (SCIn). Completion of generation of a start condition, restart condition, or stop condition (for simple IIC mode).
Loop Back function	Self-diagnosis of communication function by IP internal transmission/reception is possible
Synchronizer Bypass function	Ability to bypass synchronization circuit between bus clock (PCLK) and operation clock (TCLK)
Module-stop function	Module-stop state can be set for each channel
Reception in Software Standby mode	Only SCI0 is available

Table 38.1 SCI specifications (2 of 4)

Parameter		Specifications
Clock synchronous mode	Data length	8 bits
	Adjustment of receive sampling timing	Adjustable receive sampling timing after the default timing in master mode only when using internal clock
	Receive error detection	Overrun error
	Clock source	Selectable to internal clock (master mode) or external clock (slave mode)
	Double-speed mode	Baud rate generator double-speed mode is selectable
	Hardware flow control	Transmission and reception controllable with CTSn_RTSn pins
	Transmission and reception	Selectable to 1-stage register or 16-stage FIFO
Asynchronous mode	Data length	7, 8, or 9 bits
	Transmission stop bit	1 or 2 bits
	Adjustment of receive sampling timing	Adjustable receive sampling timing before/after the default timing
	Adjustment of transmit timing	Adjustable edge timing of transmit waveform controlled by the setting value of registers.
	Parity	Even parity, odd parity, or no parity
	Receive error detection	- Parity error - Overrun error - Framing error
	Hardware flow control	Transmission and reception controllable with CTSn_RTSn pin and CTSn pin
	Transmission and reception	Selectable to 1-stage register or 16-stage FIFO
	Address match	Interrupt request/event output can be issued upon detecting a match between received data and the value in the compare match register
	Start-bit detection	Selectable to low level or falling edge detection
	Break detection	Breaks from framing errors detectable by read from CSR register
	Clock source	Selectable to internal or external clock. Transfer rate clock input from the GPT can be used (SCI _n (n = 1, 2)).
	Double-speed mode	Baud rate generator double-speed mode is selectable
	Multi-processor communications function	Serial communication enabled among multiple processors
	RS-485 driver control function	Output DEN signal to enable external transceiver transmit mode
	Noise cancellation	Digital noise filters included on signal paths from the RXD _n pin inputs
Smart card interface mode	Error processing	Error signal can be automatically transmitted upon detecting a parity error during reception
		Data can be automatically retransmitted upon receiving an error signal during transmission
	Data type	Both direct and inverse convention supported

Table 38.1 SCI specifications (3 of 4)

Parameter		Specifications
Manchester mode	Communication format	Manchester code with the preface and the Start Bit added
	Data length	7, 8, or 9 bits
	Transmission stop bit	1 or 2 bits
	Parity function	Even parity, odd parity, or no parity
	Receive error detection	Parity, overrun, framing, Manchester errors
	Hardware flow control	Transmission and reception controllable with CTSn_RTSn pin and CTSn pin
	Clock source	Only internal clock can be used.
	Double-speed mode	Baud rate generator double-speed mode is selectable
	Multi-processor communication function	Serial communication among multiple processors
	Manchester encoding/decoding function	Function to perform Manchester encoding/decoding of transmission/reception data and communicate using Manchester code
	Noise cancellation	The signal paths from input on the RXDn pins incorporate digital noise filters
	Preface setting/detection function	The function outputs the configured the preface pattern and detects it.
	Start Bit setting/detection function	The function outputs the configured the Start Bit pattern and detects it.
	Reception retiming function	Timing correction is performed for each bit of the received signal
Simple IIC mode	Transfer format	I ² C bus format (MSB-first only)
	Operating mode	Master (single-master operation only)
	Transfer rate	Up to 400 kbps
	Noise cancellation	The signal paths from input on the SCLn and SDAn pins incorporate digital noise filters and provide an adjustable interval for noise cancellation
Simple SPI mode	Data length	8 bits
	Error detection	Overrun error
	Clock source	Selectable to internal clock (master mode) or external clock (slave mode)
	Double-speed mode	Baud rate generator double-speed mode is selectable
	Transmission/Reception	Selectable either 1 stage register or 16-stage FIFO
	Adjustment of receive sampling timing	Adjustable receive sampling timing after the default timing in master mode only when using internal clock
	SSn input pin function	High impedance state can be invoked on the output pins by driving the SSn pin high.
	Clock settings	Configurable among four clock phase and clock polarity settings

Table 38.1 SCI specifications (4 of 4)

Parameter		Specifications
Simple LIN mode	Start Frame Transmission	- Break Field output possible, Break Field output complete interrupt output possible - Bus collision detection possible, bus collision detection interrupt output possible
	Start Frame Reception	- Break Field detectable, Break Field detected interrupt output possible - Control Field 0/1 data comparison function - Control Field 1 can set two types of comparison data of primary and secondary - Priority interrupt bit can be set in Control Field 1 - Handling of Start Frames that do not include a Break Field - Handling of Start Frames that do not include a Control Field 0 - Bit rate measurement function
	Input/Output control function	- Selectable polarity for TXDn and RXDn signals - Selection of a digital filter for the RXDn signal - Half-duplex operation employing RXDn and TXDn signals multiplexed on the same pin - Selectable timing for the sampling of data received through RXDn
Bit rate modulation function		Error reduction through correction of outputs from the on-chip baud rate generator
Event link function		Error event output for receive error or error signal detection (SCIn_ERI) (n = 0 to 9)
		Receive data full event output (SCIn_RXI) (n = 0 to 9)
		Transmit data empty event output (SCIn_TXI) (n = 0 to 9)
		Address match event output (SCIn_AM) (n = 0 to 9)
		Active edge detection event output (SCIn_AED)
		Transmit end event output (SCIn_TEI) (n = 0 to 9)
TrustZone Filter		Security and Privilege attribution can be set for each channels

Table 38.2 Functions of SCI channel

Item	SCI0	SCI1	SCI2	SCI3 to SCI9
Asynchronous mode	Available			
Clock synchronous mode	Available			
Smart card interface mode	Available			
Simple I ² C mode	Available			
Simple SPI mode	Available			
FIFO mode	Available (16 stages)			
Address match	Available			
Manchester mode	Available	Available	Available	Available
Simple LIN mode	Available	Available	Available	Available
GPT clock input	Not Available	GTIOC3A GTIOC4A	GTIOC6A GTIOC7A	Not Available

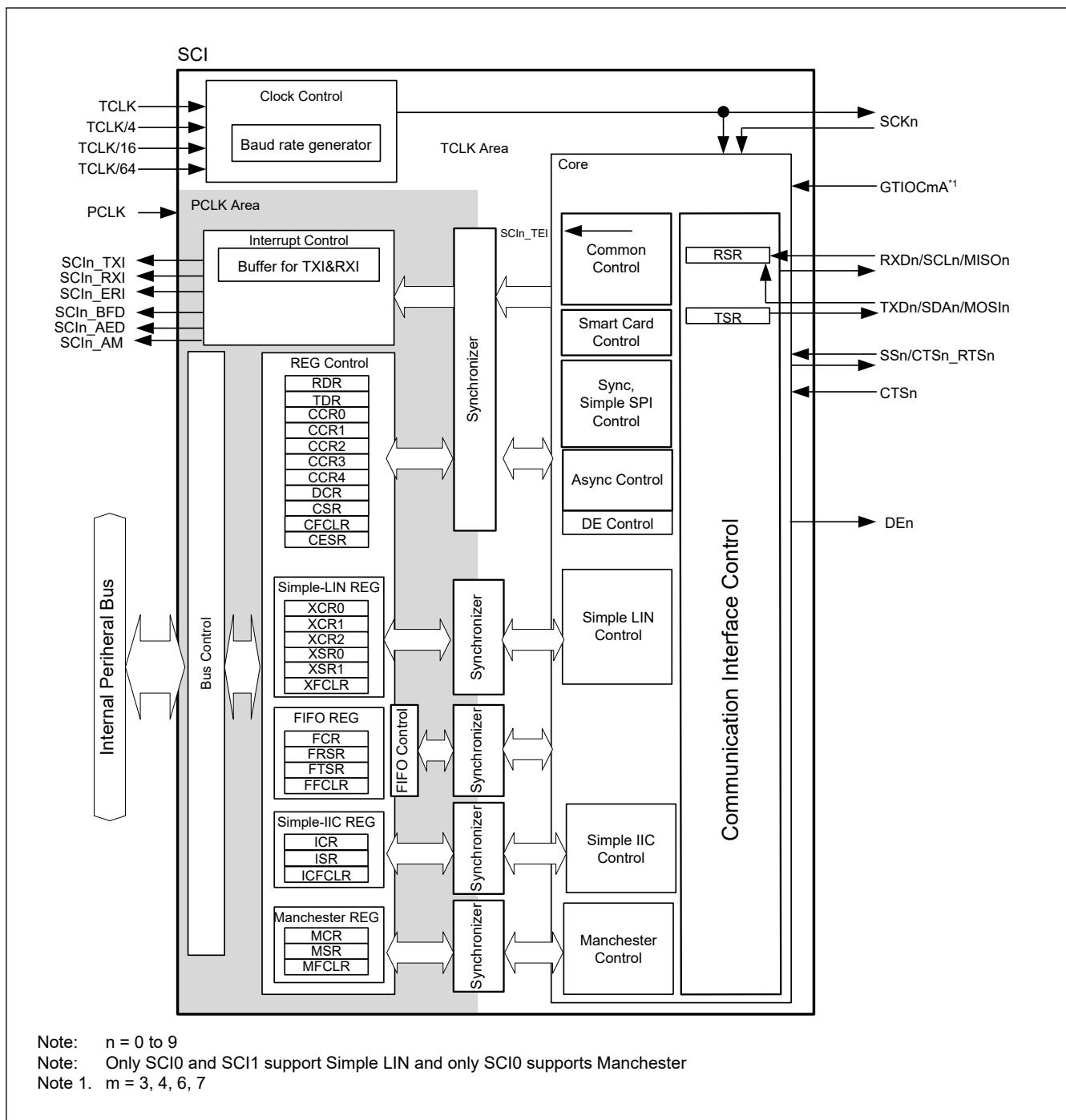


Figure 38.1 SCI block diagram

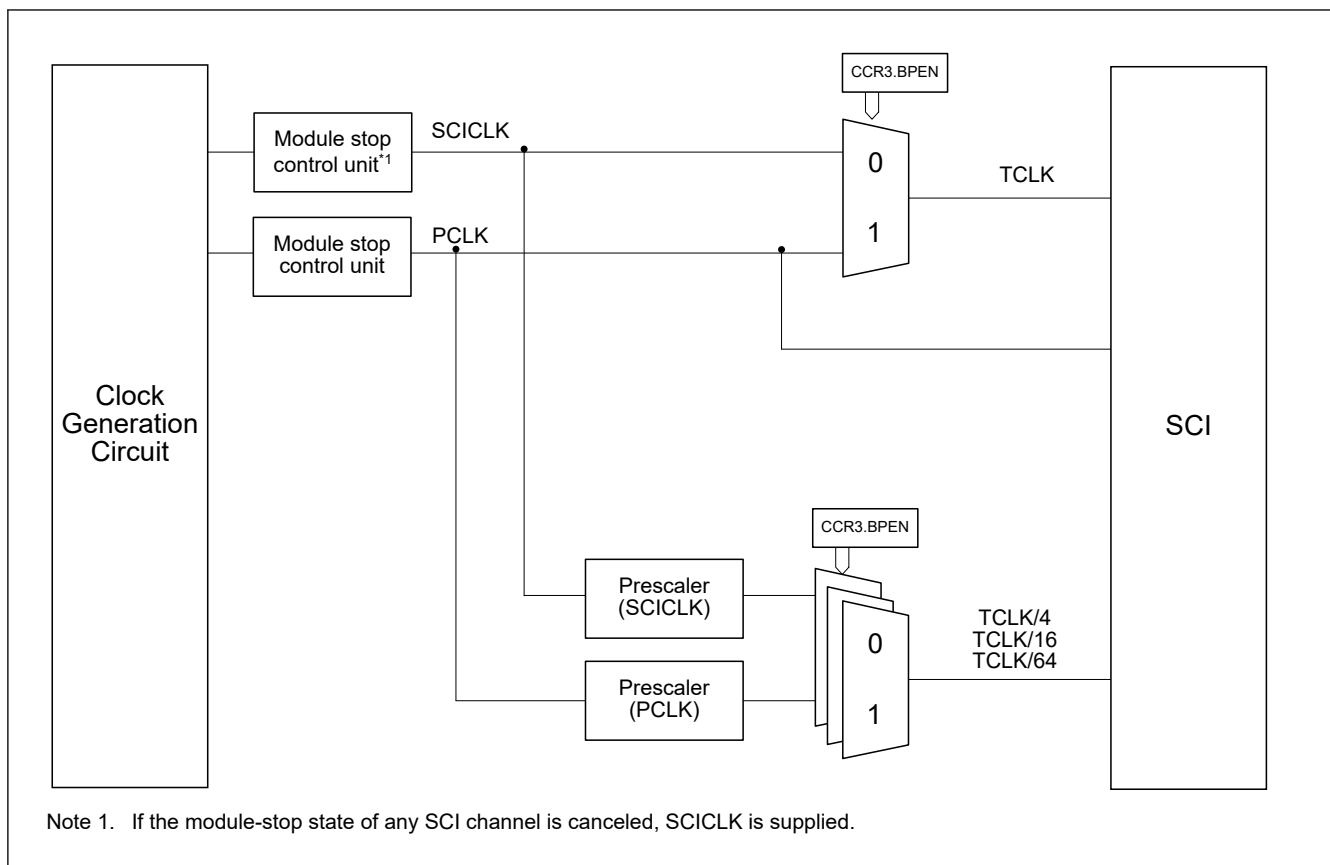


Figure 38.2 Clock source selector block diagram

Table 38.3 SCI I/O pins

Function	Pin name	Input/Output	Description
SCIn (n = 0 to 9)	RXDn/SCLn/MISO _n	Input/Output	SCIn receive data input SCIn I ² C clock input/output SCIn slave transmit data input/output
	TXDn/SDAn/MOS _{In}	Input/Output	SCIn transmit data output SCIn I ² C data input/output SCIn master transmit data input/output
	SSn/CTS _n _RTSn	Input/Output	SCIn chip select input, active-low SCIn transfer start control input/output, active-low
	CTS _n	Input	SCIn transfer start control input, active-low
	DE _n	Output	Driver Enable signal output
	SCK _n	Input/Output	SCIn clock input/output

38.2 Register Descriptions

38.2.1 RSR : Receive Shift Register

RSR is a shift register that receives serial data input from the RXD_n pin and converts it into parallel data. When one frame of data is received, the data is automatically transferred to the RDR register. The RSR register cannot be directly accessed by the CPU.

38.2.2 RDR/RDR_BY : Receive Data Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	FER	PER	—	—	ORER	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	FFER	FPER	DR	MPB	RDAT[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	RDAT[8:0]	Serial receive data RDAT is a 9-bit register for storing received data. Received data is stored in [6:0] when 7-bit data is selected, in [7:0] when 8-bit data is selected, and in [8:0] when 9-bit data is selected. And 0 is stored in the unused bit. Use RDR_BY for byte access to RDR[7:0].	R
9	MPB	Multi-processor flag 0: Data transmission cycles 1: ID transmission cycles	R
10	DR	Receive data ready flag FRSR.DR can be read.	R
11	FPER	FIFO parity error flag Valid only in Asynchronous mode 0: There is no parity error in the data read from the receive-FIFO 1: There is parity error in the data read from the receive-FIFO	R
12	FFER	FIFO framing error flag Valid only in Asynchronous mode 0: There is no framing error in the data read from the receive-FIFO 1: There is framing error in the data read from the receive-FIFO	R
23:13	—	These bits are read as 0.	R
24	ORER	Overflow Error flag CSR.ORER can be read.	R
26:25	—	These bits are read as 0.	R
27	PER	Parity error flag CSR.PER can be read.	R
28	FER	Framing error flag CSR.FER can be read.	R
31:29	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

In FIFO mode (CCR3.FM = 1), the RDR is 16-stage FIFO buffer configuration.

When using FIFO mode, use RDR for 32-bit access.

RDAT[8:0] bits (Serial receive data)

After one frame of data is received, the received data is transferred from the RSR register to this register, thus allowing the RSR register to receive the next data.

The RSR and RDR registers have a double-buffered construction to enable continuous reception.

For Non-FIFO mode, read RDR only once when a receive data full interrupt (SCIn_RXI) request is issued. Without reading received data from RDR, if the next one frame is received, an overrun error occurs.

For FIFO mode, continuous reception is executed until 16 stages are stored. If data is read when there is no received data in the receive-FIFO(RDR), the value is undefined. When the receive-FIFO (RDR) are full of received data, subsequent serial receive data is lost.

The CPU cannot write to RDR.

0 is stored in the bit position which isn't received (RDR.bit8 or RDR.bit7) at the time of 7bit or 8bit communication of Asynchronous and Manchester mode.

MPB bit (Multi-processor flag)

In Asynchronous mode and Manchester mode, during multi-processor communication (CCR3.MP = 1), the value of the multi-processor bit corresponding to the received data (RDAT[8:0]) can be read.

FPER bit (FIFO parity error flag)

Indicates whether the data read from the receive-FIFO has a parity error.

0 is stored for non-FIFO mode.

FFER bit (FIFO framing error flag)

Indicates whether the data read from receive-FIFO has a framing error.

0 is stored for non-FIFO mode.

38.2.3 TDR/TDRLL/TDRLH : Transmit Data Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x04 (TDR/TDRLL)
0x05 (TDRLH)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TSYN C	—	—	MPBT	TDAT[8:0]								
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
8:0	TDAT[8:0]	Serial transmit data TDAT is a 9-bit register for setting transmit data. Transmit data is set in [6:0] when 7-bit data is selected, in [7:0] when 8-bit data is selected, and in [8:0] when 9-bit data is selected. When byte access, write TDR [15:8] and then write TDR [7:0]. Use TDRLL for byte access to TDR[7:0], TDRLH for byte access to TDR[15:8].	R/W
9	MPBT	Multi-processor transfer bit flag Value of the multi-processor bit in the transmission frame. This bit is use in Asynchronous and Manchester mode. When writing to this bit when not used, write the initial value. 0: Data transmission cycles 1: ID transmission cycles	R/W
11:10	—	These bits are read as 1. The write value should be 1.	R/W
12	TSYNC	Transmit SYNC data It is valid when MCR.SBSEL = 1 and MCR.SYNSEL = 1 in Manchester mode. When this bit is not used, write the initial value. 0: The Start Bit is transmitted as DATA SYNC. 1: The Start Bit is transmitted as COMMAND SYNC.	R/W
31:13	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-3, P-TYPE-3

In FIFO mode (CCR3.FM = 1), the TDR is 16-stage FIFO buffer configuration.

When using FIFO mode, use TDR for 32-bit access.

TDAT[8:0] bits (Serial transmit data)

The TDR is a 9-bit register for storing transmit data.

When empty space is detected in the TSR register, the transmit data stored in the TDR registers is transferred to TSR, and transmitting is started.

The TSR and TDR registers have a double-buffered construction to realize continuous transmission. When the next data to be transmitted is stored in TDR after one frame of data has been transmitted, the transmitting operation is continued by transfer to the TSR register.

When the SCI detects that the transmit shift register (TSR) is empty, it transmits data written in the transmit-FIFO (TDR) into TSR and starts serial transmission. Continuous serial transmission is executed until there is no transmit data left in the transmit-FIFO (TDR).

For non-FIFO mode, when a transmit data empty interrupt (SCIn_TXI) request is issued and CCR0.TE is 1, write transmit data to the TDR only once.

For FIFO mode, when transmit-FIFO is full of transmit data 16 frames, no more data can be written. If writing of new data is attempted, the data is ignored.

The TDR register can always be read / written from the CPU. And when byte access, write TDR[15:8] and then write TDR[7:0].

MPBT bit (Multi-processor transfer bit flag)

Selects the multi processor bit of transmit frame.

TSYNC bit (Transmit SYNC data)

When Manchester mode and MCR.SBSEL = 1 and MCR.SYNSEL = 1, the type of SYNC selected according to the TSYNC bit becomes the Start Bit of the transmission frame.

38.2.4 TSR : Transmit Shift Register

TSR is a shift register that transmits serial data. To perform serial data transmission, the SCI first automatically transfers transmit data from TDR to TSR, then sends the data to the TXDn pin. The CPU cannot directly access the TSR.

38.2.5 CCR0 : Common Control Register 0

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	SSE	—	—	TEIE	TIE	—	—	—	RIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	IDSEL	DCME	MPIE	—	—	—	TE	—	—	—	RE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RE	Receive Enable 0: Serial reception is disabled 1: Serial reception is enabled	R/W ^{*1} *3
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	TE	Transmit Enable 0: Serial transmission is disabled 1: Serial transmission is enabled	R/W ^{*1}

Bit	Symbol	Function	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	MPIE	Multi-Processor Interrupt Enable Valid in Asynchronous mode and Manchester mode when CCR3.MP is 1. This bit should set 0 in Smart card interface mode. 0: Non-multi-processor reception 1: Multi-processor reception When the data with the multi-processor bit set to 0 is received, the data is not read, and setting the status flags to 1 is disabled. When the data with the multi-processor bit set to 1 is received, the MPIE bit is automatically cleared to 0, and non-multi-processor reception is resumed. If you want to continue receiving operation using the multiprocessor function, set this bit to 1 sufficiently earlier than receiving the STOP bit of the next received frame (consider the synchronization delay time).	R/W ²
9	DCME	Data Compare Match Enable Valid only in Asynchronous mode 0: Address match function is disabled 1: Address match function is enabled	R/W ²
10	IDSEL	ID Frame Select Valid only in Asynchronous mode with multi-processor 0: Compare data irrespective of the value of the MPB bit 1: Compare data only when the MPB bit is 1 (ID frame)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
16	RIE	Receive Interrupt Enable 0: SCIn_RXI and SCIn_ERI interrupt requests are disabled 1: SCIn_RXI and SCIn_ERI interrupt requests are enabled	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	TIE	Transmit Interrupt Enable 0: SCIn_TXI interrupt request is disabled 1: SCIn_TXI interrupt request is enabled	R/W
21	TEIE	Transmit End Interrupt Enable This bit should set 0 in Smart card interface mode. 0: SCIn_TEI interrupt request is disabled 1: SCIn_TEI interrupt request is enabled	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
24	SSE	SSn Pin Function Enable Valid in Simple SPI mode. In slave mode (CCR3.CKE[1:0] = 1x), set 1 to this bit. 0: SSn pin function is disabled 1: SSn pin function is enabled	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. In Clock synchronous mode (CCR3.MOD[2:0] = 010b), Simple SPI mode (CCR3.MOD[2:0] = 011b), and Simple IIC mode (CCR3.MOD[2:0] = 100b), 1 can be written only when TE = 0 and RE = 0. After setting TE or RE to 1, only 0 can be written in TE and RE. In other mode, writing is enabled under any condition.

Note 2. This bit is a bit that is cleared by hardware. Note that writing to a bit other than this bit with a bit manipulation instruction may cause this bit to be unintentionally set to 1 by a read-modify-write operation.

Note 3. In Clock synchronous mode or Simple SPI mode, and internal clock (master mode), receive-only setting is prohibited in the internal clock (master mode) (TE = 0 and RE = 1 setting prohibited).

RE bit (Receive Enable)

Enables or disables serial receive operation.

When the RE bit is set to 1, serial reception becomes possible after the synchronization delay time has elapsed in Asynchronous mode or the synchronous clock input in Clock synchronous mode or the neg-edge of RXDn in Manchester mode or start bit in Smart card interface mode.

Note that CCR3 should be set prior to setting the RE bit to 1 in order to designate the reception format.

Except Smart card interface mode, even if reception is halted by setting the RE bit to 0, the CSR.RDRF, FER, PER, ORER, MSR.MER, SBER, SYER, PFER, FRSR. DR flags are not affected, and the previous values is retained. In Smart card interface mode, even if reception is halted by setting the RE bit to 0, the CSR.FER, PER, ORER flags are not affected and

the previous value is retained. Also, to stop the reception operation, synchronization delay time is required from when the RE bit is set to 0 until the reception operation is stopped.

TE bit (Transmit Enable)

Enables or disables serial transmission.

When the TE bit is set to 1, serial transmission becomes possible after the synchronization delay time has elapsed. After the synchronization delay time, transmission is started by writing transmit data to TDR. Note that CCR3 should be set prior to setting the TE bit to 1 in order to designate the transmission format. In addition, the synchronization delay time is required until the transmission control circuit is stopped after the TE bit is set to 0.

MPIE bit (Multi-Processor Interrupt Enable)

When the MPIE bit is set to 1 and data with the multi-processor bit set to 0 is received, the data is not read and setting the status flags (CSR.RDRF, ORER, FER, FRSR, DR, MSR.MER, SYER, PFER, SBER) are disabled.

When data with the multi-processor bit set to 1 is received, the MPIE is automatically cleared to 0, and normal reception is resumed. For details, see [section 38.4. Multi-Processor Communication Function](#). If you want to continue receiving operation using the multi-processor function, set this bit to 1 sufficiently earlier than receiving the STOP bit of the next received frame.

When the receive data includes the MPB bit set to 0, the receive data is not transferred from the RSR to the RDR, a receive error is not detected, and setting the flags ORER, FER, MER, SYER, PFER, and SBER to 1 is disabled.

When the receive data includes the MPB bit set to 1, the MPIE bit is automatically cleared to 0, the SCIn_RXI and SCIn_ERI interrupt requests are enabled (if CCR0.RIE is set to 1), and setting the flags ORER, FER, MER, SYER, PFER, and SBER to 1 is enabled.

MPIE should be set to 0 if multi-processor communications function is not to be used.

DCME bit (Data Compare Match Enable)

The DCME bit selects whether the address match function (data compare match function) is enabled or not.

When DCME is 1, if SCI detects the match to the comparison data (CCR4.CMPD) with receive data, DCME is cleared automatically, and after that, SCI operation mode is in receive mode without data compare match function.

See [section 38.3.6. Address Match \(Receive Data Match Detection\) Function](#).

The write value should be 0 other than Asynchronous mode.

IDSEL bit (ID Frame Select)

The IDSEL bit selects whether to compare data irrespective of the value of MPB bit or to compare data only when the MPB bit = 1 (ID frame) when the address match function is valid. Set at the same time as DCME.

RIE bit (Receive Interrupt Enable)

Enables or disables SCIn_RXI and SCIn_ERI interrupt requests.

SCIn_RXI and SCIn_ERI interrupt request is disabled by setting the RIE bit to 0.

An SCIn_ERI interrupt request can be canceled by reading 1 from the CSR.ORER, FER, or PER and then setting the flag to 0 or setting the RIE bit to 0.

In Manchester mode, the MER, SYER, PFER and SBER flags are also the cause of SCIn_ERI interrupt request, so the same processing is necessary. For details of these flags, see [section 38.2.12. MCR : Manchester Control Register](#) and [section 38.2.21. MSR : Manchester Status Register](#).

TIE bit (Transmit Interrupt Enable)

Enables or disables SCIn_TXI interrupt request.

An SCIn_TXI interrupt request is disabled by setting the TIE bit to 0. At the beginning of transmission, set 1 to CCR0.TE and CCR0.TIE simultaneously. The SCIn_TXI interrupt request is then generated.

TEIE bit (Transmit End Interrupt Enable)

Enables or disables a SCIn_TEI interrupt request. A SCIn_TEI interrupt request is disabled by setting the TEIE bit to 0.

In simple IIC mode, the SCIn_TEI is allocated to the interrupt on completion of issuing a start, restart, or stop condition (STIn). In this case, the TEIE bit can be used to enable or disable the STIn.

SSE bit (SSn Pin Function Enable)

Set the SSE bit to 1 if the SSn pin is to be used in control of transmission and reception (in simple SPI mode).

Set the SSE bit to 0 in any other mode. Do not set both the SSE and CTSE bits to enabled (even if this setting is made, operation is the same as that when these bits are set to 0).

In the slave mode (CCR3.CKE[1:0] = 10 or 11), SSE should be set 1.

In the master mode (CCR3.CKE[1:0] = 00 or 01) and single-master, the SSn pin on the master side is not required to control reception and transmission, so SSE should be set 0.

38.2.6 CCR1 : Common Control Register 1

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x0C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	NFM	NFEN	—	NFCS[2:0]			—	—	—	SHAR PS	—	—	—	SPLP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RINV	TINV	—	—	PM	PE	—	—	SPB2I O	SPB2 DT	—	—	CTSP EN	CTSE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
0	CTSE	CTS Enable 0: CTS function is disabled (RTS output function is enabled) 1: CTS function is enabled	R/W
1	CTSPEN	CTS External Pin Enable 0: Alternate setting to use CTS and RTS functions as either 1 pin 1: Dedicated setting for separately using CTS and RTS functions with 2 pins	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	SPB2DT	Serial Port Break Data Select The output level of TXDn pin is selected when CCR0.TE = 0 and SPB2IO = 1.*1 0: When TINV is 0, low level is output in TXDn pin. When TINV is 1, high level is output in TXDn pin. 1: When TINV is 0, high level is output in TXDn pin. When TINV is 1, low level is output in TXDn pin.	R/W
5	SPB2IO	Serial Port Break I/O This bit selects whether the value of SPB2DT is output to TXDn pin when CCR0.TE = 0.*1 0: The value of SPB2DT bit is not output in TXDn pin 1: The value of SPB2DT bit is output in TXDn pin	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	PE	Parity Enable Valid only in Asynchronous mode and Manchester mode. In Smart Card Interface mode, set 1 to this bit. 0: When transmitting: Do not add parity bit When receiving: Do not check parity bit 1: When transmitting: Add parity bit When receiving: Check parity bit	R/W
9	PM	Parity Mode Valid only when the PE bit is 1 0: Selects even parity 1: Selects odd parity	R/W

Bit	Symbol	Function	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
12	TINV	TXD Invert 0: Transmit data is not inverted and output to TXDn*2 1: Transmit data is inverted and output to TXDn	R/W
13	RINV	RXD Invert 0: Received data from RXDn is not inverted and input*2 1: Received data from RXDn is inverted and input	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
16	SPLP	Loopback Control This bit can be used for internal clock operation in Asynchronous mode, internal mode operation in Manchester mode, internal clock operation in Clock synchronous mode. 0: Normal mode 1: Loopback mode	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	SHARPS	Half-Duplex Communication Select In Simple IIC mode, Smart Card Interface mode, or Simple SPI mode, this bit should be set to 0. 0: TXDn pin, RXDn pin independent 1: TXDn/RXDn pin combination use (half-duplex communication using TXDn pin)	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
26:24	NFCS[2:0]	Noise Filter Clock Select Valid in Asynchronous mode and Manchester mode, Simple LIN mode, and Simple IIC mode. In Simple IIC mode, 000 setting is prohibited. The on-chip baud rate generator source clock means the clock selected by CCR2.CKS [1:0]. Select for the clock source of the noise filter. 0 0 0: The base clock signal divided by 1 0 0 1: The on-chip baud rate generator source clock divided by 1 0 1 0: The on-chip baud rate generator source clock divided by 2 0 1 1: The on-chip baud rate generator source clock divided by 4 1 0 0: The on-chip baud rate generator source clock divided by 8 others: Setting prohibited	R/W
27	—	These bits are read as 0. The write value should be 0.	R/W
28	NFEN	Digital Noise Filter Function Enable Valid in Asynchronous mode, Manchester mode, Simple LIN mode, and Simple IIC mode 0: In Asynchronous, Manchester, Simple LIN modes: Disable noise cancellation function for RXDn input signal In Simple IIC mode: Disable noise cancellation function for SCLn and SDAn input signals 1: In Asynchronous, Manchester, Simple LIN modes: Enable noise cancellation function for RXDn input signal In Simple IIC mode: Enable noise cancellation function for SCLn and SDAn input signals	R/W
29	NFM	Noise Filter Mode Valid in asynchronous mode and ABCSE2 = 1 0: 3-point matching mode 1: Majority vote mode	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Use this bit in Asynchronous mode and Manchester mode only. Operation in other mode is not guaranteed.

Note 2. RINV/TINV should be set to 0 in Smart card interface mode and Simple IIC mode.

CTSE bit (CTS Enable)

Set the CTSE bit to 1 if the SSn pin is to be used for inputting of the CTSn control signal to control of transmission and reception. The RTSn signal is output when this bit is set to 0. Set this bit to 0 in smart card interface mode, simple SPI mode, Simple LIN mode, and simple IIC mode. Do not set both the CTSE and SSE bits to enabled (even if this setting is made, operation is the same as that when these bits are set to 0).

CTSPEN bit (CTS External Pin Enable)

When CTSE is 1, select the pins usage method when using the CTS and RTS functions. Set this bit to 1 when assigning the CTS/RTS function to 2 pins and using them at the same time. Set it to 0 except in Asynchronous and Manchester modes.

Table 38.4 shows the relationship between the CTSE bit and CTSPEN bit settings and the functions of the CTSn_RTSn pin and CTSn pin.

Table 38.4 CTSE bit and CTSPEN bit settings and pin functions

CTSE bit	CTSPEN bit	CTS _n _RTS _n pin	CTS _n pin
0	0	RTS _n signal output	Not use
1	0	CTS _n signal input	Not use
1	1	RTS _n signal output	CTS _n signal input

Note: Set CTSPEN bit = 0 when CTSE bit = 0.

SPB2DT bit (Serial Port Break Data Select), SPB2IO bit (Serial Port Break I/O)

The TXD_n pin status determined by combination of CCR0.TE bit, CCR1.SPB2IO bit, and CCR1.SPB2DT bit is indicated in Table 38.5.

Table 38.5 TXD_n pin status

Value of CCR0.TE	Value of CCR1.SPB2IO	Value of CCR1.SPB2DT	TXD _n pin status (when TINV is 0)
0	0	—	Hi-Z (initial value)
0	1	0	Low level output
0	1	1	High level output
1	—	—	Serial transmission data is output

Note: —: Don't care

PE bit (Parity Enable)

When PE bit to 1, the parity bit is added to transmit data before transmission, and the parity bit is checked in reception.

In the multi-processor format, the parity bit is not added or checked regardless of the PE bit setting.

PM bit (Parity Mode)

Selects the parity mode for transmission and reception (even or odd). In Multi-processor mode, the PM bit is invalid.

For details on the usage of this bit in Smart card interface mode, see [section 38.7.2. Data Format \(Except in Block Transfer Mode\)](#).

TINV bit (TXD Invert), RINV bit (RXD Invert)

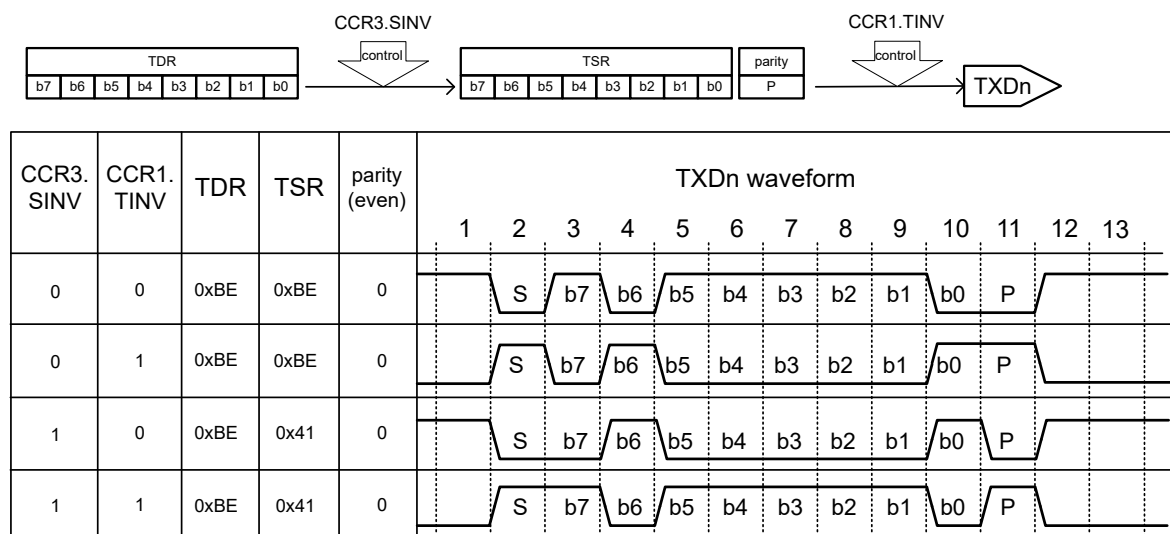
The data of RDR is controlled by RINV and CCR3.SINV. And the data from TXD_n pin is controlled by TINV and CCR3.SINV. The control by RINV/TINV are done to communication pins (RXD_n / TXD_n), so they can control not only data-bits but also other bits (start bit, stop bit, parity bit). For details, see [Figure 38.3](#).

During half-duplex communication and slave operation in Simple SPI mode, use the TXD_n pin for reception, so set the inversion control of the received data with the TINV bit.

Note: Description and a timing chart of the IP operation are provided when TINV = 0 and RINV = 0 (when the value of TINV and the value of RINV are not specified).

The receive/transmit data control (Data size = 8bits, Even parity, MSB first)

The transmit data is controlled by CCR1.TINV and CCR3.SINV.



The received data is controlled by CCR1.RINV and CCR3.SINV.

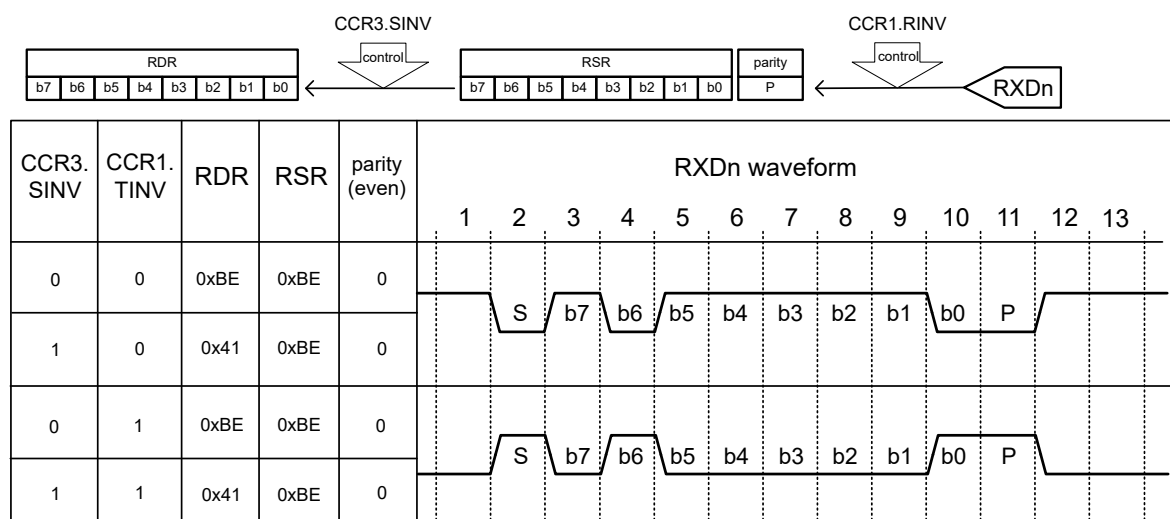


Figure 38.3 Example of receive or transmit data control

SPLP bit (Loopback Control)

When the SPLP bit is 1, SCI blocks the input path from RXDn and connects the output path to TXDn to the reception data register.

Transmit data can be inverted and received by combining it with TINV bit.

Set this bit to 0 for slave operation in Clock synchronous mode, when using an external clock in Asynchronous mode, and when in Simple LIN mode.

SHARPS bit (Half-Duplex Communication Select)

Setting the SHARPS bit to 1 enables half-duplex communication using the TXDn pin. However, the SHARPS bit cannot be used in Simple SPI mode, Simple IIC mode and Smart card interface mode.

If the SHARPS bit is set to 1 and CCR0.TE = 1, CCR0.RE = 0, the TXDn pin becomes the communication output. If this bit is set to 1 and CCR0.TE = 0, CCR0.RE = 1, the TXDn pin becomes the communication input. For details, see [section 38.17. Half-Duplex Communication Function](#).

NFCS[2:0] bits (Noise Filter Clock Select)

The NFCS[2:0] bits select the sampling clock for the digital noise filter.

To use the noise filter in Asynchronous mode, Manchester mode, and Simple LIN mode set the NFCS[2:0] bits from 000b to 100b. In Simple IIC mode, set the bits to a value in the range from 001b to 100b. In particular, if the ABCSE or ABCSE2 bit is set to 1 in asynchronous mode, set it to 000b or 001b.

NFEN bit (Digital Noise Filter Function Enable)

The NFEN bit enables or disables the digital noise filter function. When the function is enabled, noise cancellation is applied to the RXDn input signal in Asynchronous mode, Manchester mode, Simple LIN mode, and noise cancellation is applied to the SDAn and SCLn input signals in Simple IIC mode. In any mode other than above, set the NFEN bit to 0 to disable the digital noise filter function. When the function is disabled, input signals are transferred as is, as internal signals.

NFM bit (Noise Filter Mode)

Selects the operation mode of the digital noise filter function. See xxx. Noise Cancellation Function for details.

38.2.7 CCR2 : Common Control Register 2

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MDDR[7:0]								—	—	CKS[1:0]	—	—	—	—	BRME
Value after reset:	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BRR[7:0]								ABCS E2	ABCS E	ABCS	BGDM	—	BCP[2:0]		
Value after reset:	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
2:0	BCP[2:0]	Base Clock Pulse Selects the number of base clock cycles in Smart card interface mode. 0 0 0: 93 clock cycles (S = 93)*1 0 0 1: 128 clock cycles (S = 128)*1 0 1 0: 186 clock cycles (S = 186)*1 0 1 1: 512 clock cycles (S = 512)*1 1 0 0: 32 clock cycles (S = 32)*1 (Initial value) 1 0 1: 64 clock cycles (S = 64)*1 1 1 0: 372 clock cycles (S = 372)*1 1 1 1: 256 clock cycles (S = 256)*1	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	BGDM	Baud Rate Generator Double-Speed Mode Select Valid in Asynchronous/Manchester/Clock-synchronous/Simple SPI mode and when CCR3.CKE[1] = 0. 0: Baud rate generator outputs the clock with single frequency 1: Baud rate generator outputs the clock with doubled frequency	R/W
5	ABCS	Asynchronous Mode Base Clock Select Valid only in Asynchronous mode, Manchester mode, and Simple LIN mode 0: Selects 16 base clock cycles for 1-bit period 1: Selects 8 base clock cycles for 1-bit period	R/W
6	ABCSE	Asynchronous Mode Extended Base Clock Select Valid only in Asynchronous mode and when CCR3.CKE[1] = 0 0: Clock cycles for 1-bit period is determined by combination between CCR2.BGDM and CCR2.ABCS. 1: Baud rate is 6 base clock cycles for 1-bit period and the clock of a double frequency is output from the baud rate generator.	R/W

Bit	Symbol	Function	R/W
7	ABCSE2	Asynchronous Mode Extended Base Clock Select 2 Valid only in asynchronous mode and CCR3.CKE[1] = 0 0: Clock cycles for 1-bit period is decided with combination between CCR2.BGDM and CCR2.ABCS. 1: Baud rate is 4 base clock cycles for 1-bit period and the clock of a double frequency is output from the baud rate generator.	R/W
15:8	BRR[7:0]	Bit Rate Setting BRR is an 8-bit register that adjusts the bit rate.	R/W
16	BRME	Bit Rate Modulation Enable 0: Bit rate modulation function is disabled 1: Bit rate modulation function is enabled	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
21:20	CKS[1:0]	Clock Select 0 0: TCLK clock ($n = 0$)* ² 0 1: TCLK/4 clock ($n = 1$)* ² 1 0: TCLK/16 clock ($n = 2$)* ² 1 1: TCLK/64 clock ($n = 3$)* ²	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
31:24	MDDR[7:0]	Modulation Duty Setting MDDR corrects the bit rate adjusted by the BRR[7:0] bits.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. S is the value of S in BRR[7:0] bits explanation.

Note 2. n is the decimal notation of the value of n in BRR[7:0] bits explanation.

BCP[2:0] bits (Base Clock Pulse)

The BCP[2:0] bits select the number of base clock cycles in a 1-bit data transfer time in smart card interface mode.

For details, see [section 38.7.4. Receive Data Sampling Timing and Reception Margin](#).

BGDM bit (Baud Rate Generator Double-Speed Mode Select)

The BGDM bit is valid when the on-chip baud rate generator is selected as the clock source (CCR3.CKE[1] = 0) in Asynchronous mode, Manchester mode, Clock synchronous mode, Simple SPI mode. When external clock is selected (CCR3.CKE[1] = 1), set it to 0. For the clock output from the baud rate generator, either single or doubled frequency can be selected. The base clock is generated by the clock output from the baud rate generator. When the BGDM bit is set to 1, the base clock cycle is halved, and the bit rate is doubled.

Set this bit to 0 in modes other than Asynchronous mode, Manchester mode, Clock synchronous mode, or Simple SPI mode.

ABCS bit (Asynchronous Mode Base Clock Select)

Selects the clock cycles for 1-bit period.

Set the ABCS bit to 0 in modes other than Asynchronous mode, Manchester mode, or Simple LIN mode.

ABCSE bit (Asynchronous Mode Extended Base Clock Select)

The pulse number for a base clock at 1-bit period is 6 and the clock of a double frequency is output from baud rate generator. Only when the bit rate is set to 6 dividing frequency of the bus clock can this bit be used to set CCR2.CKS [1:0] = 00b and BRR[7:0] = 0x00.

Set it to 0 in modes other than Asynchronous mode. Even in Asynchronous mode, set it to 0 when using external clock.

ABCSE2 bit (Asynchronous Mode Extended Base Clock Select 2)

The pulse number for a base clock at 1-bit period is 4 and the clock of a double frequency is output from baud rate generator. Only when the bit rate is set to 4 dividing frequency of the bus clock, please use this bit and set CCR2.CKS [1:0] = 00 and BRR[7:0] = 0.

Set it to 0 in modes other than asynchronous mode. Even in asynchronous mode, set it to 0 when using external clock.

Table 38.6 Base clock cycle number per 1-bit

ABCSE2	ABCSE	ABCS	BGDM	The base clock cycles /1bit	The frequency of the baud rate generator
0	0	0	0	16	×1
0	0	0	1	16	×2
0	0	1	0	8	×1
0	0	1	1	8	×2
0	1	— (don't care)	— (don't care)	6	×2
1	0	— (don't care)	— (don't care)	4	×2

BRR[7:0] bit (Bit Rate Setting)

BRR is an 8-bit register that adjusts the bit rate.

SCI has independent baud rate generator control, different bit rates can be set for each. [Table 38.7](#) shows the relationship between the setting (N) in the BRR and the bit rate (B) for Asynchronous mode, multiprocessor transfer, Manchester mode, clock synchronous mode, smart card interface mode, simple SPI mode, and simple IIC mode.

Table 38.7 Relationship between N Setting in BRR and Bit Rate B

Mode	CCR2 settings				BRR[7:0] setting	Error*4
	BGDM bit	ABCS bit	ABCSE bit	ABCSE 2 bit		
Asynchronous, multi-processor, Manchester*2, Simple-LIN*3	0	0	0	0	$N = \frac{TCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$	$Error (\%) = \left\{ \frac{TCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	1	0	0	0	$N = \frac{TCLK \times 10^6}{32 \times 2^{2n-1} \times B} - 1$	$Error (\%) = \left\{ \frac{TCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	0	1	0	0		
	1	1	0	0	$N = \frac{TCLK \times 10^6}{16 \times 2^{2n-1} \times B} - 1$	$Error (\%) = \left\{ \frac{TCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	Don't care	Don't care	1	0	$N = \frac{TCLK \times 10^6}{12 \times 2^{2n-1} \times B} - 1$	$Error (\%) = \left\{ \frac{TCLK \times 10^6}{B \times 12 \times 2^{2n-1} \times (N+1)} - 1 \right\} \times 100$
	Don't care	Don't care	0	1	$N = 0$	—
Clock synchronous, simple SPI	0	0 (Initial value)	0 (Initial value)	0 (Initial value)	$N = \frac{TCLK \times 10^6}{8 \times 2^{2n-1} \times B} - 1$	—
	1	0 (Initial value)	0 (Initial value)	0 (Initial value)	$N = \frac{TCLK \times 10^6}{4 \times 2^{2n-1} \times B} - 1$	—
Smart card interface					$N = \frac{TCLK \times 10^6}{S \times 2^{2n+1} \times B} - 1$	$Error (\%) = \left\{ \frac{TCLK \times 10^6}{B \times S \times 2^{2n+1} \times (N+1)} - 1 \right\} \times 100$
Simple IIC*1					$N = \frac{TCLK \times 10^6}{64 \times 2^{2n-1} \times B} - 1$	—

Note: B: Bit rate (bps)

N: BRR setting for on-chip baud rate generator ($0 \leq N \leq 255$)

TCLK: Operating frequency (MHz)

n and S: Determined by the settings of the CCR2 registers as listed in [Table 38.9](#) and [Table 38.10](#). Please be careful about $2^{(2n+1)}$ is used in the expression for Smart card interface, $2^{(2n-1)}$ is used in other mode.

Note 1. Adjust the bit rate so that the widths at high and low level of the SCL output in simple IIC mode satisfy the IIC standard.

Note 2. In Manchester mode, only ABCSE = 0 and ABCSE2 = 0 can be selected.

Note 3. In Simple-LIN mode, BGDM = 0, ABCSE = 0 and ABCSE2 = 0 can be selected.

Note 4. Error means that if the N value calculated by the above formula is not an integer for the baud rate value B to be set, the BRR is set to an approximate integer value in which case it represents the error between the baud rate value and B.

Table 38.8 Calculating widths at high and low Level for SCL

Mode	SCLn	Formula (result in seconds)
IIC	Width at high level (minimum value)	$(N + 1) \times 4 \times 2^{2n-1} \times 7 \times \frac{1}{TCLK \times 10^6}$
	Width at low level (minimum value)	$(N + 1) \times 4 \times 2^{2n-1} \times 8 \times \frac{1}{TCLK \times 10^6}$

Table 38.9 Clock source settings

CCR2 setting	Clock source	n
CKS[1:0] bit		
00	TCLK clock	0
01	TCLK/4 clock	1
10	TCLK/16 clock	2
11	TCLK/64 clock	3

Table 38.10 Base clock settings in Smart card interface mode

CCR2 setting	Base clock cycles for 1-bit period	S
BCP[2:0] setting		
0 0 0	93 clock cycles	93
0 0 1	128 clock cycles	128
0 1 0	186 clock cycles	186
0 1 1	512 clock cycles	512
1 0 0	32 clock cycles	32
1 0 1	64 clock cycles	64
1 1 0	372 clock cycles	372
1 1 1	256 clock cycles	256

Table 38.11 and Table 38.12 list examples of N settings in BRR in Asynchronous mode and Manchester mode. Table 38.13 lists the maximum bit rate settable for each operating frequency. Examples of BRR (N) settings in Clock synchronous mode and Simple SPI mode are listed in Table 38.16. Examples of BRR (N) settings in Smart card interface mode are listed in Table 38.18. Examples of BRR (N) settings in Simple IIC mode are listed in Table 38.20. In Smart card interface mode, the number of base clock cycles S in a 1-bit data transfer time can be selected. For details, see section 38.7.4. [Receive Data Sampling Timing and Reception Margin](#). Table 38.14 and Table 38.17 list the maximum bit rates with external clock input. Table 38.15 lists the maximum bit rates with GPT clock input.

When either the Asynchronous mode base clock select bit (ABCS) or the baud rate generator double-speed mode select bit (BGDM) is set to 1 in Asynchronous mode and Manchester mode, the bit rate becomes twice that listed in Table 38.11 and Table 38.12. When both of those registers are set to 1, the bit rate becomes four times the listed value.

Table 38.11 Examples of BRR Settings for various Bit Rates (Asynchronous Mode and Manchester Mode)
(1) (1 of 3)

Bit rate (bps)	Operating Frequency TCLK (MHz)														
	8			9.8304			10			12			12.288		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	2	141	0.03	2	174	-0.26	2	177	-0.25	2	212	0.03	2	217	0.08
150	2	103	0.16	2	127	0	2	129	0.16	2	155	0.16	2	159	0
300	1	207	0.16	1	255	0	2	64	0.16	2	77	0.16	2	79	0
600	1	103	0.16	1	127	0	1	129	0.16	1	155	0.16	1	159	0

**Table 38.11 Examples of BRR Settings for various Bit Rates (Asynchronous Mode and Manchester Mode)
(1) (2 of 3)**

Bit rate (bps)	Operating Frequency TCLK (MHz)														
	8			9.8304			10			12			12.288		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
1200	0	207	0.16	0	255	0	1	64	0.16	1	77	0.16	1	79	0
2400	0	103	0.16	0	127	0	0	129	0.16	0	155	0.16	0	159	0
4800	0	51	0.16	0	63	0	0	64	0.16	0	77	0.16	0	79	0
9600	0	25	0.16	0	31	0	0	32	-1.36	0	38	0.16	0	39	0
19200	0	12	0.16	0	15	0	0	15	1.73	0	19	-2.34	0	19	0
31250	0	7	0	0	9	-1.7	0	9	0	0	11	0	0	11	2.4
38400	—	—	—	0	7	0	0	7	1.73	0	9	-2.34	0	9	0

**Table 38.11 Examples of BRR Settings for various Bit Rates (Asynchronous Mode and Manchester Mode)
(1) (3 of 3)**

Bit rate (bps)	Operating Frequency TCLK (MHz)														
	14			16			17.2032			18			19.6608		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	2	248	-0.17	3	70	0.03	3	75	0.48	3	79	-0.12	3	86	0.31
150	2	181	0.16	2	207	0.16	2	223	0	2	233	0.16	2	255	0
300	2	90	0.16	2	103	0.16	2	111	0	2	116	0.16	2	127	0
600	1	181	0.16	1	207	0.16	1	223	0	1	233	0.16	1	255	0
1200	1	90	0.16	1	103	0.16	1	111	0	1	116	0.16	1	127	0
2400	0	181	0.16	0	207	0.16	0	223	0	0	233	0.16	0	255	0
4800	0	90	0.16	0	103	0.16	0	111	0	0	116	0.16	0	127	0
9600	0	45	-0.93	0	51	0.16	0	55	0	0	58	-0.69	0	63	0
19200	0	22	-0.93	0	25	0.16	0	27	0	0	28	1.02	0	31	0
31250	0	13	0	0	15	0	0	16	1.2	0	17	0	0	19	-1.7
38400	—	—	—	0	12	0.16	0	13	0	0	14	-2.34	0	15	0

Note: This is an example when the CCR2.ABCS = 0, CCR2.BGDM = 0 and CCR2.ABCSE = 0.

When either ABCS bit or BGDM bit is set to 1, the bit rate doubles.

When both ABCS = 1 and BGDM = 1, the bit rate increases four times.

**Table 38.12 Examples of BRR Settings for Various Bit Rates (Asynchronous Mode and Manchester Mode)
(2) (1 of 3)**

Bit rate (bps)	Operating frequency TCLK (MHz)														
	20			25			30			33			40		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	3	88	-0.25	3	110	-0.02	3	132	0.13	3	145	0.33	3	177	-0.25
150	3	64	0.16	3	80	0.47	3	97	-0.35	3	106	0.39	3	129	0.16
300	2	129	0.16	2	162	-0.15	2	194	0.16	2	214	-0.07	3	64	0.16
600	2	64	0.16	2	80	0.47	2	97	-0.35	2	106	0.39	2	129	0.16
1200	1	129	0.16	1	162	-0.15	1	194	0.16	1	214	-0.07	2	64	0.16
2400	1	64	0.16	1	80	0.47	1	97	-0.35	1	106	0.39	1	129	0.16
4800	0	129	0.16	0	162	-0.15	0	194	0.16	0	214	-0.07	1	64	0.16
9600	0	64	0.16	0	80	0.47	0	97	-0.35	0	106	0.39	0	129	0.16
19200	0	32	-1.36	0	40	-0.76	0	48	-0.35	0	53	-0.54	0	64	0.16

**Table 38.12 Examples of BRR Settings for Various Bit Rates (Asynchronous Mode and Manchester Mode)
(2) (2 of 3)**

Bit rate (bps)	Operating frequency TCLK (MHz)														
	20			25			30			33			40		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
31250	0	19	0	0	24	0	0	29	0	0	32	0	0	39	0
38400	0	15	1.73	0	19	1.73	0	23	1.73	0	26	−0.54	0	32	−1.36

**Table 38.12 Examples of BRR Settings for Various Bit Rates (Asynchronous Mode and Manchester Mode)
(2) (3 of 3)**

Bit rate (bps)	Operating frequency TCLK (MHz)														
	50			60			100			120			125		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
110	3	221	-0.02	—	—	—	—	—	—	—	—	—	—	—	—
150	3	162	-0.15	3	194	0.16	—	—	—	—	—	—	—	—	—
300	3	80	0.47	3	97	-0.35	3	162	-0.15	3	194	0.16	3	202	0.22
600	2	162	-0.15	2	194	0.16	3	80	0.47	3	97	-0.35	3	101	-0.27
1200	2	80	0.47	2	97	-0.35	2	162	-0.15	2	194	0.16	2	202	0.22
2400	1	162	-0.15	1	194	0.16	2	80	0.47	2	97	-0.35	2	101	-0.27
4800	1	80	0.47	1	97	-0.35	1	162	-0.15	1	194	0.16	1	202	0.22
9600	0	162	-0.15	0	194	0.16	1	80	0.47	1	97	-0.35	1	101	-0.27
19200	0	80	0.47	0	97	-0.35	0	162	-0.15	0	194	0.16	0	202	0.22
31250	0	49	0	0	59	0	1	24	0	0	119	0	0	124	0.00
38400	0	40	-0.76	0	48	-0.35	0	80	0.47	0	97	-0.35	0	101	-0.27

Note: This is an example when the CCR2.ABCS = 0, CCR2.BGDM = 0, CCR2.ABCSE = 0 and CCR2.ABCSE2 = 0.

When either ABCS bit or BGDM bit is set to 1, the bit rate doubles.

When both ABCS = 1 and BGDM = 1, the bit rate increases four times.

Table 38.13 Maximum Bit Rate for Each Operating Frequency (Asynchronous Mode and Manchester mode) (1 of 3)

TCLK (MHz)	CCR2 settings				Maximum bit rate (bps)	TCLK (MHz)	CCR2 settings				Maximum bit rate (bps)
	BGDM	ABCS	ABCSE	ABCSE2			BGDM	ABCS	ABCSE	ABCSE2	
8	0	0	0	0	250000	16	0	0	0	0	500000
		1	0	0	500000			1	0	0	1000000
	1	0	0	0			1	0	0	0	
		1	0	0	1000000			1	0	0	2000000
	Don't care	Don't care	1	0	1333333		Don't care	Don't care	1	0	2666666
	Don't care	Don't care	0	1	2000000		Don't care	Don't care	0	1	4000000
9.8304	0	0	0	0	307200	17.2032	0	0	0	0	537600
		1	0	0	614400			1	0	0	1075200
	1	0	0	0			1	0	0	0	
		1	0	0	1228800			1	0	0	2150400
	Don't care	Don't care	1	0	1638400		Don't care	Don't care	1	0	2867200
	Don't care	Don't care	0	1	2457600		Don't care	Don't care	0	1	4300800

Table 38.13 Maximum Bit Rate for Each Operating Frequency (Asynchronous Mode and Manchester mode) (2 of 3)

TCLK (MHz)	CCR2 settings				Maximum bit rate (bps)	TCLK (MHz)	CCR2 settings				Maximum bit rate (bps)
	BGDM	ABCS	ABCSE	ABCSE2			BGDM	ABCS	ABCSE	ABCSE2	
10	0	0	0	0	312500	18	0	0	0	0	562500
		1	0	0	625000			1	0	0	1125000
	1	0	0	0	1250000		1	0	0	0	2250000
		1	0	0				1	0	0	
	Don't care	Don't care	1	0	1666666		Don't care	Don't care	1	0	3000000
	Don't care	Don't care	0	1	2500000		Don't care	Don't care	0	1	4500000
12	0	0	0	0	375000	19.6608	0	0	0	0	614400
		1	0	0	750000			1	0	0	1228800
	1	0	0	0	1500000		1	0	0	0	2457600
		1	0	0				1	0	0	
	Don't care	Don't care	1	0	2000000		Don't care	Don't care	1	0	3276800
	Don't care	Don't care	0	1	3000000		Don't care	Don't care	0	1	4915200
12.288	0	0	0	0	384000	20	0	0	0	0	625000
		1	0	0	768000			1	0	0	1250000
	1	0	0	0	1536000		1	0	0	0	2500000
		1	0	0				1	0	0	
	Don't care	Don't care	1	0	2048000		Don't care	Don't care	1	0	3333333
	Don't care	Don't care	0	1	3072000		Don't care	Don't care	0	1	5000000
14	0	0	0	0	437500	25	0	0	0	0	781250
		1	0	0	875000			1	0	0	1562500
	1	0	0	0	1750000		1	0	0	0	3125000
		1	0	0				1	0	0	
	Don't care	Don't care	1	0	2333333		Don't care	Don't care	1	0	4166666
	Don't care	Don't care	0	1	3500000		Don't care	Don't care	0	1	6250000
30	0	0	0	0	937500	50	0	0	0	0	1562500
		1	0	0	1875000			1	0	0	3125000
	1	0	0	0	3750000		1	0	0	0	6250000
		1	0	0				1	0	0	
	Don't care	Don't care	1	0	5000000		Don't care	Don't care	1	0	8333333
	Don't care	Don't care	0	1	7500000		Don't care	Don't care	0	1	12500000

Table 38.13 Maximum Bit Rate for Each Operating Frequency (Asynchronous Mode and Manchester mode) (3 of 3)

TCLK (MHz)	CCR2 settings				Maximum bit rate (bps)	TCLK (MHz)	CCR2 settings				Maximum bit rate (bps)
	BGDM	ABCS	ABCSE	ABCSE2			BGDM	ABCS	ABCSE	ABCSE2	
33	0	0	0	0	1031250	60	0	0	0	0	1875000
		1	0	0	2062500			1	0	0	3750000
	1	0	0	0			1	0	0	0	
		1	0	0	4125000			1	0	0	7500000
	Don't care	Don't care	1	0	5500000		Don't care	Don't care	1	0	10000000
	Don't care	Don't care	0	1	8250000		Don't care	Don't care	0	1	15000000
40	0	0	0	0	1250000	120	0	0	0	0	3750000
		1	0	0	2500000			1	0	0	7500000
	1	0	0	0			1	0	0	0	
		1	0	0	5000000			1	0	0	15000000
	Don't care	Don't care	1	0	6666666		Don't care	Don't care	1	0	20000000
	Don't care	Don't care	0	1	10000000		Don't care	Don't care	0	1	30000000
						125	0	0	0	0	3906250
								1	0	0	7812500
							1	0	0	0	
								1	0	0	15625000
							Don't care	Don't care	1	0	20833333
							Don't care	Don't care	0	1	31250000

Note: For maximum bit rate, the CKS and BRR register values should be n = 0 and N = 0, respectively. In Manchester mode, use ABCSE = ABCSE2 = 0.

Table 38.14 Maximum Bit Rate with External Clock Input (Asynchronous Mode)

TCLK (MHz)	External clock (MHz)	Maximum bit rate (bps)		TCLK (MHz)	External clock (MHz)	Maximum bit rate (bps)	
		CCR2.ABCS bit = 0	CCR2.ABCS bit = 1			CCR2.ABCS bit = 0	CCR2.ABCS bit = 1
8	2	125000	250000	25	6.25	390625	781250
9.8304	2.4576	153600	307200	30	7.5	468750	937500
10	2.5	156250	312500	33	8.25	515625	1031250
12	3	187500	375000	40	10	625000	1250000
12.288	3.072	192000	384000	50	12.5	781250	1562500
14	3.5	218750	437500	60	15	937500	1875000
16	4	250000	500000	120	30	1875000	3750000
17.2032	4.3008	268800	537600	125	31.25	1953125	3906250
18	4.5	281250	562500				
19.6608	4.9152	307200	614400				
20	5	312500	625000				

Table 38.15 Maximum Bit Rate with GPT Clock Input (Asynchronous Mode)

TCLK (MHz)	GPT clock (MHz)	Maximum bit rate (bps)		TCLK (MHz)	GPT clock (MHz)	Maximum bit rate (bps)	
		CCR2.ABCS bit = 0	CCR2.ABCS bit = 1			CCR2.ABCS bit = 0	CCR2.ABCS bit = 1
8	4	250000	500000	25	12.5	781250	1562500
9.8304	4.9152	307200	614400	30	15	937500	1875000
10	5.0	312500	625000	33	16.5	1031250	2062500
12	6	375000	750000	40	20	1250000	2500000
12.288	6.144	384000	768000	50	25	1562500	3125000
14	7.0	437500	875000	60	30	1875000	3750000
16	8	500000	1000000	120	60	3750000	7500000
17.2032	8.6016	537600	1075200	125	62.5	3906250	7812500
18	9.0	562500	1125000				
19.6608	9.8304	614400	1228800				
20	10	625000	1250000				

Table 38.16 BRR Settings for Various Bit Rates (Clock Synchronous Mode, Simple SPI Mode)

Bit rate (bps)	Operating frequency TCLK (MHz)																	
	8			10			30			60			120			125		
	BGD M	n	N	BGD M	n	N	BGD M	n	N	BGD M	n	N	BGD M	n	N	BGD M	n	N
250	0	3	124	0	3	177	—	—	—	—	—	—	—	—	—	—	—	—
500	0	2	249	0	3	77	0	3	233	—	—	—	—	—	—	—	—	—
1k	0	2	124	0	3	38	0	3	116	0	3	233	—	—	—	—	—	—
2.5k	0	2	49	0	1	249	0	3	46	0	3	93	0	3	187	(0)	(3)	(194)
5k	0	2	24	0	1	124	0	2	93	0	3	46	0	3	93	(1)	(3)	(194)
10k	0	1	49	0	0	249	0	2	46	0	2	93	0	3	46	(0)	(2)	(194)
25k	0	2	4	0	1	24	0	1	74	0	1	149	0	2	74	(0)	(2)	(77)
50k	0	1	9	0	0	49	0	0	149	0	1	74	0	1	149	(0)	(2)	(38)
100k	0	1	4	0	0	24	0	0	74	0	0	149	0	1	74	(0)	(1)	(77)
250k	0	1	1	0	0	9	0	0	29	0	1	14	0	1	29	0	0	124
500k	0	1	0	0	0	4	0	0	14	0	0	29	0	1	14	1	0	124
1M	0	0	1	1	0	4	1	0	14	0	0	14	0	0	29	(1)	(0)	(62)
2.5M	—	—	—	0	0	0	0	0	2	0	0	5	0	1	2	1	0	24
5M	—	—	—	1	0	0	1	0	2	0	0	2	0	0	5	(1)	(0)	(12)
7.5M	—	—	—	—	—	—	0	0	0	0	0	1	1	1	0	(0)	(1)	(0)
60M	—	—	—	—	—	—	—	—	—	—	—	—	1	0	0	(1)	(0)	(0)

Note: —: Can be set, but an error over 10% will occur.

Table 38.17 Maximum Bit Rate with External Clock Input (Clock Synchronous Mode, Simple SPI Mode) (1 of 2)

TCLK (MHz)	External clock (MHz)	MAX Bit rate (Mbps)	TCLK (MHz)	External clock (MHz)	MAX Bit rate (Mbps)
8	4	4	25	12.5	12.5
10	5	5	30	15	15
12	6	6	33	16.5	16.5
14	7	7	40	20	20

Table 38.17 Maximum Bit Rate with External Clock Input (Clock Synchronous Mode, Simple SPI Mode) (2 of 2)

TCLK (MHz)	External clock (MHz)	MAX Bit rate (Mbps)	TCLK (MHz)	External clock (MHz)	MAX Bit rate (Mbps)
16	8	8	50	25	25
18	9	9	60	30	30
20	10	10	120	60	60
			125	62.5	62.5

Table 38.18 BRR Settings for Various Bit Rates (Smart Card Interface Mode, n = 0, S = 372) (1 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	7.1424			10.00			10.7136			13.00		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
9600	0	0	0.00	0	1	-30	0	1	-25	0	1	-8.99

Table 38.18 BRR Settings for Various Bit Rates (Smart Card Interface Mode, n = 0, S = 372) (2 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	14.2848			16.00			18.00			20.00		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
9600	0	1	0.00	0	1	12.01	0	2	-15.99	0	2	-6.66

Table 38.18 BRR Settings for Various Bit Rates (Smart Card Interface Mode, n = 0, S = 372) (3 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	25.00			30.00			33.00			40.00		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
9600	0	3	-12.49	0	3	5.01	0	4	-7.59	0	5	-6.66

Table 38.18 BRR Settings for Various Bit Rates (Smart Card Interface Mode, n = 0, S = 372) (4 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	50.00			60.00			120.00			125		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
9600	0	6	0.01	0	7	5.01	0	16	-1.17	0	17	-2.71

Table 38.19 Maximum Bit Rate for Each Operating Frequency (Smart Card Interface Mode, S = 32)

TCLK (MHz)	MAX Bit rate (bps)	n	N	TCLK (MHz)	MAX Bit rate (bps)	n	N
10	156250	0	0	30	468750	0	0
10.7136	167400	0	0	33	515625	0	0
13	203125	0	0	40	625000	0	0
16	250000	0	0	50	781250	0	0
18	281250	0	0	60	937500	0	0
20	312500	0	0	120	1875000	0	0
25	390625	0	0	125	1953125	0	0

Table 38.20 BRR Settings for Various Bit Rates (Simple IIC Mode) (1 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	8			10			16			20		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
10k	0	24	0	0	31	-2.3	1	12	-3.8	1	15	-2.3
25k	0	9	0	0	12	-3.8	1	4	0	1	6	-10.7
50k	0	4	0	0	6	-10.7	1	2	-16.7	1	3	-21.9
100k	0	2	-16.7	0	3	-21.9	0	4	0	0	6	-10.7
250k	0	0	0	0	1	-37.5	0	1	0	0	2	-16.7
350k										0	1	-10.7
400k										0	1	-21.9

Table 38.20 BRR Settings for Various Bit Rates (Simple IIC Mode) (2 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	25			30			33			40		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
10k	1	19	-2.3	1	23	-2.3	1	25	-0.8	0	124	0
25k	1	7	-2.3	1	9	-6.3	1	10	-6.3	0	40	0
50k	1	3	-2.3	1	4	-6.3	1	5	-14.1	0	24	0
100k	1	1	-2.3	1	2	-21.9	1	2	-14.1	0	12	-3.85
250k	0	3	-21.9	0	3	-6.3	0	4	-17.5	0	4	0
350k	0	2	-25.6	0	2	-10.7	0	2	-1.8	0	3	-10.71
400k	0	1	-2.3	0	1	17.2	0	2	-14.1	0	2	4.17

Table 38.20 BRR Settings for Various Bit Rates (Simple IIC Mode) (3 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	50			60			120			125		
	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)	n	N	Error (%)
10k	2	9	-2.3	1	46	-0.27	1	93	-0.27	1	97	-0.35
25k	2	3	-2.3	0	74	0	0	149	0	1	38	0.16
50k	2	1	-2.3	0	37	-1.32	0	74	0	0	77	0.16
100k	1	3	-2.3	0	18	-1.32	0	37	-1.31	0	38	0.16
250k	0	6	-10.7	0	7	-6.25	0	14	0	2	0	-2.34
350k	0	4	-10.7	0	4	7.14	0	10	-2.6	0	10	1.46
400k	0	3	-2.34	0	4	-6.25	0	8	4.17	0	10	-11.22

Table 38.21 Minimum Widths at High and Low Level for SCL at Various Bit Rates (Simple IIC Mode) (1 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	8			10			16			20		
	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)
10k	0	24	43.75 / 50.00	0	31	44.80 / 51.20	1	12	45.50 / 52.00	1	15	44.80 / 51.20
25k	0	9	17.50 / 20.00	0	12	18.20 / 20.80	1	4	17.50 / 20.00	1	6	19.60 / 22.40
50k	0	4	8.75 / 10.00	0	6	9.80 / 11.20	1	2	10.50 / 12.00	1	3	11.20 / 12.80
100k	0	2	5.25 / 6.00	0	3	5.60 / 6.40	0	4	4.37 / 5.00	0	6	4.90 / 5.60
250k	0	0	1.75 / 2.00	0	1	2.80 / 3.20	0	1	1.75 / 2.00	0	2	2.10 / 2.40
350k										0	1	1.40 / 1.60
400k										0	1	1.40 / 1.60

Table 38.21 Minimum Widths at High and Low Level for SCL at Various Bit Rates (Simple IIC Mode) (2 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	25			30			33			40		
	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)
10k	1	19	44.80 / 51.20	1	23	44.80 / 51.20	1	25	44.12 / 50.42	1	32	46.20 / 52.80
25k	1	7	17.92 / 20.48	1	9	18.66 / 21.33	1	10	18.66 / 21.33	1	12	18.20 / 20.80
50k	1	3	8.96 / 10.24	1	4	9.33 / 10.66	1	5	10.18 / 11.63	1	6	9.80 / 11.20
100k	1	1	4.48 / 5.12	1	2	5.60 / 6.40	1	2	5.09 / 5.81	0	13	4.90 / 5.60
250k	0	3	2.24 / 2.56	0	3	1.86 / 2.13	0	4	2.12 / 2.42	0	4	1.75 / 2.00
350k	0	2	1.68 / 1.92	0	2	1.40 / 1.60	0	2	1.27 / 1.45	0	3	1.40 / 1.60
400k	0	1	1.12 / 1.28	0	1	0.93 / 1.07	0	2	1.27 / 1.45	0	2	1.05 / 1.20

Table 38.21 Minimum Widths at High and Low Level for SCL at Various Bit Rates (Simple IIC Mode) (3 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)											
	50			60			120			125		
	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)	n	N	Minimum width of High/Low Level (μs)
10k	2	9	44.80 / 51.20	1	47	44.80 / 51.20	1	93	43.87 / 50.13	1	97	43.90/50.18
25k	2	3	17.92 / 20.48	0	74	17.50 / 20.00	0	149	17.50 / 20.00	1	38	17.47/19.97
50k	2	1	8.96 / 10.24	0	37	8.87 / 10.13	0	74	8.75 / 10.00	0	77	8.74/9.98
100k	1	3	4.48 / 5.12	0	18	4.43 / 5.07	0	37	4.43 / 5.07	0	38	4.37/4.99
250k	0	6	1.96 / 2.24	0	7	1.87 / 2.13	0	15	1.87 / 2.13	2	0	1.79/2.05
350k	0	4	1.40 / 1.60	0	5	1.40 / 1.60	0	10	1.28 / 1.47	0	10	1.23/1.41
400k	0	3	1.12 / 1.28	0	4	1.17 / 1.33	0	9	1.17 / 1.33	0	10	1.23/1.41

BRME bit (Bit Rate Modulation Enable)

Enables and disables the bit rate modulation function. The bit rate generated by on-chip baud rate generator is evenly corrected when this function is enabled.

Set to 0 in Clock-synchronous mode, Simple SPI mode, Smart Card Interface mode, Manchester mode and Simple LIN mode.

Even in asynchronous mode, set this bit to 0 when ABCSE = 1 or ABCSE2 = 1.

CKS[1:0] bits (Clock Select)

The CKS[1:0] bits select the clock source for the on-chip baud rate generator.

For the relationship between the settings of the CKS[1:0] bits and the baud rate, see BRR[7:0] bits explanation.

MDDR[7:0] bits (Modulation Duty Setting)

When the BRME bit is set to 1, the bit rate generated by the on-chip baud rate generator is evenly corrected according to the settings of MDDR (M/256). The relationship between the MDDR setting (M) and the bit rate (B) is given in [Table 38.22](#).

The initial value of MDDR is 0xFF. Bit 7 in the CCR2 is fixed to 1.

Table 38.22 Relationship between MDDR Setting (M) and Bit Rate (B) When Bit Rate Modulation Function is Used

Mode*1	CCR2 settings			BRR setting	Error
	BG DM bit	AB CS bit	AB CS E bit		
Asynchronous, Multiprocessor transfer	0	0	0	$N = \frac{TCLK \times 10^6}{64 \times 2^{2n-1} \times (256/M) \times B} - 1$	$\text{Error (\%)} = \left\{ \frac{TCLK \times 10^6}{B \times 64 \times 2^{2n-1} \times (256/M) \times (N+1)} - 1 \right\} \times 100$
	1	0	0	$N = \frac{TCLK \times 10^6}{32 \times 2^{2n-1} \times (256/M) \times B} - 1$	$\text{Error (\%)} = \left\{ \frac{TCLK \times 10^6}{B \times 32 \times 2^{2n-1} \times (256/M) \times (N+1)} - 1 \right\} \times 100$
	0	1	0	$N = \frac{TCLK \times 10^6}{16 \times 2^{2n-1} \times (256/M) \times B} - 1$	$\text{Error (\%)} = \left\{ \frac{TCLK \times 10^6}{B \times 16 \times 2^{2n-1} \times (256/M) \times (N+1)} - 1 \right\} \times 100$
	1	1	0	$N = \frac{TCLK \times 10^6}{12 \times 2^{2n-1} \times (256/M) \times B} - 1$	$\text{Error (\%)} = \left\{ \frac{TCLK \times 10^6}{B \times 12 \times 2^{2n-1} \times (256/M) \times (N+1)} - 1 \right\} \times 100$
Simple IIC*2	x (Arbitrarily)	x (Arbitrarily)	1*2	$N = \frac{TCLK \times 10^6}{64 \times 2^{2n-1} \times (256/M) \times B} - 1$	—

Note: B: Bit rate (bps)
M: MDDR setting ($128 \leq M \leq 255$)
N: BRR setting for bound rate generator ($0 \leq N \leq 255$)
TCLK: Operating frequency (MHz)
n: Determined by the settings of the CKS[1:0] as listed in [Table 38.9](#).

Note 1. Do not use this function in Clock-synchronous mode, Simple SPI mode, Smart card Interface mode, Manchester mode and Simple LIN mode.

Note 2. Adjust the bit rate so that the widths at high and low level of the SCL output in simple IIC mode satisfy the I2C standard.

[Table 38.23](#) and [Table 38.24](#) list examples of N settings in BRR and M settings in MDDR in Asynchronous mode.

Table 38.23 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (1) (1 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)														
	8					9.8304					10				
	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)
38400	0	5	236	0	0.03	0	7	(256)*1	0	0	0	10	173	1	-0.01
57600	0	3	236	0	0.03	0	4	240	0	0	0	4	236	0	0.03

Table 38.23 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (1) (2 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)														
	8					9.8304					10				
	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)
115200	0	1	236	0	0.03	0	1	192	0	0	0	4	236	1	0.03
230400	0	0	236	0	0.03	0	0	192	0	0	0	1	189	1	0.14
460800	0	0	236	1	0.03	0	0	192	1	0	0	0	189	1	0.14

Table 38.23 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (1) (3 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)														
	12					12.288					14				
	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)
38400	0	8	236	0	0.03	0	9	(256) *1	0	0	0	16	191	1	0
57600	0	5	236	0	0.03	0	4	192	0	0	0	13	236	1	0.03
115200	0	2	236	0	0.03	0	4	192	1	0	0	6	236	1	0.03
230400	0	2	236	1	0.03	0	2	230	1	-0.17	0	2	202	1	-0.11
460800	0	0	157	1	-0.18	0	0	154	1	0.26	0	0	135	1	0.14

Table 38.23 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (1) (4 of 4)

Bit rate (bps)	Operating frequency TCLK (MHz)														
	16					17.2032					18				
	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)
38400	0	11	236	0	0.03	0	13	(256) *1	0	0	0	18	166	1	-0.01
57600	0	7	236	0	0.03	0	6	192	0	0	0	18	249	1	-0.01
115200	0	3	236	0	0.03	0	6	192	1	0	0	8	236	1	0.03
230400	0	1	236	0	0.03	0	3	219	1	-0.2	0	1	210	0	0.14
460800	0	1	236	1	0.03	0	1	219	1	-0.2	0	0	210	0	0.14

Note: This is an example when the CCR2.ABCS = 0 and CCR2.ABCSE = 0.

Note 1. It means that the bit rate modulation function is disable. (CCR2.BRME = 0, M = 256)

Table 38.24 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (2) (1 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)														
	19.6608					20					25				
	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)
38400	0	15	(256) *1	0	0	0	10	173	0	-0.01	0	11	151	0	0
57600	0	9	240	0	0	0	9	236	0	0.03	0	7	151	0	0
115200	0	4	240	0	0	0	4	236	0	0.03	0	3	151	0	0
230400	0	1	192	0	0	0	4	236	1	0.03	0	1	151	0	0
460800	0	0	192	0	0	0	0	189	0	0.14	0	0	151	0	0

Table 38.24 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (2) (2 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)														
	30					33					40				
	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)	n	N	M	BGD M bit	Error (%)
38400	0	36	194	1	0.01	0	14	143	0	0.01	0	21	173	0	-0.01
57600	0	10	173	0	-0.01	0	9	143	0	0.01	0	38	230	1	-0.01
115200	0	10	173	1	-0.01	0	4	143	0	0.01	0	9	236	0	0.03
230400	0	6	220	1	-0.09	0	4	143	1	0.01	0	4	236	0	0.03
460800	0	3	252	1	0.14	0	1	229	0	0.1	0	4	236	1	0.03

Table 38.24 Examples of BRR and MDDR Settings for Various Bit Rates (Asynchronous Mode) (2) (3 of 3)

Bit rate (bps)	Operating frequency TCLK (MHz)																			
	50					60					120					125				
	n	N	M	BG DM bit	Error (%)	n	N	M	BG DM bit	Error (%)	n	N	M	BG DM bit	Error (%)	n	N	M	BG DM bit	Error (%)
38400	0	23	151	0	0	0	36	194	0	0.01	0	73	194	0	0.01	0	90	229	0	0.00
57600	0	15	151	0	0	0	21	173	0	-0.01	0	58	232	0	0	0	132	251	1	-0.01
115200	0	7	151	0	0	0	10	173	0	-0.01	0	21	173	0	-0.01	0	19	151	0	0.00
230400	0	3	151	0	0	0	10	173	1	-0.01	0	10	173	0	-0.01	0	9	151	0	0.00
460800	0	1	151	0	0	0	6	220	1	-0.09	0	10	173	1	-0.01	0	4	151	0	0.00

Note: This is an example when the CCR2.ABCS = 0 and CCR2.ABCSE = 0.

Note 1. It means that the bit rate modulation function is disable. (CCR2.BRME = 0, M = 256)

38.2.8 CCR3 : Common Control Register 3

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x14

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	BLK	GM	—	ACS0	CKE[1:0]	—	—	DEN	FM	MP	MOD[2:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RXDE SEL	STP	SINV	LSBF	—	—	CHR[1:0]	BPEN	—	—	—	—	—	—	CPOL	CPHA
Value after reset:	0	0	0	1	0	0	1	0	0	0	0	0	0	0	1	1

Bit	Symbol	Function	R/W
0	CPHA	Clock Phase Select Valid in Clock synchronous mode and Simple SPI mode. Set this bit only when CCR0.TE = 0 and RE = 0. 0: Data is sampled at an odd edge and changes at an even edge (clock is delayed) 1: Data changes at an odd edge and is sampled at an even edge (clock is not delayed)	R/W

Bit	Symbol	Function	R/W
1	CPOL	Clock Polarity Select Valid in Clock synchronous mode and Simple SPI mode. Set this bit only when CCR0.TE = 0 and RE = 0. 0: SCKn in idle state is 0 1: SCKn in idle state is 1	R/W
6:2	—	These bits are read as 0. The write value should be 0.	R/W
7	BPEN	Synchronizer Bypass Enable This bit controls whether to bypass the synchronizer circuit between the bus clock and operation clock. 0: Synchronizer circuit is not bypassed 1: Synchronizer circuit is bypassed	R/W
9:8	CHR[1:0]	Character Length Valid in Asynchronous mode and Manchester mode* ¹ Select the data length for transmission and reception. 0 0: Transmit/receive in 9-bit data length 0 1: Transmit/receive in 9-bit data length 1 0: Transmit/receive in 8-bit data length (initial value) 1 1: Transmit/receive in 7-bit data length* ²	R/W
11:10	—	These bits are read as 0. The write value should be 0.	R/W
12	LSBF	LSB First select Set this bit to 0 in Simple IIC mode. Set this bit to 1 in Simple LIN mode. 0: MSB-first 1: LSB-first	R/W
13	SINV	Transmitted/Received Data Invert Set this bit to 0 in Simple IIC mode. The level of communication pins (RXDn/TXDn) are controlled by combination of this bit and CCR1.TINV/RINV. For details, see Figure 38.3 . 0: TDR contents are transmitted to TSR as is. RSR contents are stored to RDR as is. 1: TDR contents are inverted before being transmitted to TSR. RSR contents are inverted and stored to RDR.	R/W
14	STP	Stop Bit Length Valid in Asynchronous mode, Manchester mode, and Simple LIN mode 0: 1 stop bit/break delimiter length is 1-bit 1: 2 stop bits/break delimiter length is 2-bit	R/W
15	RXDESEL	Asynchronous Start Bit Edge Detection Select Valid only in Asynchronous mode Set this bit to 1 in Simple LIN mode. 0: The low level on the RXDn pin is detected as the start bit 1: A falling edge on the RXDn pin is detected as the start bit	R/W
18:16	MOD[2:0]	Communication Mode Select Select the SCI communication mode. 0 0 0: Asynchronous mode (multi-processor mode) 0 0 1: Smart card interface mode 0 1 0: Clock synchronous mode 0 1 1: Simple SPI mode 1 0 0: Simple IIC mode 1 0 1: Manchester mode 1 1 0: Simple LIN mode 1 1 1: Setting prohibited	R/W
19	MP	Multi-Processor Mode Valid in Asynchronous mode and Manchester mode 0: Multi-processor communications function is disabled 1: Multi-processor communications function is enabled	R/W
20	FM	FIFO Mode Select Valid in Asynchronous mode (including Multi-processor mode), Clock synchronous mode, and Simple SPI mode 0: TDR register and RDR register are non-FIFO buffer configuration 1: TDR register and RDR register are FIFO buffer configuration	R/W

Bit	Symbol	Function	R/W
21	DEN	Driver Enable 0: RS-485 driver control function disabled 1: RS-485 driver control function enabled	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
25:24	CKE[1:0]	Clock Enable 0 0: In Asynchronous mode: On-chip baud rate generator The SCKn pin is available for use as an I/O port in accord with the I/O port settings. In Manchester mode and Simple LIN mode: On-chip baud rate generator The SCKn pin functions as I/O port. In Clock synchronous mode and Simple SPI mode: Internal clock (master operation) The SCKn pin functions as the clock output pin. In Smart card interface mode and when CCR3.GM = 0: Output disabled (the SCKn pin is available for use as an I/O port in accord with the I/O port settings). In Smart card interface mode and when CCR3.GM = 1: Output fixed low 0 1: In Asynchronous mode: On-chip baud rate generator The clock with the same frequency as the bit rate is output from the SCKn pin. In Manchester mode and Simple LIN mode: Prohibited In Clock synchronous mode and Simple SPI mode: Internal clock (master operation) The SCKn pin functions as the clock output pin. In Smart card interface mode and when CCR3.GM = 0: Clock output In Smart card interface mode and when CCR3.GM = 1: Clock output 1 0: In Asynchronous mode: External clock or GPT clock - When using the external clock 16 times the bit rate should be input from the SCKn pin when CCR2.ABCS bit is 0. Input a clock signal with a frequency 8 times the bit rate when the CCR2.ABCS bit is 1. - When using the GPT clock^{*3} The SCKn pin is available for use as an I/O port in accord with the I/O port settings when the GPT clock is used. In Manchester mode and Simple LIN mode: Prohibited In Clock synchronous mode and Simple SPI mode: External clock (slave operation) The SCKn pin functions as the clock input pin. In Smart card interface mode and when CCR3.GM = 0: Prohibited In Smart card interface mode and when CCR3.GM = 1: Output fixed high 1 1: In Asynchronous mode: External clock or GPT clock - When using the external clock 16 times the bit rate should be input from the SCKn pin when CCR2.ABCS bit is 0. Input a clock signal with a frequency 8 times the bit rate when the CCR2.ABCS bit is 1. - When using the GPT clock^{*3} The SCKn pin is available for use as an I/O port in accord with the I/O port settings when the GPT clock is used. In Manchester mode and Simple LIN mode: Prohibited In Clock synchronous mode and Simple SPI mode: External clock (slave operation) The SCKn pin functions as the clock input pin. In Smart card interface mode and when CCR3.GM = 0: Prohibited In Smart card interface mode and when CCR3.GM = 1: Clock output	R/W

Bit	Symbol	Function	R/W
26	ACS0	Asynchronous Mode Clock Source Select Valid only in Asynchronous mode. These bits for SCI channels other than SCIn (n = 1, 2) are reserved. 0: External clock input 1: Logical AND of compare matches output from the internal GPT	R/W
27	—	These bits are read as 0. The write value should be 0.	R/W
28	GM	GSM Mode Valid only in Smart card interface mode. 0: Non-GSM mode operation 1: GSM mode operation	R/W
29	BLK	Block Transfer Mode Valid only in Smart card interface mode. 0: Non-block transfer mode operation 1: Block transfer mode operation	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. In other than Asynchronous mode and Manchester mode, this bit setting is invalid and a fixed data length of 8 bits is used. In the Simple LIN mode, only the data length of 8 bits can be used, so set it to the initial value.

Note 2. LSB first is fixed and the MSB (bit 7) in TDR is not transmitted in transmission.

Note 3. GPT clock is selectable for SCI1 and SCI2.

CPHA bit (Clock Phase Select)

The CPHA bit selects the phase of the clock signal output through the SCKn pin. For details, see [Figure 38.99](#).

Set the bit to 1 in other than Simple SPI mode and Clock synchronous mode.

CPOL bit (Clock Polarity Select)

The CPOL bit selects the polarity of the clock signal output through the SCKn pin. For details, see [Figure 38.99](#).

Set the bit to 1 in other than Simple SPI mode and Clock synchronous mode.

BPEN bit (Synchronizer Bypass Enable)

The BPEN bit selects whether to enable or disable the synchronization bypass function. Set this bit to 1 when the bus clock (PCLK) is also used as operation clock (TCLK). For details, see [Figure 38.2](#) and [section 38.18. Synchronizer Bypass Function](#).

Note: For details of this bit setting, see [section 38.19.17. Notes on CCR3.BPEN Bit Setting](#).

CHR[1:0] bits (Character Length)

Selects the data length for transmission and reception.

Except for Asynchronous mode and Manchester mode, a fixed data length of 8 bits is used.

LSBF bit (LSB First select)

Select whether to transmit/receive data in MSB-first or LSB-first.

SINV bit (Transmitted/Received Data Invert)

SINV can invert the transmit data bit from TDR to TSR, and can invert the received data from RSR to RDR. The SINV bit does not affect the logic level of the parity bit. To invert the parity bit, invert the CCR1.PM.

STP bit (Stop Bit Length)

Selects the stop bit length in transmission.

In reception, only the first stop bit is checked regardless of the STP bit setting. If the second stop bit is 0, it is treated as the start bit of the next transmit frame.

In addition, it is used as the break delimiter length setting when sending start frame in Simple LIN mode.

RXDESEL bit (Asynchronous Start Bit Edge Detection Select)

Selects the detection method of the start bit for reception in Asynchronous mode. When a break occurs, data receiving operation depends on the settings of the RXDESEL bit. Set this bit to 1 when reception should be stopped while a break occurs or when reception should be started without retaining the RXDn pin input at high level for the period of one data frame or longer after completion of the break.

Set this bit to 1 in Simple LIN mode. Set this bit to 0 in modes except of Asynchronous mode and Simple LIN mode.

MOD[2:0] bits (Communication Mode Select)

Selects the SCI communication mode.

Table 38.25 Relationship between communication mode selection bits (MOD[2:0]) and other operation mode setting bits

Communication mode	Asynchronous								Smart card interface	Clock synchronous	Simple SPI		Simple IIC	Manchester		Simple LIN
CCR3.MOD[2:0]	000b								001b	010b	011b		100b	101b		110b
CCR3.MP	0		1						—	—	—		—	0	1	—
CCR3.FM	0	1	0	1	0	1	0	1	—	0	1	0	1	—	—	—
CCR3.DEN	0	1	0	1	0	1	0	1	—	—	—		—	—	—	—
CCR0.SSE	—								—	—	0	1	0	1	—	—

Note: — is prohibited setting.

MP bit (Multi-Processor Mode)

Disables/enables multi-processor communications function. The settings of the PE bit and PM bit are invalid in multi-processor mode.

FM bit (FIFO Mode Select)

When the FM bit is set to 1, the TDR register / RDR register switches to FIFO configuration, and transmit FIFO (TDR register) / receive FIFO (RDR register) can be used for serial transmission/reception.

DEN bit (Driver Enable)

Select RS-485 driver control function disabled or enabled.

CKE[1:0] bits (Clock Enable)

The CKE[1:0] bits select the clock source and SCKn pin function.

The combination of these bit settings and of the CCR3.ACS0 bit sets the internal GPT clock.

In Smart card interface mode, these bits control the clock output from the SCKn pin.

In GSM mode, clock output can be dynamically switched. For details, see [section 38.7.8. Clock Output Control](#).

GM bit (GSM Mode)

Setting the GM bit to 1 allows GSM mode operation.

In GSM mode, the CSR.TEND flag set timing is put forward to 11.0 etu (elementary time unit = 1-bit transfer time) from the start and the clock output control function is appended. For details, see [section 38.7.6. Serial Data Transmission \(Except in Block Transfer Mode\)](#), [section 38.7.8. Clock Output Control](#).

ACS0 bit (Asynchronous Mode Clock Source Select)

Selects the clock source in the Asynchronous mode.

The ACS0 bit is valid in Asynchronous mode (CCR3.MOD[2:0] = 000b) and when an external clock input is selected (CCR3.CKE[1:0] = 10b or 11b). This bit is used to select an external clock input or the logical AND of compare matches output from the internal GPT.

Set the ACS0 bit to 0 in other than Asynchronous mode.

For SCI1 and SCI2, the GTIOCmA output (m = 3, 4, 6, 7) of GPT can be set as the serial transfer base clock. See [Table 38.26](#) for details.

These bits for SCI channels other than SCI1 and SCI2 are reserved. The write values to these bits for other than SCI1 and SCI2 should be 0.

Table 38.26 Association between SCI channels and compare match outputs

SCI	GPT	Compare match output
SCI1	GPT3	GTIOC3A
	GPT4	GTIOC4A
SCI2	GPT6	GTIOC6A
	GPT7	GTIOC7A

[Figure 38.4](#), [Figure 38.5](#) show a setting example of when GTIOC3A and GTIOC4A are selected for output. This figure shows an example when GPT clock is input to SCI1.

When generating 187.5 kbps of GPT average transfer rate for PCLKD = 32 MHz:

1. Generate a frequency of 4 MHz using GTIOC3A as the base clock.
2. Generate 3/4 clock enable of base clock to set an average transfer rate of $3 \text{ MHz}/16 = 187.5 \text{ kbps}$ using GTIOC4A.

Setting examples of GPT and SCI

- GPT3.GTSSR = 0x80000000, GPT4.GTSSR = 0x80000000 (enable software counter start)
- GPT3.GTPR = 0x00000007, GPT4.GTPR = 0x0000001F (maximum count value of GTCNT)
- GPT3.GTCCRA = 0x00000003, GPT4.GTCCRA = 0x00000007 (compare match value)
- GPT3.GTCR = 0x00000000, GPT4.GTCR = 0x00000000 (Saw-wave PWM mode, timer prescaler is PCLKD)
- GPT3.GTIOR = 0x00000306, GPT4.GTIOR = 0x00000306
(initial output low, output high at GTCCRA compare match, output low at cycle end)
- CCR3.CKE[1:0] = 10b (external clock input or GPT clock input is selected)
- CCR3.ACS0 = 1, CCR2.ABCSE = 1, CCR2.ABCS=0 (GPT clock input is selected with 16 base clock cycles for 1-bit period)
- GPT3.GTSTR = 0x00000018 (software start GTCNT counter)

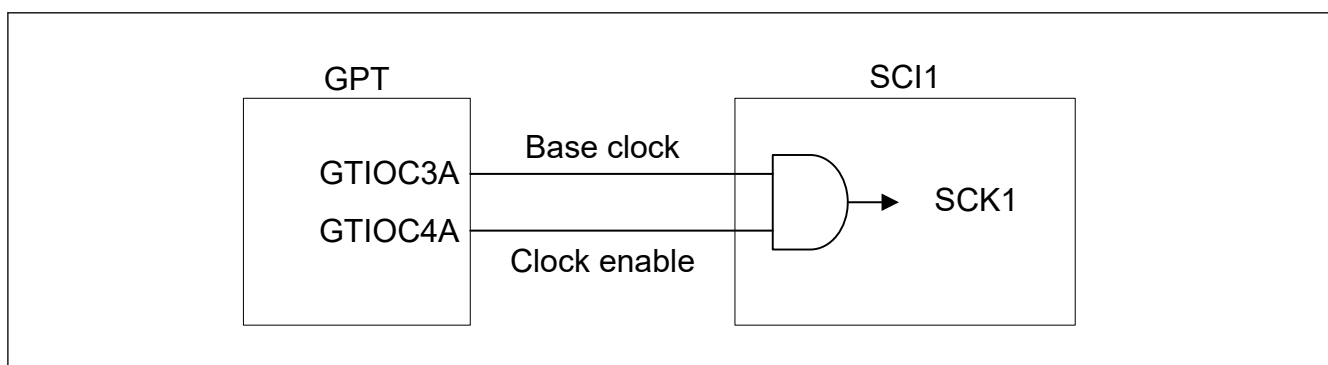


Figure 38.4 Example of GPT and SCI connection

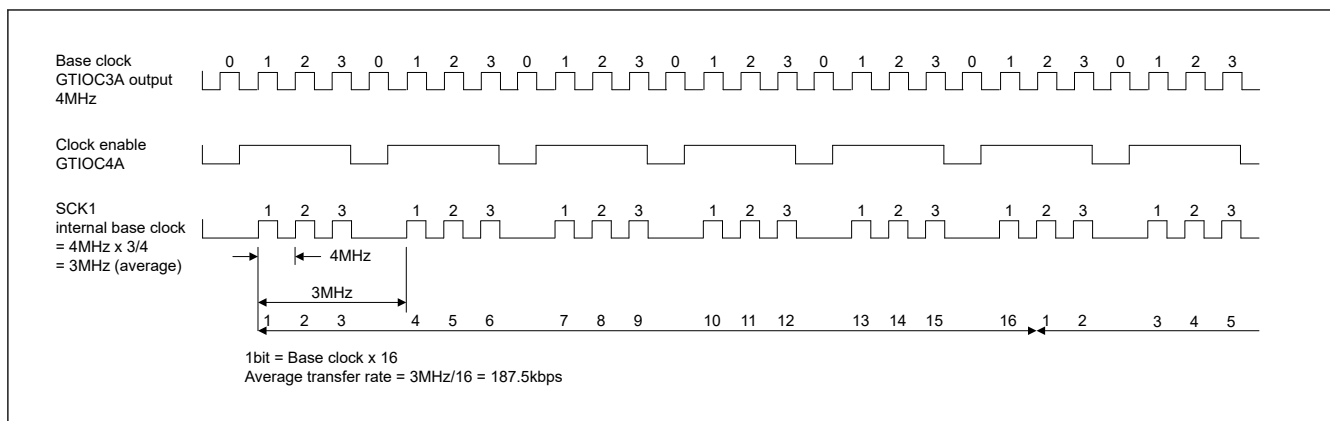


Figure 38.5 Example of average transfer rate setting when GPT clock is input

BLK bit (Block Transfer Mode)

Setting the BLK bit to 1 allows block transfer mode operation.

For details, see [section 38.7.3. Block Transfer Mode](#).

38.2.9 CCR4 : Common Control Register 4

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x18

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	AET	ATT[2:0]			AJD	AST[2:0]			—	—	—	—	—	—	ATEN	ASEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CMPD[8:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	CMPD[8:0]	Compare Match Data Valid only in Asynchronous mode. Set the compare data pattern for address match wake-up function.	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
16	ASEN	Adjust Receive Sampling Timing Enable Valid in Asynchronous mode using internal clock, Simple LIN mode using internal clock, Clock-Asynchronous mode operating as master, and Simple SPI mode operating as master. 0: Adjust sampling timing disabled 1: Adjust sampling timing enabled	R/W
17	ATEN	Adjust Transmit Timing Enable Valid only in Asynchronous mode using internal clock. 0: Adjust transmit timing disabled 1: Adjust transmit timing enabled	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
26:24	AST[2:0]	Adjustment Value for Receive Sampling Timing This bit enables only when ASEN is 1. In Asynchronous mode and Simple LIN mode using internal clock: The sampling timing of RXDn pin is adjusted from the middle of bit by the following formula. Adjustment sampling timing = base clock × the setting value of AST[2:0]. In Clock synchronous mode and Simple SPI mode using internal clock: The RXDn sampling timing can be adjusted by delaying 1 to 4 TCLK 0 0 0: 1-TCLK delay 0 0 1: 2-TCLK delay 0 1 0: 3-TCLK delay 0 1 1: 4-TCLK delay Others: Setting prohibited	R/W
27	AJD	Adjustment Direction for Receive Sampling Timing Valid in Asynchronous mode using internal clock and Simple LIN mode using internal clock. This bit enables only when ASEN is 1. Adjustment direction for RXDn receive sampling timing is determined by this bit. For details, see section 38.3.10. The Function of Adjust Receive Sampling Timing (Asynchronous Mode) . 0: The sampling timing is adjusted backward to the middle of bit. 1: The sampling timing is adjusted forward to the middle of bit.	R/W
30:28	ATT[2:0]	Adjustment Value for Transmit Timing Valid in Asynchronous mode using internal clock and Simple LIN mode using internal clock. This bit enables only when ATEN is 1. The selected edge timing of TXDn is adjusted by the following formula. Adjustment edge timing = base clock × the setting value of ATT[2:0] This setting timing is limited by setting the base clock cycles. For details, see section 38.3.11. The Function of Adjust Transmit Timing (Asynchronous Mode) .	R/W
31	AET	Adjustment Edge for Transmit Timing Valid in Asynchronous mode using internal clock and Simple LIN mode using internal clock. The adjustable edge is set by this bit. This bit enables only when ATEN is 1. For details, see section 38.3.11. The Function of Adjust Transmit Timing (Asynchronous Mode) . 0: When CCR1.TINV is 0, adjust the rising edge timing. When CCR1.TINV is 1, adjust the falling edge timing. 1: When CCR1.TINV is 0, adjust the falling edge timing. When CCR1.TINV is 1, adjust the rising edge timing.	R/W

Note: S-TYPE-3, P-TYPE-3

CMPD[8:0] bits (Compare Match Data)

Set the comparison data for receive data when address match function is enabled (CCR0.DCME = 1). CCR4.CMPD[8:0] should be written while CCR0.DCME is 0.

For the comparison data, it can select length from three types, they are CMPD[6:0] with 7-bit length, CMPD[7:0] with 8-bit length, and CMPD[8:0] with 9-bit length.

Note: If the description in this document and the timing chart do not specify the ASEN / ATEN setting value, it means that the reception sampling adjustment function/transmission timing adjustment function are OFF (CCR4.ASEN = 0, CCR4.ATEN = 0).

ASEN bit (Adjust Receive Sampling Timing Enable)

When the ASEN bit is 1, the receive sampling timing adjustment function is enabled. Control is different in Asynchronous mode, Simple LIN mode, Clock-synchronous mode, and Simple SPI mode.

For details of using internal clock in Asynchronous mode, see [section 38.6.7. Reception Sampling Timing Adjustment Function in Clock Synchronous Mode with Internal Clock Used](#). The operation when the Simple LIN mode internal clock is selected is the same as when the Asynchronous mode internal clock is selected.

For details of Clock synchronous mode operating as master and Simple SPI mode operating as master, see [section 38.6.6. Simultaneous Serial Data Transmission and Reception in Clock Synchronous Mode](#). Only the digital delay of the master mode receive sampling clock (MRCLK) can be controlled by this bit. MRCLK analog delay cannot be controlled.

ATEN bit (Adjust Transmit Timing Enable)

When the ATEN bit is 1, the transmission timing adjustment function is enabled. The transmission timing adjustment function can adjust the edge timing of the waveform output from the TXDn pin. For details, see [section 38.3.11. The Function of Adjust Transmit Timing \(Asynchronous Mode\)](#).

AST[2:0] bits (Adjustment Value for Receive Sampling Timing)

When the ASEN bit is 1, the receive sampling timing can be adjusted according to this bit setting value.

In Asynchronous mode and Simple LIN mode using internal clock.

The sampling timing of RXDn pin is adjusted from the middle of bit by the following formula. This setting value is limited by setting the base clock cycles. For details, see [section 38.3.10. The Function of Adjust Receive Sampling Timing \(Asynchronous Mode\)](#).

Adjustment sampling timing = base clock × the setting value of AST[2:0].

In Clock-synchronous mode and Simple SPI mode using internal clock

The sampling timing of RXDn pin can be adjusted by delaying 1 to 4 TCLK. For details, see [section 38.6.7. Reception Sampling Timing Adjustment Function in Clock Synchronous Mode with Internal Clock Used](#).

000: 1-TCLK delay

001: 2-TCLK delay

010: 3-TCLK delay

011: 4-TCLK delay

1xx: Setting prohibited

AJD bit (Adjustment Direction for Receive Sampling Timing)

Set the RXDn pin sampling timing adjustment direction from the bit center to the rear or front. For details, see [section 38.3.10. The Function of Adjust Receive Sampling Timing \(Asynchronous Mode\)](#).

ATT[2:0] bits (Adjustment Value for Transmit Timing)

The edge timing of the TXDn pin specified by the AET bit is adjusted by the base clock × ATT[2:0] setting value. The upper limit of the adjustment time that can be set is limited by the number of base clock cycles. For details, see [section 38.3.11. The Function of Adjust Transmit Timing \(Asynchronous Mode\)](#).

AET bit (Adjustment Edge for Transmit Timing)

Set the TXDn pin edge for timing adjustment. For details, see [section 38.3.11. The Function of Adjust Transmit Timing \(Asynchronous Mode\)](#).

38.2.10 ICR : Simple IIC Control Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	IICSDAS[1:0]	IICSDAS[1:0]	—	IICST PREQ	IICRS TARE Q	IICSTA REQ		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	IICAC KT	—	—	—	IICCS C	IICINT M	—	—	—	IICDL[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	IICDL[4:0]	SDA Delay Output Select The following cycles are of the clock signal from the on-chip baud rate generator: 0x00: No output delay 0x01: 0 to 1 cycle 0x02: 1 to 2 cycles 0x03: 2 to 3 cycles 0x04: 3 to 4 cycles 0x05: 4 to 5 cycles ⋮ 0x1E: 29 to 30 cycles 0x1F: 30 to 31 cycles	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	IICINTM	IIC Interrupt Mode Select 0: Use ACK/NACK interrupts 1: Use reception and transmission interrupts	R/W
9	IICSCS	Clock Synchronization 0: No synchronization with the clock signal 1: Synchronization with the clock signal	R/W
12:10	—	These bits are read as 0. The write value should be 0.	R/W
13	IICACKT	ACK Transmission Data 0: ACK transmission 1: NACK transmission and reception of ACK/NACK	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
16	IICSTAREQ	Start Condition Generation 0: A start condition is not generated 1: A start condition is generated *1 *3 *4 *5	R/W
17	IICRSTAREQ	Restart Condition Generation 0: A restart condition is not generated 1: A restart condition is generated *2 *3 *4 *5	R/W
18	IICSTPREQ	Stop Condition Generation 0: A stop condition is not generated. 1: A stop condition is generated *2 *3 *4 *5	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
21:20	IICSDAS[1:0]	SDA Output Select 0 0: Serial data output 0 1: Generate a start, restart, or stop condition 1 0: Output the low level on the SDA _n pin 1 1: Place the SDA _n pin in the high-impedance state	R/W
23:22	IICSCLS[1:0]	SCL Output Select 0 0: Serial clock output 0 1: Generate a start, restart, or stop condition 1 0: Output the low level on the SCL _n pin 1 1: Place the SCL _n pin in the high-impedance state	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. In bus free state, perform the start condition generation.

Note 2. In bus busy state, perform restart or stop condition generation when the SCL_n pin after acknowledgment described in [Figure 38.86](#) and [Figure 38.87](#) is low level.

Note 3. Do not set more than one from among the IICSTAREQ, IICRSTAREQ, and IICSTPREQ bits to 1 at a given time.

Note 4. Execute the generation of a condition after the value of the IICSTIF flag is 0.

Note 5. Do not write 0 to this bit while it is 1. Generation of a condition is suspended by writing 0 to this bit while it is 1.

IICDL[4:0] bits (SDA Delay Output Select)

The IICDL[4:0] bits are used to set a delay for output on the SDA_n pin relative to the falling edge of the output on the SCL_n pin. The available delay settings range from no delay to 31 cycles, with the clock signal from the on-chip baud rate generators the base. The signal obtained by frequency-dividing TCLK by the divisor set in CCR2.CKS[1:0] is supplied as the clock signal from the on-chip baud rate generator.

Set these bits to 00000b unless operation is in simple IIC mode. In simple IIC mode, set the bits to a value in the range from 00001b to 11111b.

IICINTM bit (IIC Interrupt Mode Select)

The IICINTM bit selects the sources of interrupt requests in Simple IIC mode.

IICCSCL bit (Clock Synchronization)

Set the IICCSCL bit to 1 if the internally generated SCLn clock signal is to be synchronized when the SCLn pin is placed at the low level and when a wait is inserted by other device, for example.

The SCLn clock signal is not synchronized if the IICCSCL bit is 0. The SCLn clock signal is generated in accord with the rate selected in the BRR regardless of the level being input on the SCLn pin.

Set the IICCSCL bit to 1 except during debugging.

IICACKT bit (ACK Transmission Data)

Transmitted data contains ACK bits. Set this bit to 1 when ACK and NACK bits are received.

IICSTAREQ bit (Start Condition Generation)

When a start condition is to be generated, set both the IICSDAS[1:0] and IICSCSL[1:0] bits to 01b as well as setting the IICSTAREQ bit to 1.

To generate the start condition after generating the stop condition, start the generation of the start condition with a half cycle period of the bit rate from the stop condition generation interrupt (STI) request output.

[Setting condition]

- Writing 1 to the bit

[Clearing condition]

- Completion of generation of the start condition

IICRSTAREQ bit (Restart Condition Generation)

When a restart condition is to be generated, set both the IICSDAS[1:0] and IICSCSL[1:0] bits to 01b as well as setting the IICRSTAREQ bit to 1.

[Setting condition]

- Writing 1 to the bit

[Clearing condition]

- Completion of generation of the restart condition

IICSTPREQ bit (Stop Condition Generation)

When a stop condition is to be generated, set both the IICSDAS[1:0] and IICSCSL[1:0] bits to 01b as well as setting the IICSTPREQ bit to 1.

[Setting condition]

- Writing 1 to the bit

[Clearing condition]

- Completion of generation of the stop condition

IICSDAS[1:0] bits (SDA Output Select)

The IICSDAS[1:0] bits control output from the SDA_n pin.

IICSCSL[1:0] bits (SCL Output Select)

The IICSCSL[1:0] bits control output from the SCL_n pin.

38.2.11 FCR : FIFO Control Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x24

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	RSTRG[4:0]					RFRS T	—	—	RTRG[4:0]				
Value after reset:	0	0	0	1	1	1	1	1	0	0	0	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TFRS T	—	—	TTRG[4:0]					—	—	—	—	—	—	—	DRES
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DRES	Receive Data Ready Error Select Bit Valid in Asynchronous mode. This bit selects the interrupt request for a reception data ready detection. 0: Reception data full interrupt (SCIn_RXI) 1: Receive error interrupt (SCIn_ERI)	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
12:8	TTRG[4:0]	Transmit FIFO Data Trigger Number Valid in Asynchronous mode (including Multiprocessor mode), Clock synchronous mode, and Simple SPI mode. Trigger number must be set to 15 or less. 0x00: Trigger number 0 ⋮ 0x1F: Trigger number 31	R/W
14:13	—	These bits are read as 0. The write value should be 0.	R/W
15	TFRST	Transmit FIFO Data Register Reset This bit enables only when CCR3.FM is 1. The read value is always 0. 0: When set, this bit is invalid and does not affect operation 1: The number of data stored in Transmit-FIFO (TDR register) is 0	W
20:16	RTRG[4:0]	Receive FIFO Data Trigger Number Valid in Asynchronous mode (including Multi processor mode), Clock synchronous mode, and Simple SPI mode. Trigger number must be set to 15 or less. 0x00: Trigger number 0 ⋮ 0x1F: Trigger number 31	R/W
22:21	—	These bits are read as 0. The write value should be 0.	R/W
23	RFRST	Receive FIFO Data Register Reset This bit enables only when CCR3.FM is 1. The read value is always 0. 0: When set, this bit is invalid and does not affect operation 1: The number of data stored in Receive-FIFO(RDR register) is 0	W
28:24	RSTRG[4:0]	RTS Output Active Trigger Number Select Valid in Asynchronous mode (including Multi processor mode) and Clock synchronous mode. This bit enables only when CCR3.FM = 1 and CCR1.CTSE = 0 and CCR0.SSE = 0. Trigger number must be set to 15 or less. 0x00: Trigger number 0 ⋮ 0x1F: Trigger number 31	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DRES bit (Receive Data Ready Error Select Bit)

Select whether the detection of receive data ready (FRSR.DR flag = 1) is the cause of SCIn_RXI interrupt request or the cause of SCIn_ERI interrupt request.

TTRG[4:0] bits (Transmit FIFO Data Trigger Number)

The TDFE flag is set to 1 when the quantity of transmit data in the transmit-FIFO(TDR register) is equal to or less than the specified transmit triggering number. If CCR0.TIE = 1, SCIn_TXI interrupt request occurs.

Note: Trigger number must be set to 15. When the trigger number is set to 16 or more, unexpected SCIn_TXI interrupt occurs.

TFRST bit (Transmit FIFO Data Register Reset)

When the TFRST bit is set to 1, the number of the transmission data stored in Transmit-FIFO(TDR register) is 0.

RTRG[4:0] bits (Receive FIFO Data Trigger Number)

The CSR.RDRF flag is set to 1 when the quantity of receive data in the receive-FIFO(RDR register) is equal to or greater than the specified receive triggering number. If CCR0.RIE = 1, SCIn_RXI interrupt request occurs. When FCR.RTRG is set to 0, RDRF bit is set if the quantity of data in receive-FIFO is greater than or equal to 1.

Note: Trigger number must be set to 15. When the trigger number is set to 16 or more, unexpected SCIn_RXI interrupt occurs.

RFRST bit (Receive FIFO Data Register Reset)

When the RFRST bit is set to 1, the number of the reception data stored in Receive-FIFO(RDR register) is 0.

RSTRG[4:0] bits (RTS Output Active Trigger Number Select)

When the quantity of receive data stored in the receive-FIFO (RDR register) is equal to or greater than this number, the RTSn signal is in the High state. When FCR.RSTRG is set to 0, RTSn is in the high state if the quantity of data in receive FIFO is greater than or equal to 1.

Note: Trigger number must be set to 15. When the trigger number is set to 16 or more, RTSn goes to a high state at unexpected timing.

38.2.12 MCR : Manchester Control Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x2C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	SBER EN	SYER EN	PFER EN	—	—	RPPAT[1:0]	RPLEN[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	TPPAT[1:0]	TPLEN[3:0]				—	SBSE L	SYNS EL	SYNV AL	—	ERTE N	TMPO L	RMPO L	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RMPOL	Polarity of Received Manchester Code Sets the polarity of the received Manchester code. 0: Logic 0 is coded as a zero-to-one transition in Manchester code Logic 1 is coded as a one-to-zero transition in Manchester code 1: Logic 0 is coded as a one-to-zero transition in Manchester code Logic 1 is coded as a zero-to-one transition in Manchester code	R/W

Bit	Symbol	Function	R/W
1	TMPOL	Polarity of Transmit Manchester Code Sets the polarity of the transmit Manchester code 0: Logic 0 is coded as a zero-to-one transition in Manchester code Logic 1 is coded as a one-to-zero transition in Manchester code 1: Logic 0 is coded as a one-to-zero transition in Manchester code Logic 1 is coded as a zero-to-one transition in Manchester code	R/W
2	ERTEN	Manchester Edge Retiming Enable Sets the receive retiming function. 0: Disables the receive retiming function 1: Enables the receive retiming function	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	SYNVAL	SYNC value Setting Sets the SYNC type of the start bit(s) in the Manchester code. When the start bit area consists of one bit (SBSEL = 0) - When transmitting: 0: The start bit is added as a zero-to-one transition 1: The start bit is added as a one-to-zero transition - When receiving: 0: Only when the start bit is a zero-to-one transition, the data is received. Other cases are determined as an error. 1: Only when the start bit is a one-to-zero transition, the data is received. Other cases are determined as an error. When the start bit area consists of three bits (SBSEL = 1) - When transmitting: 0: The start bits are added as a zero-to-one transition (DATA SYNC) 1: The start bits are coded as a one-to-zero transition (COMMAND SYNC) - When receiving: When the start bit area consists of three bits, data is received regardless of the value of this bit.	R/W
5	SYNSEL	SYNC Select 0: The start bit pattern is set with the SYNVAL bit 1: The start bit pattern is set with the TSYNC bit	R/W
6	SBSEL	Start Bit Select 0: The start bit area consists of one bit 1: The start bit area consists of three bits (COMMAND SYNC or DATA SYNC)	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
11:8	TPLEN[3:0]	Transmit Preface Length Set the preface length of the transmit data in Manchester mode. 0x0: Disables the transmit preface generation Others: Transmit preface length (bit length)	R/W
13:12	TPPAT[1:0]	Transmit Preface Pattern Set the preface pattern of the transmit data. 0 0: ALL ZERO 0 1: ZERO ONE 1 0: ONE ZERO 1 1: ALL ONE	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
19:16	RPLEN[3:0]	Receive Preface Length Set the preface length in received frames when Manchester mode is enabled. 0x0: Disables the receive preface generation Others: Receive preface length (bit length)	R/W
21:20	RPPAT[1:0]	Receive Preface Pattern Set the preface pattern of received frames 0 0: ALL ZERO 0 1: ZERO ONE 1 0: ONE ZERO 1 1: ALL ONE	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
24	PFEREN	Preface Error Enable Specifies whether to handle a preface error as an interrupt source. 0: Does not handle a preface error as an interrupt source 1: Handles a preface error as an interrupt source	R/W
25	SYEREN	Receive SYNC Error Enable Specifies whether to handle a receive SYNC error as an interrupt source. 0: Does not handle a receive SYNC error as an interrupt source 1: Handles a receive SYNC error as an interrupt source	R/W
26	SBEREN	Start Bit Error Enable Specifies whether to handle a start bit error as an interrupt source. 0: Does not handle a start bit error as an interrupt source 1: Handles a start bit error as an interrupt source	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

RMPOL bit (Polarity of Received Manchester Code)

The RMPOL bit sets the polarity of the received Manchester code. For details, see [section 38.5.7. Serial Data Reception in Manchester Mode](#).

TMPOL bit (Polarity of Transmit Manchester Code)

The TMPOL bit sets the polarity of the transmit Manchester code. For details, see [section 38.5.6. Serial Data Transmission in Manchester Mode](#).

ERTEN bit (Manchester Edge Retiming Enable)

The ERTEN bit sets the receive retiming function in Manchester mode.

For information on the receive retiming function, see [section 38.5.9. Receive Retiming](#).

SYNVAL bit (SYNC value Setting)

The SYNVAL bit is valid when the SYNSEL bit of the MCR is set to 0.

The SYNC type can be set by combining this bit and the SBSEL bit.

For the start bit area determined by the combination of this bit and the SBSEL bit, see [Figure 38.54](#) and [Figure 38.55](#).

SYNSEL bit (SYNC Select)

The SYNSEL bit is valid when the SBSEL bit of the MCR is set to 1. This bit determines the destination to be referred to for setting the SYNC type of the start bit area added to Manchester frames.

When this bit is set to 0, the SYNVAL bit of the MCR is referred to.

When this bit is set to 1, the TSYNC bit in the TDR register is referred to.

SBSEL bit (Start Bit Select)

The SBSEL bit sets the start bit area in Manchester frames.

When this bit is set to 1, the start bit area added to each frame consists of three bits, and the SYNSEL and SYNVAL bits in this register are valid.

When this bit is set to 0, the start bit area added to each frame consists of one bit.

TPLEN[3:0] bits (Transmit Preface Length)

The TPLEN[3:0] bits set the preface bit length of the transmit data in Manchester mode.

The settable range is 0x0 to 0xF (0d to 15d). The setting of 0x0 disables the transmit preface, which is not added.

TPPAT[1:0] bits (Transmit Preface Pattern)

The TPPAT[1:0] bits set one of the four preface patterns in Manchester mode. For the transmit data when the TPPAT[1:0] bits are set, see [Figure 38.53](#).

When these bits are set to 00b, the preface area is set to all 0s.

When these bits are set to 01b, the preface area is set to the zero-one-zero-one pattern.

When these bits are set to 10b, the preface area is set to the one-zero-one-zero pattern.

When these bits are set to 11b, the preface area is set to all 1s.

RPLEN[3:0] bits (Receive Preface Length)

The RPLEN[3:0] bits set the preface bit length of the received frames in Manchester mode.

The settable range is 0x0 to 0xF (0d to 15d). The setting of 0x0 disables the receive preface, which is not added. When 0x1 to 0xF is set, the set value is handled as the receive preface bit length.

RPPAT[1:0] bits (Receive Preface Pattern)

The RPPAT[1:0] bits set one of the four preface patterns in Manchester mode. For the transmit and receive data when the RPPAT[1:0] bits are set, see [Figure 38.53](#).

When these bits are set to 00b, the preface area is handled as all 0s.

When these bits are set to 01b, the preface area is handled as the zero-one-zero-one pattern.

When these bits are set to 10b, the preface area is handled as the one-zero-one-zero pattern.

When these bits are set to 11b, the preface area is handled as all 1s.

PFEREN bit (Preface Error Enable)

The PFEREN bit specifies whether to handle a preface error as an interrupt source.

When it is set to 0, a preface error is not handled as an interrupt source. When it is set to 1, a preface error is handled as an interrupt source.

SYEREN bit (Receive SYNC Error Enable)

The SYEREN bit specifies whether to handle a receive SYNC error as an interrupt source.

When it is set to 0, a receive SYNC error is not handled as an interrupt source. When it is set to 1, a receive SYNC error is handled as an interrupt source.

SBEREN bit (Start Bit Error Enable)

The SBEREN bit specifies whether to handle a start bit error as an interrupt source.

When it is set to 0, a start bit error is not handled as an interrupt source. When it is set to 1, a start bit error is handled as an interrupt source.

38.2.13 DCR : Driver Control Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x30

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	DENG T[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DEAST[4:0]				—	—	—	—	—	—	—	—	DEPOL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DEPOL	Driver Effective Polarity Select Valid only in Asynchronous mode. 0: The DEN signal is active-high 1: The DEN signal is active-low	R/W

Bit	Symbol	Function	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
12:8	DEAST[4:0]	Driver Assertion Time Valid only in Asynchronous mode. Set the driver assertion time. When DEN = 1, the driver assertion time is inserted in addition to the normal transmission waiting time. Setting DEAST[4:0] = 0x00 is prohibited.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
20:16	DENG[4:0]	Driver Negate Time Valid only in Asynchronous mode. Set the driver negation time. When DEN = 1, the driver negate time is inserted after STOP bit transmission end. Setting DENG[4:0] = 0x00 is prohibited.	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DEPOL bit (Driver Effective Polarity Select)

Select the active level of the DEN signal.

DEAST[4:0] bits (Driver Assertion Time)

Set the driver assertion time (= time from the activation of the DEN (Driver Enable) signal to the start of the start bit). It is expressed in base clock period.

Driver assertion time

= DEAST [4:0] set value × base clock period + transmission waiting time

DENG[4:0] bits (Driver Negate Time)

Set the driver negation time (= time from the end of the last stop bit of the transmitted message until the DEN (Driver Enable) signal is disabled). It is expressed in base clock period.

Driver negate time

= DENG[4:0] set value × base clock period

If the transmission data is written during the driver negate time, transmit starting operation is different depends on the writing timing. The DEN signal remains valid, and transmission of the start bit may start after the transmission wait time has elapsed. Also, the DEN signal may become invalid once, and start bit transmission may start after the Driver assertion time has elapsed.

38.2.14 XCR0 : Simple LIN Control Register 0

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x34

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	BCCS[1:0]	—	AEDIE	COFIE	BFDIE	—	—	BCDIE	BFOIE	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PIBS[2:0]			PIBE	CF1DS[1:0]		CF0RE	BFE	—	—	—	—	—	—	TCSS[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	TCSS[1:0]	Timer Count Clock Source Selection Valid in Simple LIN mode. Select the clock source of the timer in Simple LIN module. 0 1: TCLK/4 1 0: TCLK/16 1 1: TCLK/64	R/W ^{*1}
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	BFE	Break Field Enable Set the presence or absence of Break Field of Start Frame. 0: No Break Field 1: With Break Field	R/W ^{*3}
9	CF0RE	Control Field 0 Enable Set the presence or absence of Control Field 0 of Start Frame 0: No Control Field 0 1: With Control Field 0	R/W ^{*3}
11:10	CF1DS[1:0]	Control Field1 Compare Data Select Select the compare data for Control Field 1. 0 0: Select XCR1.PCF1D[7:0] as the compare data 0 1: Select XCR1.SCF1D[7:0] as the compare data 1 0: Select both XCR1.PCF1D[7:0] and XCR1.SCF1D[7:0] as the compare data 1 1: Setting prohibited	R/W ^{*3}
12	PIBE	Priority Interrupt Bit Enable 0: Priority interrupt bit disabled 1: Priority interrupt bit enabled	R/W ^{*3}
15:13	PIBS[2:0]	Priority Interrupt Bit Select Specify one of bits 0 to 7 of Control Field 1 as the priority interrupt bit. 0 0 0: Bit 0 of Control Field 1 0 0 1: Bit 1 of Control Field 1 0 1 0: Bit 2 of Control Field 1 0 1 1: Bit 3 of Control Field 1 1 0 0: Bit 4 of Control Field 1 1 0 1: Bit 5 of Control Field 1 1 1 0: Bit 6 of Control Field 1 1 1 1: Bit 7 of Control Field 1	R/W ^{*3}
16	BFOIE	Break Field Output Completion Interrupt Enable Select whether to include Break Field output completion as a SCIn_TXI interrupt factor. 0: Break Field output completion is not included in SCIn_TXI interrupt factor 1: Break Field output completion is included in SCIn_TXI interrupt factor	R/W
17	BCDIE	Bus Conflict Detection Interrupt Enable Select whether to output an SCIn_ERI interrupt when a bus collision is detected. 0: Bus conflict detection is not included in SCIn_ERI interrupt factor 1: Bus conflict detection is included in SCIn_ERI interrupt factor	R/W
19:18	—	These bits are read as 0. The write value should be 0.	R/W
20	BFDIE	Break Field Detection Interrupt Enable Select whether to output a SCIn_BFD interrupt when a Break Field is detected. 0: Break Field detection interrupt disabled 1: Break Field detection interrupt enabled	R/W
21	COFIE	Counter Overflow Interrupt Enable Select whether to include counter overflow as an SCIn_ERI interrupt factor. 0: Counter overflow is not included in SCIn_ERI interrupt factor 1: Counter overflow is included in SCIn_ERI interrupt factor	R/W
22	AEDIE	Active Edge Detection Interrupt Enable Select whether to output an SCIn_AED interrupt when a valid edge is detected. 0: Active edge detection interrupt disabled 1: Active edge detection interrupt enabled	R/W
23	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
25:24	BCCS[1:0]	Bus Conflict Detection Clock Selection Select the sampling clock for the bus conflict detection circuit. When CCR2.ABCS = 1, setting BCCS[1] = 1 is prohibited. 0 0: Base clock*2 0 1: Base clock/2 1 0: Base clock/4 1 1: Setting prohibited	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. You can rewrite TCSS[1:0] only when the timer is stopped (TCST = 0 and SDST = 0 and BMEN = 0)

Note 2. Base clock: 1/16 period of 1 bit period when CCR2.ABCS = 0, 1/8 period of 1 bit period when CCR2.ABCS = 1.

Note 3. This bit is a setting bit required for the start frame reception operation. Rewrite this bit when the start frame reception or transmission is not in progress (XCR1.SDST = 0 and XCR1.TCST = 0).

TCSS[1:0] bits (Timer Count Clock Source Selection)

Select clock source of timer in Simple LIN module.

BFE bit (Break Field Enable)

Set the presence or absence of Break Field of Start Frame.

CF0RE bit (Control Field 0 Enable)

Set the presence or absence of Control Field 0 of Start Frame.

CF1DS[1:0] bits (Control Field1 Compare Data Select)

Select the compare data for Control Field 1.

PIBE bit (Priority Interrupt Bit Enable)

Select whether to enable priority interrupt bit comparison of Control Field 1. When the PIBE bit is 1, regardless of the XCR1.CF1CE [7:0] setting value, the bit specified in PIBS [2:0] is compared with the primary comparison data for Control Field 1 (XCR1.PCF1D [7:0]).

PIBS[2:0] bits (Priority Interrupt Bit Select)

Specify bit N (N = 0-7) of Control Field 1 as the priority interrupt bit.

BFOIE bit (Break Field Output Completion Interrupt Enable)

Select whether to include Break Field output completion as a SCIn_TXI interrupt factor. Set CCR0.TIE = 1 and CCR3.MOD [1:0] = 110b, to output SCIn_TXI upon completion of Break Field output.

BCDIE bit (Bus Conflict Detection Interrupt Enable)

Select whether to output an SCIn_ERI interrupt when a bus collision is detected. In Simple LIN mode (CCR3.MOD [1:0] = 110b), SCIn_ERI output control is performed with the BCDIE bit. When CCR3.MOD [1:0] = 110b and BCDIE = 1, an SCIn_ERI interrupt is issued when a bus collision is detected even if CCR0.RIE = 0.

COFIE bit (Counter Overflow Interrupt Enable)

Select whether to include counter overflow as an SCIn_ERI interrupt factor. Set CCR0.RIE = 1 and CCR3.MOD [1:0] = 110b are required to output SCIn_ERI upon counter overflow.

AEDIE bit (Active Edge Detection Interrupt Enable)

Select whether to output an SCIn_AED interrupt when a valid edge is detected. To output SCIn_AED with valid edge detection, XCR1.BMEN = 1 and CCR3.MOD [1:0] = 110b must be set.

BCCS[1:0] bits (Bus Conflict Detection Clock Selection)

Select the sampling clock for the bus conflict detection circuit.

38.2.15 XCR1 : Simple LIN Control Register 1

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x38

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CF1CE[7:0]								SCF1D[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PCF1D[7:0]								—	—	BMEN	SDST	—	—	—	TCST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCST	Break Field Output Timer Count Start Trigger 0: Break Field output timer count stopped 1: Break Field output timer count start	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	SDST	Start Frame Detection Enable 0: Start Frame/Break Field detection disabled 1: Start Frame/Break Field detection enabled Do not set this bit and TCST bit to 1 at the same time.	R/W
5	BMEN	Bit Rate Measurement Enable Set this bit to 1 simultaneously with the SDST bit. This bit can be set to 0 at any timing. 0: Bit rate measurement disabled 1: Bit rate measurement enabled	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
15:8	PCF1D[7:0]	Priority Compare Data for Control Field 1 The priority compare data for Control Field 1.	R/W
23:16	SCF1D[7:0]	Secondary Compare Data for Control Field 1 The secondary compare data for Control Field 1.	R/W
31:24	CF1CE[7:0]	Control Field 1 Compare Bit Enable Select whether to compare bit N of Control Field 1. (N = 0 -7) 0: Control Field 1 bit N compare disabled 1: Control Field 1 bit N compare enabled	R/W

Note: S-TYPE-3, P-TYPE-3

TCST bit (Break Field Output Timer Count Start Trigger)

[Clearing condition]

- When 0 is written to TCST. Break Field output timer count is stopped and TXDn output becomes idle level.
- When Break Field output for the period set in XCR2.BFLW [15: 0] is completed.

[Setting condition]

- When 1 is written to TCST. Start Break Field output from TXDn. Holds 1 during Break Field output.

SDST bit (Start Frame Detection Enable)

When 1 is written to the SDST bit, Start Frame detection starts. When XCR0.BFE = 1 is set, Break Field can be detected during Start Frame is detected and after Start Frame is detected. When XCR0.BFE = 0 is set, Break Field is not detected.

When 0 is written to the SDST bit, Start Frame detection and Break Field detection are stopped. However, if XSR0.RXDSF = 0 at the time of stop, it is not possible to stop data reception of the SCI core with this bit. Write 0 to CCR0.RE to stop the reception operation or perform reception completion processing (CSR.RDRF clear or RDR read) after reception is completed.

BMEN bit (Bit Rate Measurement Enable)

Set this bit to 1 simultaneously with the SDST bit. When the BMEN bit is set to 1, the valid edge interval of Control Field 0 and Control Field 1 data is measured.

PCF1D[7:0] bits (Priority Compare Data for Control Field 1)

Set the priority compare data for Control Field 1.

SCF1D[7:0] bits (Secondary Compare Data for Control Field 1)

Set the secondary compare data for Control Field 1.

CF1CE[7:0] bits (Control Field 1 Compare Bit Enable)

Select whether to compare bit N of Control Field 1. (N = 0 - 7)

When all of the CF1CE[7:0] bits are set to 0 (CF1CE[7:0] = 0x00), it is always determined that Control Field 1 matches when reception is completed, and XSR0.CF1MF is set. This bit is a comparison enable with PCF1D[7:0] or SCF1D[7:0], and is not a priority interrupt bit comparison enable.

38.2.16 XCR2 : Simple LIN Control Register 2

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x3C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	BFLW[15:0]															
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CF0CE[7:0]								CF0D[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CF0D[7:0]	Control Field 0 Compare Data The compare data for Control Field 0.	R/W
15:8	CF0CE[7:0]	Control Field 0 Compare Bit Enable Select whether to compare bit N of Control Field 0. (N = 0 - 7) 0: Control Field 0 bit N compare disabled 1: Control Field 0 bit N compare enabled	R/W
31:16	BFLW[15:0]	Break Field Length Setting This register sets the Break Field length. The Break Field length is (BFLW [15:0] setting value + 1) × clock of the timer. The upper limit for setting this register is 0xFFFF. Setting is prohibited for 0xFFFF.	R/W

Note: S-TYPE-3, P-TYPE-3

CF0D[7:0] bits (Control Field 0 Compare Data)

The compare data for Control Field 0.

CF0CE[7:0] bits (Control Field 0 Compare Bit Enable)

Select whether to compare bit N of Control Field 0. (N = 0 - 7)

When all of the CF0CE[7:0] bits are set to 0 (CF0CE[7:0] = 0x00), it is always determined that Control Field 0 matches when reception is completed, and XSR0.CF0MF is set.

BFLW[15:0] bits (Break Field Length Setting)

BFLW[15:0] are 16-bit Break Field length setting bits and the initial value is 0xFFFFE.

Set the Break Field length to 1 frame or more. The LIN standard stipulates that the Break Field length is 13 bits or more.

When the Break Field is sending, writing 1 to TCST causes SCI to output the Break Field on TXDn. Up-counting is performed with the clock of the timer selected by XCR0.TCSS[1:0]. When the count value matches the value set in this register, up-counting is stopped and Break Field output from TXDn is also stopped.

When the Break Field is receiving, writing 1 to SDST enables Start Frame detection. SCI starts counting from the negative edge of RXDn. The clock of the timer is selected by XCR0.TCSS[1:0].

When the count value matches the value set in the XCR2, a Break Field has been detected. Up-counting continues until the next valid edge or counter overflow.

38.2.17 CSR : Common Status Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x48

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	RDRF	TEND	TDRE	FER	PER	MFF	—	ORER	—	—	—	—	—	DFER	DPER	DCMF
Value after reset:	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RXDMON	—	—	—	—	—	—	—	—	—	—	ERS	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0.	R
4	ERS	Error Signal Status Flag Valid only in Smart card interface mode. 0: Error signal low not responded 1: Error signal low responded	R
14:5	—	These bits are read as 0.	R
15	RXDMON	Serial Input Data Monitor Bit The state of the RXDn pin without synchronizing by bus clock is shown. 0: When RINV is 0, RXDn pin is the low level. When RINV is 1, RXDn pin is the high level. 1: When RINV is 0, RXDn pin is the high level. When RINV is 1, RXDn pin is the low level.	R
16	DCMF	Data Compare Match Flag Valid only in Asynchronous mode. 0: No matched 1: Matched	R
17	DPER	Data Compare Match Parity Error Flag Valid only in Asynchronous mode. 0: No parity error occurred at address match detection 1: A parity error has occurred at address match detection	R
18	DFER	Data Compare Match Framing Error Flag Valid only in Asynchronous mode. 0: No framing error occurred at address match detection 1: A framing error has occurred at address match detection	R
23:19	—	These bits are read as 0.	R
24	ORER	Overflow Error Flag 0: No overflow error occurred 1: An overflow error has occurred	R
25	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
26	MFF	Mode Fault Error Flag Valid only in Simple SPI mode. 0: No mode fault error 1: Mode fault error	R
27	PER	Parity Error Flag 0: Non-FIFO selected (CCR3.FM = 0): No parity error occurred FIFO selected (CCR3.FM = 1): No parity error in all received data in receive-FIFO 1: Non-FIFO selected (CCR3.FM = 0): A parity error has occurred FIFO selected (CCR3.FM = 1): One or more parity errors occurred in received data in receive-FIFO	R
28	FER	Framing Error Flag 0: Non-FIFO selected (CCR3.FM = 0): No framing error occurred FIFO selected (CCR3.FM = 1): No framing error in all received data in receive-FIFO 1: Non-FIFO selected (CCR3.FM = 0): A framing error has occurred FIFO selected (CCR3.FM = 1): One or more framing errors occurred in received data in receive-FIFO	R
29	TDRE	Transmit Data Empty Flag 0: Non-FIFO selected (CCR3.FM = 0): Transmit data is in TDR register FIFO selected (CCR3.FM = 1): The quantity of transmit data written in transmit-FIFO exceeds the specified transmit triggering number. 1: Non-FIFO selected (CCR3.FM = 0): No transmit data is in TDR register FIFO selected (CCR3.FM = 1): The quantity of transmit data written in transmit-FIFO is equal to or less than the specified transmit triggering number.	R
30	TEND	Transmit End Flag 0: A character is being transmitted or standing by for transmission. 1: Character transfer has been completed, or sending Break Field.	R
31	RDRF	Receive Data Full Flag 0: Non-FIFO selected (CCR3.FM = 0): No received data is in RDR register FIFO selected (CCR3.FM = 1): The quantity of receive data written in receive-FIFO falls below the specified receive triggering number. 1: Non-FIFO selected (CCR3.FM = 0): Received data is in RDR register FIFO selected (CCR3.FM = 1): The quantity of receive data written in receive-FIFO is equal to or greater than the specified receive triggering number.	R

Note: S-TYPE-3, P-TYPE-3

ERS bit (Error Signal Status Flag)

[Setting condition]

- When an error signal low is sampled.

[Clearing condition]

- When write 1 to CFCLR.ERSC.

DCMF bit (Data Compare Match Flag)

The DCMF bit indicates that SCI detects the match to the comparison data (CCR4.CMPD) with receive data.

Clearing the CCR0.RE bit to 0 does not affect the DCMF flag, which retains its previous state.

[Setting condition]

- Matched to the comparison data (CCR4.CMPD) with receive data, while CCR0.DCME = 1.

[Clearing condition]

- When write 1 to CFCLR.DCMFC.

DPER bit (Data Compare Match Parity Error Flag)

The DPER bit indicates that a parity error occurred at address match detection (reception data match detection).

Clearing the CCR0.RE bit to 0 does not affect the DPER flag, which retains its previous state.

[Setting condition]

- When a parity error is detected by the frame in which an address match was detected.

[Clearing condition]

- When writing 1 to CFCLR.DPERC.

DFER bit (Data Compare Match Framing Error Flag)

The DFER bit indicates that a framing error occurred at address match detection (reception data match detection).

Clearing the CCR0.RE bit to 0 does not affect the DFER flag, which retains its previous state.

[Setting condition]

- When a stop bit of the frame in which an address match was detected is 0.
When it is a 2-stop mode, the 1st bit of the stop bit only determines whether it is 1 and does not check the 2nd bit of the stop bit.

[Clearing condition]

- When writing 1 to CFCLR.DFERC.

ORER bit (Overrun Error Flag)

The ORER bit indicates that an overrun error has occurred during reception and the reception ends abnormally.

Clearing the CCR0.RE bit to 0 does not affect the ORER flag, which retains its previous state. In Simple IIC mode, the ORER bit is not used.

[Setting condition with non-FIFO mode (CCR3.FM = 0)]

- When the next data is received before reading out RDR with no-error reception data stored in RDR.
In RDR, receive data prior to an overrun error occurrence is retained, but data received after the overrun error occurrence is lost. When the ORER flag is set to 1, reception data is not forwarded to RDR register.

Note: In Clock synchronous mode and Simple SPI mode, serial reception is stopped.

[Setting condition with FIFO mode (CCR3.FM = 1)]

- When the next serial reception is completed while the receive FIFO is full of 16 receive data.

[Clearing condition]

- When write 1 to CFCLR.ORERC.

MFF bit (Mode Fault Error Flag)

The MFF bit indicates mode fault errors. In a multi-master configuration, determine the mode fault error occurrence by reading the MFF flag.

[Setting condition]

- Input on the SSn pin being at the low level during master operation (CCR3.CKE[1:0] = 00 or 01) in Simple SPI mode.

[Clearing condition]

- When writing 1 to CFCLR.MFFC.

PER bit (Parity Error Flag)

The PER bit indicates that a parity error has occurred during reception and the reception ends abnormally.

Clearing the CCR0.RE bit to 0 does not affect the PER flag, which retains its previous state.

In Clock synchronous mode, Simple SPI mode, and Simple IIC mode, the PER bit is not used.

[Setting condition]

- When a parity error is detected during reception. In FIFO select mode, when one or more parity error is detected in receive-FIFO data.
In non-FIFO mode, although receive data when the parity error occurs is transferred to RDR, no SCIn_RXI interrupt request occurs. Note that when the PER flag is set to 1, the subsequent receive data is not transferred to RDR.

[Clearing condition]

- When writing 1 to CFCLR.PERC.

FER bit (Framing Error Flag)

The FER bit indicates that a framing error has occurred during reception and the reception ends abnormally.

Clearing the CCR0.RE bit to 0 does not affect the FER flag, which retains its previous state.

In Clock synchronous mode, Simple SPI mode, and Simple IIC mode, the FER bit is not used.

[Setting condition]

- When 0 is sampled as the stop bit during reception. In FIFO select mode, when one or more framing error is detected in receive-FIFO data. In Manchester mode, when both sampling results (1/4 and 3/4 sampling points) are not 1 for 1 stop bit. In Simple LIN mode, even if a condition that changes to 1 occurs when XCR1.SDST = 1, the FER set timing is delayed up to the Break Field evaluation timing at the longest, since it may be a Break Field. If an edge is detected on the RXDn signal before the Break Field evaluation timing, FER is detected. If no edge is detected in the RXDn signal before the Break Field evaluation timing, Break Field is detected.
In 2-stop-bit mode, only the first stop bit is checked whether it is 1, but the second stop bit is not checked. In non-FIFO mode, although receive data when the framing error occurs is transferred to RDR, no SCIn_RXI interrupt request occurs. In addition, when the FER flag is being set to 1, the subsequent receive data is not transferred to RDR.

[Clearing condition]

- When writing 1 to CFCLR.FERC.

TDRE bit (Transmit Data Empty Flag)

[Non-FIFO selected (CCR3.FM = 0)]

The TDRE bit indicates the presence of transmit data in the TDR register.

The condition of CCR0.TE = 0 has priority over the condition of 0.

If other conditions that become 1 and conditions that become 0 are satisfied at the same time, the TDRE flag is set to 0.

[Setting condition]

- When CCR0.TE is 0.
- When data is transmitted from the TDR register to the TSR register.

[Clearing condition]

- When writing 1 to CFCLR.TDREC.
- When the transmission data is written to the TDR register during CCR0.TE = 1.

[FIFO selected (CCR3.FM = 1)]

The TDRE bit indicates that data has been transferred from the transmit-FIFO (TDR) into the transmit shift register (TSR), the quantity of data in transmit-FIFO has fallen below the specified transmit triggering number.

When the condition which becomes 1 and the condition which becomes 0 are formed at the same time, TDRE flag is 0. After that, when the number of data stored in transmit-FIFO is evaluated, and if that is equal to or less than TTRG value, TDRE is set to 1 after 1 PCLK.

[Setting condition]

- When the quantity of transmit data written in transmit-FIFO is equal to or less than the specified transmit triggering number^{*1}.

Note 1. The transmit-FIFO is FIFO register of 16 stages, the maximum number of data that can be written when the TDRE flag is 1 is indicated in 0x10 - FTSR.T[5:0]. Even if more data is written, data is discarded.

[Clearing condition]

- When writing 1 to CFCLR.TDREC.
- When the transmission data is written to transmit-FIFO by the DTC or DMAC (the last block transfer of block transfer).

TEND bit (Transmit End Flag)

[Non-FIFO selected (CCR3.FM = 0), and not Smart card interface mode (CCR3.MOD[2:0] ≠ 001)]

The TEND bit indicates completion of transmission.

[Setting condition]

- When CCR0.TE is 0.
- When the CCR0.TE bit is changed from 0 to 1, the TEND flag is not affected and retains the value 1.
- When the TDR register is not updated at the time of transmission of the tail-end bit of a character being transmitted.
- When the TDR register is not updated at the end of DE negate time with DE control function enable (CCR3.DEN = 1).
- When Break Field is sending.

[Clearing condition]

- After the synchronization delay time has elapsed since the transmission data was written to the TDR register during CCR0.TE = 1.
- When writing 1 to CFCLR.TDREC during CCR0.TE = 1.

[Non-FIFO selected (CCR3.FM = 0), and Smart card interface mode (CCR3.MOD [2:0] = 001)]

With no error signal from the receiving side, this bit is set to 1 when additional data for transfer is ready to be transferred to the TDR register.

[Setting condition]

- When CCR0.TE is 0.
- When the CCR0.TE bit is changed from 0 to 1, the TEND flag is not affected and retains the value 1.
- When a specified period has elapsed after the latest transmission of 1 byte, the ERS flag is 0, and the TDR register is not updated. The set timing is determined by register settings as listed below.1
 When GM = 0 and BLK = 0, 12.5etu after the start of transmission
 When GM = 0 and BLK = 1, 11.5etu after the start of transmission
 When GM = 1 and BLK = 0, 11.0etu after the start of transmission
 When GM = 1 and BLK = 1, 11.0etu after the start of transmission

[Clearing condition]

- After the synchronization delay time has elapsed since the transmission data was written to the TDR register during CCR0.TE = 1.
- When writing 1 to CFCLR.TDREC during CCR0.TE = 1.

[FIFO selected (CCR3.FM = 1)]

The TEND bit indicates that the transmit-FIFO does not contain valid data when transmitting the last bit of a serial character, so the transmission is halted.

[Setting condition]

- TEND is set to 1 when transmit-FIFO does not contain transmit data when the last bit of a one-byte serial character is transmitted.
- When the TDR register is not updated at the end of DE negate time with DE control function enable (CCR3.DEN = 1).

[Clearing condition]

- After the synchronization delay time has elapsed since the transmission data was written to the TDR register during CCR0.TE = 1.

RDRF bit (Receive Data Full Flag)

[Non-FIFO selected (CCR3.FM = 0)]

The RDRF bit indicates the presence of receive data in the RDR register.

[Setting condition]

- When the reception ends normally, and receive data is forwarded from the RSR register to the RDR register.

[Clearing condition]

- When writing 1 to CFCLR.RDRFC.
- When the read data is read from the RDR register.

[FIFO selected (CCR3.FM = 1)]

The RDRF bit indicates that receive data has been transferred to the receive FIFO data register (RDR), and the quantity of data in receive-FIFO is equal to or greater than the specified receive triggering number. When FCR.RTRG is set to 0, RDRF is set if the quantity of data in receive-FIFO is greater than or equal to 1.

[Setting condition]

- RDRF is set to 1 when the quantity of receive data in receive-FIFO is equal to or greater than the specified receive triggering number^{*1}.

Note 1. Since the receive-FIFO is 16 stage FIFO register, the maximum quantity of data that can be read when RDRF is 1 is equivalent to the specified receive data count number (FDR.R[5:0]). If an attempt is made to read after all the data in receive-FIFO has been read, the data is undefined.

[Clearing condition]

- When writing 1 to CFCLR.RDRFC.
- When the reception data is read from receive-FIFO by the DTC or DMAC (the last block transfer of a block transfer).

When the condition which becomes 1 and the condition which becomes 0 are formed at the same time, RDRF flag is 0. After that, when the number of stored data in receive-FIFO is evaluated, and if that is equal to or greater than RTRG value, RDRF is set to 1 after 1 PCLK.

Note: When Non-FIFO is selected, except when interruption communication, RDRF and TDRE should not be cleared by CFCLR register.

38.2.18 ISR : Simple IIC Status Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x4C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	IICSTI F	—	—	IICAC KR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IICACKR	ACK Reception Data Flag 0: ACK received 1: NACK received	R
1	—	This bit is read as 0.	R
2	—	The read value is undefined.	R
3	IICSTIF	Issuing of Start, Restart, or Stop Condition Completed Flag 0: There are no requests for generating conditions or a condition is being generated. 1: A start, restart, or stop condition is completely generated.	R
5:4	—	The read value is undefined.	R

Bit	Symbol	Function	R/W
31:6	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

IICACKR bit (ACK Reception Data Flag)

Received ACK and NACK bits can be read from this bit.

The IICACKR flag is updated at the rising of SCLn clock for the ACK/NACK receiving bit.

IICSTIF bit (Issuing of Start, Restart, or Stop Condition Completed Flag)

After generating a condition, this bit indicates that the generation is completed. When using the IICSTAREQ, IICRSTAREQ, or IICSTPREQ bit to cause generation of a condition, do so after setting the IICSTIF flag to 0.

When the IICSTIF flag is 1 while an interrupt request is enabled by setting the CCR0.TEIE bit, an STIn request is output.

[Setting condition]

- Completion of the generation of a start, restart, or stop condition (however, in cases where this conflicts with any of the conditions for the flag becoming 0 listed below, the clearing condition takes precedence)

[Clearing condition]

- Writing 1 to ICFCLR.IICSTIFC bit
- When operation is not in Simple IIC mode
- Writing 0 to CCR0.TE bit

38.2.19 FRSR : FIFO Receive Status Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x50

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	FNUM[5:0]						—	—	PNUM[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	R[5:0]						—	—	—	—	—	—	—	DR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0

Bit	Symbol	Function	R/W
0	DR	Receive Data Ready Flag 0: Receiving is in progress, or no received data remains in the receive-FIFO after a successful reception (receive-FIFO is empty) 1: The following receive data is not received for a fixed period after storing data under the threshold in the receive-FIFO	R
1	—	The read value is undefined.	R
7:2	—	These bits are read as 0.	R
13:8	R[5:0]	Receive-FIFO Data Count Valid in Asynchronous mode (including Multiprocessor mode), Clock synchronous mode, and Simple SPI mode, when CCR3.FM is 1. Indicate the quantity of receive data stored in receive-FIFO	R
15:14	—	These bits are read as 0.	R
21:16	PNUM[5:0]	Parity Error Count Valid only in Asynchronous mode Indicates the quantity of data with a parity error for the receive data stored in the Receive FIFO Data register.	R

Bit	Symbol	Function	R/W
23:22	—	These bits are read as 0.	R
29:24	FNUM[5:0]	Framing Error Count Valid only in Asynchronous mode. Indicates the quantity of data with a framing error for the receive data stored in the Receive FIFO Data register.	R
31:30	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

DR bit (Receive Data Ready Flag)

The DR bit indicates that the quantity of data stored in the Receive FIFO Data register (RDR) falls below the specified receive triggering number, and that no next data has been received yet after the elapse of 15 etu from the last stop bit in Asynchronous mode. This bit is valid only Asynchronous mode (including Multiprocessor mode) and FIFO selected. In other mode, this bit does not set to 1.

[Setting conditions]

- DR is set to 1 when the following conditions are met:
 - After receive FIFO Data register (RDR) receives less data than the specified receive triggering number, no next data has been received yet after the elapse of 15 etu^{*1} from the last stop bit
 - CSR.FER, PER flags are 0.

Note 1. This is equivalent to one and a half (1.5) frames in the 8-bit format with one stop bit (etu: elementary time unit).

[Clearing conditions]

- When all receive data in the receive FIFO Data (RDR register) is read and 1 is written to FFCLR.DRC.
- When CCR3.FM bit is 0.

R[5:0] bits (Receive-FIFO Data Count)

The R[5:0] bits indicate the quantity of receive data stored in the Receive FIFO Data register.

0x00 means no receive data. 0x10 means receive-FIFO is full.

PNUM[5:0] bits (Parity Error Count)

The value indicates the quantity of data stored in the Receive FIFO Data register with a parity error.

FNUM[5:0] bits (Framing Error Count)

The value indicates the quantity of data stored in the Receive FIFO Data registers with a framing error.

38.2.20 FTSR : FIFO Transmit Status Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x54

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	T[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	T[5:0]	Transmit-FIFO Data Count Valid in Asynchronous mode (including Multiprocessor mode), Clock synchronous mode, and Simple SPI mode, when CCR3.FM is 1. Indicate the quantity of non-transmit data stored in transmit-FIFO	R
31:6	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

T[5:0] bits (Transmit-FIFO Data Count)

The T[5:0] bits indicate the quantity of non-transmitted data stored in transmit-FIFO.

0x00 means no non-transmit data. 0x10 means transmit-FIFO is full.

38.2.21 MSR : Manchester Status Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x58

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	RSYN C	—	MER	—	SBER	SYER	PFER
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PFER	Preface Error Flag This bit is set when a preface error (pattern mismatch) is detected 0: No preface error detected 1: Preface error detected	R
1	SYER	SYNC Error Flag This bit is set when no edge is detected in the adjustable range during receive retiming 0: No receive SYNC error detected 1: Receive SYNC error detected	R
2	SBER	Start Bit Error Flag This bit is set when a pattern mismatch in the start bit area is detected 0: No start bit error detected 1: Start bit error detected	R
3	—	This bit is read as 0.	R
4	MER	Manchester Error Flag Valid for Manchester mode only 0: No Manchester error occurred 1: Manchester error has occurred	R
5	—	This bit is read as 0.	R
6	RSYNC	Receive SYNC Data Bit It is valid when MCR.SBSEL = 1 in Manchester mode, 0 is read otherwise. 0: The received start bit is DATA SYNC 1: The received start bit is COMMAND SYNC	R
31:7	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

PFER bit (Preface Error Flag)

The PFER bit indicates that a preface error was detected when receiving frames in Manchester mode.

Even when the RE bit in CCR0 is set to 0 (serial reception is disabled), the PFER flag is not affected and retains its previous value.

[Setting condition]

- When detecting a preface error when receiving frames in Manchester mode
The following operations are performed when a preface error occurs.
 - When MCR.PFEREN = 1
The received data is not transferred to the RDR register and no SCIn_RXI interrupt request occurs. Instead, an SCIn_ERI interrupt request occurs. Note that when the PFER flag is set to 1, the subsequently received data is not transferred to the RDR register.
 - When MCR.PFEREN = 0
The received data is transferred to the RDR register and an SCIn_RXI interrupt request occurs. An SCIn_ERI interrupt request is not generated. The subsequent receive operations are not affected even with the PFER flag is set to 1.

[Clearing condition]

- Writing 1 to MFCLR.PFERC.

SYER bit (SYNC Error Flag)

The SYER bit indicates that a receive SYNC error was detected when receiving frames in Manchester mode with MCR.ERTEN = 1 (Manchester edge retiming enabled).

Even when the RE bit in CCR0 is set to 0 (serial reception is disabled), the SYER flag is not affected and retains its previous value.

[Setting condition]

- When detecting a receive SYNC error when receiving frames in Manchester mode
The following operations are performed when a receive SYNC error occurs.
 - When MCR.SYEREN = 1
Although the received data is transferred to the RDR register, no SCIn_RXI interrupt request occurs. Instead, an SCIn_ERI interrupt request occurs. Note that when the SYER flag is set to 1, the subsequently received data is not transferred to the RDR register.
 - When MCR.SYEREN = 0
The received data is transferred to the RDR register and an SCIn_RXI interrupt request occurs. An SCIn_ERI interrupt request is not generated. The subsequent receive operations are not affected even with the SYER flag is set to 1.

[Clearing condition]

- Writing 1 to MFCLR.SYERC.

SBER bit (Start Bit Error Flag)

The SBER bit indicates that a start bit error was detected when receiving frames in Manchester mode.

Even when the RE bit in CCR0 is set to 0 (serial reception is disabled), the SBER flag is not affected and retains its previous value.

[Setting condition]

- When detecting a start bit error when receiving frames in Manchester mode
The following operations are performed when a start bit error occurs.
 - When MCR.SBEREN = 1
The received data is not transferred to the RDR register and no SCIn_RXI interrupt request occurs. Instead, an SCIn_ERI interrupt request occurs. Note that when the SBER flag is set to 1, the subsequently received data is not transferred to the RDR register.
 - When MCR.SBEREN = 0

The received data is transferred to the RDR register and an SCIn_RXI interrupt request occurs. An SCIn_ERI interrupt request is not generated. The subsequent receive operations are not affected even with the SBER flag is set to 1.

[Clearing condition]

- Writing 1 to MFCLR.SBERC.

MER bit (Manchester Error Flag)

When data is received in Manchester mode, Manchester error is detected and displayed. Even when the RE bit in CCR0 is set to 0 (serial reception is disabled), the MER flag is not affected and retains its previous value.

[Setting conditions]

- When receiving in Manchester mode and detecting Manchester code error in data area of received frame.
Received data when an error occurs is transferred to the RDR register, but the SCIn_RXI interrupt request is not generated and the SCIn_ERI interrupt request is generated.
When the Manchester error flag is set to 1, subsequent receive data is not transferred to the RDR register.
For details on Manchester error, [section 38.5.11. Errors in Manchester Mode](#).

[Clearing condition]

- Writing 1 to MFCLR.MERC.

RSYNC bit (Receive SYNC Data Bit)

When Manchester mode (CCR3.MOD[2:0] = 101b) and MCR.SBSEL = 1, this bit indicates the type of SYNC of the received start bit. For other settings, it is fixed to 0.

38.2.22 XSR0 : Simple LIN Status Register 0

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x5C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CF1RD[7:0]								CF0RD[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AEDF	COF	PIBDF	CF1M F	CF0M F	BDF	BCDF	BFOF	—	—	—	—	—	—	RXDS F	SFSF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SFSF	Start Frame Status Flag 0: Start Frame detection disabled or Start Frame detection complete 1: Before Start Frame detection or during detection	R ^{*1}
1	RXDSF	RxDn Input Status Flag 0: RxDn input to SCI is enabled 1: RxDn input to SCI is disabled	R ^{*1}
7:2	—	These bits are read as 0.	R
8	BFOF	Break Field Output Completion Flag 0: Break Field is being output or is not being output 1: Completed BF output	R
9	BCDF	Bus Conflict Detection Flag 0: No bus collision detected 1: Detected a bus collision	R

Bit	Symbol	Function	R/W
10	BDFD	Break Field Detection Flag 0: Break Field is not detected 1: Break Field is detected	R
11	CF0MF	Control Field 0 Compare Match Flag 0: Control Field 0 received data does not match the setting data 1: Control Field 0 received data matches the setting data	R
12	CF1MF	Control Field 1 Compare Match Flag 0: Control Field 1 received data does not match the setting data 1: Control Field 1 received data matches the setting data	R
13	PIBDF	Priority Interrupt Bit Detection Flag 0: Priority interrupt bit is not detected 1: Priority interrupt bit is detected	R
14	COF	Counter Overflow Flag 0: Break Field detection counter has not overflowed 1: Break Field detection counter overflowed	R
15	AEDF	Active Edge Detection Flag 0: No valid edge detected 1: Detected a valid edge	R
23:16	CF0RD[7:0]	Control Field 0 received data Control Field 0 received data.	R
31:24	CF1RD[7:0]	Control Field 1 received data Control Field 1 received data.	R

Note: S-TYPE-3, P-TYPE-3

Note 1. When PCLK is faster than TCLK, the flag set timing is delayed from the receive data full interrupt (SCIn_RXI) output. To see this flag under these conditions, wait at least 1 TCLK cycle after the receive data full interrupt (SCIn_RXI) before reading this register.

SFSF bit (Start Frame Status Flag)

Indicates whether detect Start Frame is being detected.

[Setting condition]

- When 1 is written to XCR1.SDST.
- When a Break Field is detected in the Control Field 0/Control Field 1/Information Field phase and the transition to the Control Field 0 or Control Field 1 reception state occurs.

[Clearing condition]

- When XCR1.SDST is 0.
- When Start Frame detection is completed.

RXDSF bit (RXDn Input Status Flag)

Indicates the RXDn input status to the SCI core. When the RXDSF bit is 1, RXDn input is received only by the simple LIN module and the Break Field is detected and is not input to the SCI core.

BFOF bit (Break Field Output Completion Flag)

Indicates the completion of Break Field output.

The BFOF bit can be cleared to 0 by writing 1 to XFCLR.BFOC.

BCDF bit (Bus Conflict Detection Flag)

Indicates the detection of bus conflict in Simple LIN transmit operation.

The BCDF bit can be cleared to 0 by writing 1 to XFCLR.BCDC.

BFDF bit (Break Field Detection Flag)

Indicates Break Field detection.

The BFDF bit can be cleared to 0 by writing 1 to XFCLR.BFDC.

CF0MF bit (Control Field 0 Compare Match Flag)

Indicates compare match of Control Field 0 and compare data.

The CF0MF bit can be cleared to 0 by writing 1 to XFCLR.CF0MC.

CF1MF bit (Control Field 1 Compare Match Flag)

Indicates compare match detection of Control Field 1 and compare data.

The CF1MF bit can be cleared to 0 by writing 1 to XFCLR.CF1MC.

PIBDF bit (Priority Interrupt Bit Detection Flag)

Indicates compare match detection of Control Field 1 and priority interrupt bit.

The PIBDF bit can be cleared to 0 by writing 1 to XFCLR.PIBDC.

COF bit (Counter Overflow Flag)

Indicates that the 16-bit counter in the simple LIN module has overflowed.

The COF bit can be cleared to 0 by writing 1 to XFCLR.COFC.

AEDF bit (Active Edge Detection Flag)

Indicates active edge detection.

The AEDF bit can be cleared to 0 by writing 1 to XFCLR.AEDC and when read out XSR1.TCNT[15:0].

CF0RD[7:0] bits (Control Field 0 received data)

Stores the received data with a Control Field 0 match detected.

CF1RD[7:0] bits (Control Field 1 received data)

Stores the received data with a Control Field 1 match detected.

38.2.23 XSR1 : Simple LIN Status Register 1

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x60

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TCNT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TCNT[15:0]	Timer Count Capture Value Stores the 16-bit counter capture value. The initial value is 0000.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

TCNT[15:0] bits (Timer Count Capture Value)

Stores the capture value of the 16-bit counter of the Simple LIN module.

- When sending Start Frame
The XSR1 holds the previous value.
- When receiving Start Frame with bit rate measurement disabled

If a Break Field is detected in the Break Field detection state (see [Figure 38.81](#)), the Break Field length is captured and held (counter value is captured at the rising edge of RXDn). If a Break Field is detected in a state other than the Break Field detection state, the previous value is retained.

If the counter overflows, it is not captured.

- When receiving Start Frame with bit rate measurement enabled

The count value is captured and held at the valid edge (both RXDn edges). However, in the Break Field detection state, the count value is not captured even if a valid edge occurs. Counter capture value retention is canceled by the XSR1 read. Even if a valid edge occurs before reading, the counter value is not captured.

38.2.24 CFCLR : Common Flag Clear Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x68

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	RDRF C	—	TDRE C	FERC	PERC	MFFC	—	ORER C	—	—	—	—	—	DFER C	DPER C	DCMF C
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	ERSC	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	The write value should be 0.	W
4	ERSC	ERS Clear Bit Setting this bit to 1 clears the CSR.ERS bit. The read value is always 0.	W
15:5	—	The write value should be 0.	W
16	DCMFC	DCMF Clear Bit Setting this bit to 1 clears the CSR.DCMF bit. The read value is always 0.	W
17	DPERC	DPER Clear Bit Setting this bit to 1 clears the CSR.DPER bit. The read value is always 0.	W
18	DFERC	DFER Clear Bit Setting this bit to 1 clears the CSR.DFER bit. The read value is always 0.	W
23:19	—	The write value should be 0.	W
24	ORERC	ORER Clear Bit Setting this bit to 1 clears the CSR.ORER bit. The read value is always 0.	W
25	—	The write value should be 0.	W
26	MFFC	MFF Clear Bit Setting this bit to 1 clears the CSR.MFF bit. The read value is always 0.	W
27	PERC	PER Clear Bit Setting this bit to 1 clears the CSR.PER bit. The read value is always 0.	W
28	FERC	FER Clear Bit Setting this bit to 1 clears the CSR.FER bit. The read value is always	W
29	TDREC	TDRE Clear Bit Setting this bit to 1 clears the CSR.TDRE bit. The read value is always 0.	W
30	—	The write value should be 0.	W
31	RDRFC	RDRF Clear Bit Setting this bit to 1 clears the CSR.RDRF bit. The read value is always 0.	W

Note: S-TYPE-3, P-TYPE-3

38.2.25 ICFCLR : Simple IIC Flag Clear Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x6C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	IICSTI FC	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	—	The write value should be 0.	W
3	IICSTIFC	IICSTIF Clear Bit Setting this bit to 1 clears the ISR.IICSTIF bit. The read value is always 0.	W
31:4	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

38.2.26 FFCLR : FIFO Flag Clear Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x70

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DRC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DRC	DR Clear Bit Setting this bit to 1 clears the FRSR.DR bit. The read value is always 0.	W
31:1	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

38.2.27 MFCLR : Manchester Flag Clear Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x74

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	MERC	—	SBER C	SYER C	PFER C
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PFERC	PFER Clear Bit Setting this bit to 1 clears the MSR.PFER bit. The read value is always 0.	W
1	SYERC	SYER Clear Bit Setting this bit to 1 clears the MSR.SYER bit. The read value is always 0.	W
2	SBERC	SBER Clear Bit Setting this bit to 1 clears the MSR.SBER bit. The read value is always 0.	W
3	—	The write value should be 0.	W
4	MERC	MER Clear Bit Setting this bit to 1 clears the MSR.MER bit. The read value is always 0.	W
31:5	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

38.2.28 XFCLR : Simple LIN Flag Clear Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x78

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AEDC	COFC	PIBDC	CF1M C	CF0M C	BFDC	BCDC	BFOC	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	The write value should be 0.	W
8	BFOC	BFOF Clear Bit Setting this bit to 1 clears the XSR0.BFOF bit. The read value is always 0.	W
9	BCDC	BCDF Clear Bit Setting this bit to 1 clears the XSR0.BCDF bit. The read value is always 0.	W
10	BFDC	BFDF Clear Bit Setting this bit to 1 clears the XSR0.BFDF bit. The read value is always 0.	W
11	CF0MC	CF0MF Clear Bit Setting this bit to 1 clears the XSR0.CF0MF bit. The read value is always 0.	W

Bit	Symbol	Function	R/W
12	CF1MC	CF1MF Clear Bit Setting this bit to 1 clears the XSR0.CF1MF bit. The read value is always 0.	W
13	PIBDC	PIBDF Clear Bit Setting this bit to 1 clears the XSR0.PIBDF bit. The read value is always 0.	W
14	COFC	COFF Clear Bit Setting this bit to 1 clears the XSR0.COF bit. The read value is always 0.	W
15	AEDC	AEDF Clear Bit Setting this bit to 1 clears the XSR0.AEDF bit and cancels holding the XSR1 register. The read value is always 0.	W
31:16	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

38.2.29 CESR : Communication Enable Status Register

Base address: SCIn_B = 0x4035_8000 + 0x0100 × n (n = 0 to 9)
 SCIn_B_NS = 5035_8000 + 0x0100 × n (n = 0 to 9)

Offset address: 0x1C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TIST	—	—	—	RIST

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RIST	RE Internal Status 0: RE signal internal state value 0 1: RE signal internal state value 1	R
3:1	—	These bits are read as 0.	R
4	TIST	TE Internal Status 0: TE signal internal state value 0 1: TE signal internal state value 1	R
7:5	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The operation clocks of the communication module and control register can be used asynchronously. Since some control register values are transmitted internally through the synchronization circuit so that they operate correctly even if they are asynchronous, it takes some time for the state to be reflected internally after rewriting the control register.

Communication enable CCR0.TE and CCR0.RE correspond to this register, and when these control bits change from 1 to 0 to rewrite the control bits for the next communication, the TE and RE bits signal it is necessary to rewrite the next control bit after the internal state becomes 0. If a very slow clock is used for the communication module clock, the TE and RE bit states are not reflected internally. At this time, you can check the internal status using this register.

38.3 Operation in Asynchronous Mode

Figure 38.6 shows the general format for asynchronous serial communications. One frame consists of a start bit (low level), transmit or receive data, a parity bit, and stop bits (high level). In asynchronous serial communications, the communications line is usually held in the mark state (high level).

The SCI monitors the communications line. When the SCI detects a start bit, it starts serial communication. The detection condition of the start bit changes according to the CCR3.RXDESEL setting. SCI regards space (Low level) as a start bit when CCR3.RXDESEL is 0. SCI regards a fall edge as a start bit when RXDESEL is 1.

Inside the SCI, the transmitter and receiver are independent units, enabling full-duplex communications. Both the transmitter and receiver have a double-buffered structure in addition to FIFO mode, so that data can be read or written during transmission or reception, enabling continuous data transmission and reception.

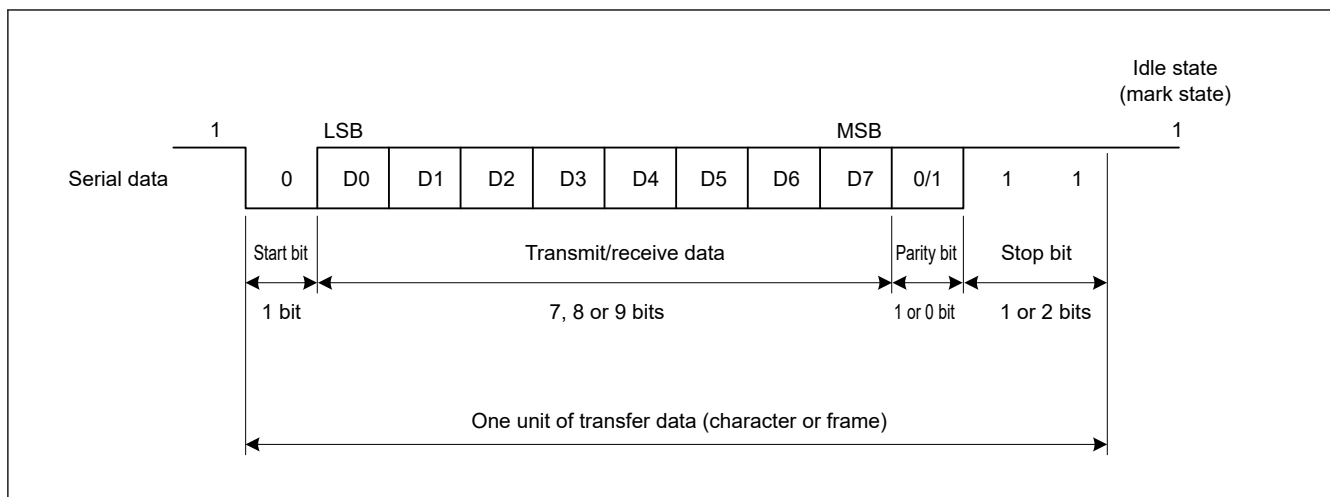


Figure 38.6 Data format in asynchronous serial communications with 8-bit data, parity bit, and 2 stop bits

38.3.1 Serial Data Transfer Format

Table 38.27 lists the serial data transfer formats that can be used in Asynchronous mode. Any of 18 transfer formats can be selected with the CCR1 and CCR3 settings. For details on the multi-processor function, see [section 38.4. Multi-Processor Communication Function](#).

Table 38.27 Serial transfer formats in Asynchronous mode (1 of 2)

CCR3		CCR1		CCR3	Serial transfer format and frame length												
CHR[1:0]		PE	MP	STP	1	2	3	4	5	6	7	8	9	10	11	12	13
0	0	0	0	0	ST	9-bit data									SP		
0	0	0	0	1	ST	9-bit data									SP	SP	
0	0	1	0	0	ST	9-bit data									P	SP	
0	0	1	0	1	ST	9-bit data									P	SP	SP
1	0	0	0	0	ST	8-bit data								SP			
1	0	0	0	1	ST	8-bit data								SP	SP		
1	0	1	0	0	ST	8-bit data								P	SP		
1	0	1	0	1	ST	8-bit data								P	SP	SP	
1	1	0	0	0	ST	7-bit data							SP				
1	1	0	0	1	ST	7-bit data							SP	SP			
1	1	1	0	0	ST	7-bit data							P	SP			
1	1	1	0	1	ST	7-bit data							P	SP	SP		

Table 38.27 Serial transfer formats in Asynchronous mode (2 of 2)

CCR3		CCR1		CCR3	Serial transfer format and frame length												
CHR[1:0]		PE	MP	STP	1	2	3	4	5	6	7	8	9	10	11	12	13
0	0	—	1	0													
0	0	—	1	1													
1	0	—	1	0													
1	0	—	1	1													
1	1	—	1	0													
1	1	—	1	1													

ST: Start bit
 SP: Stop bit
 P: Parity bit
 MPB: Multi-processor bit

38.3.2 Receive Data Sampling Timing and Reception Margin in Asynchronous Mode

In Asynchronous mode, the SCI operates on a base clock with a frequency of 16 times^{*1} the bit rate.

In reception, the SCI samples the falling edge of the start bit using the base clock, and performs internal synchronization.^{*2}

Because receive data is sampled on the rising edge of the 8th pulse^{*1} of the base clock, data is latched at the middle of each bit (when sampling timing does not adjust (CCR4.ASEN = 0 or CCR4.ASEN = 1 and CCR4.AST[2:0] = 000b)), as shown in [Figure 38.7](#) The reception margin in Asynchronous mode is determined by the following formula (1):

$$M = \left| \left(0.5 - \frac{1}{2N} \right) - (L - 0.5)F - \frac{|D - 0.5|}{N} (1 + F) \right| \times 100 [\%] \quad \dots \text{Formula (1)}$$

Note: M: Reception margin

N: Ratio of bit rate to clock

See [Table 38.6](#) for bit rate to basic clock ratios.

D: Duty cycle of clock (D = 0.5 to 1.0)

L: Frame length (L = 9 to 13)

F: Absolute value of clock frequency deviation

Assuming values of F = 0 and D = 0.5 in formula (1), the reception margin is determined using the following formula:

$$M = \{0.5 - 1/(2 \times 16)\} \times 100 (\%) = 46.875 \%$$

This represents the computed value. Renesas recommends a margin of 20% to 30% in system design.

Note 1. This is an example when CCR2.ABCS is 0, CCR2.ABCSE is 0 and CCR2.ABCSE2 is 0. When CCR2.ABCS is 1, CCR2.ABCSE is 0 and CCR2.ABCSE2 is 0, a frequency of 8 times the bit rate is used as a base clock and receive data is sampled at the rising edge of the 4th pulse of the base clock. When CCR2.ABCSE is 1 and CCR2.ABCSE2 is 0, a frequency of 6 times the bit rate is used as a base clock and receive data is sampled at the rising edge of the 3rd pulse of the base clock. When CCR2.ABCSE is 0 and CCR2.ABCSE2 is 1, a frequency of 4 times the bit rate is used as a base clock and receive data is sampled at the rising edge of the 2nd pulse of the base clock.

Note 2. The determination condition of the start bit is as follows.

The function of adjust sampling timing is OFF (ASEN = 0):

The determination condition of a start bit is that Low beyond half bit length continues. It is same as the sampling timing. In [Figure 38.7](#), Low period should be kept over 8-cycles to detect a start bit. If Low period does not keep over 8-cycles, the IP judges this as a noise. So, the IP does not start reception and wait start bit.

The function of adjust sampling timing is ON (ASEN = 1):

The determination condition of a start bit is that Low keeps up until the sampling timing. Adjusting the sampling timing forward (AJD = 1) increases the possibility of erroneously determining a noise as the start bit.

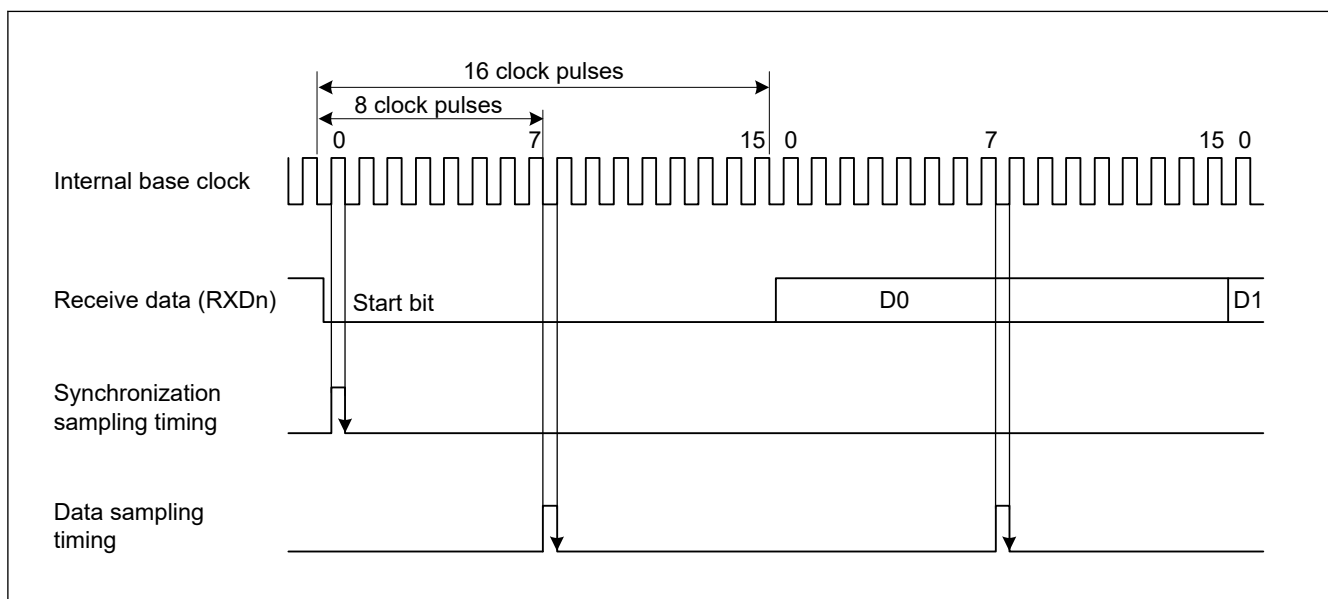


Figure 38.7 Receive data sampling timing in Asynchronous mode

38.3.3 Clock

Either an internal clock generated by the on-chip baud rate generator or an external clock input to the SCKn pin can be selected as the transfer clock of the SCI, based on the CCR3.CKE[1:0] settings.

When an external clock is input to the SCKn pin, the clock frequency must be 16 times the bit rate (when CCR2.ABCS = 0) or 8 times the bit rate (when CCR2.ABCS = 1).

In addition, when an external clock is specified, the base clock of GPTn (n = 3, 4, 6, 7) can be selected by the CCR2.ACS0.

When the SCI uses its internal clock, the clock can be output from the SCKn pin. The frequency of the clock output in this case is equal to the bit rate, and the phase is configured so that the rising edge of the clock is in the middle of the transmit data, as shown in [Figure 38.8](#).

When the internal clock is selected, the SCKn pin is outputted after setting the CCR0.TE or CCR0.RE bit to 1.

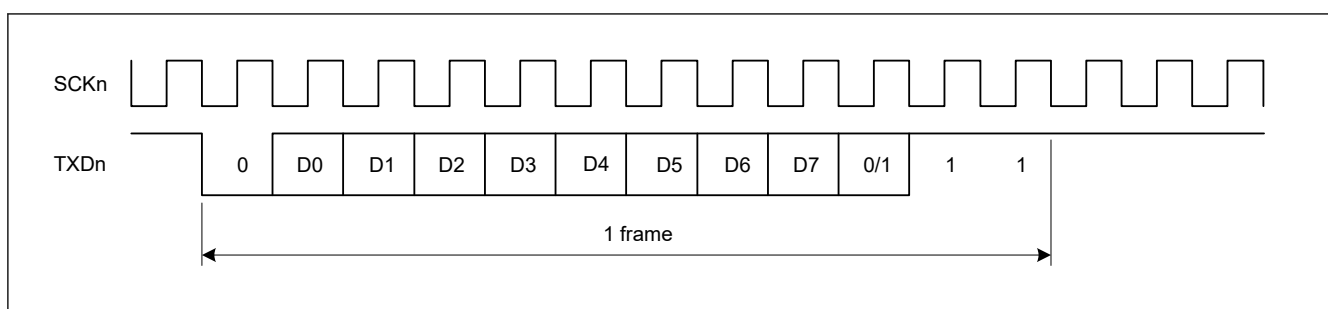


Figure 38.8 Phase relationship between output clock and transmit data in Asynchronous mode when CCR1.PE = 1, CCR3.CHR[1:0] = 10b, MP = 0, and STP = 1

38.3.4 Double-Speed Operation and Frequency of 6 or 4 Times the Bit Rate

When the CCR2.ABCS bit is set to 1, the SCI operates on the bit rate twice that of when ABCS is set to 0. When the CCR2.BGDM bit is set to 1, the cycle of the base clock is half and the bit rate is double that of when BGDM is set to 0. When the CCR3.CKE[1] bit is set to 0 and the on-chip baud rate generator is selected, setting the ABCS and BGDM bits to 1 allows the SCI to operate at a bit rate four times that when the ABCS and BGDM bits are set to 0.

When CCR2.ABCSE is set to 1 and CCR2.ABCSE2 is set to 0, the number of base clock pulses are 6 during a period of 1 bit, and the base clock frequency is half. And RSCI works 16/3 times of bit rate compared with a case of CCR2.ABCS = 0, CCR2.BGDM = 0, CCR2.ABCSE = 0 and CCR2.ABCSE2 = 0.

When CCR2.ABCSE is set to 1 and CCR2.ABCSE2 is set to 1, the number of base clock pulses are 4 during a period of 1 bit, and the base clock frequency is half. And RSCI works 8 times of bit rate compared with a case of CCR2.ABCS = 0, CCR2.BGDM = 0, CCR2.ABCSE = 0 and CCR2.ABCSE2 = 0.

As shown by Formula (1) in [section 38.3.2. Receive Data Sampling Timing and Reception Margin in Asynchronous Mode](#), the reception margin decreases when CCR2.ABCS is set to 1, CCR2.ABCSE is set to 1 or CCR2.ABCSE2 is set to 1. Therefore, if the desired bit rate can be obtained, it is recommended to use the RSCI with CCR2.ABCS set to 1, CCR2.ABCSE set to 1 and CCR2.ABCSE2 set to 1.

38.3.5 CTS and RTS Functions

The CTS function controls transmission using the CTSn pin input. Setting the CCR1.CTSE bit to 1 enables the CTS function. For the functions of CTS and RTS, you can select the alternate setting to set CTSn_RTSn pin as a multiplexed pin that uses either function with one pin or the dedicated setting that uses each function independently with two pins at CTSn pin for CTSn signal and CTSn_RTSn pin for RTSn signal. This setting is done with the CCR1.CTSPEN bit.

When the CTS function is enabled, placing a low level on the CTSn_RTSn pin causes transmission to start.

If FIFO is used and CTSn_RTSn signal is held high before transmission, transmission will not start, so the number of TDR registers written, and the number of data stored are the same (unlike using clock synchronous FIFO).

Driving the CTSn_RTSn pin high while transmission is in progress does not affect transmission of the current frame.

In the RTS function, which uses output on the CTSn_RTSn pin, a low level is output when reception becomes possible. Conditions for output of the low and high levels are shown in this section.

[Conditions for low level output]

Satisfaction of all conditions are listed in this section.

Non-FIFO selected

- The value of the CCR0.RE bit is 1
- When the next reception is possible
 - There are no received data yet to be read and not receiving.
 - CSR.ORER, FER, PER flags are all 0

FIFO selected

- The value of the CCR0.RE bit is 1
- When the next reception is possible
 - When the quantity of receive data written in receive-FIFO(RDR) are less than the setting value of FCR.RSTRG[4:0]
 - CSR.ORER(RDR.ORER) is 0

[Condition for high level output]

- The conditions for low-level output are not satisfied

38.3.6 Address Match (Receive Data Match Detection) Function

The address match function can be used only in Asynchronous mode.

If the CCR0.DCME bit is set to 1^{*2}, when one frame of data is received, the SCI compares that received data with the data set in CCR4.CMPD. If the SCI detects a match to the comparison data (CCR4.CMPD^{*1}) with the received data, the SCI can issue the SCIn_RXI interrupt request.

If the CCR3.MP bit is set to 0, comparison occurs only for valid data in receive format. In multi-processor mode (CCR3.MP bit = 1), if the CCR0.IDSEL bit is set to 1, receive data where the MPB bit is 1 is subject to comparison for address match and receive data where the MPB bit is 0 is always treated as a non-match.

If the CCR0.IDSEL bit is set to 0, SCI performs address match detection regardless of the MPB bit value of the received data.

Until SCI detects a match to the comparison data (CCR4.CMPD^{*1}) with receive data, received data is skipped (discarded), and the SCI cannot detect a parity error or framing error.

When SCI detects a match, the CCR0.DCME bit is automatically cleared, and the CSR.DCMF flag is set to 1. If the CCR0.IDSEL bit is set to 1, the CCR0.MPIE bit is automatically cleared. If CCR0.IDSEL is set to 0, the value of the CCR0.MPIE bit is retained. If the CCR0.RIE bit is set to 1, the SCI issues an SCI_{IN}_RXI interrupt request.

If the SCI detects a framing error in the receive data for which a match is detected, the CSR.DFER flag is set to 1, and if the SCI detects a parity error in that frame, the CSR.DPER flag is set to 1. The compared receive data and MPB bit is not stored in the RDR register, and CSR.RDRF remains 0.

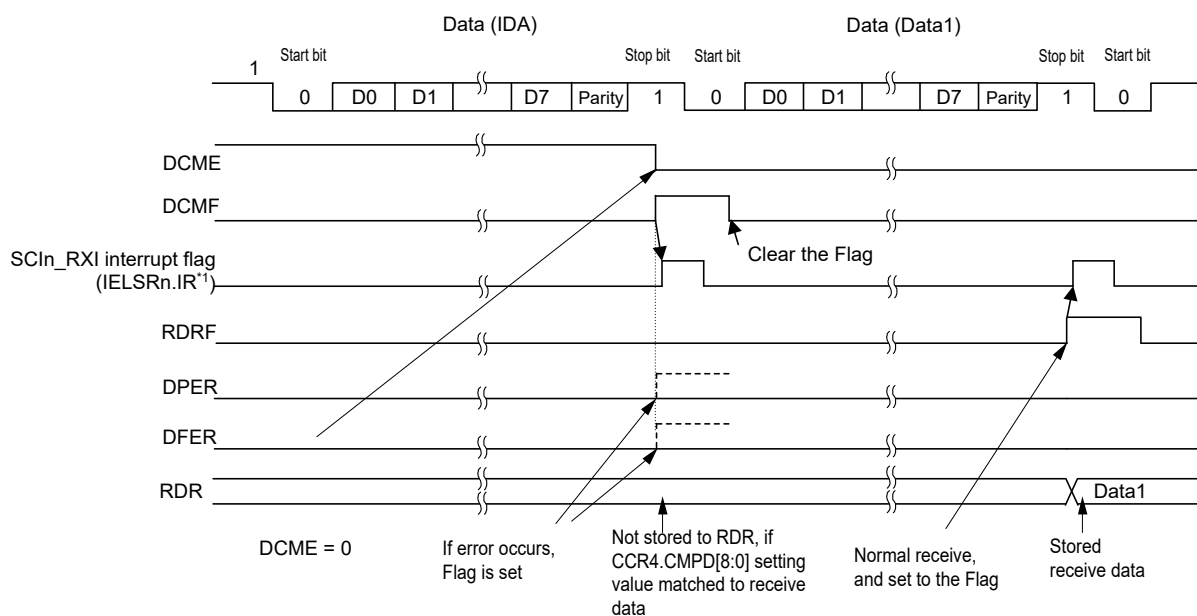
After the SCI detects a match, and CCR0.DCME is automatically cleared, the SCI receives the next data continuously based on the current register setting.

When the CSR.DFER or CSR.DPER flag is set, the address match is not performed. Before enabling the address match function, set the DCCR.DFER and DCCR.DPER flags to 0.

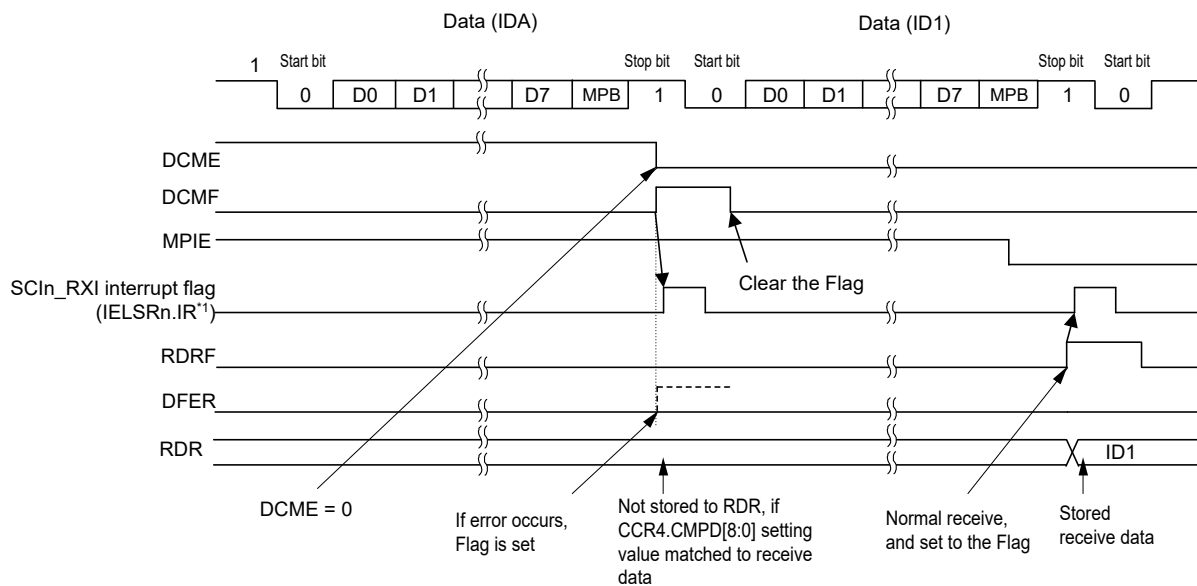
Examples of the address match function are shown in [Figure 38.9](#) and [Figure 38.10](#).

Note 1. This comparative target can select one length of 3 types: CMPD[6:0] with 7-bit length, CMPD[7:0] with 8-bit length, and CMPD[8:0] with 9-bit length.

Note 2. Set the CCR0.DCME bit to 1 before receiving the start bit of the received frame that performs address matching.



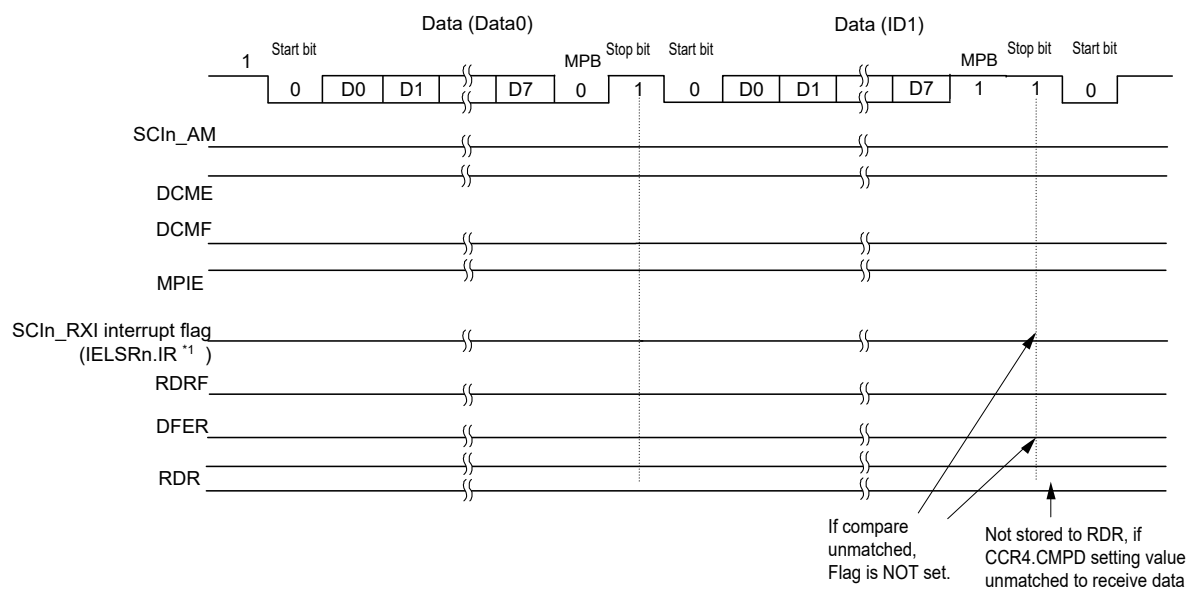
(a) Example of Compare matched between receive data and CCR4.CMPD[8:0] (8bit length/parity/Normal mode)



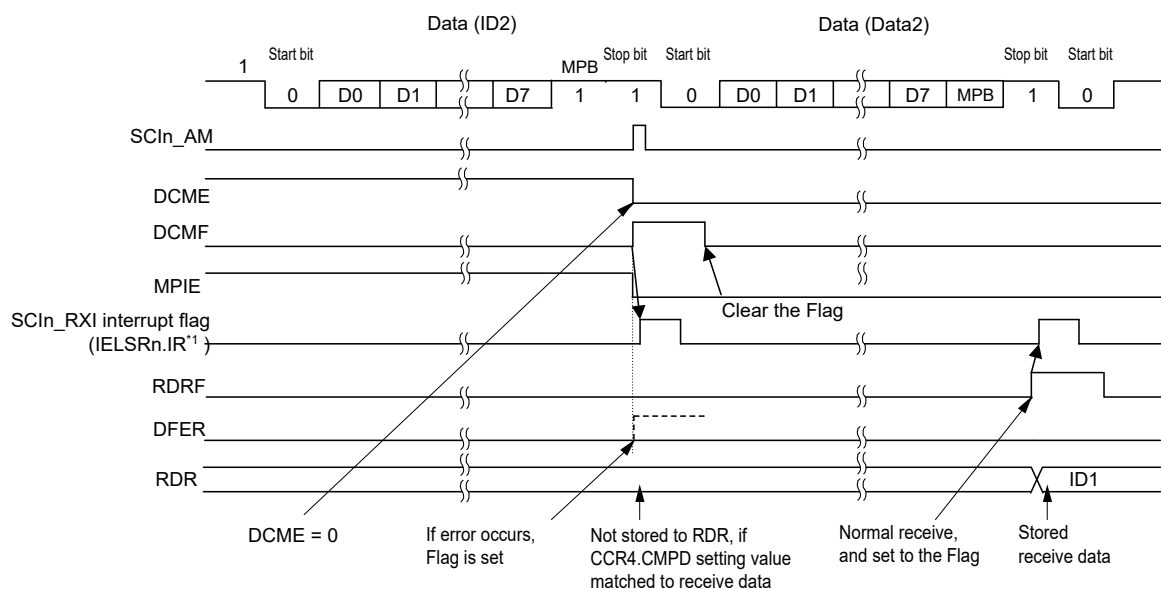
(b) Example of Compare matched between receive data and CCR4.CMPD[8:0] (8bit length/Multi-processor mode)

Note 1. See [section 14, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

Figure 38.9 Example of address match (1) normal mode and multi-processor mode



(a) Example of Compare unmatched between receive data and CCR4.CMPD (8bit length/IDSEL = 1/Multi-processor mode)



(b) Example of Compare matched between receive data and CCR4.CMPD (8bit length/IDSEL = 1/Multi-processor mode)

Note 1. See [section 14, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

Figure 38.10 Example of address match (2) multi-processor mode

38.3.7 SCI Initialization in Asynchronous Mode

Before transmitting and receiving data, start by writing the initial value 0 to the CCR0.TE and CCR0.RE (or writing the initial value to CCR0), then continue through the SCI initialization procedure (select non-FIFO or FIFO) shown in [Table 38.28](#) and [Table 38.29](#). Whenever the operating mode or transfer format is to be changed, the CCR0.TE and CCR0.RE bits must be initialized before the change is made.

When the external clock is used in Asynchronous mode, ensure that the clock signal is supplied during initialization.

Note: Setting the CCR0.RE bit to 0 initializes neither the ORER, FER, RDRF, RDAT, PER, and DR flags in CSR nor RDR. When FIFO selected, even if the TE bit is set as 0, the TEND flag for the selected FIFO buffer is not initialized. Please be also careful at the time of change in the operation mode.

Note: Switching the value of the CCR0.TE bit from 0 to 1 while the CCR0.TIE bit is 1 leads to the generation of an SCIn_TXI interrupt request.

Table 38.28 Example flow of SCI initialization in Asynchronous mode with non-FIFO selected

No.	Step Name	Description
1	Start initialization	—
2	Set CCR0	Set CCR0.TEIE, TIE, RIE, TE, RE to 0. If you have not changed from the initial settings, you can skip this step.
3	Set CCR3	Set up following function and communication mode. Driver control function for RS-485, FIFO no-use, Multi-Processor mode, Communication mode (MOD[2:0] = 000b) Transmission / reception format Clock enable (Leave the initial value when outputting the clock) Leave unused bits at their initial values.
4	Set CCR2	Set up the bit-rate-modulation function ^{*1 *2} , select clock, set bit rate ^{*2}
5	Set CCR1	Set up the Noise-filter function, the loop-back function, communication pin status, the parity check function, and the CTSn_RTSn function.
6	Set CCR4	Set up the adjust sampling timing function and the adjust transmit timing function.
7	Set the I/O port functions	Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.
8	Set CCR3	Set clock enable bit (CKE[1:0]) at this step when outputting the clock. After this register setting, the clock pin will be in the output state immediately.
9	Set CFCLR	Write 1 to the following bits and clear the corresponding flag. CFCLR.RDRFC, FER, PER, MFFC, ORERC, DFERC, DPERC, DCMFC, ERSC If it's initialization flow after a reset, you can skip this step.
10	Set CCR0	Set the TE or RE bit to 1. To enable interrupts, set the TE bit and TIE bit, and the RE bit and RIE bit to 1 with one instruction at the same time. Setting the TE and RE bits allows TXDn and RXDn to be used.
11	Initialization completed	—

Note 1. If you do not use the bit rate modulation function, you do not need to set it.

Note 2. If you use an external clock, you do not need to set it.

Table 38.29 Example flow of SCI initialization in Asynchronous mode with FIFO selected (1 of 2)

No.	Step Name	Description
1	Start initialization	—
2	Set CCR0	Set CCR0.TEIE, TIE, RIE, TE, RE to 0. If you have not changed from the initial settings, you can skip this step.
3	Set CCR3	Set up following function and communication mode. Driver control function for RS-485, FIFO use, Multi-Processor mode, Communication mode (MOD [2:0] = 000b) Transmission / reception format Clock enable (Leave the initial value when outputting the clock) Leave unused bits at their initial values.
4	Set CCR2	Set up the bit-rate-modulation function ^{*1 *2} , select clock, set bit rate ^{*2}
5	Set CCR1	Set up the Noise-filter function, the loop-back function, communication pin status, the parity check function, and the CTSn_RTSn function.
6	Set CCR4	Set up the adjust sampling timing function and the adjust transmit timing function.
7	Set FCR	Set the TFRST and RFRST to 1 to empty FIFO. Set the DRES, TTRG[4:0], RTRG[4:0], and RSTRG[4:0] bits

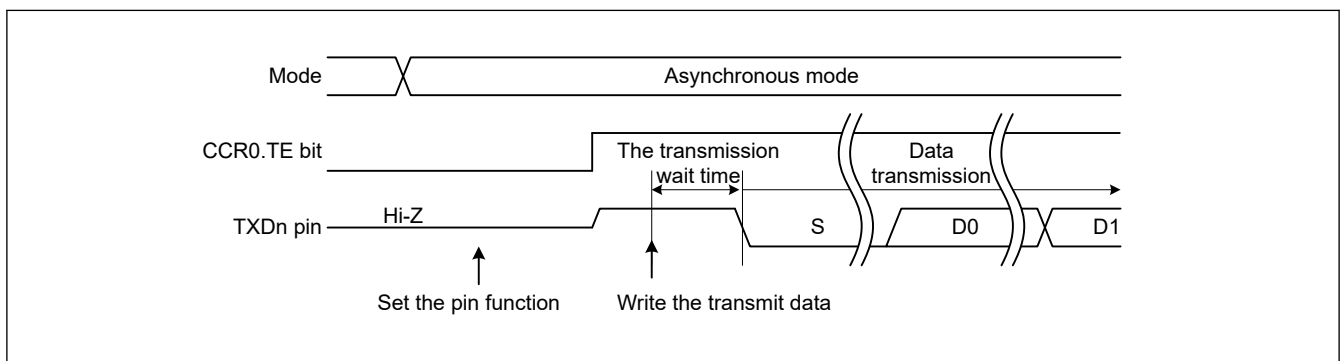
Table 38.29 Example flow of SCI initialization in Asynchronous mode with FIFO selected (2 of 2)

No.	Step Name	Description
8	Set the I/O port functions	Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.
9	Set CCR3	If you select a clock output in Asynchronous mode, set the CKE[1:0] bit here. After setting the register, the clock pin will be in the output state immediately. But the clock operates after setting TE or RE to 1.
10	Set CFCLR	Write 1 to the following bits and clear the corresponding flag. CFCLR.RDRFC, FERC, PERC, MFFC, ORERC, DFERC, DPERC, DCMFC, ERSC If it's initialization flow after a reset, you can skip this step.
11	Set FFCLR	Write 1 to The FFCLR.BRKC, DRC and clear the corresponding flag. If it's initialization flow after a reset, you can skip this step.
12	Set CCR0	Set the TE or RE bit to 1. To enable interrupts, set the TE bit and TIE bit, and the RE bit and RIE bit to 1 with one instruction at the same time. Setting the TE and RE bits allows TXDn and RXDn to be used.
13	Initialization completed	—

Note 1. If you do not use the bit rate modulation function, you do not need to set it.

Note 2. If you use an external clock, you do not need to set it.

Figure 38.11 shows an example of the timing when data is transmitted after reset is released, and SCI is set to Asynchronous mode according to Table 38.28 or Table 38.29. As shown in the figure, when the pin function is set to the TXDn pin, the CCR0.TE bit is 0, so the pin is high impedance. When transmit data is written after setting the CCR0.TE bit to 1, data transmission starts. There is a transmission wait time from writing TDR to data transmission starts. In Asynchronous mode, TXDn is high during this period.

**Figure 38.11 Data transmission timing example in Asynchronous mode**

38.3.8 Serial Data Transmission in Asynchronous Mode

(1) Non-FIFO selected

Figure 38.12, Figure 38.13, Figure 38.14 and Figure 38.15 show examples of serial transmission in Asynchronous mode.

In serial transmission, the SCI operates as described in this section.

- The SCI transfers data from the TDR register to the TSR register when data is written to TDR in the SCIn_TXI interrupt handling routine.
The SCIn_TXI interrupt request at the beginning of transmission is generated when the CCR0.TE and CCR0.TIE bits are set to 1 simultaneously by a single instruction.
- Transmission starts after the CCR1.CTSE bit is set to 0 (CTS function is disabled) or a low level on the CTSn_RTsn pin causes data transfer from the TDR register to the TSR register. If the CCR0.TIE bit is 1, an SCIn_TXI interrupt request is generated. Continuous transmission is possible by writing the next transmit data to the TDR register in the SCIn_TXI interrupt handling routine before transmission of the current transmit data is complete. When SCIn_TEI interrupt requests are in use, set the CCR0.TIE bit to 0 (SCIn_TXI interrupt requests are disabled) and the CCR0.TEIE bit to 1 (an SCIn_TEI interrupt request is enabled) after the last of the data to be transmitted is written to the TDR register from the handling routine for SCIn_TXI requests.
- Data is sent from the TXDn pin in the following order:
 - Start bit

- Transmit data
 - Parity bit or multi-processor bit (can be omitted depending on the format)
 - Stop bit
- The SCI checks for update of the TDR register on output of the stop bit.
 - When the TDR register is updated, setting the CCR1.CTSE bit to 0 (CTS function is disabled) or a low level input on the CTSn_RTSn pin causes transfer of the next transmit data from the TDR register to the TSR register and transmission of the stop bit, after which serial transmission of the next frame starts.
 - If the TDR register is not updated, the CSR.TEND flag is set to 1, the stop bit is sent, and the mark state is entered, in which 1 is output. If the CCR0.TEIE bit is 1, the CSR.TEND flag is set to 1 and an SCIn_TEI interrupt request is generated.

Figure 38.12, Figure 38.13, Figure 38.14 and Figure 38.15 show examples of serial transmission in Asynchronous mode.

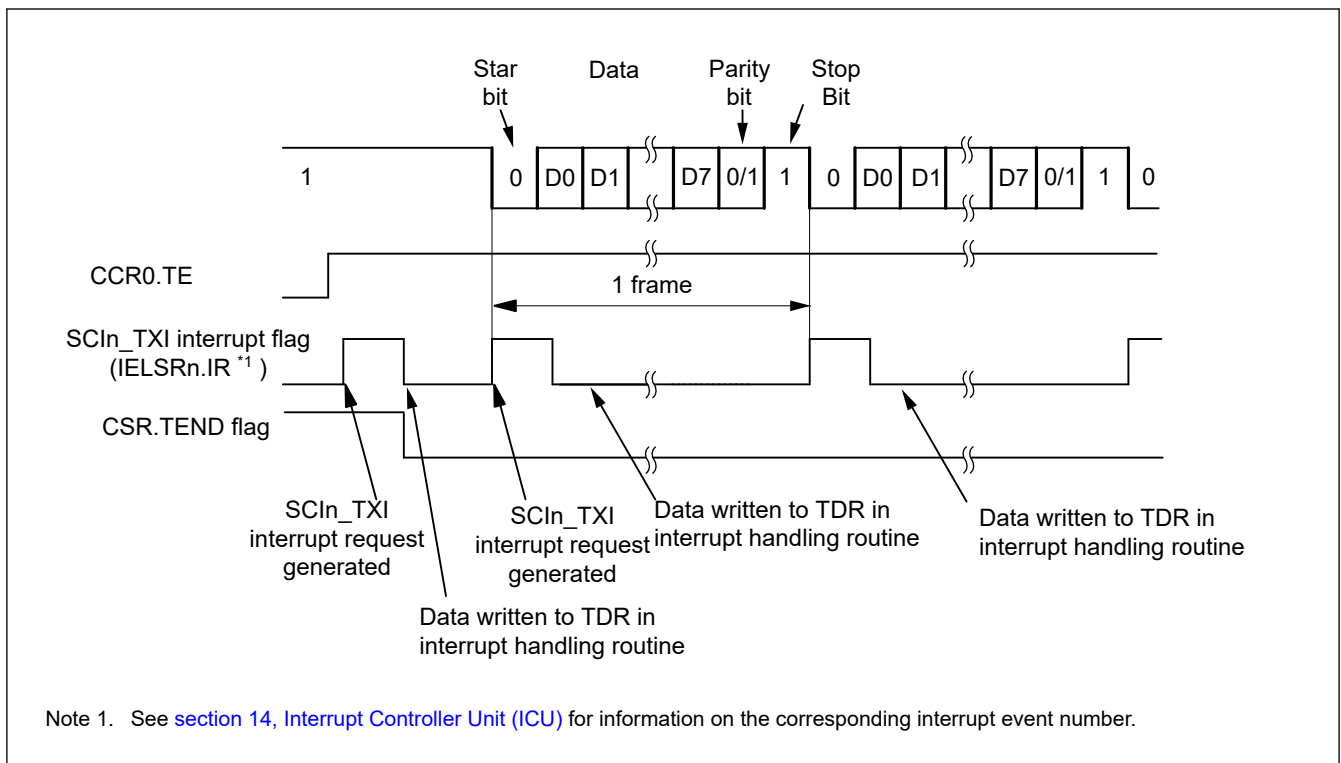


Figure 38.12 Example operation for serial transmission in Asynchronous mode (1) with 8-bit data, parity bit, 1 stop bit, CTS function not used, and at the beginning of transmission

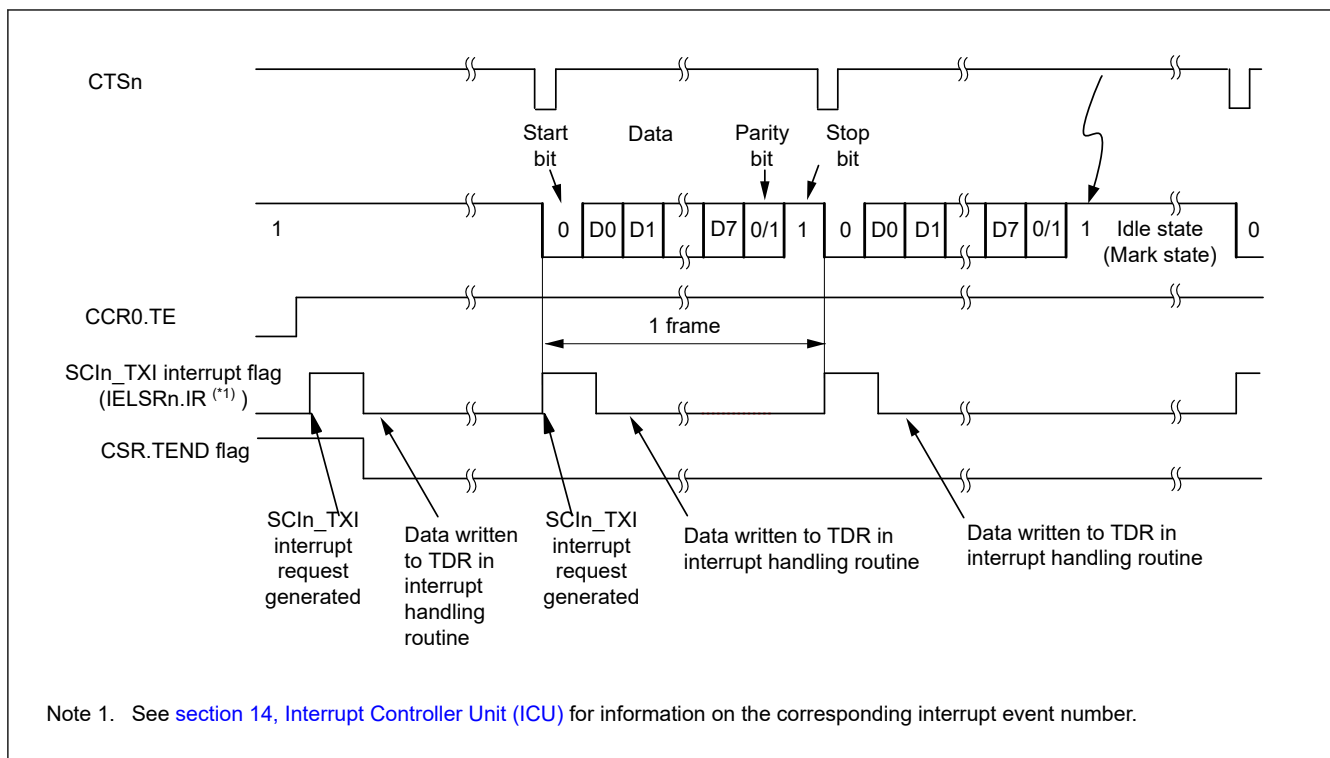


Figure 38.13 Example operation for serial transmission in Asynchronous mode (2) with 8-bit data, parity bit, one stop bit, CTS function used, and at the beginning of transmission

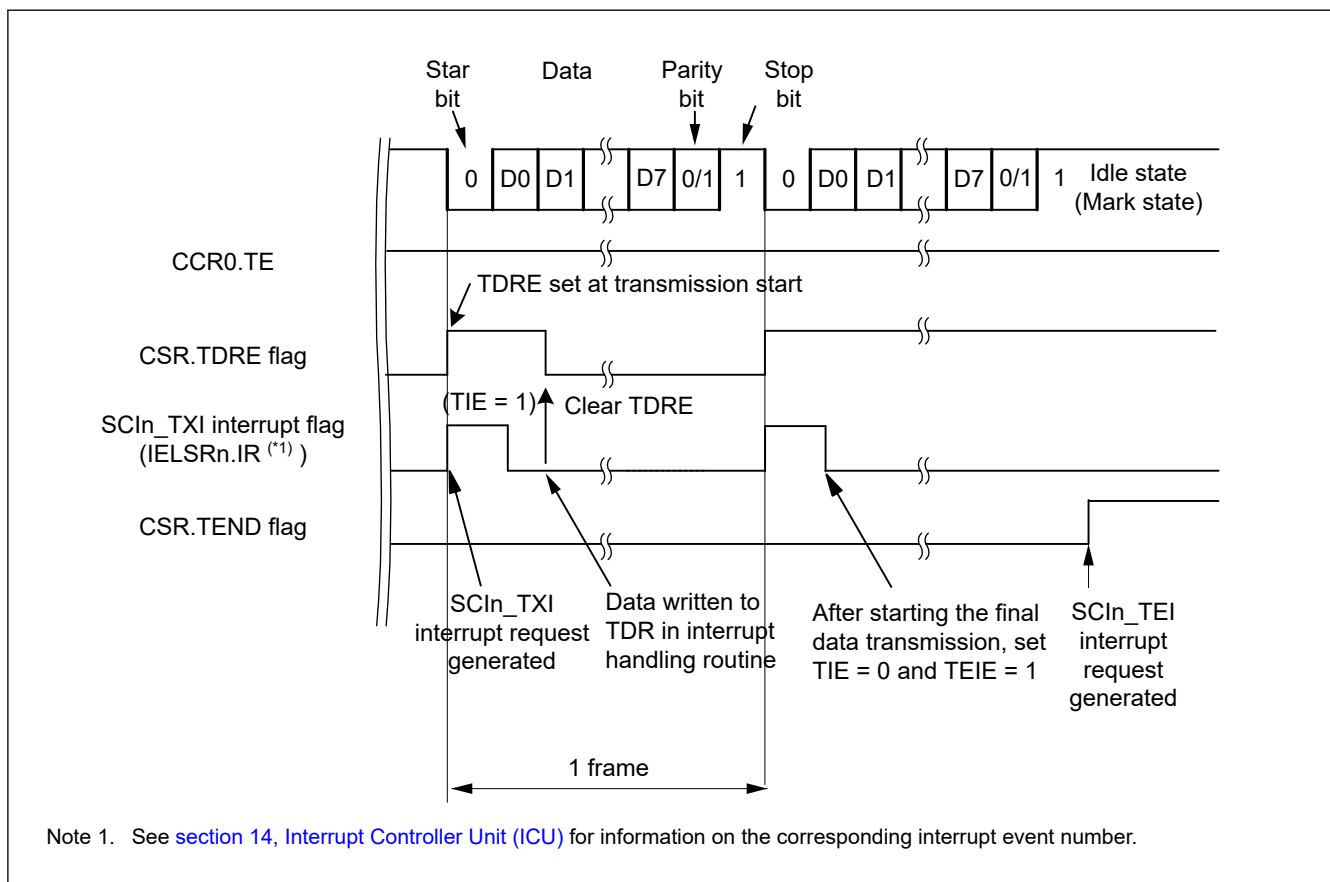


Figure 38.14 Example operation for serial transmission in Asynchronous mode (3) with 8-bit data, parity bit, one stop bit, CTS function not used, and from the middle of transmission until transmission completion

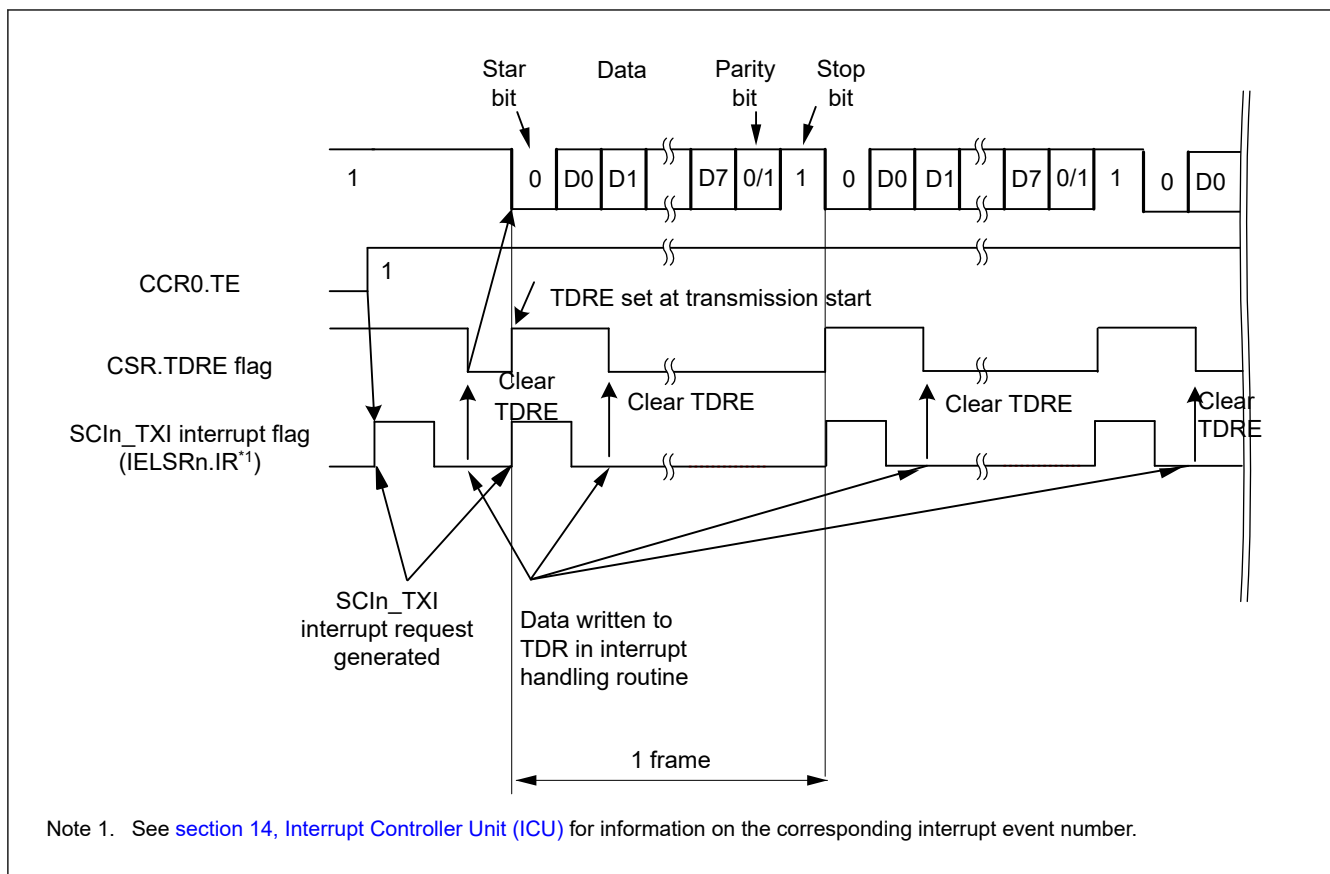


Figure 38.15 Example of Operation for Serial Transmission in Asynchronous Mode (4)(with 8-Bit Data, Parity, 1 Stop Bit, CTS Function Not Used, from the Middle of Transmission until Transmission Completion)

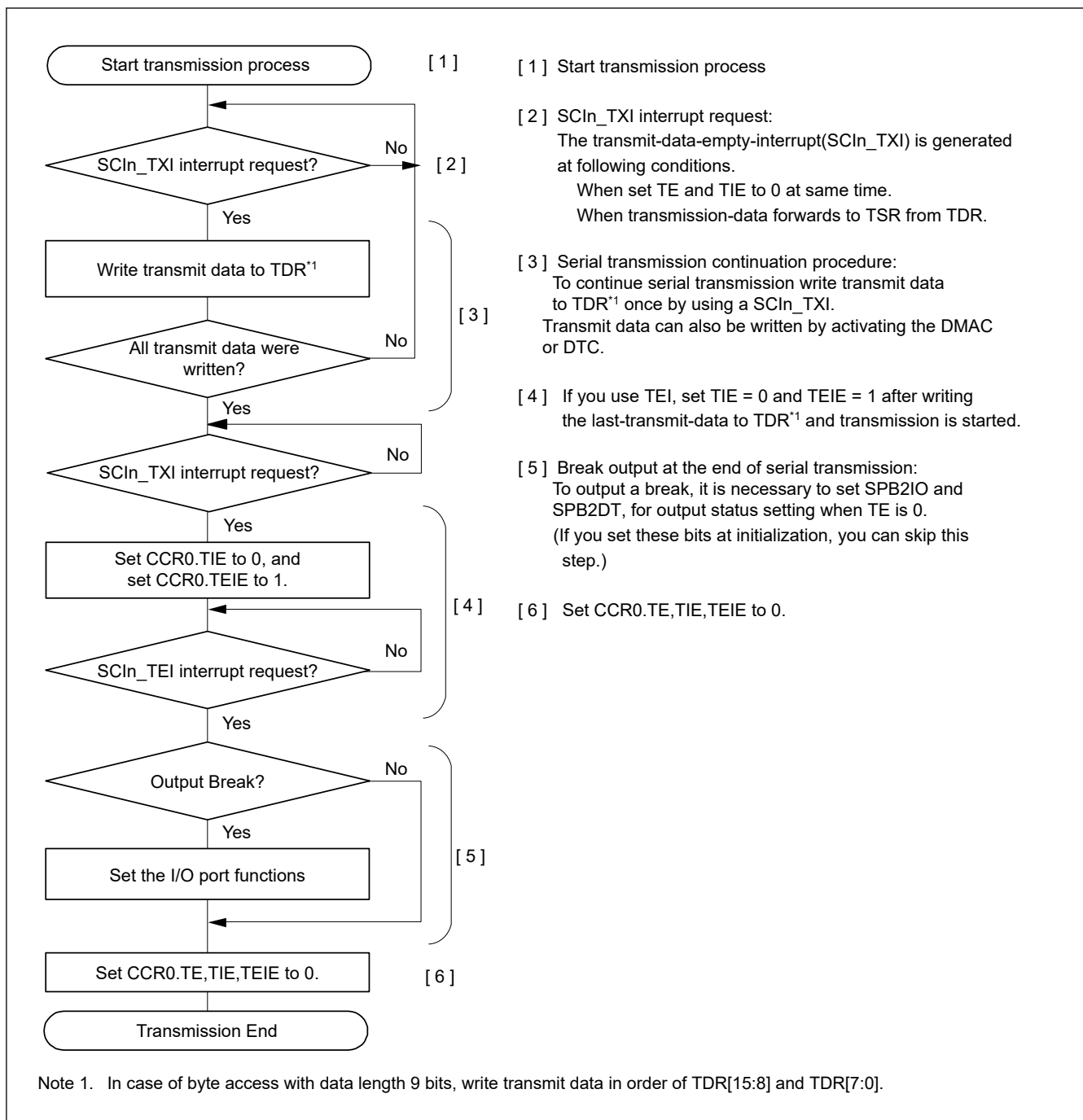


Figure 38.16 Example flow of serial transmission in Asynchronous mode with non-FIFO selected

(2) FIFO selected

Figure 38.17 shows an example of a data format that is written to TDR register in asynchronous mode with FIFO selected.

Data corresponding to the data length is set to TDR[8:0]. Write 0 for unused bits. Write in order from TDR[15:8] to TDR[7:0].

Data Length	Register setting		Transmit data in TDR[15:0]															
	CCR3. CHR[1:0]		b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
7 bit	1	1	—	—	—	—	—	—	MPB T	—	—							
8 bit	1	0	—	—	—	—	—	—	MPB T	—								
9 bit	0	Don't Care	—	—	—	—	—	—	MPB T									

Note: —: Invalid. The write value should be 0.

Figure 38.17 Data format written to transmit-FIFO(TDR) with FIFO selected

In serial transmission, the SCI operates as described in this section.

1. The SCI transfers data from the TDR register to the TSR register when data is written to TDR in the SCIn_TXI interrupt handling routine. The amount of data that can be written to TDR is 16 minus FTSR.T[5:0] bytes. The SCIn_TXI interrupt request at the beginning of transmission is generated when the CCR0.TE and CCR0.TIE bits are set to 1 simultaneously by a single instruction.
2. Transmission starts after the CCR1.CTSE bit is set to 0 (CTS function is disabled) or a low level on the CTSn_RTSn pin causes data transfer from the TDR register to the TSR register. When the amount of transmit data written in TDR is equal to or less than the specified transmit triggering number, CSR.TDFE is set to 1. If the CCR0.TIE bit is 1, an SCIn_TXI interrupt request is generated. Continuous transmission is possible by writing the next transmit data to TDR in the SCIn_TXI interrupt handling routine before transmission of the current transmit data is complete. When SCIn_TEI interrupt requests are in use, set the CCR0.TIE bit to 0 (SCIn_TXI interrupt requests are disabled) and the CCR0.TEIE bit to 1 (an SCIn_TEI interrupt request is enabled) after the last of the data to be transmitted is written to the TDR register from the handling routine for SCIn_TXI requests.
3. Data is sent from the TXDn pin in the following order:
 - Start bit
 - Transmit data
 - Parity bit or multi-processor bit (can be omitted depending on the format)
 - Stop bit
4. On output of the stop bit, the SCI checks whether non-transmitted data remains in the TDR register.
5. When data is set to transmit-FIFO (TDR), setting the CCR1.CTSE bit to 0 (CTS function is disabled) or a low level input on the CTSn_RTSn pin causes transfer of the next transmit data from TDR to TSR and transmission of the stop bit, after which serial transmission of the next frame starts.
6. If data is not set in transmit-FIFO (TDR), the TEND flag in CSR is set to 1, the stop bit is sent, and the mark state is entered in which 1 is output. If the CCR0.TEIE bit is 1, an SCIn_TEI interrupt request is generated.

Figure 38.18 shows an example flow of serial transmission in Asynchronous mode with FIFO selected.

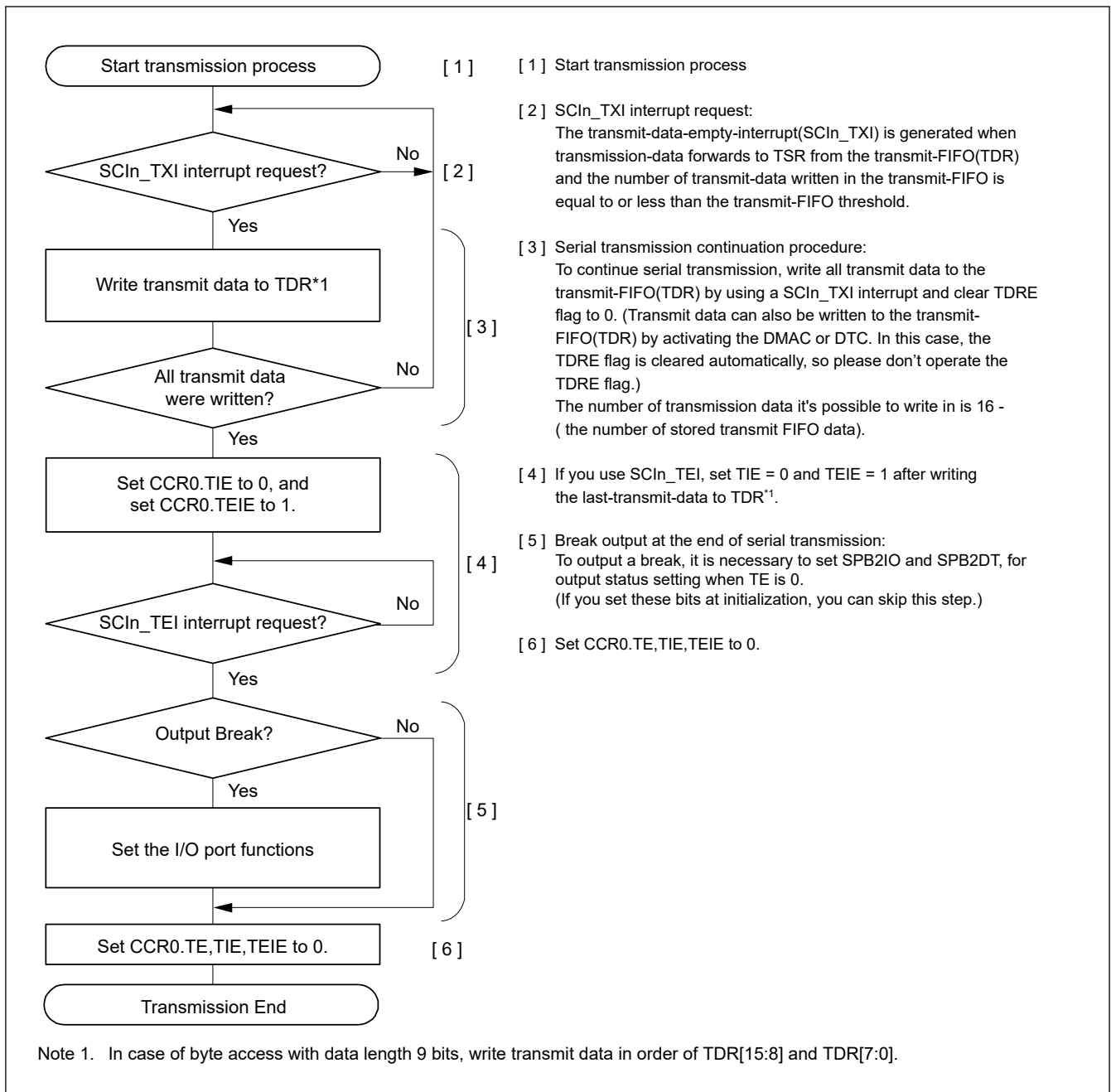


Figure 38.18 Example flow of serial transmission in Asynchronous mode with FIFO selected

38.3.9 Serial Data Reception in Asynchronous Mode

(1) Non-FIFO selected

Figure 38.19 and Figure 38.20 show an example of the operation for serial data reception in Asynchronous mode.

In serial data reception, the SCI operates as follows:

1. When the value of the CCR0.RE bit becomes 1, the output signal on the CTSn_RTSn pin goes low.
2. The SCI monitors the communications line and when it detects a start bit, the SCI performs internal synchronization, stores receive data in RSR, and checks the parity bit and stop bit.
3. If an overrun error occurs, the CSR.ORER flag is set to 1. If the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated. Receive data is not transferred to the RDR register.
4. If a parity error is detected, the CSR.PER flag is set to 1 and receive data is transferred to the RDR register. If the SCR.RIE bit is 1, an SCIn_ERI interrupt request is generated.

5. If a framing error is detected, the CSR.FER flag is set to 1 and receive data is transferred to the RDR register. If the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated.
6. When reception finishes successfully, receive data is transferred to the RDR register. If the CCR0.RIE bit is 1, an SCIn_RXI interrupt request is generated. Continuous reception is enabled by reading the receive data transferred to the RDR register in the SCIn_RXI interrupt handling routine before reception of the next receive data is complete. Reading the received data that was transferred to the RDR register causes the CTSn_RTSn pin to output low.

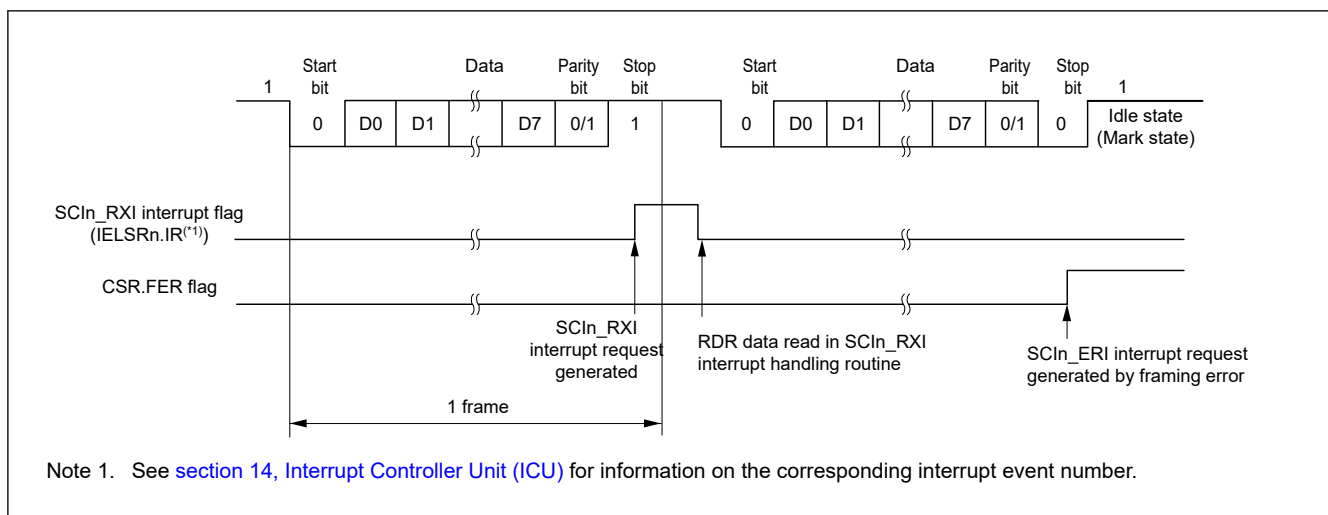


Figure 38.19 Example of SCI operation for serial reception in Asynchronous mode (1) when the RTS function is not used, and with 8-bit data, parity bit, and 1 stop bit

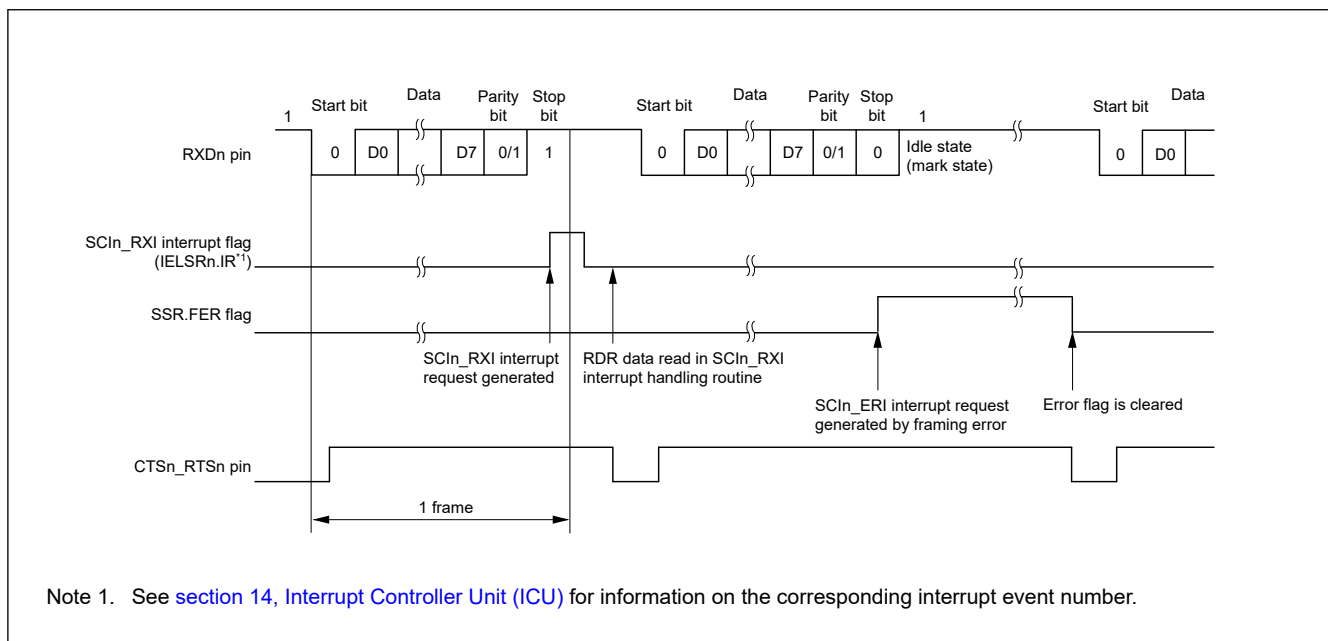


Figure 38.20 Example of SCI operation for serial reception in Asynchronous mode (2) when RTS function is used, and with 8-bit data, parity bit, and 1 stop bit

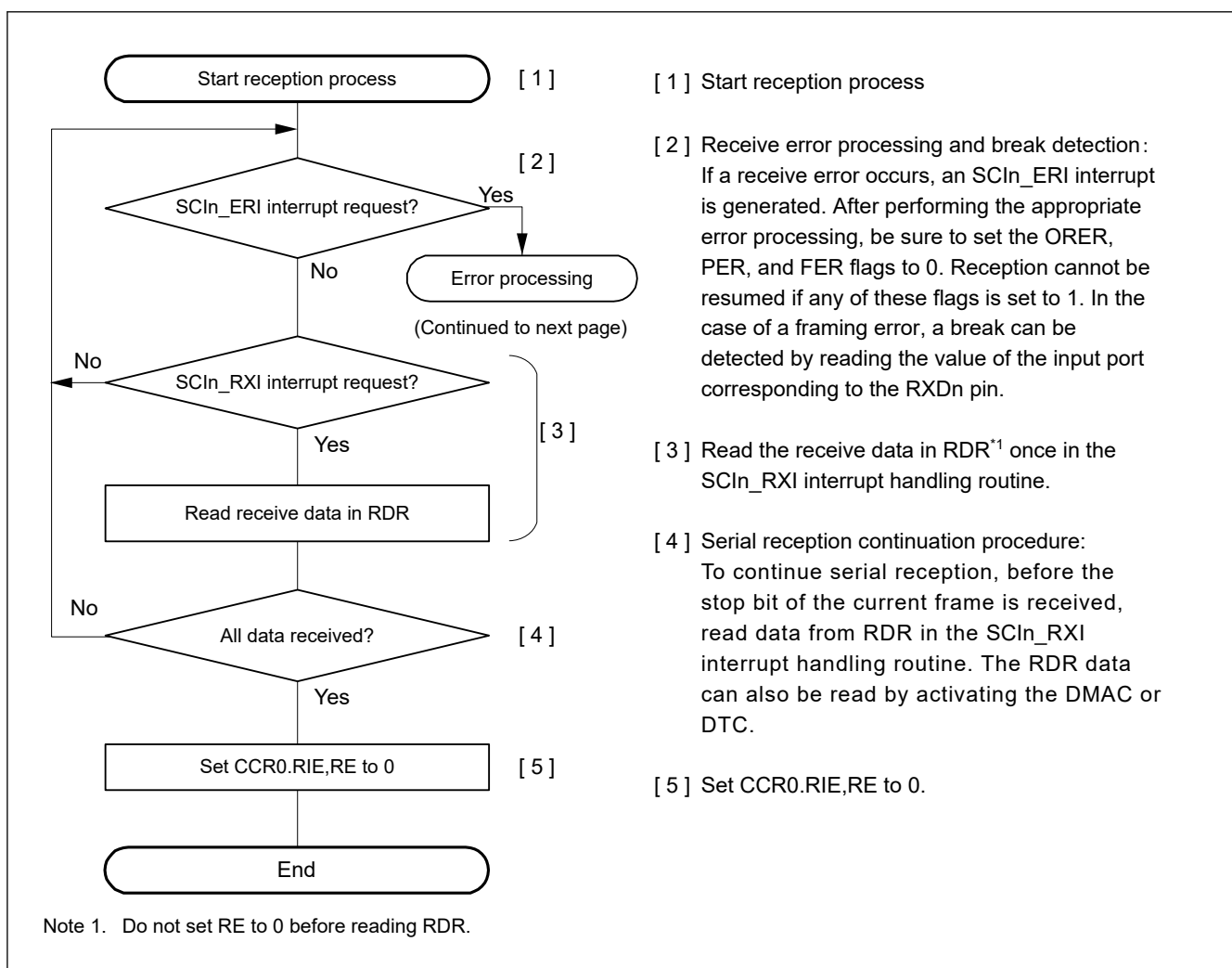
[Table 38.30](#) lists the states of the flags in the CSR status register and receive data handling when a receive error is detected.

If a receive error is detected, an SCIn_ERI interrupt request is generated but an SCIn_RXI interrupt request is not generated. Data reception cannot be resumed while the receive error flag is 1. Accordingly, set the ORER, FER, and PER bits to 0 before resuming reception. In addition, be sure to read the RDR register during overrun error processing. When a reception is forced to terminate by setting the CCR0.RE bit to 0 during operation, read the RDR register because received data that is not yet read might be left in the RDR.

[Figure 38.21](#) and [Figure 38.22](#) show example flows of serial data reception.

Table 38.30 Flags in CSR Status Register and receive data handling

Flags in the CSR Status Register			Receive data	Receive error type
ORER	FER	PER		
1	0	0	Lost	Overrun error
0	1	0	Transferred to RDR	Framing error
0	0	1	Transferred to RDR	Parity error
1	1	0	Lost	Overrun error + framing error
1	0	1	Lost	Overrun error + parity error
0	1	1	Transferred to RDR	Framing error + parity error
1	1	1	Lost	Overrun error + framing error + parity error

**Figure 38.21 Example flow of serial reception in Asynchronous mode with non-FIFO selected and Address Matching Disabled (1)**

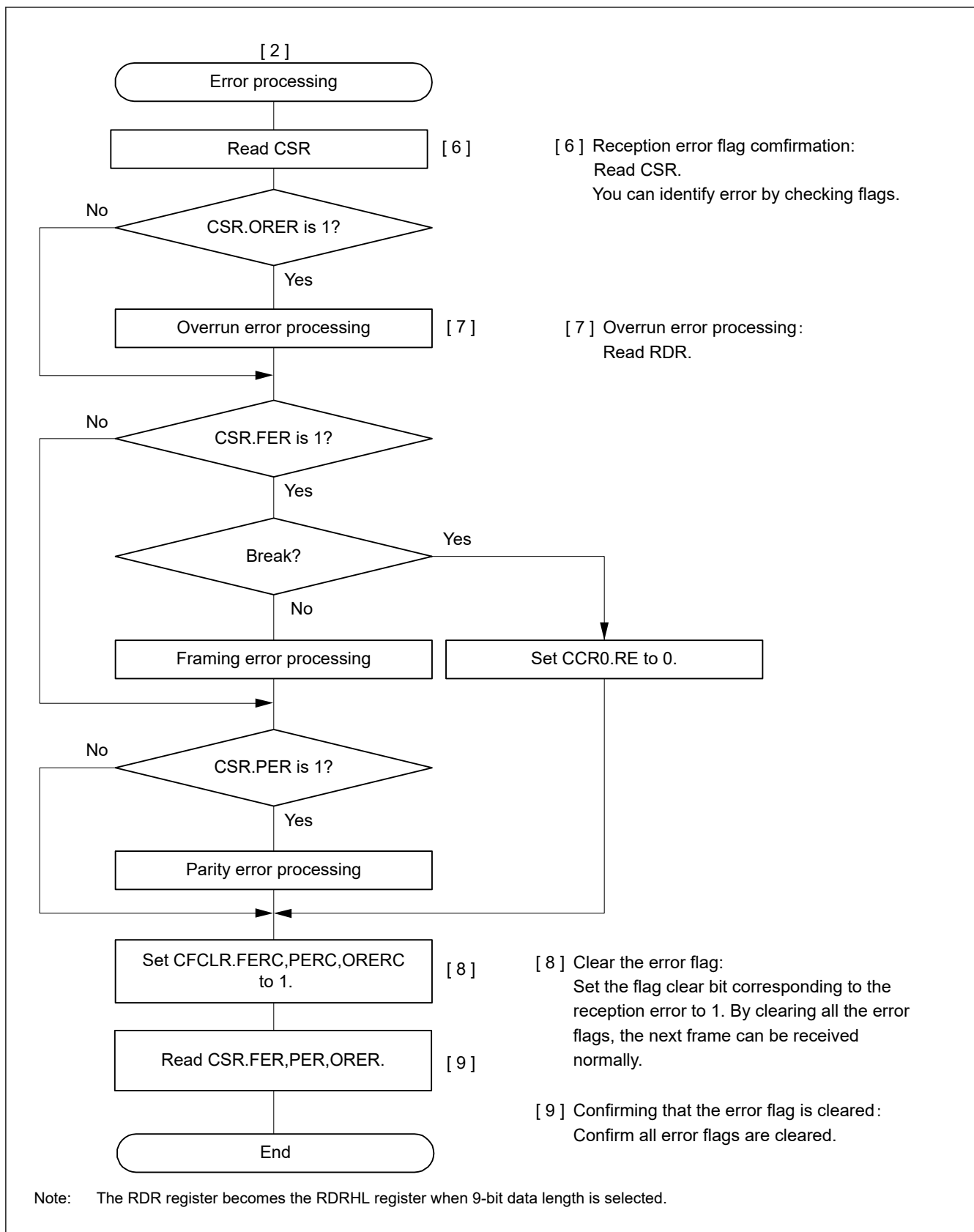


Figure 38.22 Example flow of serial reception in Asynchronous mode with non-FIFO selected and Address Matching Disabled (2)

(2) FIFO selected

Figure 38.23 shows an example of a data format that is written to Receive-FIFO(RDR) register in Asynchronous mode.

In Asynchronous mode, 0 is written to the MPB bit in the RDR register. Data that corresponds to the data length is written to RDR. Unused bits are written as 0. If software reads RDR, the SCI updates FER, PER, and receive data (RDAT[8:0]) in the RDR register with the next data. The flags RDF, ORER, and DR in the RDR register always reflect the associated flags in the CSR register.

Data Length	Register Setting		Receive flag in RDR[31:0], MPB, RDAT[8:0]															
	CCR3.CHR[1:0]		b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
7bit	1	1	-	-	-	FFER	FPER	DR	MPB	0	0							
8bit	1	0	-	-	-	FFER	FPER	DR	MPB	0								
9bit	0	Don't care	-	-	-	FFER	FPER	DR	MPB									
	CCR3.CHR[1:0]		b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
7bit	1	1	-	-	-	FER	PER	-	-	ORER	-	-	-	-	-	-	-	-
8bit	1	0	-	-	-	FER	PER	-	-	ORER	-	-	-	-	-	-	-	-
9bit	0	Don't care	-	-	-	FER	PER	-	-	ORER	-	-	-	-	-	-	-	-

Note: 0 is always read from the MPB flag (RDR [9] bit).
 When a 7-bit data length is selected, 0 is read from the RDAT [8:7] bits.
 When 8-bit data length is selected, 0 is read from the RDAT [8] bit.

Figure 38.23 Data format stored in receive-FIFO(RDR) with FIFO selected

Table 38.31 lists the states of the flags in CSR status register and receive data handling when a receive error is detected with FIFO selected. Figure 38.24 and Figure 38.25 show samples of flowcharts for serial data reception with FIFO selected.

In serial data reception, the SCI operates as follows:

1. When the value of the CCR0.RE bit becomes 1, the output signal on the CTSn_RTSn pin goes low.
2. The SCI monitors the communications line and, when it detects a start bit, the SCI performs internal synchronization, stores receive data in the RSR register.
3. If an overrun error occurs during normal communications, the CSR.ORER flag is set to 1. If the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated. Receive data is not transferred to the RDR register.
4. If a parity error is detected, the PER flag and receive data are transferred to the RDR register. If the CCR0.RIE bit is set to 1, an SCIn_ERI interrupt request is generated.
5. If a framing error is detected, the FER flag and receive data are transferred to the RDR register. If the CCR0.RIE bit is set to 1, an SCIn_ERI interrupt request is generated.
6. After a framing error is detected and when SCI detects that the continuous receive data is zero for one frame, reception stops.
7. When the amount of data stored in the RDR register falls below the specified receive triggering number, and the next data is not received after 15 etu from the last stop bit in Asynchronous mode, the FRSR.DR flag is set to 1. When the CCR0.RIE bit is 1 and the FCR.DRES bit is 0, the SCI generates an SCIn_RXI interrupt request. When the FCR.DRES bit is 1, SCI generates an SCIn_ERI interrupt request.
8. When reception finishes successfully, receive data is transferred to the RDR register. The RDRF bit is set to 1 when the amount of receive data written to RDR is equal to or greater than the specified receive triggering number. If the CCR0.RIE bit is 1, an SCIn_RXI interrupt request is generated. Continuous reception is enabled by reading the receive data transferred to the RDR register in the SCIn_RXI interrupt handling routine, before an overrun error occurs. If the received data that is transferred to RDR is less than the RTS trigger number, the CTSn_RTSn pin outputs low.

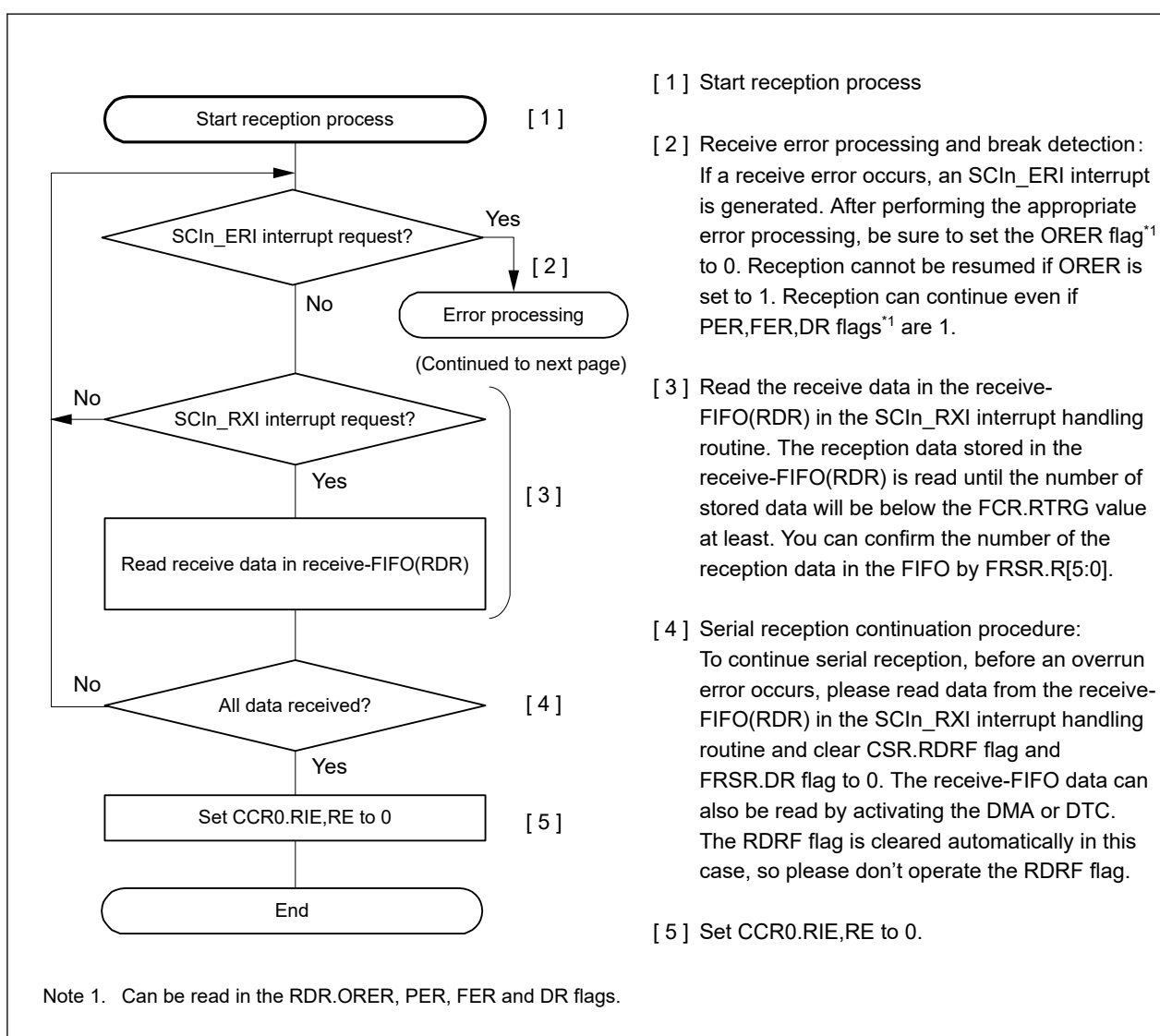
Table 38.31 Flags in the CSR Status Register and Receive Data Handling (FIFO selected) (1 of 2)

CSR value			Receive-FIFO (RDR)	Receive Error Type
ORER	FER*1	PER*1	RDAT[8:0]	
1	0	0	Lost	Overrun error
0	1	0	Transferred RDR	Framing error

Table 38.31 Flags in the CSR Status Register and Receive Data Handling (FIFO selected) (2 of 2)

CSR value			Receive-FIFO (RDR)	Receive Error Type
ORER	FER ^{*1}	PER ^{*1}	RDAT[8:0]	
0	0	1	Transferred RDR	Parity error
1	1	0	Lost	Overrun error + Framing error
1	0	1	Lost	Overrun error + Parity error
0	1	1	Transferred RDR	Framing error + Parity error
1	1	1	Lost	Overrun error + Framing error + Parity error

Note 1. This flag indicates whether there is an error in received data when reception is completed.

**Figure 38.24 Example flow of serial reception in Asynchronous mode with FIFO selected and Address Matching Enabled (1)**

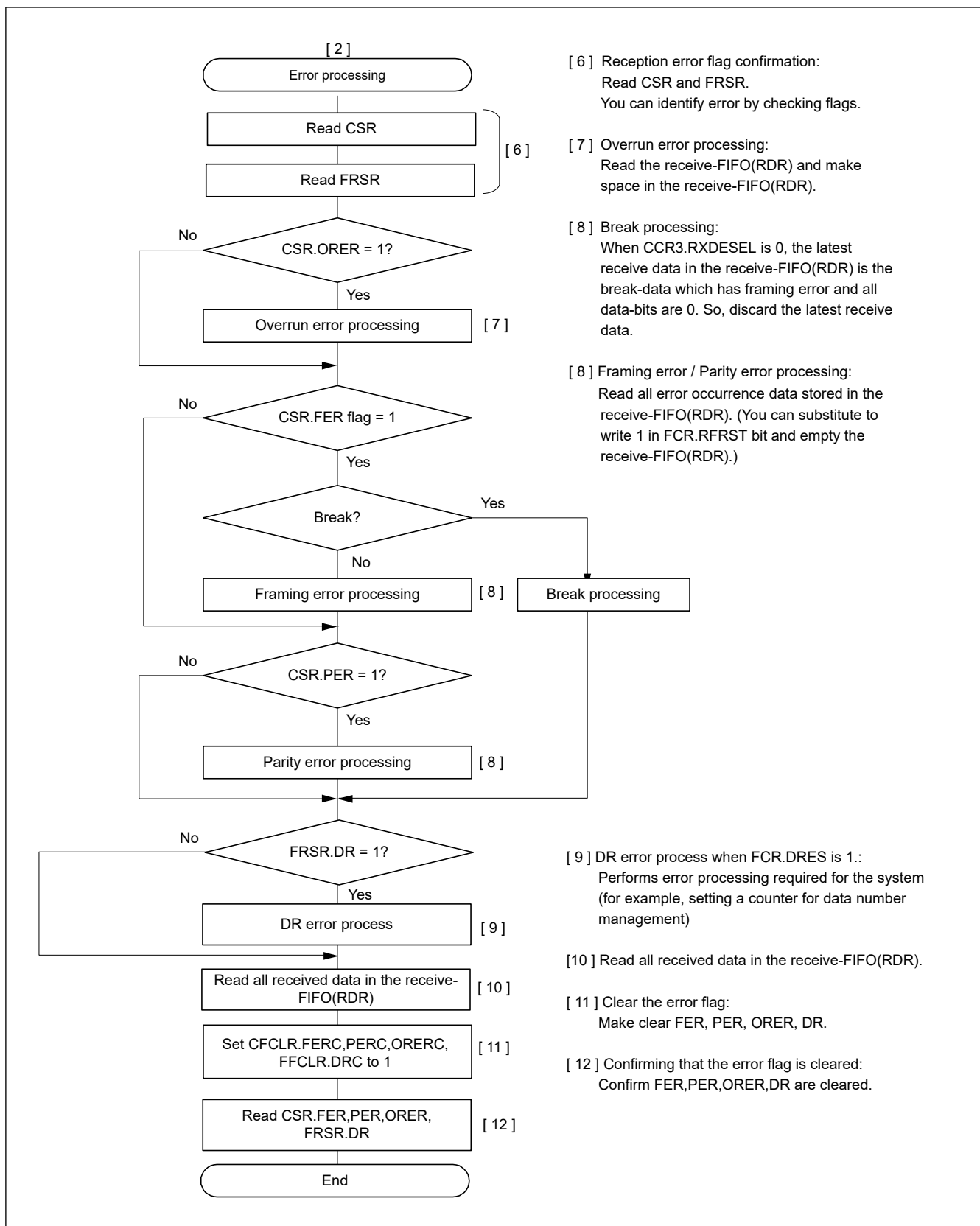


Figure 38.25 Example flow of serial reception in Asynchronous mode with FIFO selected Address Matching Disabled (2)

38.3.10 The Function of Adjust Receive Sampling Timing (Asynchronous Mode)

When there is the difference between the rising transfer time and the falling transfer time through a photo coupler, the receive sampling timing at middle of bit affects the reception margin. In this case, the receive sampling timing is able to adjust from the middle of bit to the optimum timing by using this function.

The receive sampling timing is adjusted from the middle of bit by following formula. The adjustable direction is set by CCR4.AJD. When adjusting backward (CCR4.AJD = 0), substitute AJD = +1 and substitute AJD = -1 when adjusting forward (CCR4.AJD = 1).

Adjusted sampling timing = the middle of bit + AJD × (base clock × the setting value of CCR4.AST[2:0])

The setting timing is limited by base clock cycles per 1 bit. For details, see [Table 38.32](#).

An overview of reception operation of the communication through a photo coupler with this function is shown in [Figure 38.26](#), [Figure 38.27](#) and [Figure 38.28](#), the explanation of operation with this function is shown in [Figure 38.29](#).

Do not use this function when there is no difference between the rising transfer time and the falling transfer time, because there is a possibility of deteriorating the reception margin.

Table 38.32 The acceptable value of CCR4 register (Asynchronous mode using internal clock)

CCR2.ABCSE	CCR2.ABCS	The number of base clock cycles/1bit	The acceptable value of CCR4	
			CCR4.AJD	CCR4.AST
1	x	6	0	000b – 010b ^{*1}
			1	
0	1	8	0	000b – 011b ^{*1}
			1	
0	0	16	0	000b – 111b
			1	

Note: x: Don't care

Note 1. When the value of CCR4.AST exceeds the acceptable value, sampling is done at default timing. (Adjustment of sampling is not done.)

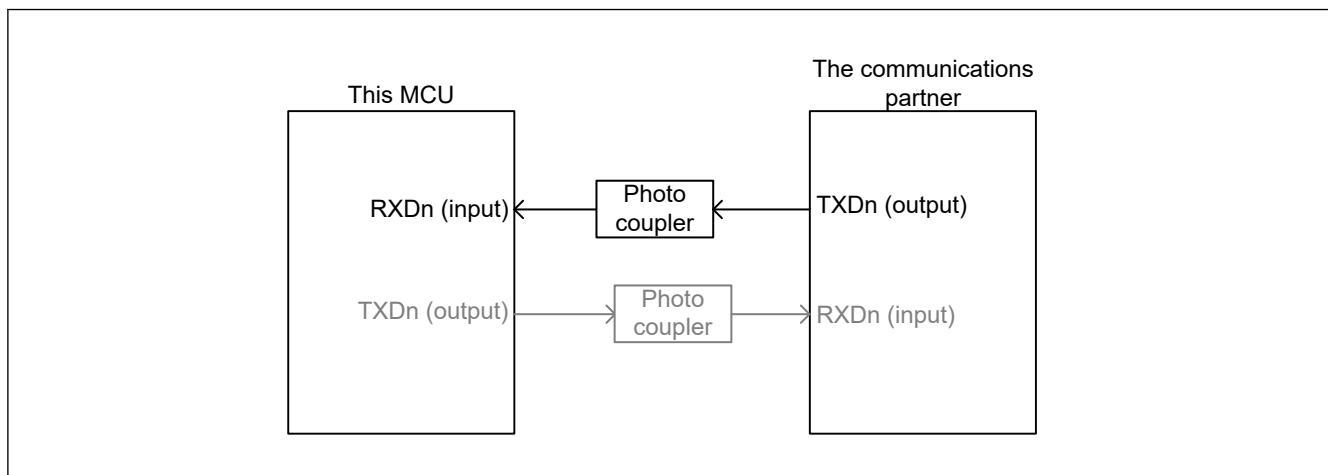
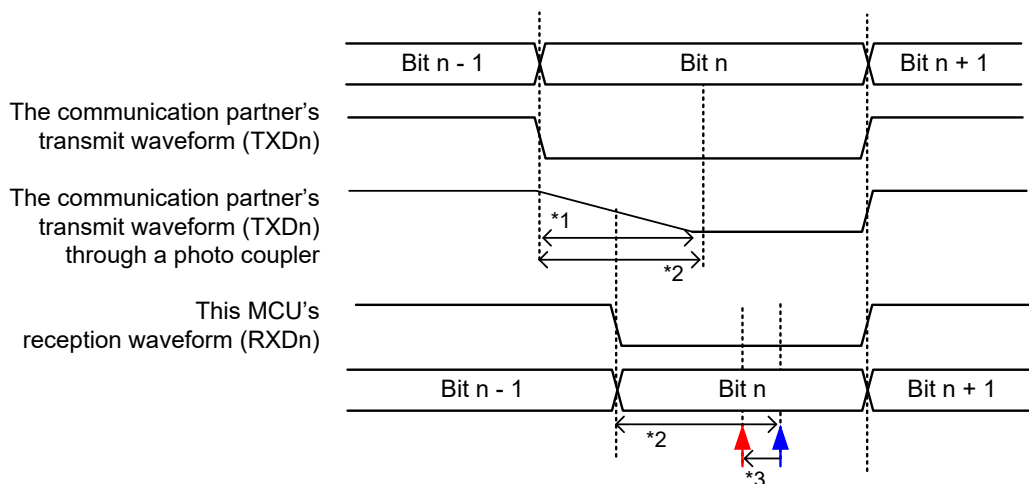


Figure 38.26 block diagram image of the reception through a photo coupler

(a) In the case of the falling transfer time >> rising transfer time

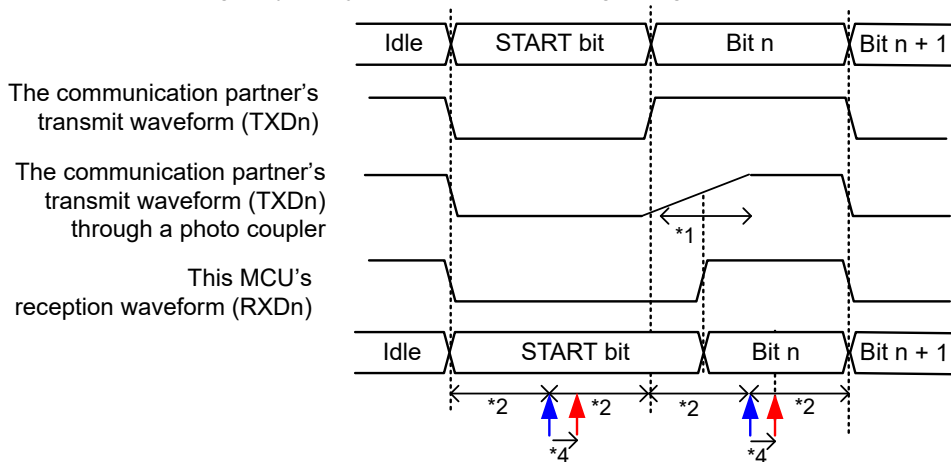
The falling edge of reception waveform is made dull like following chart.

In this case, you can sampling at the middle of bit if you adjust the receive sampling timing to forward (AJD = 1).

**(b) In the case of the falling transfer time << rising transfer time**

The rising edge of reception waveform is made dull like following chart.

So, the reception margin of communications partner will be bad. In this case, you can improve the reception margin if you adjust the receive sampling timing to back.



↑ The receive sampling timing when unadjusted (middle of bit)

↑ The adjusted receive sampling timing

Note: This waveform shows the operation image of adjustment in reception sampling timing.

Note 1. The dull period by a photo coupler

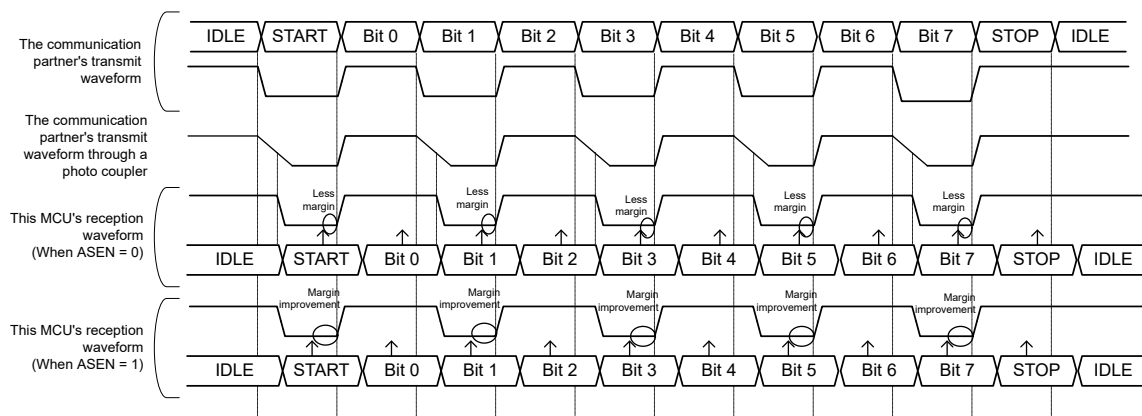
Note 2. Bit center timing at set communication rate

Note 3. When CCR4.AJD is 1, the receive sampling timing is shifted to forward by the setting value of CCR4.AST[2:0].

Note 4. When CCR4.AJD is 0, the Receive sampling timing is shifted to backward by the setting value of CCR4.AST[2:0].

Figure 38.27 Overview of reception operation of the communication through a photo coupler

In the case of the falling transfer time $\gg$ the rising transfer time by a photo coupler.



In the case of the falling transfer time $\ll$ the rising transfer time by a photo coupler.

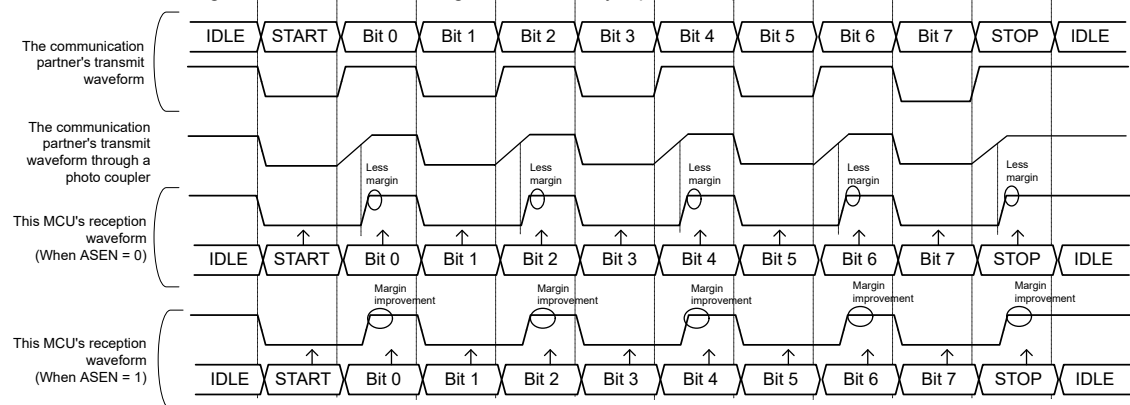


Figure 38.28 Example of improvement in reception margin by the reception sampling timing adjustment function

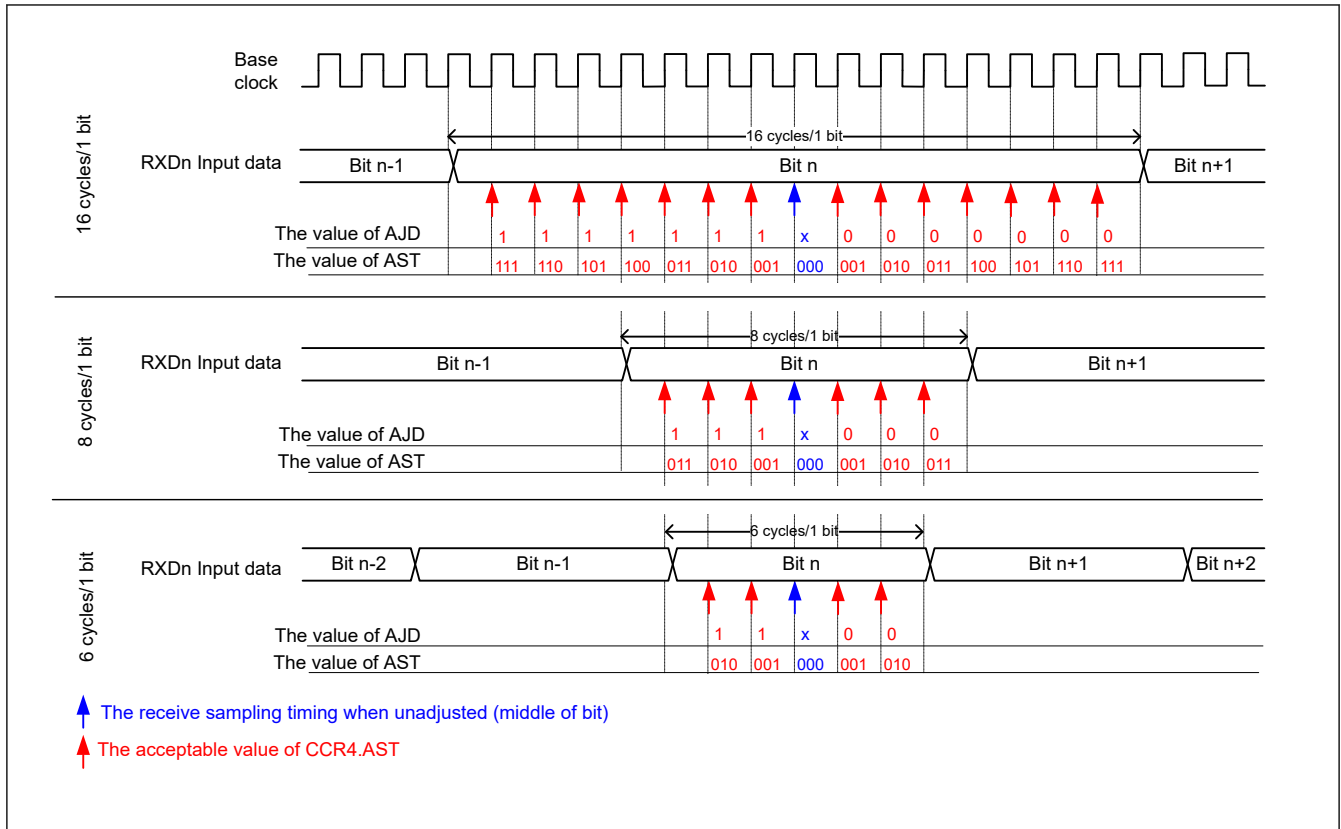


Figure 38.29 Overview of the adjustment operation for the reception sampling timing (Asynchronous mode using internal clock)

38.3.11 The Function of Adjust Transmit Timing (Asynchronous Mode)

In communication via a photo coupler or the like, when either the rising or falling transition time of the TXDn output signal is long, then a communication partner receive dulled waveform. In this case, the reception margin may be affected.

In these cases, make a communication partner to be sampling at middle of bit using the function of adjust transmit timing.

When CCR4.ATEN is 1, this function can adjust the edge timing at the timing calculated by the following formula for the edge set with CCR4.AET.

The adjustment edge timing = the base clock × CCR4.ATT[2:0]

In addition, the upper limit of the adjustment edge timing is limited by setting the base clock cycles. For details, see [Table 38.33](#).

A transmission movement image figure of the communication through a photo coupler with this function is shown in [Figure 38.30](#), [Figure 38.31](#) and [Figure 38.32](#), the overview of operation with this function is shown in [Figure 38.33](#) and [Figure 38.34](#).

Do not use this function when there is not the difference between the rising transfer time and the falling transfer time, there is a possibility of deteriorating the reception margin of a communication partner.

Table 38.33 The acceptable value of CCR4.AET and CCR4.ATT (Asynchronous mode using internal clock) (1 of 2)

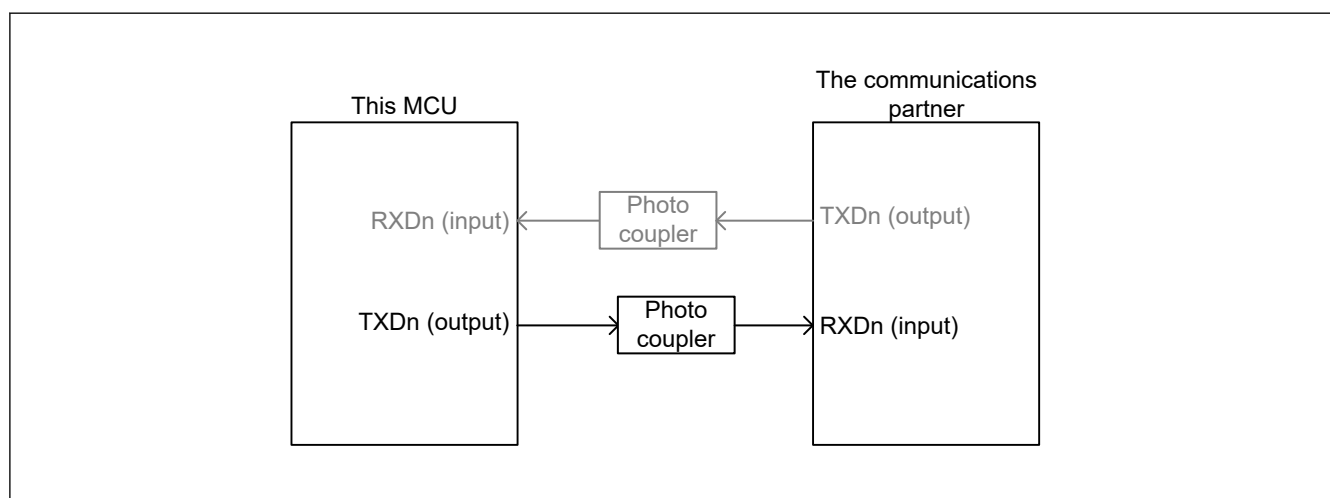
ABCSE2	ABCSE	ABCS	The number of base clock cycles/1bit	The acceptable value	
				AET	ATT [2:0]
1	0	x	4	0	000 to 011
				1	
0	1	x	6	0	000 to 101
				1	

Table 38.33 The acceptable value of CCR4.AET and CCR4.ATT (Asynchronous mode using internal clock) (2 of 2)

ABCSE2	ABCSE	ABCS	The number of base clock cycles/1bit	The acceptable value	
				AET	ATT [2:0]
0	0	1	8	0	000 to 111
				1	
0	0	0	16	0	000 to 111
				1	

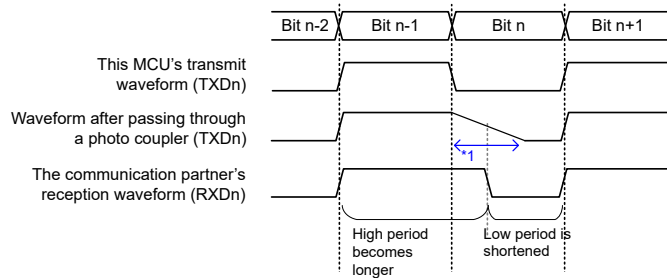
Note: x: Don't care

Note: When the value of ACTR.AET/ATT is out of the acceptable value, this SCI module does not adjust transmit timing.

**Figure 38.30** block diagram image of the transmission through a photo coupler

(a) In the case of the falling transfer time $\gg$ rising transfer time

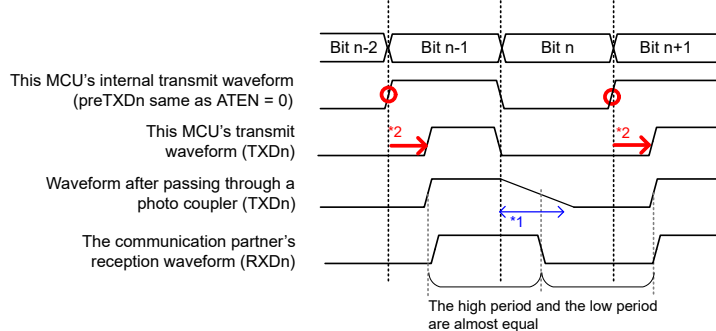
(a-1) The adjust transmit waveform function OFF (ATEN = 0)



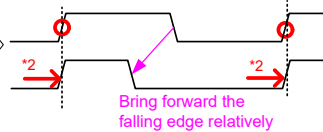
The falling edge of Transmission waveform is made dull like the figure on the left.
For the reception waveform of the communication partner, High of Bit n-1 becomes longer and Bit n Low becomes shorter.

As this kind of communication waveform collapses, there is a concern that Low value cannot be sampled depending on the sampling timing of the communication partner.

(a-2) The adjust transmit waveform function ON (ATEN = 1), and adjust the rising edge (AET = 0)



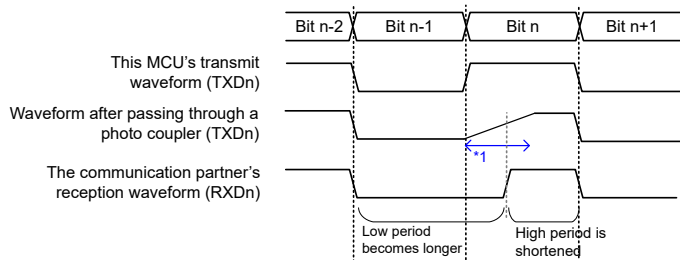
Thus, adjust the rising edge timing of TXDn waveform.
This IP can relatively bring forward the falling edge.



Then, the communication partner can receive the ideal waveform. The communication partner can sample data certainly and secure the reception margin.

(b) In the case of the falling transfer time $\ll$ rising transfer time

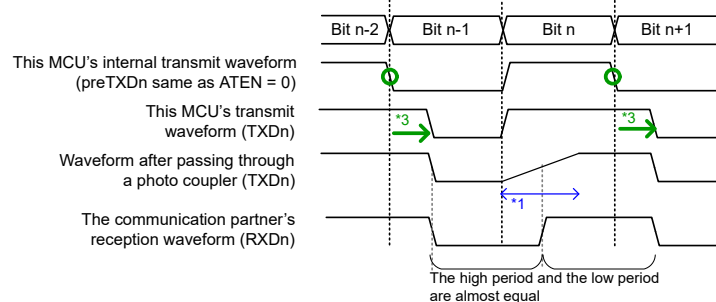
(b-1) The adjust transmit waveform function OFF (ATEN = 0)



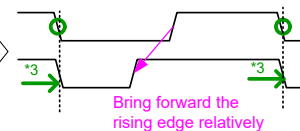
The rising edge of transmission waveform is made dull like the figure on the left.
For the reception waveform of the communication partner, Low of Bit n becomes longer and Bit n-1 High becomes shorter.

As this kind of communication waveform collapses, there is a concern that High value cannot be sampled depending on the sampling timing of the communication partner.

(b-2) The adjust transmit waveform function ON (ATEN = 1), and adjust the falling edge (AET = 1)



Thus, adjust the falling edge timing of TXDn waveform.
This IP can relatively bring forward the rising edge.



Then, the communication partner can receive the ideal waveform. The communication partner can sample data certainly and secure the reception margin.

Note: This waveform shows the operation image of adjustment transmit waveform.

Note 1. The dull period by a photo coupler

Note 2. When CCR4.AET is 0, the rising edge of transmit waveform timing is shifted to back by the setting value of CCR4.ATT[2:0]. This MCU transmit waveform bringing forward the falling edge relatively.

Note 3. When CCR4.AET is 1, the falling edge of transmit waveform timing is shifted to back by the setting value of CCR4.ATT[2:0]. This MCU transmit waveform bringing forward the rising edge relatively.

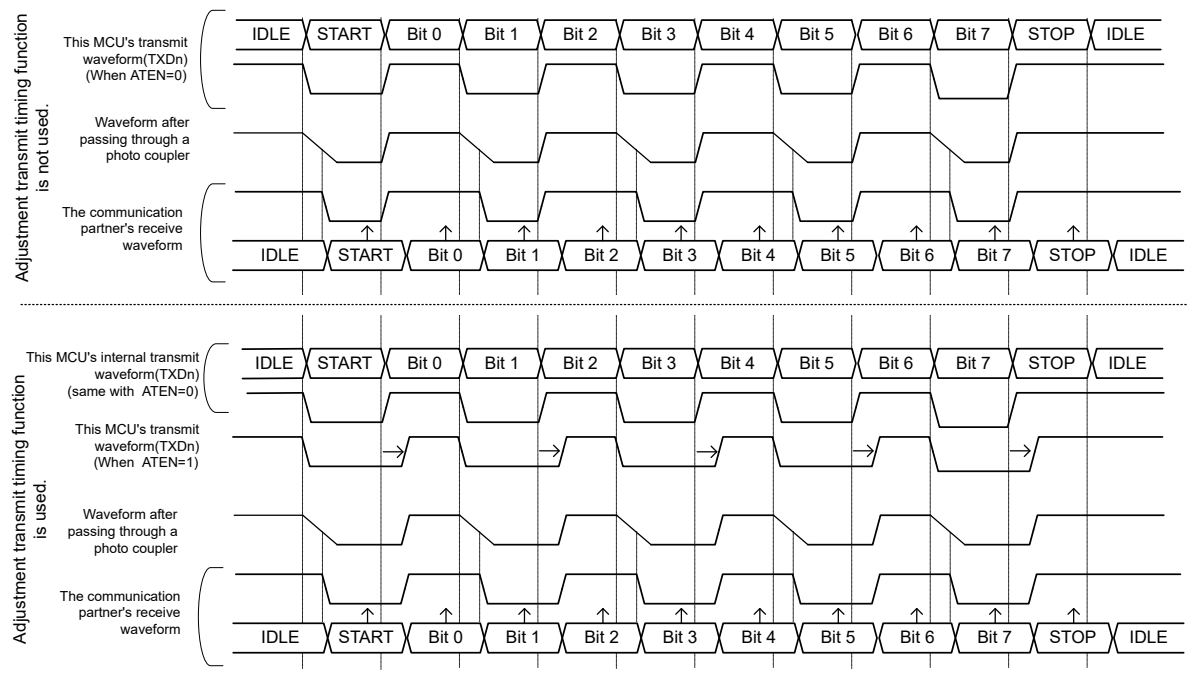
Figure 38.31 The overview of transmission operation in the communication through a photo coupler

The explanation of transmit waveforms of the communication through a photo coupler using adjust transmit timing function

When using the transmission timing adjustment function, adjust the edge timing of the transmission waveform and correct the reception waveform of the communication partner

The following example is 8 bit long data.

(a) In the case of the falling edge transfer time >> the rising transfer time



(b) In the case of the falling edge transfer time << the rising transfer time

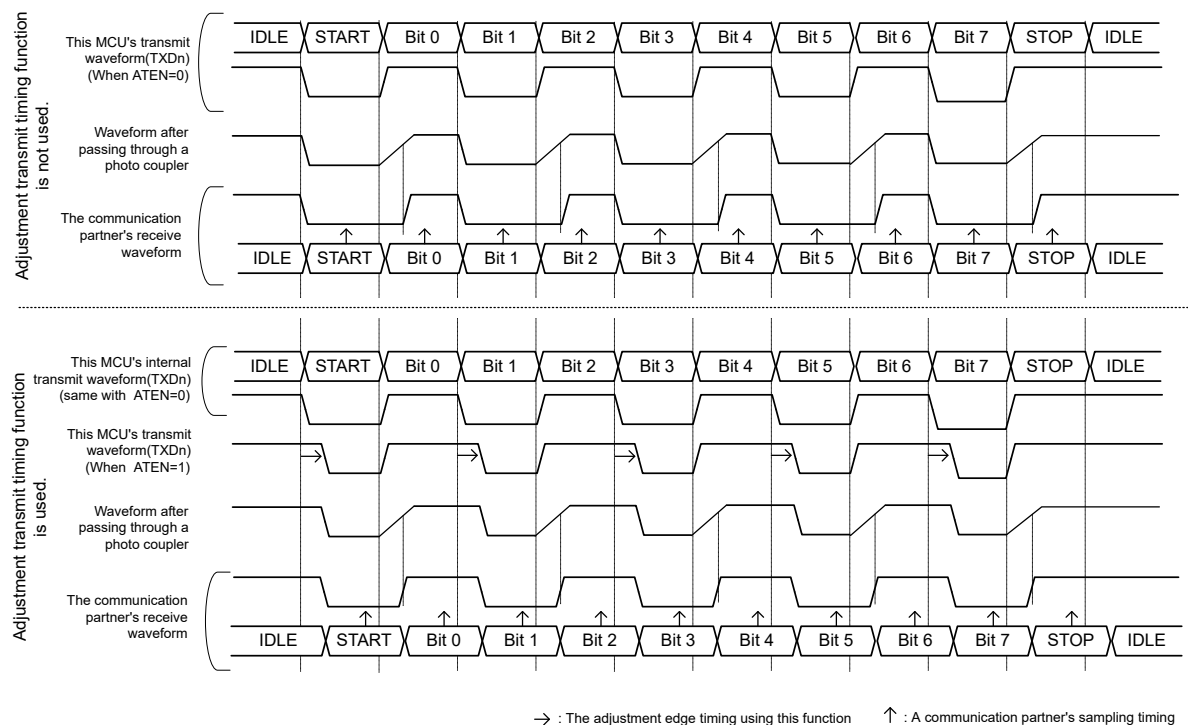


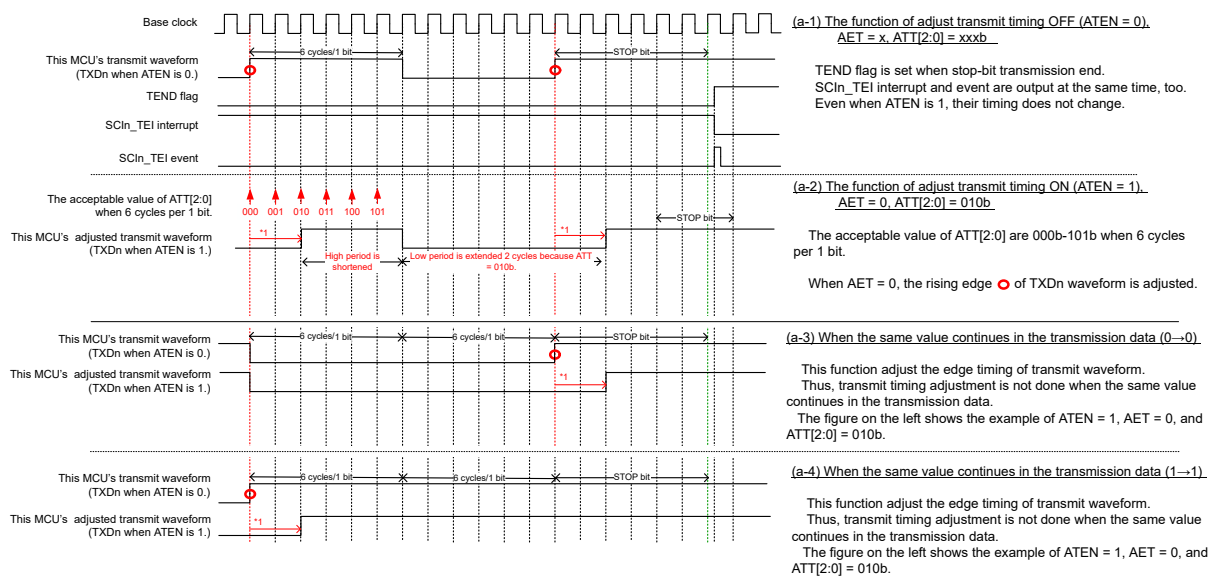
Figure 38.32 The explanation for the transmit waveform through a photo coupler

The operation explanation of adjustment the transmit timing

(a) In the case of the falling transfer time >> rising transfer time

In this case, the high period of a communication partner's reception waveform is made long, and the low period is made short. Therefore, this MCU transmits the waveform with the edge relatively brought forward by adjusting the falling edge timing. Then adjust value (ATT[2:0]) should be set to make equal the low-period/1 bit and high-period/1 bit for a communication partner.

This function's operation is explained as an example of 6 cycles/1 bit.



Note 1. The rising edge of transmit timing is shifted to back by the setting value of CCR4.ATT[2:0].

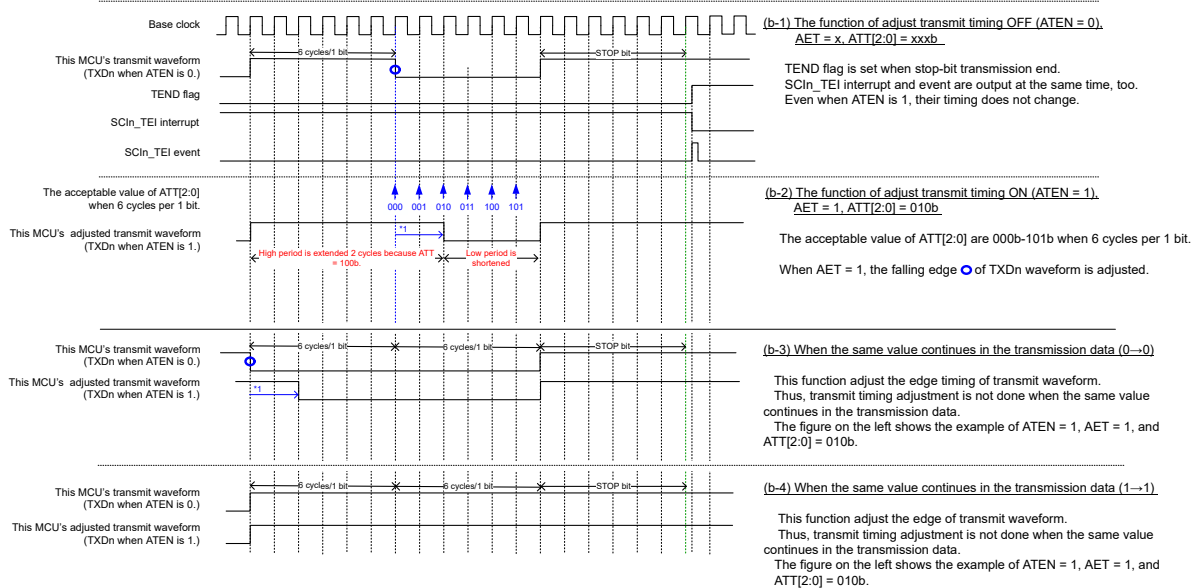
Figure 38.33 The adjustment operation explanation for the transmit timing when AET is 0

The operation explanation of adjustment the transmit timing

(b) In the case of the falling transfer time << rising transfer time

In this case, the low period of a communication partner's reception waveform is made long, and the high period is made short. Therefore, this MCU transmits the waveform with the edge relatively brought forward by adjusting the falling edge timing. The adjust value (ATT[2:0]) should be set to make equal the low-period/1 bit and high-period/1 bit for a communication partner.

This function's operation is explained as an example of 6 cycles/1 bit.



Note 1. The falling edge of transmit timing is shifted to back by the setting value of CCR4.ATT[2:0].

Figure 38.34 The adjustment operation explanation for the transmit timing when AET is 1

38.4 Multi-Processor Communication Function

The multi-processor communication function enables the SCI to transmit and receive data between multiple processors by sharing an asynchronous serial communication line that has an added multi-processor bit. In multi-processor communication, a unique ID code is allocated to each receiving station. Serial communication cycles consist of an ID transmission cycle to specify the receiving station and a data transmission cycle to transmit data to the specified receiving station.

The multi-processor bit is used to distinguish between the ID transmission cycle and the data transmission cycle:

- When the multi-processor bit is set to 1, the transmission cycle is the ID transmission cycle
- When the multi-processor bit is set to 0, the transmission cycle is the data transmission cycle

Figure 38.35 shows an example of communication between processors using a multi-processor format. First, a transmitting station transmits communication data in which the multi-processor bit set to 1 is added to the ID code of the receiving station. Next, the transmitting station transmits communication data in which the multi-processor bit set to 0 is added to the transmit data. After receiving communication data with the multi-processor bit set to 1, the receiving station compares the received ID with the ID of the receiving station itself. If the two match, the receiving station receives communication data that is subsequently transmitted. If the received ID does not match with the ID of the receiving station, the receiving station skips the communication data until it receives data in which the multi-processor bit is set to 1.

RTS control cannot be used at the time of multi-processor communication function use because this is a function corresponding to one-to-many communications.

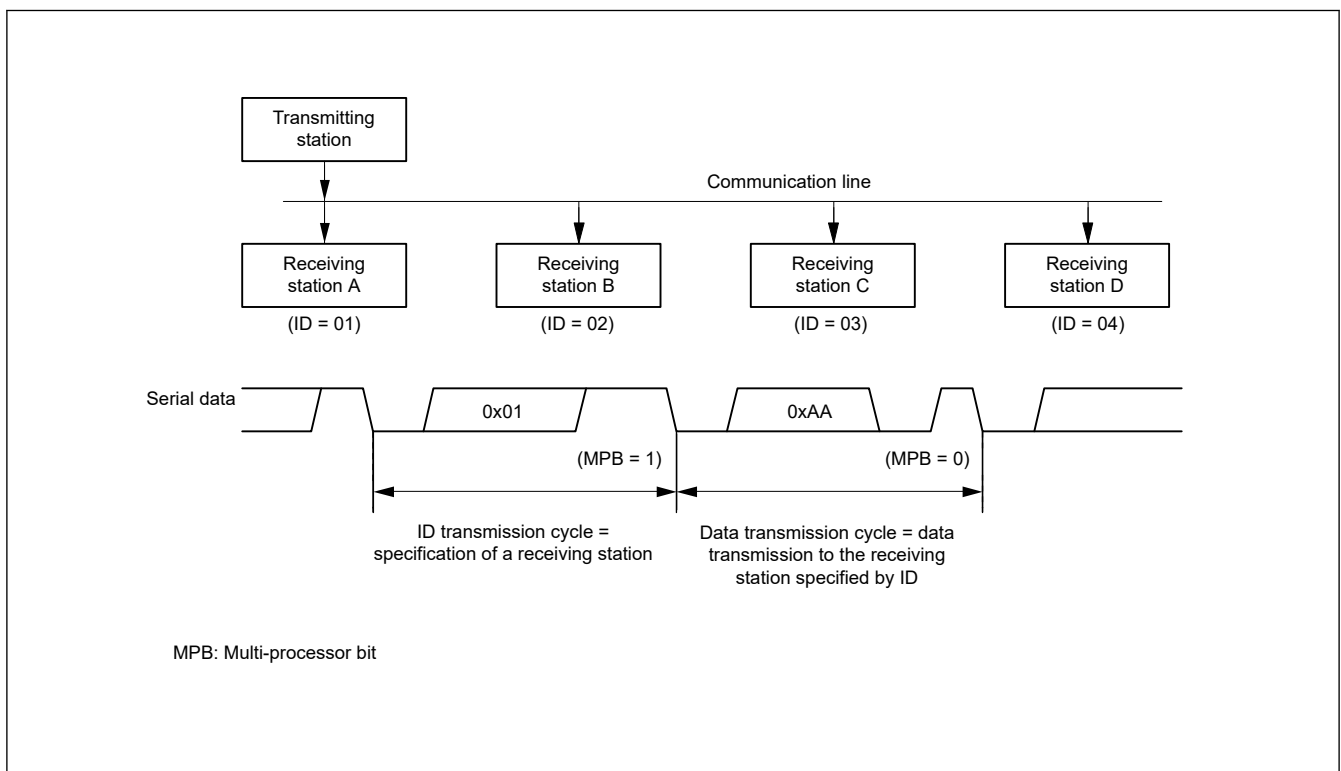


Figure 38.35 Example of communication using multi-processor format with transmission of data 0xAA to receiving station A

(1) Non-FIFO selected

To support this function, the SCI provides the CCR0.MPIE bit. When the MPIE bit is set to 1, the following operations are disabled until the reception of data in which the multi-processor bit is set to 1:

- Transfer of receive data from the RSR register to the RDR register
- Detection of a receive error
- Setting of the respective RDRF, ORER, and FER status flags in the CSR register

When the SCI receives a character in which the multi-processor bit is set to 1, the RDR.MPB bit is set to 1 and the CCR0.MPIE bit is automatically cleared, returning the SCI to non-multi-processor reception operation. If the CCR0.RIE bit is set to 1, an SCIn_RXI interrupt is generated.

When the multi-processor format is specified, the parity bit function is disabled. Apart from this, there is no difference from operation in non-multi-processor Asynchronous mode. The clock used for the multi-processor communication is the same as the clock used in non-multi-processor Asynchronous mode.

(2) FIFO selected

For data transmission, software must write data to TDR.MPBT (Multi-Processor Bit Transfer) that corresponds to transmit data in TDR.TDAT. For data reception, the multi-processor bit that is part of the receive data is written to RDR.MPB and receive data is written to RDR.RDAT.

When the MPIE bit is set to 1, the following operations are disabled until reception of data in which the multi-processor bit is set to 1:

- Transfer of receive data from the RSR register to the RDR.RDAT register
- Detection of a receive error
- Detection of DR
- Setting of the respective RDRF, ORER, and FER status flags in the CSR register

When the SCI receives a character in which the multi-processor bit is set to 1, the RDR.MPB bit is set to 1 and receive data is written to receive-FIFO(RDR.RDAT). The CCR0.MPIE bit is automatically cleared, returning the SCI to normal reception operation. If the CCR0.RIE bit is set to 1, an SCIn_RXI interrupt is generated.

When the multi-processor format is specified, the parity bit function is disabled. Apart from this, there is no difference from operation in normal Asynchronous mode with FIFO selected.

38.4.1 Multi-Processor Serial Data Transmission

(1) Non-FIFO selected

Figure 38.36 shows an example flow of multi-processor data transmission. In the ID transmission cycle, the ID must be transmitted with the TDR.MPBT bit set to 1. In the data transmission cycle, the data must be transmitted with the MPBT bit set to 0. The rest of the operations are the same as operations in Asynchronous mode.

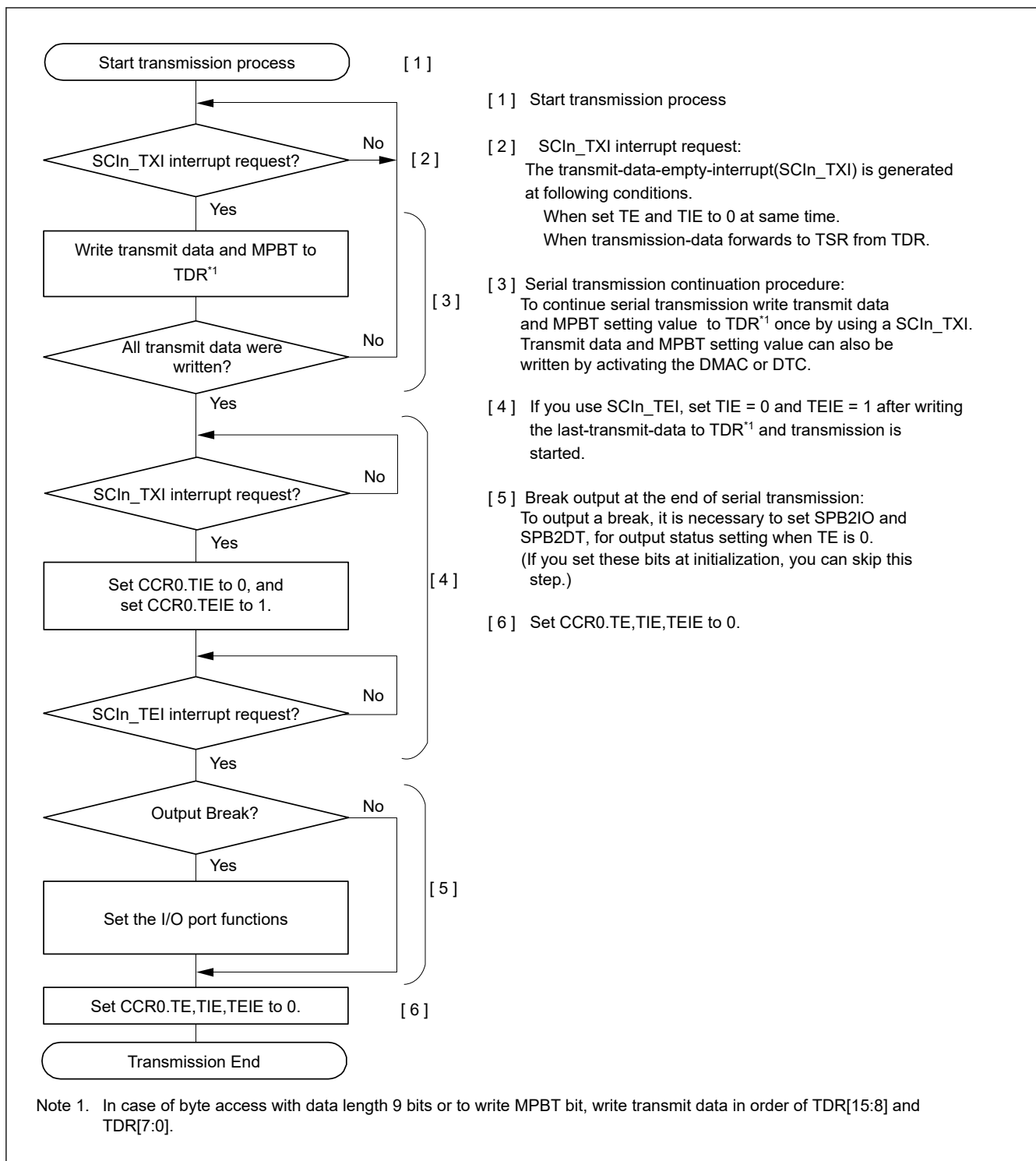


Figure 38.36 Example flow of multi-processor serial transmission with non-FIFO selected

(2) FIFO selected

Figure 38.37 shows an example of data format that is written to transmit-FIFO (TDR) in multi-processor mode. The TDR.MPBT bit is set to 1. Data is set to transmit-FIFO (TDR) with the correct data length. Write 0 for unused bits.

Data Length	Register setting		Transmit data in TDR[15:0]																	
	CCR3. CHR[1:0]		b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0		
7 bit	1	1	—	—	—	—	—	—	MPBT	—	—	TDAT[6:0]								
8 bit	1	0	—	—	—	—	—	—	MPBT	—	TDAT[7:0]									
9 bit	0	Don't Care	—	—	—	—	—	—	MPBT	TDAT[8:0]										

Note: —: Invalid. The write value should be 0.

Figure 38.37 Data format written to transmit-FIFO (TDR) in multi-processor mode with FIFO selected

Figure 38.38 shows an example flow of multi-processor serial transmission with FIFO selected. In the ID transmission cycle, the ID must be transmitted with the TDR.MPBT bit set to 1. In the data transmission cycle, the data must be transmitted with the MPBT bit set to 0. The rest of the operations are the same as operations in Asynchronous mode with FIFO selected.

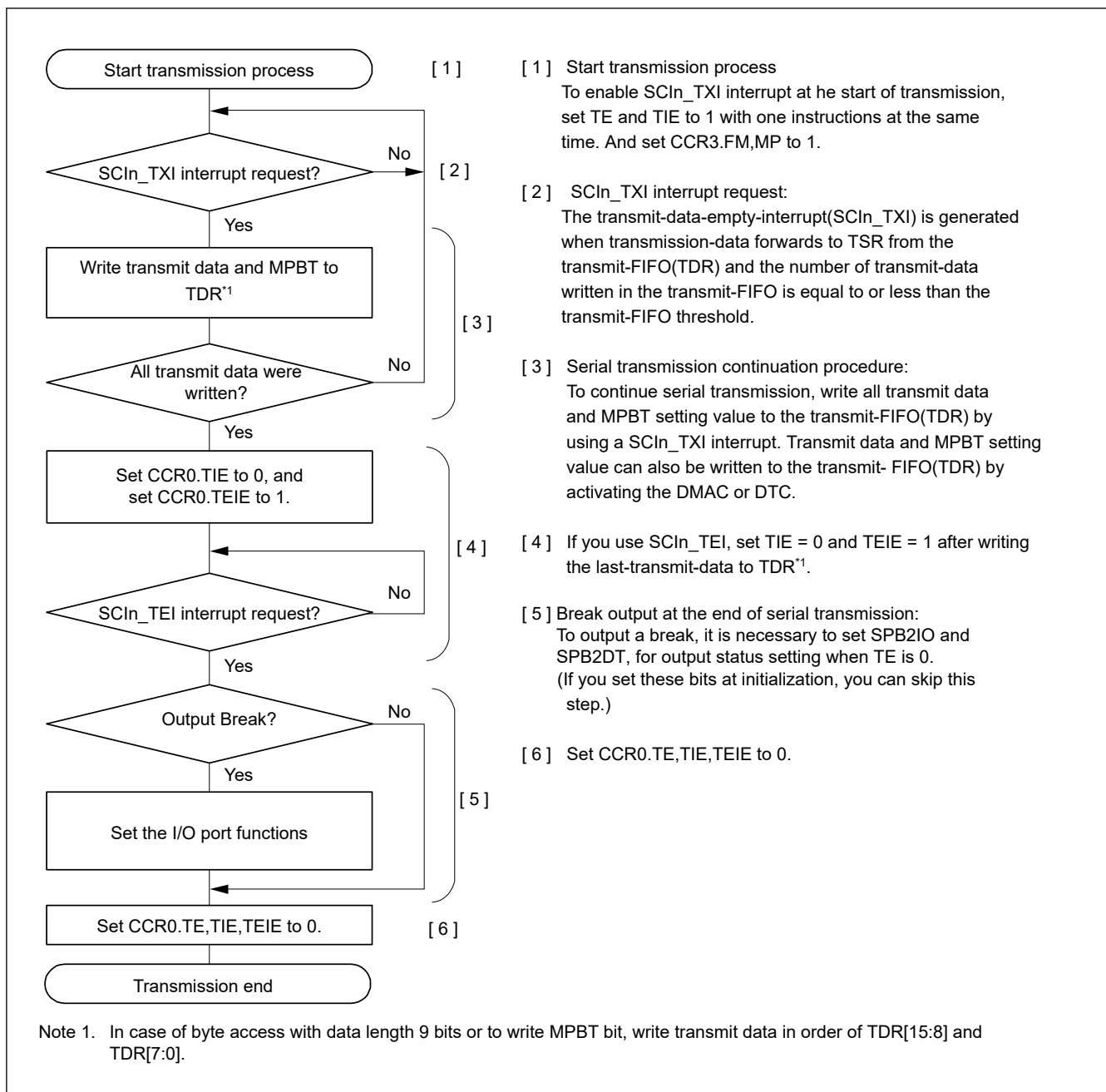


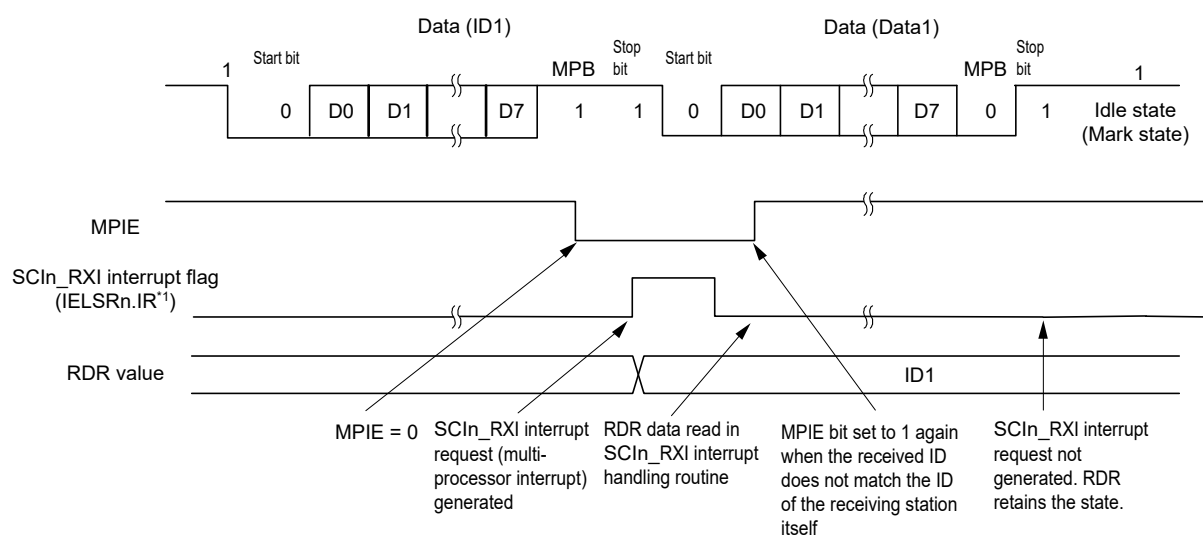
Figure 38.38 Example flow of serial transmission in multi-processor mode with FIFO selected

38.4.2 Multi-Processor Serial Data Reception

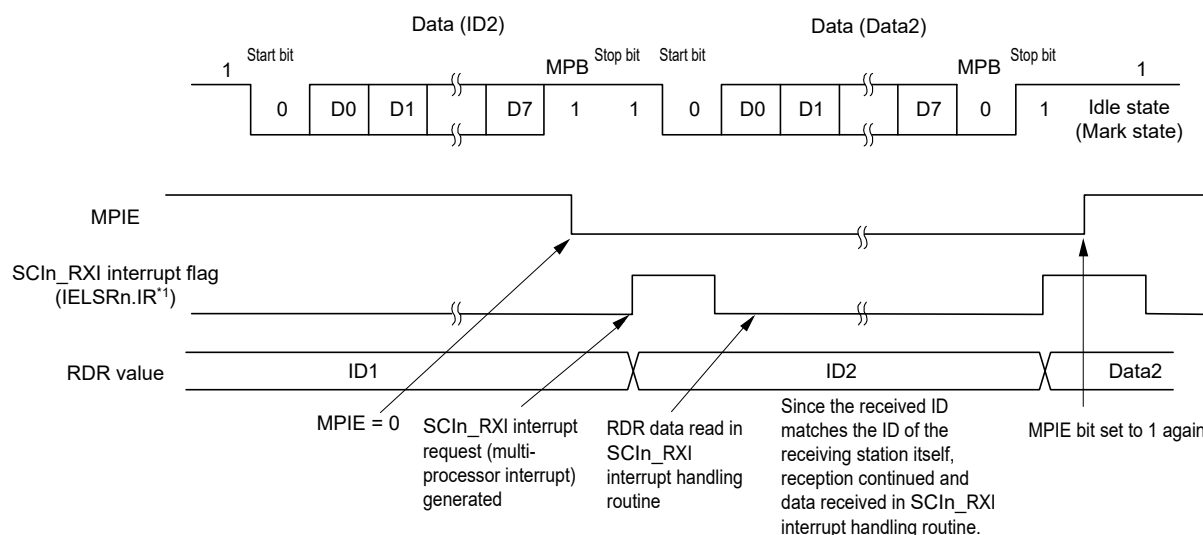
(1) Non-FIFO selected

Figure 38.40 and Figure 38.41 are example flows of multi-processor serial reception. When the CCR0.MPIE bit is set to 1, reading communication data is skipped until reception of communication data in which the multi-processor bit is set to 1. When communication data in which the multi-processor bit is set to 1 is received, the received data is transferred to the RDR register, and the SCIn_RXI interrupt request is generated. The rest of the operations are the same as operations in Asynchronous mode.

Figure 38.39 shows an example operation for data reception.



(a) When the received ID does not match the ID of the receiving station itself



(b) When the received ID matches the ID of the receiving station itself

Note 1. See [section 14, Interrupt Controller Unit \(ICU\)](#) for information on the corresponding interrupt event number.

Figure 38.39 Example of SCI reception with 8-bit data, multi-processor bit, and 1 stop bit

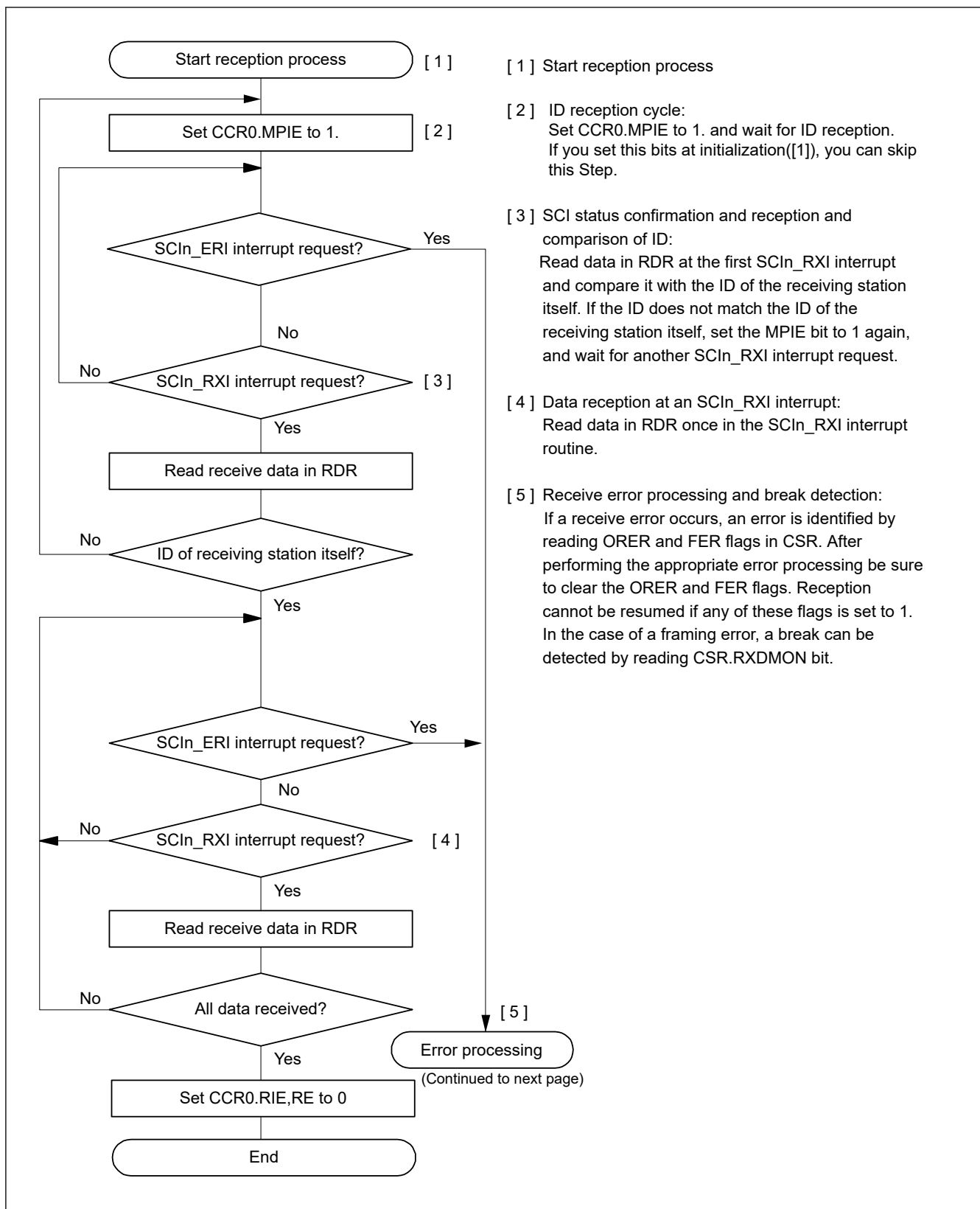


Figure 38.40 Example flow of multi-processor serial reception with non-FIFO selected (1)

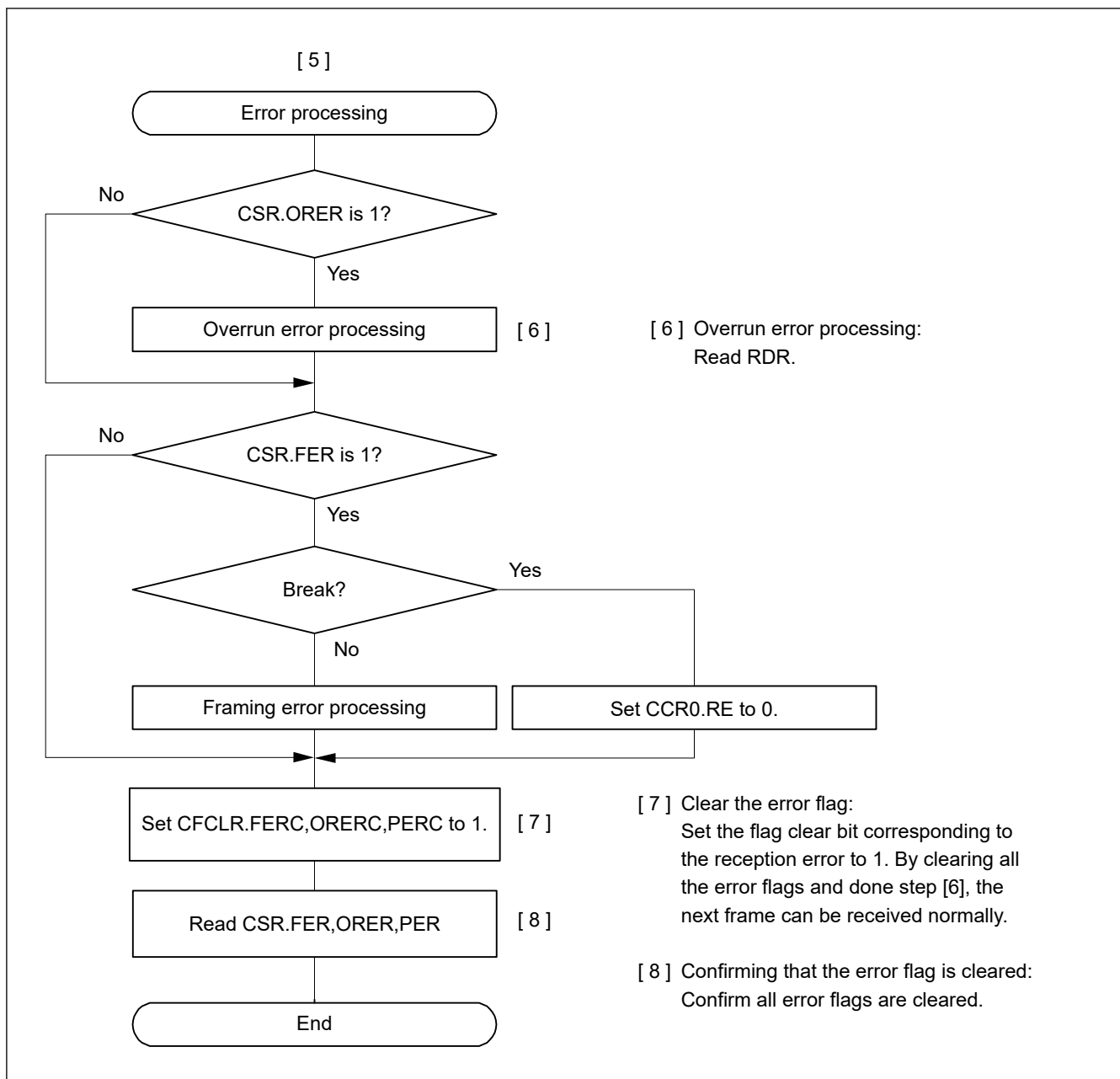


Figure 38.41 Example flow of multi-processor serial reception with non-FIFO selected (2)

(2) FIFO selected

Figure 38.42 shows an example of a data format that is written to receive-FIFO (RDR) in multi-processor mode.

In multi-processor mode, the MPB value that is a part of the receive data is written to the RDR.MPB bit. A value of 0 is written to the RDR.FPER and PER flags. Data is written to receive-FIFO (RDR) with the correct data length. Unused bits are written with 0. When software reads the receive-FIFO (RDR) register, the SCI updates RDR.FFER, FPER, and MPB flags, and receive data (RDAT[8:0]) in receive-FIFO (RDR) with the next data. The FER, PER, and ORER flags in the receive-FIFO (RDR) register always reflect the associated flags in the CSR and FRSR registers.

Data Length	Register Setting		Receive data in RDR[31:0]																		
	CCR3.CHR[1:0]		b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0			
7bit	1	1	0	0	0	FFER	FPER	DR	MPB	0	0	RDAT[6:0]									
8bit	1	0	0	0	0	FFER	FPER	DR	MPB	0	RDAT[7:0]										
9bit	0	Don't Care	0	0	0	FFER	FPER	DR	MPB	RDAT[8:0]											
Data Length	CCR3.CHR[1:0]		b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16			
7bit	1	1	0	0	0	FER	PER	0	0	ORER	0	0	0	0	0	0	0	0			
8bit	1	0	0	0	0	FER	PER	0	0	ORER	0	0	0	0	0	0	0	0			
9bit	0	Don't Care	0	0	0	FER	PER	0	0	ORER	0	0	0	0	0	0	0	0			

Note: When data length is 7bit, it can read always 0 in RDAT[8:7].
When data length is 8bit, it can read always 0 in RDAT[8].

Figure 38.42 Data format stored in receive-FIFO (RDR) in multi-processor mode with FIFO selected

Figure 38.43 shows an example flow of multi-processor data reception with FIFO selected. When the CCR0.MPIE is set to 1, reading communication data is skipped until reception of communication data in which the multi-processor bit is set to 1. When communication data in which the multi-processor bit is set to 1 is received, the received data, MPB and associated errors are transferred to the receive-FIFO (RDR) register. The CCR0.MPIE bit is automatically cleared and normal reception continues.

If a framing error occurs and the CSR.FER flag is set to 1, the SCI continues data reception. The rest of the operations are the same as operations in Asynchronous mode with FIFO selected.

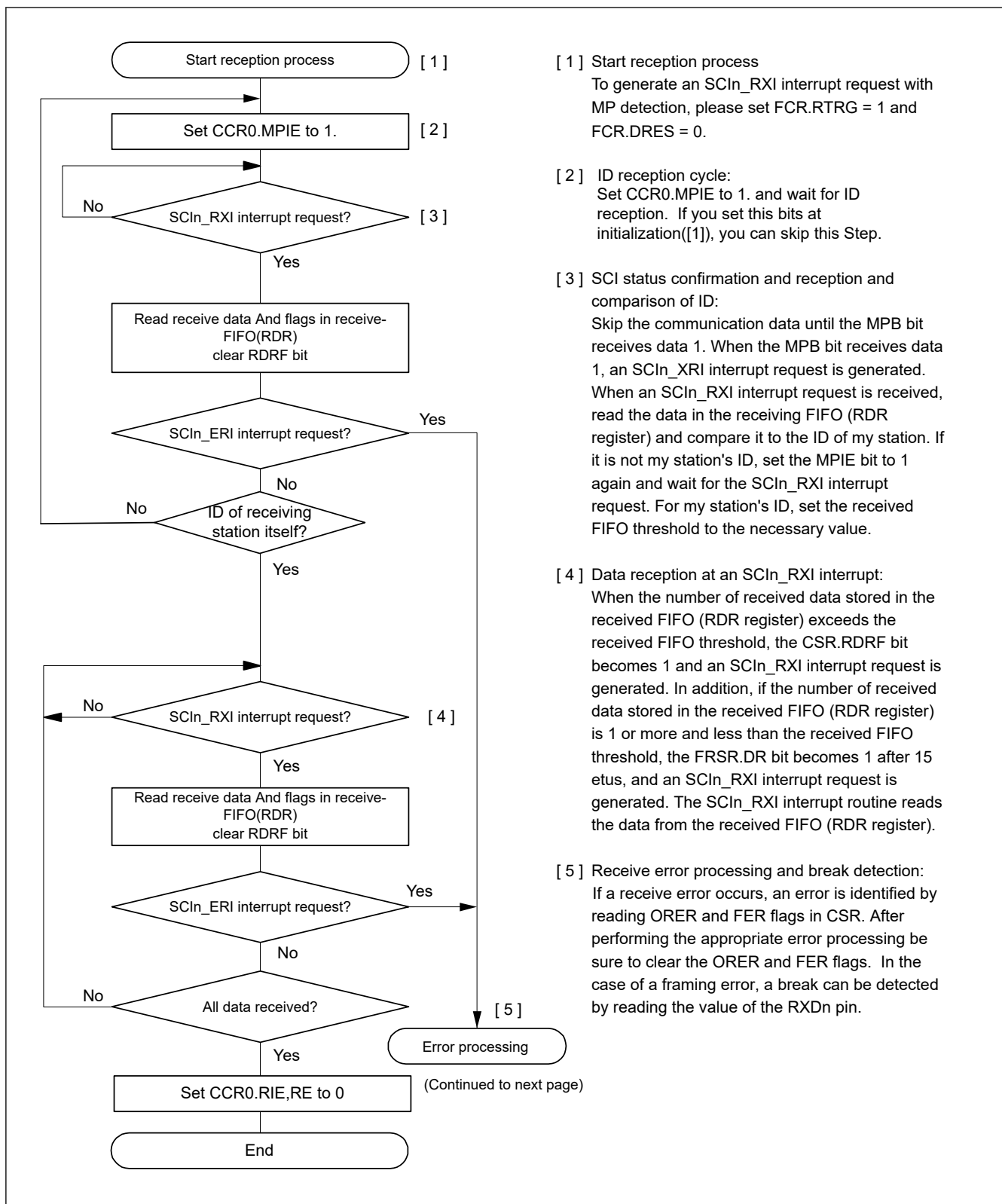


Figure 38.43 Example flow of serial reception in multi-processor mode with FIFO selected

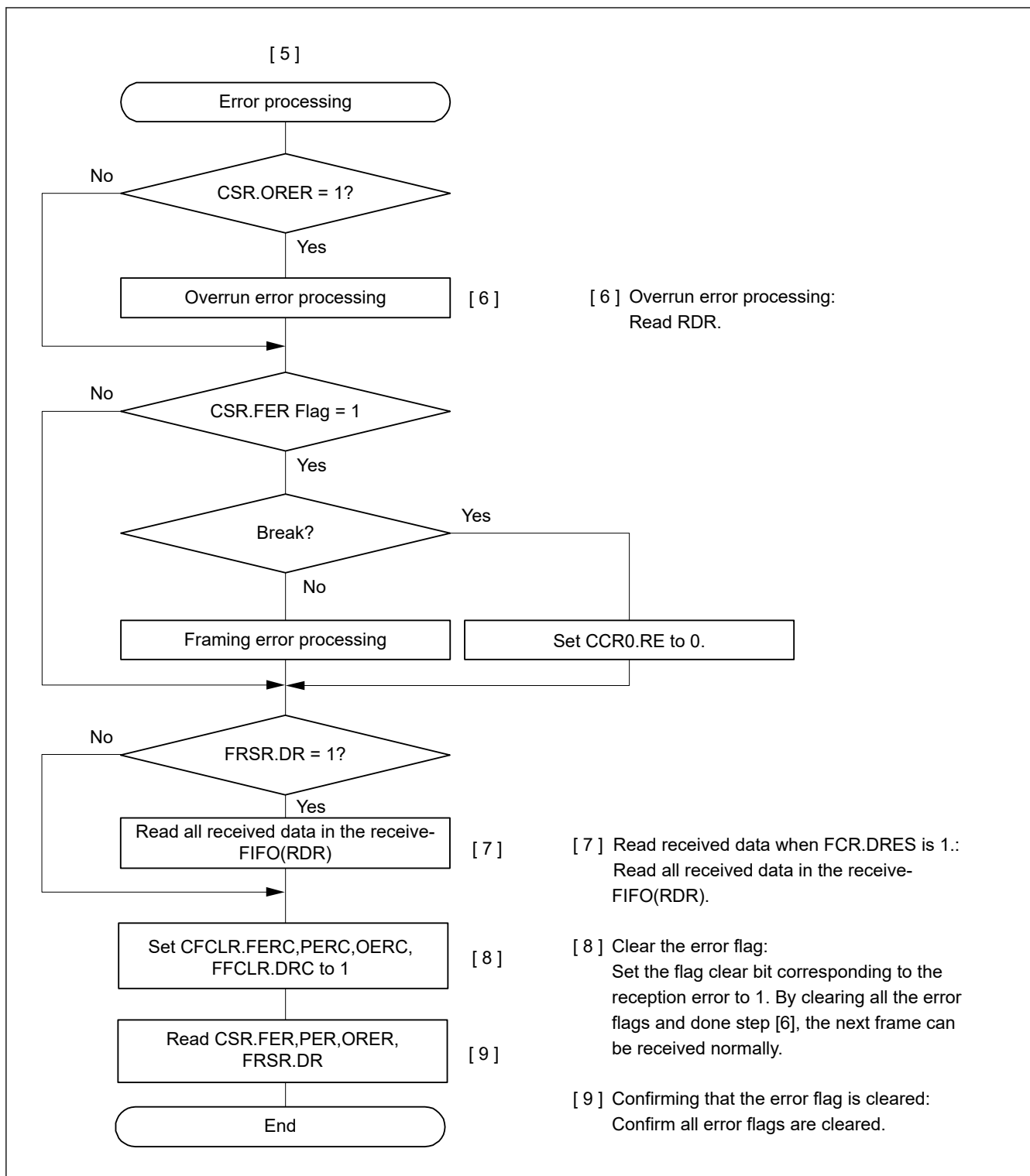


Figure 38.44 Example Flowchart of Serial Reception in Multi-Processor Mode (2) (FIFO selected)

38.5 Operation in Manchester Mode

In Manchester mode, the transmit or receive serial data is coded in Manchester encoding.

Figure 38.45 shows the conceptual image of Manchester encoding.

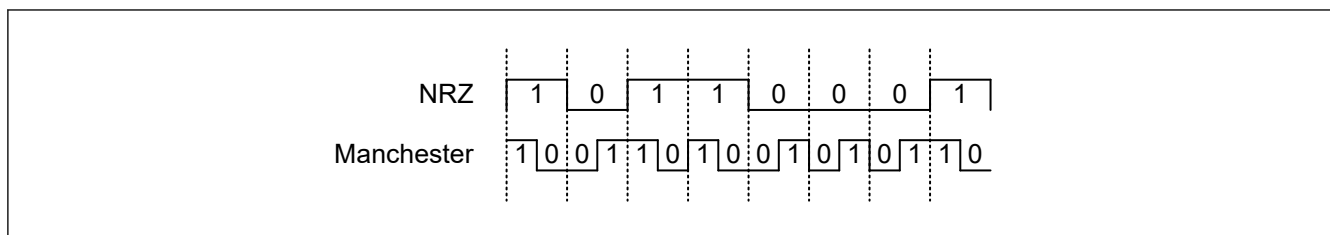


Figure 38.45 Example of Manchester Encoding

In Manchester mode, a preface and a start bit area are added to the transmit data in the register to configure a transmit frame. For transmission, data is encoded in Manchester encoding. When data is received, frames having the same format as transmitted frames are detected and Manchester decoding is performed.

For details on the frame format, see [section 38.5.1. Frame Format](#).

38.5.1 Frame Format

[Figure 38.46](#) shows the frame format in Manchester mode.

In the upper half of the figure, relevant setting registers are shown.

The preface area and the data area are encoded in Manchester encoding.

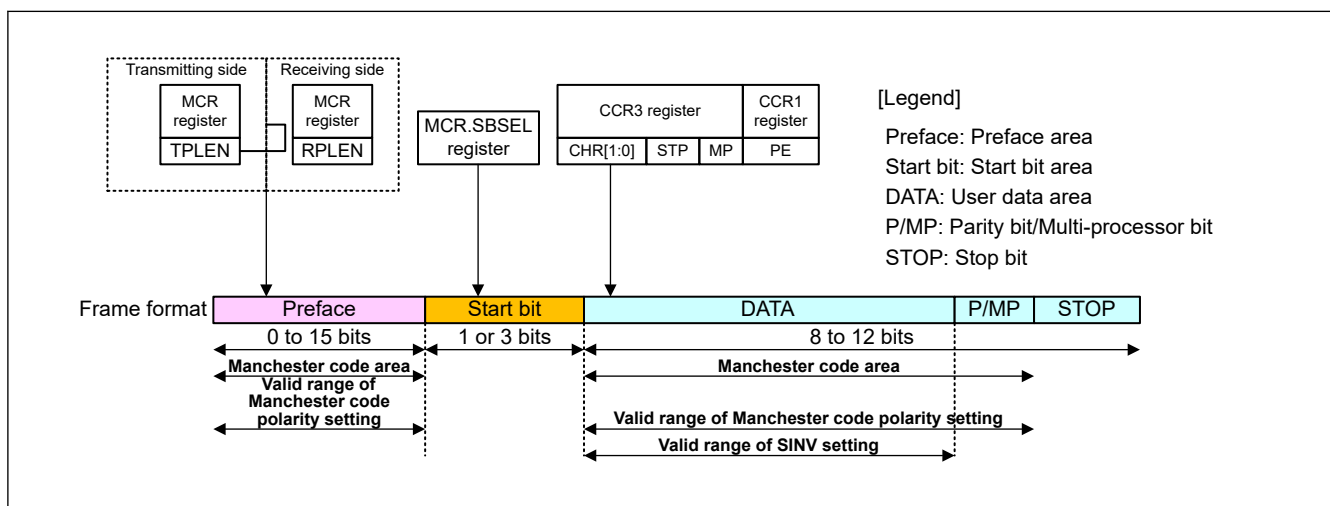


Figure 38.46 Frame Format in Manchester Mode

(1) Preface area

This is a fixed pattern area located at the beginning of each frame.

Different registers are used to set the preface area for transmission and reception. The preface length is determined by setting MCR.TPLEN[3:0] for transmission. It is determined by setting MCR.RPLEN[3:0] for reception.

If it is set to 0, the transmit preface is disabled and is not added.

If it is set to 1d to 15d, a preface whose length is determined by this setting is added.

(For example, if it is set to 1d, a 1-bit preface is added. If it is set to 15d, a 15-bit preface is added.)

The preface pattern is set with MCR.TPPAT[1:0] for transmission and MCR.RPPAT[1:0] for reception, and is selected from four types of patterns.

[Figure 38.47](#) shows how the preface pattern is set. The preface area and the start bit area are added for each communication frame.

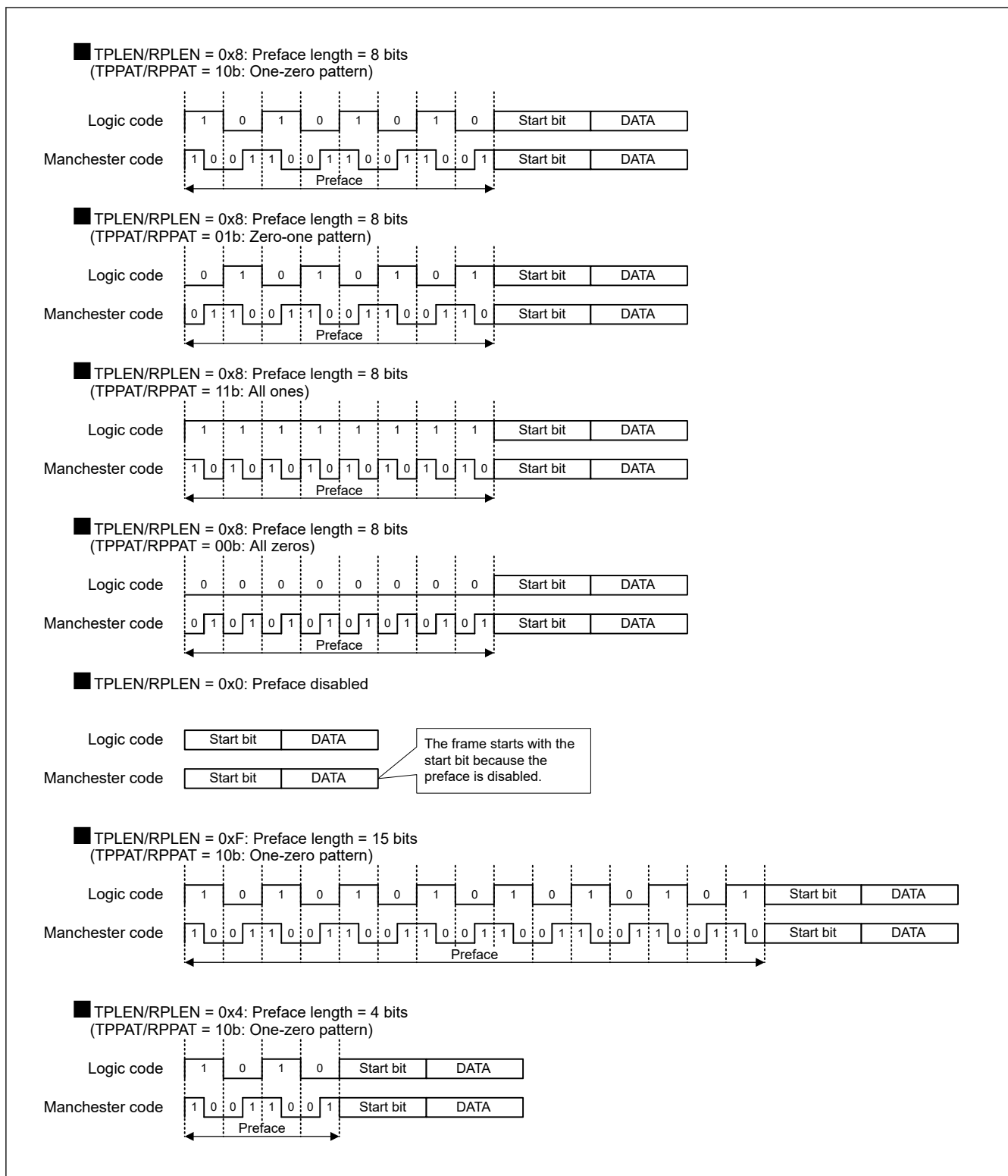


Figure 38.47 Preface Pattern Setting Example

(2) Start bit area

This is an area indicating the start of valid data in a frame. It is added after the preface area.

The start bit length is determined by MCR.SBSEL setting. When MCR.SBSEL = "0", the start bit length is 1 bit.

When MCR.SBSEL = "1", the start bit length is 3 bits.

When MCR.SBSEL = "1", the SYNC type can be selected from command SYNC and data SYNC.

Command SYNC means the three start bits are added as a one-to-zero transition.

Data SYNC means the three start bits are added as a zero-to-one transition.

The SYNC type is determined by the MCR.SYNSEL, MCR.SYNVAL and TDR.TSYNC settings.

(When receiving, the received result is applied to MSR.RSYNC.)

When MCR.SBSEL = "0", the start bit is added as a zero-to-one or one-to-zero transition.

The selection is determined by the MCR.SYNVAL setting.

The MCR.SYNSEL bit specifies the destination to be referred to when setting for transmission.

When the MCR.SYNSEL bit is set to 1, the MCR.SYNVAL setting is referred to. When the MCR.SYNSEL bit is set to 0, the TDR.TSYNC setting is referred to.

Figure 38.48 shows the state of the start bit area according to the settings in the MCR.SYNSEL, MCR.SYNVAL and TDR.TSYNC registers in the case of transmission. Figure 38.49 shows that in the case of reception.

The start bit(s) is not affected by the MCR.TMPOL or MCR.RMPOL setting.

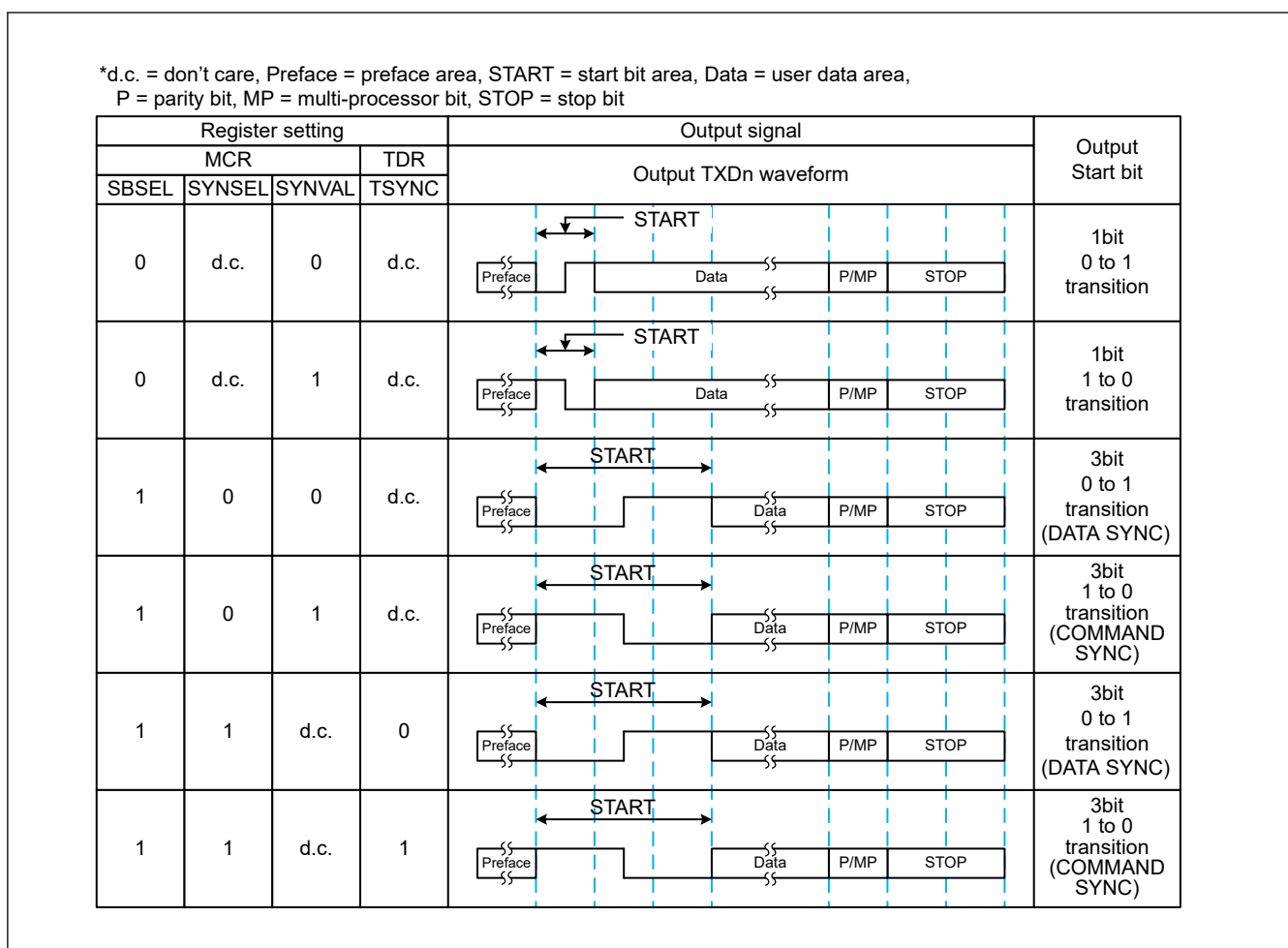


Figure 38.48 Settings Related to and Format of the Start Bit Area at Transmission

d.c. = don't care, Preface = Preface area, START = Start bit area, Data = Data area
P = Parity bit, MP = Multi-processor bit, STOP = Stop bit

Register setting				Input signal	Start bit detection result ¹	Register indication
MCR			TDR			
SBSEL	SYNSEL	SYNVAL	TSYNC	RXDn input waveform		MSR.RSYNC
0	d.c.	0	d.c.		Normal start bit (1 bit: 0-to-1 transition)	0
					Start bit error	0
					Start bit error	0
					Start bit error	0
0	d.c.	1	d.c.		Start bit error	0
					Normal start bit (1 bit: 1-to-0 transition)	0
					Start bit error	0
					Start bit error	0
1	d.c.	d.c.	d.c.		Start bit error	0
					Start bit error	0
					Data SYNC	0
					Command SYNC	1

Note 1. Data other than the start bit is assumed to be normal.

Figure 38.49 Settings Related to and Judgment of the Start Bit Area at Reception

(3) DATA

Since the format of the data area is the same as that of the Asynchronous mode, see [section 38.3.1. Serial Data Transfer Format](#).

As shown in [Figure 38.45](#), Frame Format in Manchester Mode, the stop bit is not included in the Manchester encoding range.

38.5.2 Clock

As the transfer clock in Manchester mode, the clock generated by the on-chip baud rate generator is used by setting the CCR2.CKS[1:0] bit.

Also it is possible to set the oversampling (transfer rate of one-bit period) by CCR2.ABCS bit.

When the CCR2.ABCS bit is set to 0, oversampling x16 is selected with the one-bit period being 16 cycles of the base clock. When the CCR2.ABCS bit is set to 1, oversampling x8 is selected with the one-bit period being 8 cycles of the base clock.

38.5.3 Initialization of the SCI in Manchester Mode

Before transferring data, write 0 to CCR0.TE and CCR0.RE (or write the initial value to CCR0 register) and initialize the SCI following the example of flowchart shown in [Figure 38.50](#).

Whenever the operating mode or transfer format is changed, the CCR0 register must be initialized before the change is made.

Note that setting the CCR0.RE bit to 0 initializes none of the ORER, FER, PER, RDRF, and RDF flags in the CSR register, the SYER, PFER, MER and SBER flags in the MSR register, and the RDR registers.

Note also that switching the value of CCR0.TE from 0 to 1 when CCR0.TIE is 1 generates a SCIn_TXI interrupt request.

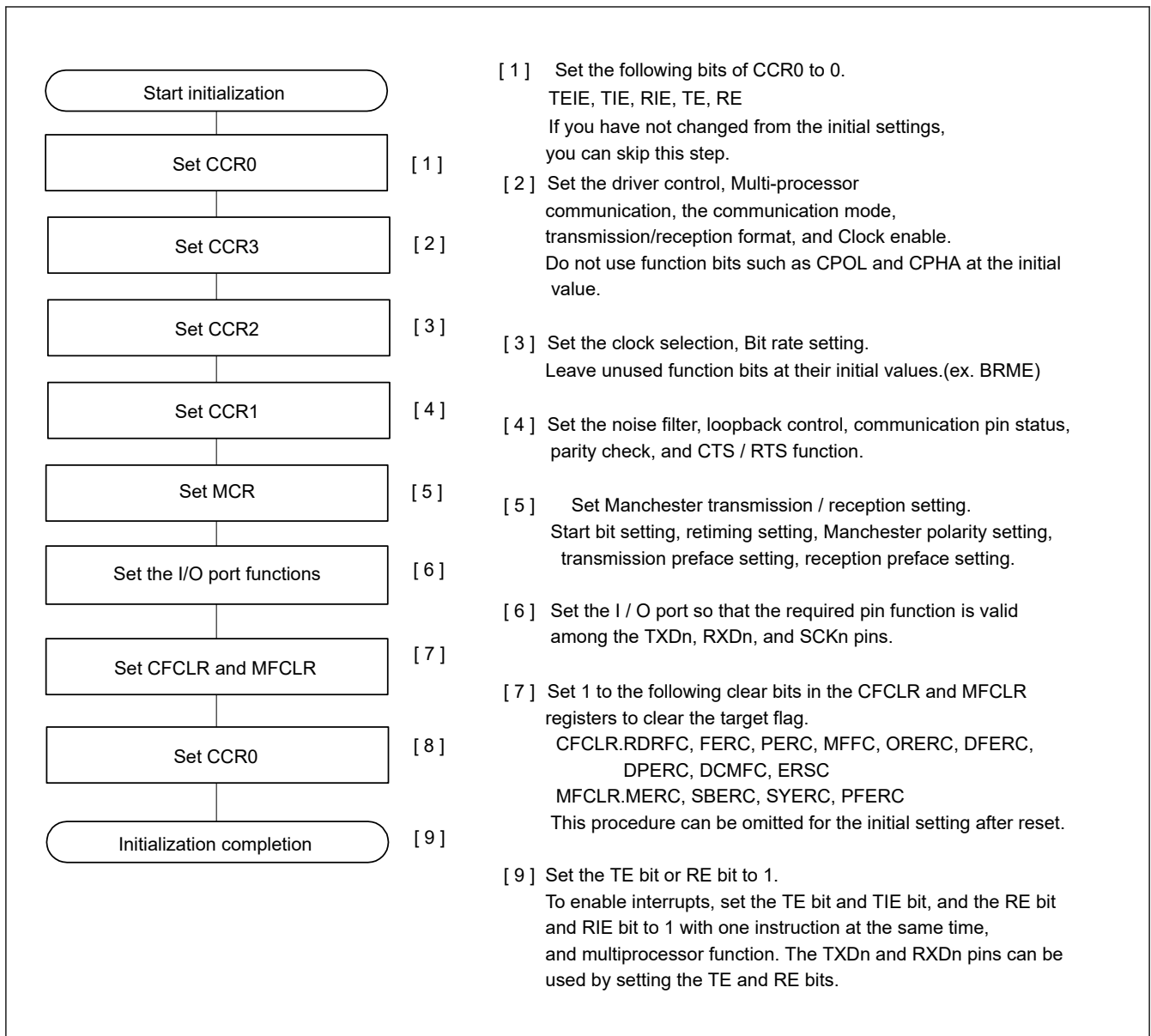


Figure 38.50 SCI Initialization Flow in Manchester Mode

38.5.4 Double-speed Operation

When the ABCS bit in CCR2 is set to 1 and eight pulses of the base clock for a 1-bit period is selected, the SCI operates on the bit rate twice that of when ABCS is set to 0.

When the BGDM bit in CCR2 is set to 1, the cycle of the base clock is reduced to half and the SCI operates on the bit rate twice that of when ABCS is set to 0.

When the ABCS and the BGDM bits in CCR2 are set to 1, the SCI operates on the bit rate four times that of when the ABCS and the BGDM bits in CCR2 are set to 0.

38.5.5 CTS and RTS Functions

The CTS function uses input on the CTSn pin in transmission control. Setting the CTSE bit in CCR1 to 1 enables the CTS function. The CTSn_RTSn pin can be set as a multiplexed pin which allows one pin to be used for either CTS or RTS function, or as dedicated pins with each pin at CTSn pin for CTS function and CTSn_RTSn pin for RTS function. Use the CTSPEN bit in CCR1 for this setting.

When the CTS function is enabled, reception starts only when the CTSn pin is at the low level.

Applying a high level to the CTSn pin after transmission starts does not affect transmission of the current frame, which continues.

The RTS function uses output on the CTSn_RTSn pin to request transmission. When the SCI is ready to receive, it outputs a low level to the CTSn_RTSn pin. Conditions for output of the low level and high level are as follows:

[Conditions for low-level output]

When all conditions listed below are satisfied:

- The value of the RE bit in CCR0 is 1.
- When The SCI is ready to receive next.
 - When there is no received data yet to be read and not receiving
 - All of the following flags are set to 0: CSR.ORER, FER, and PER, MSR.MER, SYER (when SYEREN = 1), PFER (when PFEREN = 1) and SBER flags (when SBEREN = 1).

[Conditions for high-level output]

- When the conditions for low output are not satisfied

38.5.6 Serial Data Transmission in Manchester Mode

The SCI encodes data in Manchester encoding and sends the resultant data in Manchester mode.

When the polarity setting (MCR.TMPOL) set to 0, logic 0 is coded as a zero-to-one transition in Manchester code and logic 1 is coded as a one-to-zero transition in Manchester code.

When the polarity setting (MCR.TMPOL) set to 1, logic 0 is coded as a one-to-zero transition in Manchester code and logic 1 is coded as a zero-to-one transition in Manchester code.

For this reason, a level transition occurs with the Manchester encoded data in the middle of individual logic data. (See [Figure 38.45](#)).

The transmitter constructs transmit frames in a specific format by adding a preface area to data and setting the start bit(s) according to the polarity setting and sends resultant serial data.

For details on the frame format, see [section 38.5.1. Frame Format](#).

[Figure 38.51](#) shows the flowchart in transmission. At transmission starts, set the CCR0.TIE and CCR0.TE bits to 1 simultaneously with one instruction. Then, a SCIn_TXI interrupt request is generated. [Figure 38.52](#), [Figure 38.53](#), and [Figure 38.54](#) show examples of the operation for serial transmission in Manchester mode.

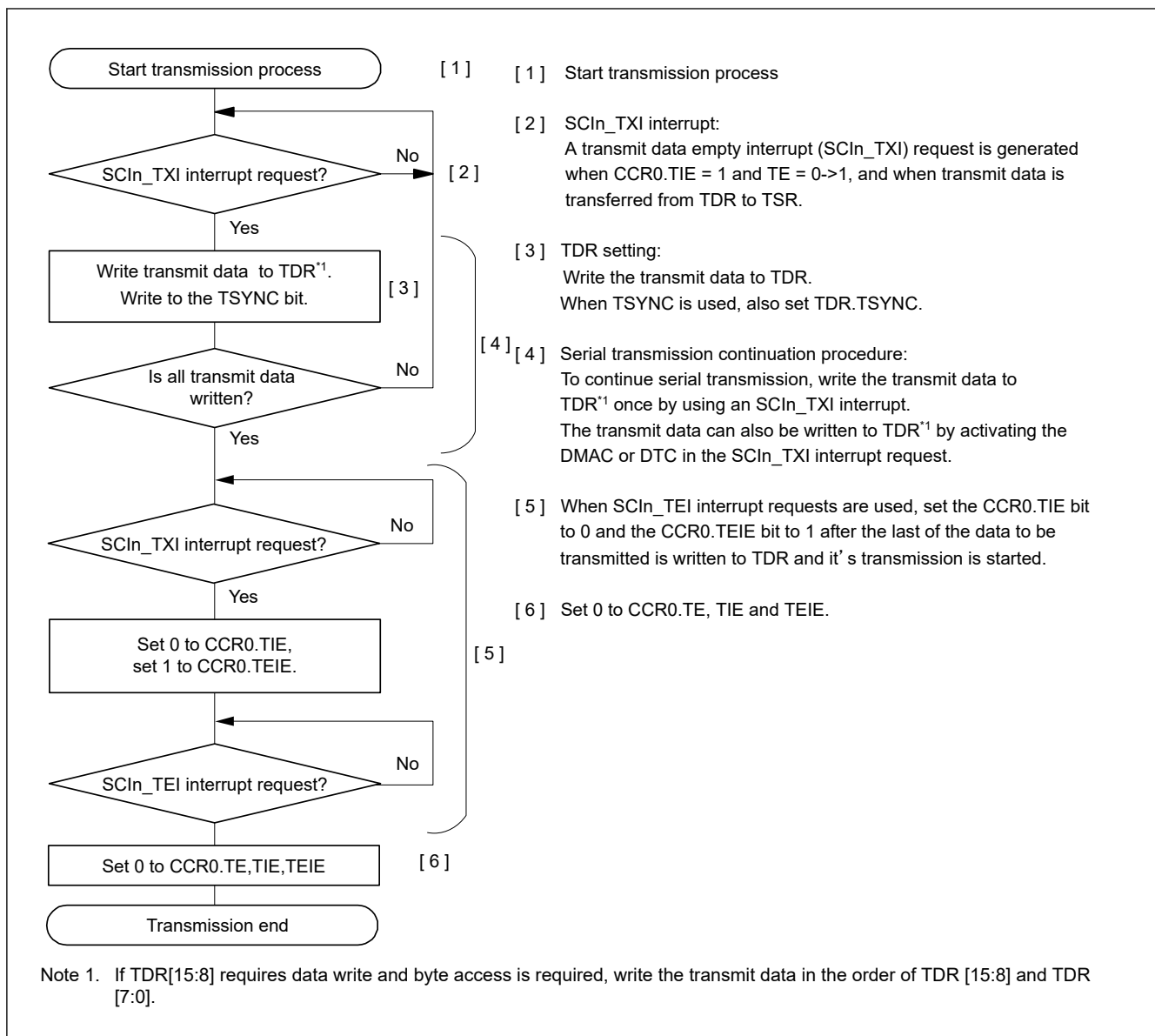


Figure 38.51 Example of Serial Transmission Flowchart in Manchester Mode

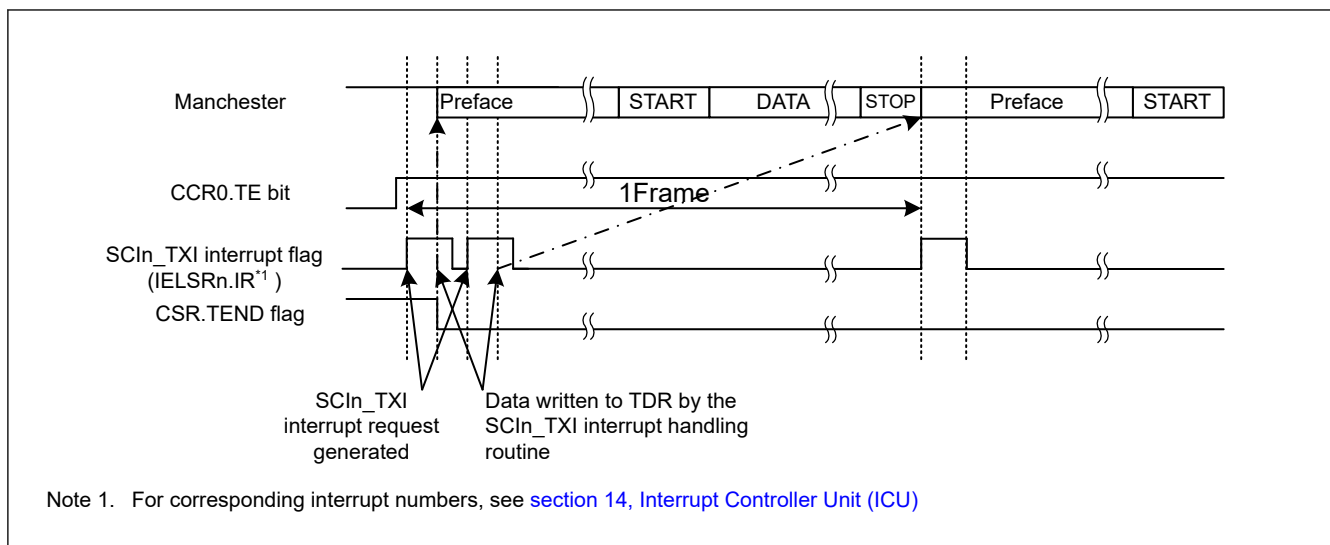


Figure 38.52 Example of Start-of-Transmission Operation for Serial Transmission in Manchester mode (with Preface but Without the CTS Function)

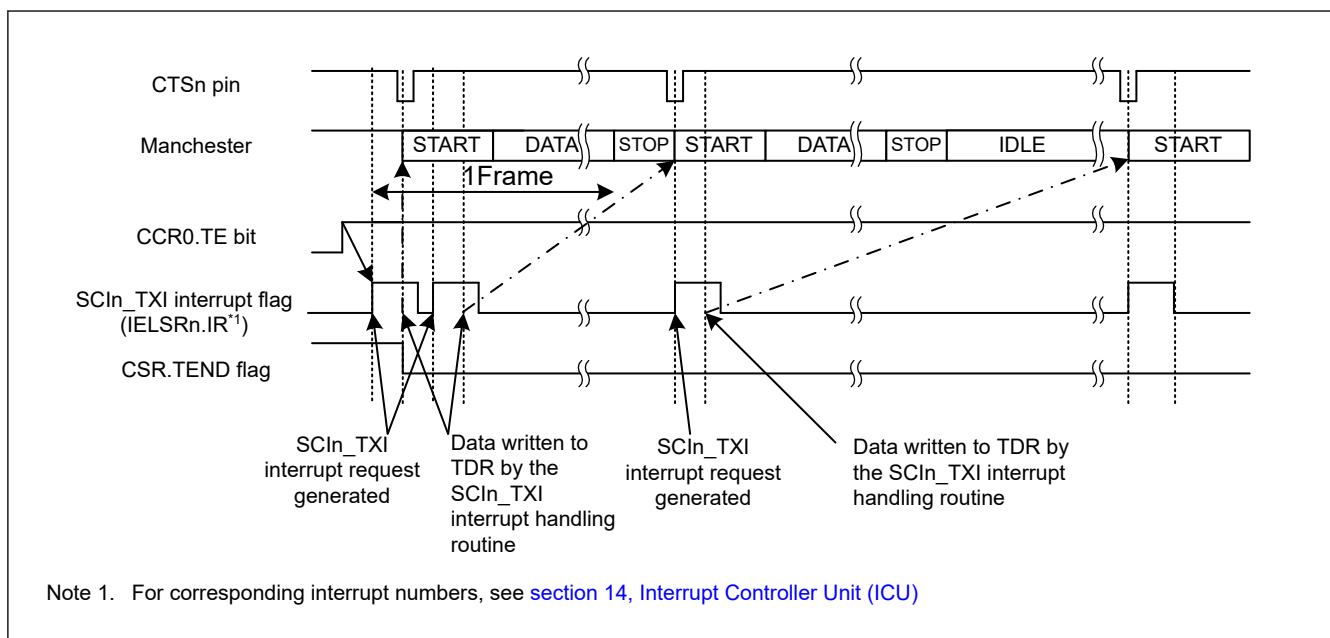


Figure 38.53 Example of Start-of-Transmission Operation for Serial Transmission in Manchester Mode (Without Preface but with the CTS Function)

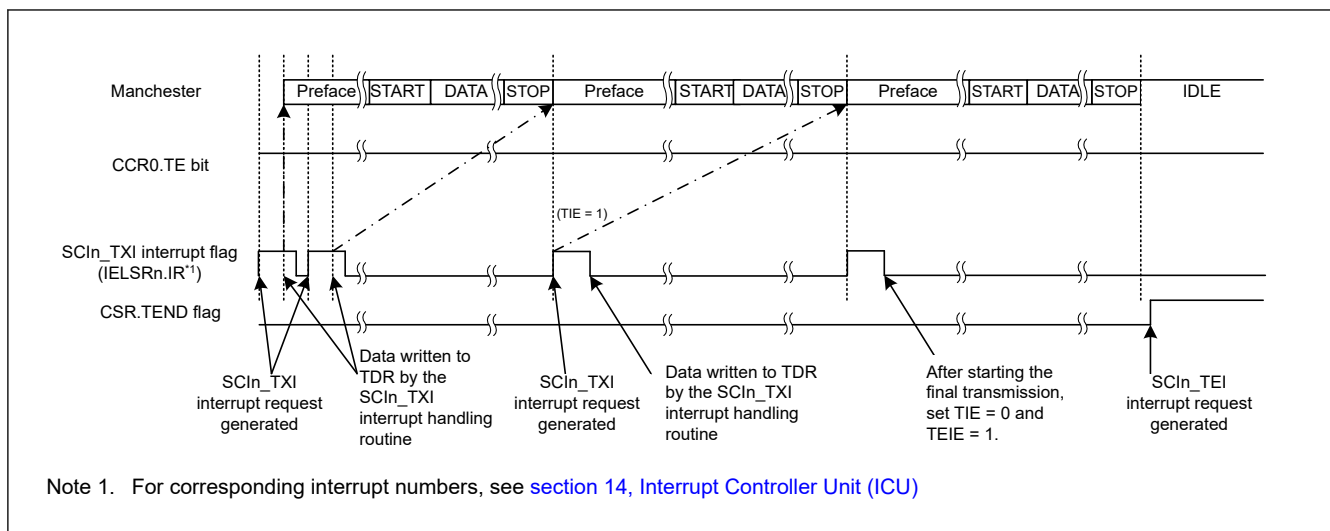


Figure 38.54 Example of End-of-Transmission Operation for Serial Transmission in Manchester Mode (with Preface but Without the CTS Function)

38.5.7 Serial Data Reception in Manchester Mode

In Manchester mode, the SCI operates on a base clock with a frequency of 16 times^{*1} the bit rate. Reception starts by sampling the falling edges of received data at the base clock. As shown in Figure 38.55, reception starts at a falling edge of the received data and it continues if the received data keeps low for the duration of 1/4 bit. If the received data goes high within the duration of 1/4 bit, the SCI judges it as an error and waits for a falling edge again.

If a high level is expected in the first half of a bit in the received data, the SCI judges a low level that continues for one base clock cycle as an error and ignores the change to the low level.

Note 1. This is the case when CCR2.ABCS = 0. When CCR2.ABCS = 1, the SCI operates on a base clock with a frequency of 8 times the bit rate.

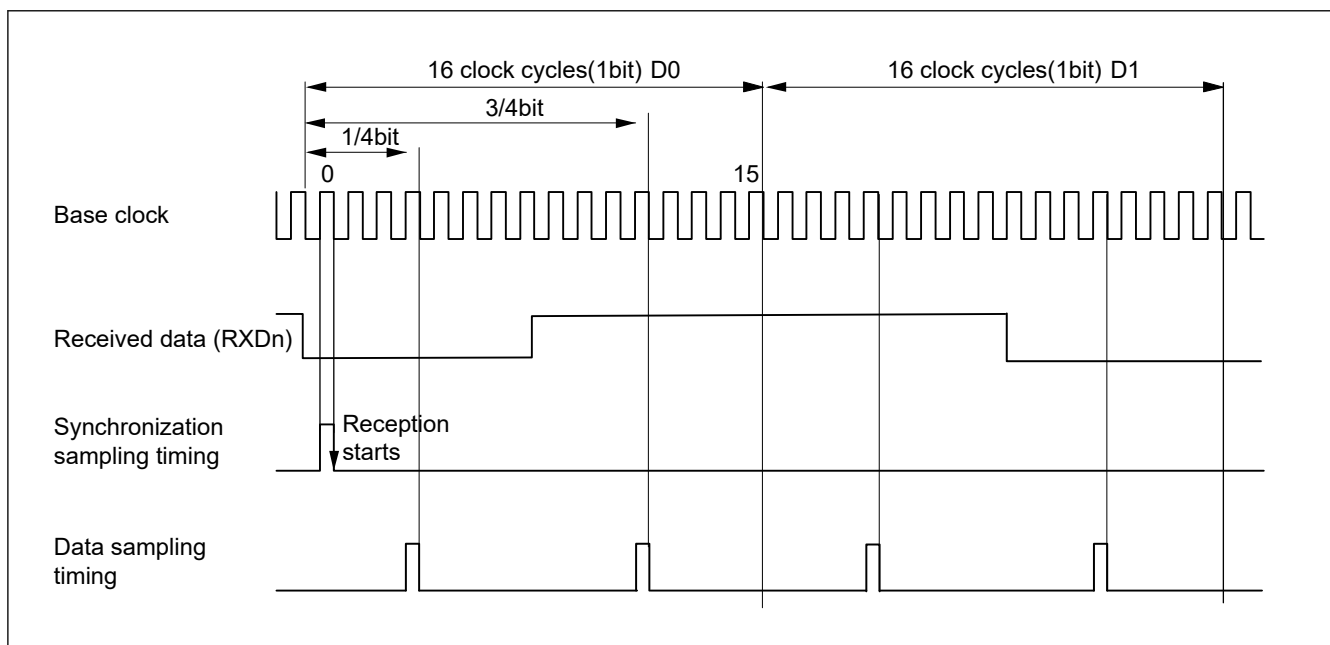


Figure 38.55 Data Reception Sampling Timing in Manchester Mode

In Manchester mode, data reception starts with detection of a preface and start bit area.

The SCI checks the input from the RXDn pin to see whether a preface is added based on the value of MCR.RPLEN.

If the preface is disabled (MCR.RPLEN = 0), it moves on to the detection of a start bit area without detecting a preface.

When a preface is enabled, it identifies a preface pattern setting according to the set value in MCR.RPPAT, and compares it with the RXDn input for a pattern match to detect a preface pattern.

Upon detection of a preface pattern match, it judges it as a normal preface and moves on to the detection of a start bit area.

If detecting a preface pattern mismatch or a Manchester code error in the preface area, it judges it as a preface error and asserts a preface error (PFER).

For start bit detection, the SCI selects an expected value based on the register settings (MCR.SBSEL and SYNVAL), compares it with the RXDn input for a pattern match to detect a start bit area. Upon detection of a start bit pattern match, it judges it as a normal start bit area and moves on to the data processing.

Only when a preface and a start bit area are detected normally, it moves on to the next phase of data reception.

Upon detection of a start bit pattern mismatch, it asserts a start bit error flag (SBER).

In data processing, the SCI shifts the data by the expected received data length based on the register settings (CCR3.CHR[1:0]) through the RSR register. If two sampling points in a bit of the received data are identical, the SCI judges this as a Manchester code error.

For details, see [section 38.5.11. Errors in Manchester Mode](#) (4).

When the parity function is disabled (CCR1.PE = 0), the SCI moves on to the next phase of stop bit detection. When the parity function is enabled (CCR1.PE = 1), the SCI performs parity checking. If detecting a parity error, it asserts a parity error flag (PER), and then moves on to stop bit detection.

In stop bit detection, the SCI checks the following in the stop bit area of the received frame:

It has two sampling points in a bit. If both points are at the high level, the bit is recognized as a normal stop bit and the data is stored in the RDR register. At least one low-level point is judged as an abnormal stop bit, causing a framing error flag (FER) to be set. Even when an error is detected, the received data is stored in the RDR register as abnormal data.

[Figure 38.56](#) shows an example of the operation for serial data reception in Manchester mode.

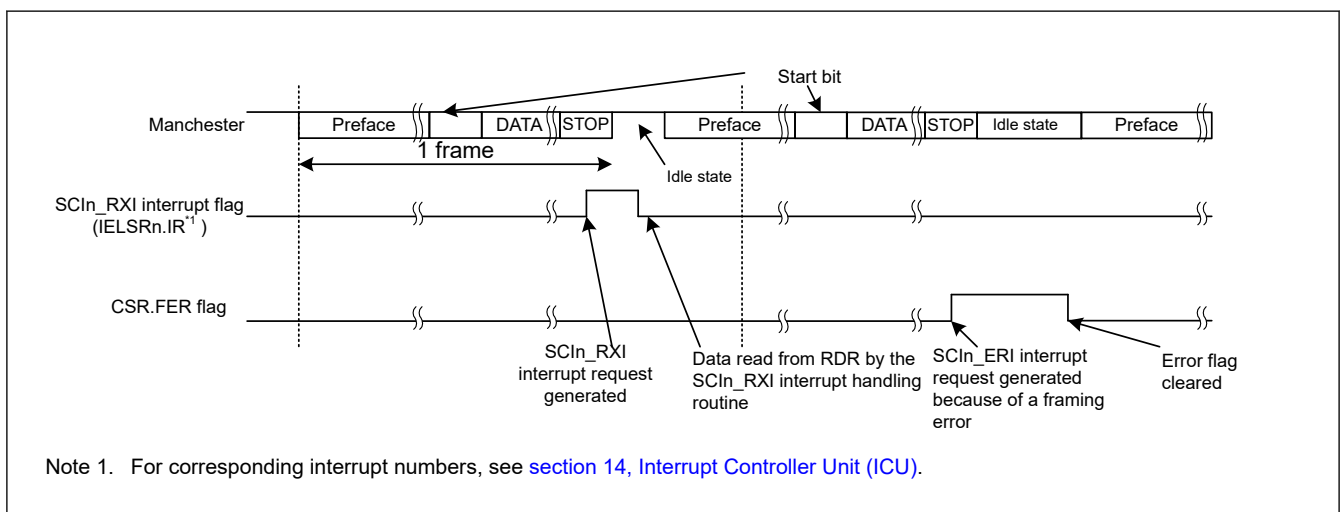


Figure 38.56 Example of Operation for Serial Data Reception in Manchester mode (with a Preface)

For the state of each status flag in the CCR0 register and RXDn input processing when a receive error is detected, see [section 38.5.11. Errors in Manchester Mode](#).

If a receive error is detected, an SCI_ERI interrupt request is generated but an SCI_RXI interrupt request is not generated.

Data reception cannot be resumed while the receive error flag is 1. Accordingly, set the ORER, FER, PER, MER, SYER^{*1}, PFER^{*1}, and SBER^{*1} flags to 0 before resuming reception. Also, be sure to read the RDR register during overrun error processing. When a reception is forcibly terminated by setting the CCR0.RE bit to 0 during operation, read the RDR register because received data which has not yet been read may be left in the RDR register.

[Figure 38.57](#) and [Figure 38.58](#) show examples of serial data reception flowchart in Manchester mode.

Note 1. Effective when the corresponding bit is enabled.

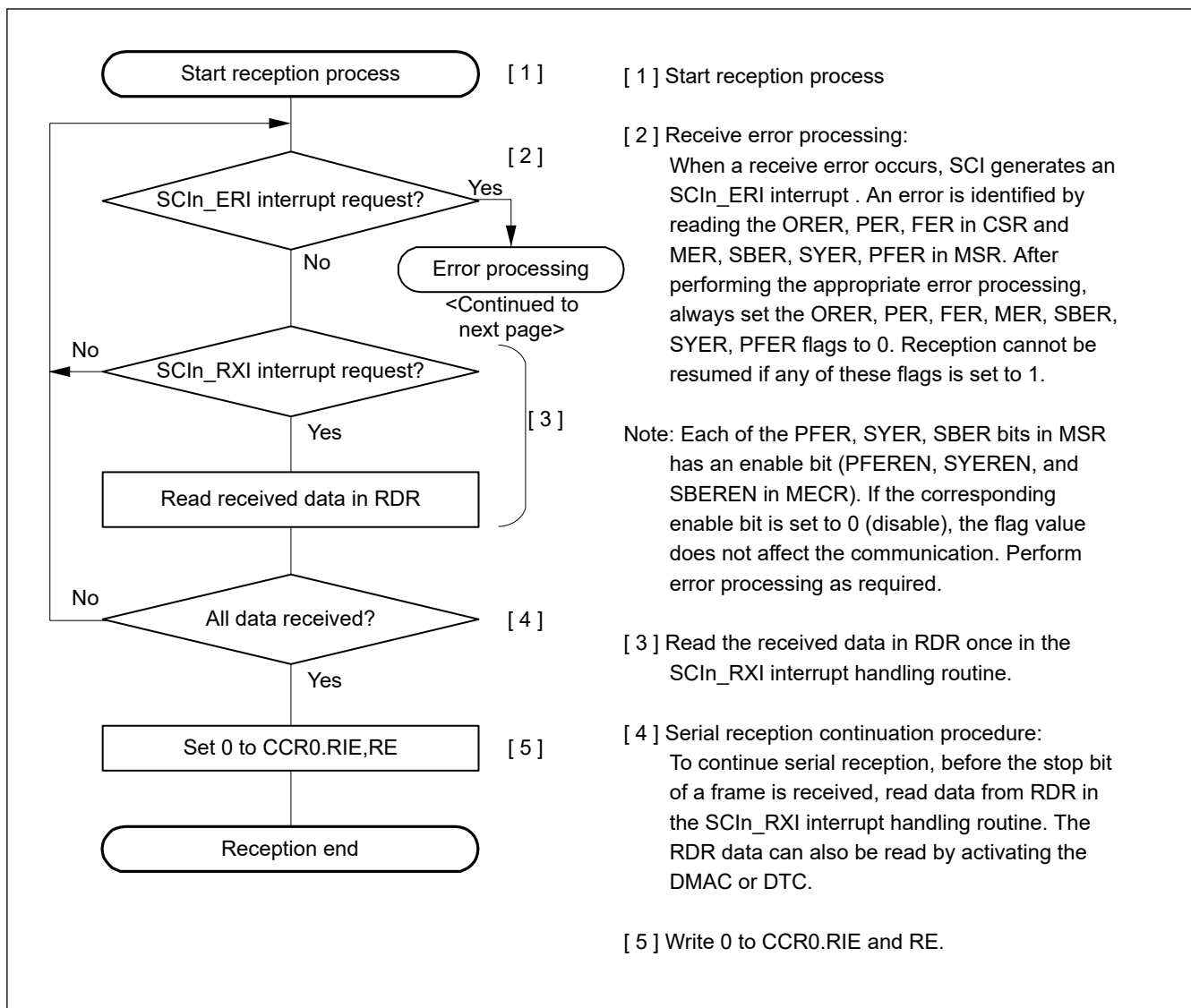


Figure 38.57 Example of Serial Data Reception Flowchart in Manchester Mode (Normal Reception)

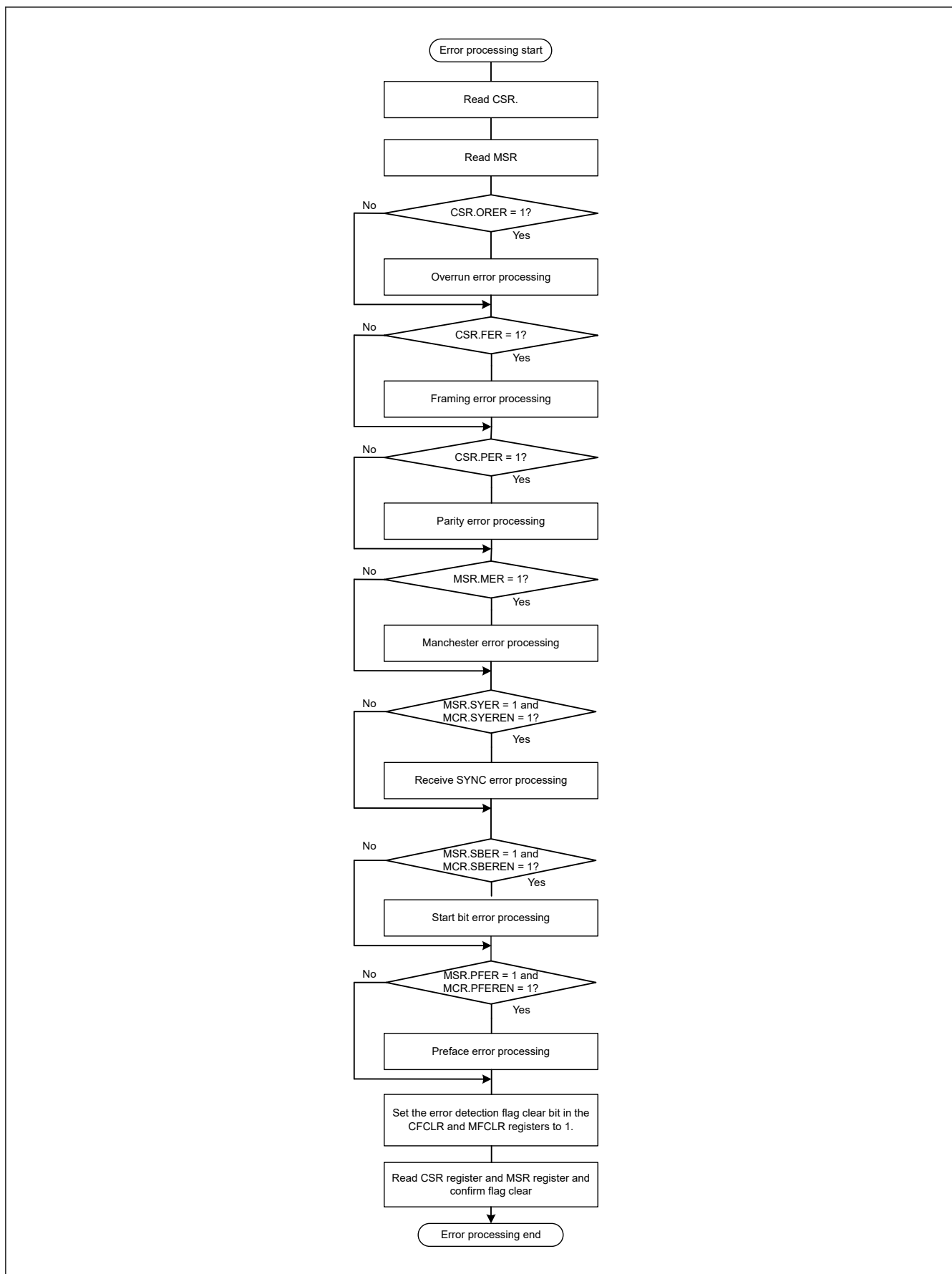


Figure 38.58 Example of Serial Reception Flowchart in Manchester Mode (Error Processing)

38.5.8 Operation When Multi-Processor Bit Is Used

See [section 38.4. Multi-Processor Communication Function](#) (1) for the operation in Manchester mode when using multi-processor mode because the operation is the same.

A preface and a start bit area are added to the frame format in Manchester mode. See [Figure 38.58](#) for error processing in Manchester mode for the reception flowchart ([Figure 38.41](#)). See [Table 38.36](#) for the operation status when detecting various errors.

38.5.9 Receive Retiming

This function corrects the timing for each central edge of the bit, taking advantage of the fact that each bit has an edge in the center in Manchester code.

The receive retiming function can be turned on or off by setting the ERTEN bit in the MCR register.

When the receive retiming function is turned off ($\text{MCR.ERTEN} = 0$), retiming is not performed, causing misalignment between the internal clock and the RXDn input to be accumulated and the receive margin to be reduced.

When the receive retiming function is turned on ($\text{MCR.ERTEN} = 1$), retiming is performed for the preface area, the start bit area^{*1}, and the data area (excluding the stop bit).

Note 1. Retiming is not performed for the start bit area if the preface length is 0 and the start bit length is 3.

As an example, the receive retiming when oversampling x16 is selected is shown below.

When detecting an RXDn input edge two to four cycles before the expected receive cycle, the receive processing is shortened by one sampling CLK cycle.

When detecting a RXDn input edge two to three cycles after the expected receive cycle, the receive processing is extended by one sampling CLK cycle.

(Even if the clock is misaligned with the data by more than two cycles, one cycle is corrected for each bit.)

[Figure 38.59](#) shows the conceptual image of receive retiming range.

When detecting an edge in the tolerance area in the figure, data is received as is without making correction.

When detecting an edge in the SyncJump area in the figure, data is corrected for reception.

When detecting an edge in the SyncError area in the figure, data is received as abnormal data with no correction made.

For a Manchester code error (data matches at the 1/4-phase and 3/4-phase sampling points), the SCI reports a code error.

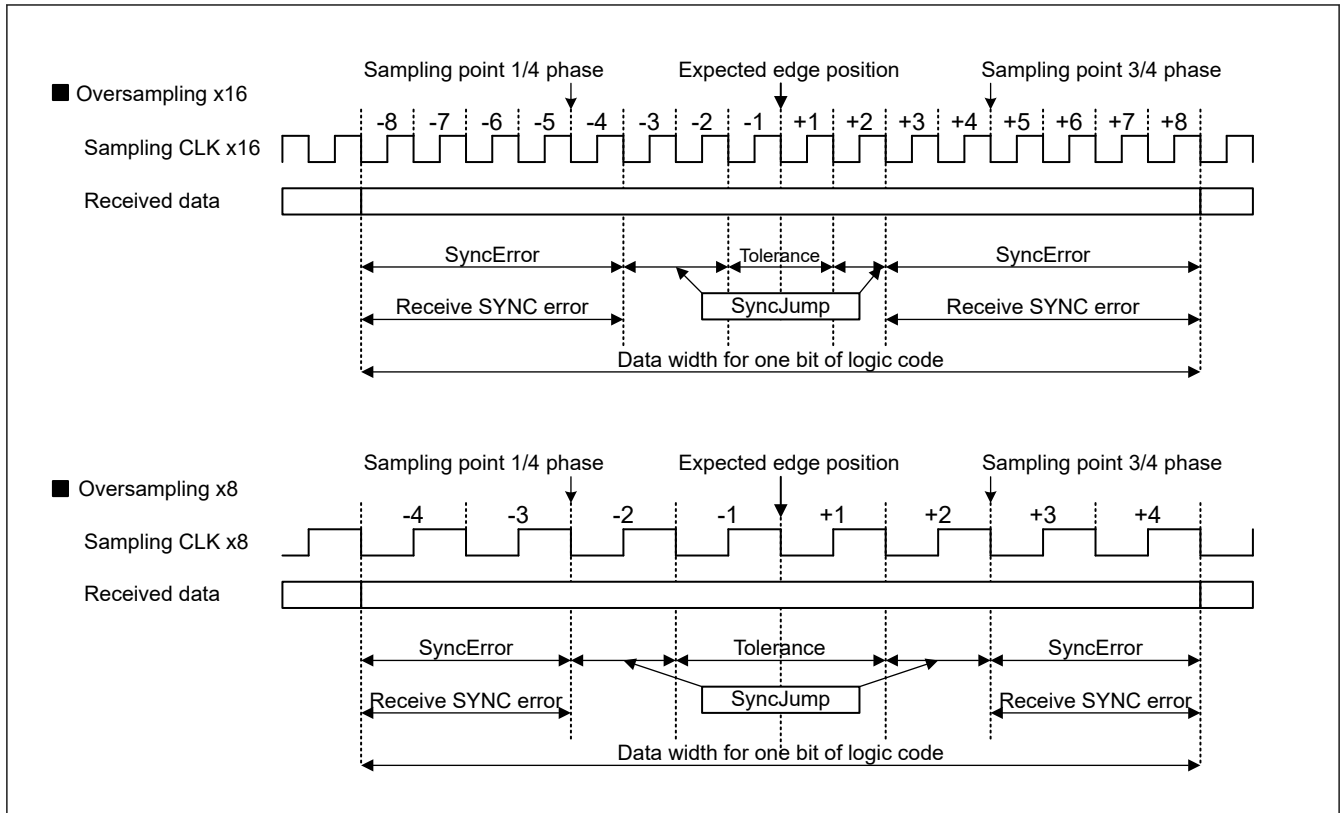


Figure 38.59 Conceptual Image of Reception Retiming Range

38.5.10 Polarity Setting for Manchester Code

The polarity of the Manchester code can be set with the Manchester Control Register (MCR).

It can be set separately for transmission and reception. Use the MCR.TMPOL bit to set the polarity for transmission and the MCR.RMPOL bit to set the polarity for reception.

The Manchester code polarity setting is valid for the preface area, the data area, and the parity or multi-processor area.

When the initial settings (TMPOL/RMPOL = 0) are used for the polarity of Manchester code, logic 0 is encoded as a zero-to-one transition in Manchester code and logic 1 is encoded as a one-to-zero transition in Manchester code. If the settings are changed to TMPOL/RMPOL = 1, logic 0 is encoded as a one-to-zero transition in Manchester code and logic 1 is encoded as a zero-to-one transition in Manchester code. [Figure 38.60](#) shows the conceptual image of the settings and operation.

Separately from the function above, the transmitted and received data in the data area can be inverted by the transmitted/received data inversion function (CCR3.SINV). Since the polarity of Manchester code (MCR.TMPOL/RMPOL) can be set separately from the transmitted/received data invert function (CCR3.SINV), if both are set to inversion (MCR.TMPOL/RMPOL = 1 and CCR3.SINV = 1), the transmitted and received data are set to initial state (inversion + inversion = normal).

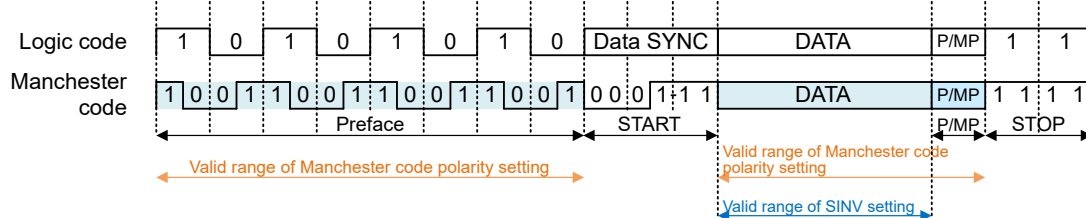
The polarity of the start bit area can be set by a register different from the ones mentioned above.

Since a different register is used, the polarity of the start bit area is not affected by the polarity setting for Manchester code mentioned above.

For details on the setting for the start bit area, see [section 38.5.1. Frame Format \(2\)](#).

■ TMPOL/RMPOL = 0: Normal

(TPLEN/RPLEN = 0x8, TPPAT/RPPAT = 01: One-zero pattern setting, SYNSEL = 0, SYNVAL = 0: Data SYNC)



■ TMPOL/RMPOL = 1: Inverted

(TPLEN/RPLEN = 0x8, TPPAT/RPPAT = 01: One-zero pattern setting, SYNSEL = 0, SYNVAL = 0: Data SYNC)

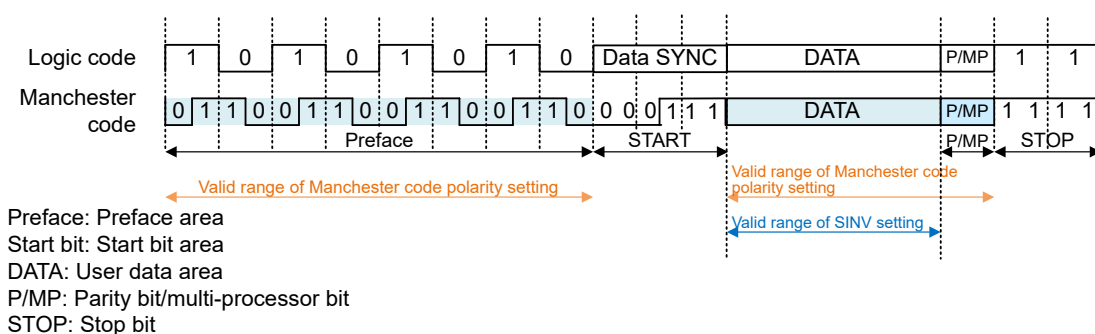


Figure 38.60 Valid Range of the Manchester Code Polarity Setting

38.5.11 Errors in Manchester Mode

There are the following errors in Manchester mode:

1. Parity error
2. Over run error
3. Framing error
4. Manchester error
5. Preface error
6. Start Bit error
7. Receive SYNC error

For errors (1) to (3), see [section 38.3.9. Serial Data Reception in Asynchronous Mode](#) (1) because they are the same as in Asynchronous mode.

Each error is judged in each area, but they are reflected on flags and operations at the timing of 3/4-bit sampling of the STOP bit area. If a preface error or start bit error is detected, subsequent data will not be received. Therefore, no other error detection is performed, and the error flag holds the previous information.

[Table 38.34](#) lists the states of the serial status register when detecting errors and judgment about whether to store data in the RDR.

[Table 38.35](#) lists the errors that can be detected in each area of a Manchester frame.

If a Preface error or Start bit error is detected, subsequent data will not be received. Therefore, no other error detection is performed, and the error flag holds the result of the previous frame reception. Also, if an error is detected in the previous

frame, data will not be received, but errors in the pre-face area and start bit area will update that flag. [Table 38.36](#) shows the flags and actions in this case.

(4) Manchester error

A Manchester error is generated when a Manchester code error is detected.

In Manchester code, there must be an edge (transition) in the center of the bit.

In the data area of a received frame (including the parity/multi-processor bit), the values of the 1/4-bit and 3/4-bit sampling points are checked in each received 1-bit data, and a Manchester code error is determined if these two values match.

If a Manchester code error is detected, the Manchester error flag (MSR.MER) is asserted.

If a Manchester error occurs, it is handled as an interrupt source and event source. If a Manchester error is detected, the next reception is not performed until the corresponding error flag is cleared.

(5) Preface error

A preface error is generated when the preface pattern does not match or a Manchester code error is detected in the preface area. If a preface error is detected, the preface error flag (MSR.PFER) is asserted.

It is possible to set whether to use this error flag as an interrupt source with the setting of the MCR register.

When MCR.PFEREN = 1, a preface error is handled as an interrupt source or event source. If a preface error is detected, the next reception is not performed until the corresponding error flag is cleared.

When MCR.PFEREN = 0, a preface error is not handled as an interrupt source or event source, and the next reception is not halted. However, a preface error is notified to MSR.PFER.

(6) Start bit error

A start bit error is generated when a mismatch is detected between the start bit area in the received frame and the preset start bit pattern. Upon detection of a start bit error, a start bit error flag (MSR.SBER) is asserted.

It is possible to set whether to use the start bit error as an interrupt source with the setting of the MCR register.

When MCR.SBEREN = 1, a start bit error is handled as an interrupt source or event source. If a start bit error is detected, the next reception is not performed until the corresponding error flag is cleared.

When MCR.SBEREN = 0, a start bit error is not handled as an interrupt source or event source, and the next reception is not halted. However, a start bit error is notified to MSR.SBER.

(7) Receive SYNC error

When the receive retiming function described in [section 38.5.9. Receive Retiming](#) is enabled, the receive retiming operation is performed.

If no edges are detected within the receive retiming range (SyncError area in [Figure 38.59](#)) when receive timing operation is being performed, a receive SYNC error is generated. Upon detection of a receive SYNC error, a receive SYNC error flag (MSR.SYER) is asserted. In areas not subject to retiming, receive SYNC errors are not detected.

The preface area^{*1}, the start bit area^{*1,*2}, and the data area (excluding the stop bit) for which receive retiming operation is performed are checked.

It is possible to set whether to use the receive SYNC error as an interrupt source with the setting of the MCR register.

When MCR.SYEREN = 1, a receive SYNC error is handled as an interrupt source or event source. If a receive SYNC error is detected, the next reception is not performed until the corresponding error flag is cleared.

When MCR.SYEREN = 0, a receive SYNC error is not handled as an interrupt source or event source, and the next reception is not halted. However, a receive SYNC error is notified to MSR.SYER.

Note 1. In the case of a frame that starts with a pattern that expects the first half of the bit to be High, it is excluded from retiming.

Note 2. In the start bit area, when there is no preface length and 3 bit start bit is set, it is not subject to retiming.

Also, the 1st bit and the 2nd bit in the start bit area when 3 bit start bit is set are not subject to retiming.

Table 38.34 Flags in the CSR Register and Receive Data Handling in Manchester Mode

Flag in the CSR register			Flag in the MRS register				Received data	Received error status (SCIn_ERI interrupt / event generation)
ORE R	FER	PER	MER	SBER ^{*1}	PFER ^{*1}	SYER		
0	0	0	0	0	0	0	Transfer to RDR	No error
0	1	0	0	0	0	0	Transfer to RDR	Framing error
0	0	1	0	0	0	0	Transfer to RDR	Parity error
0	1	1	0	0	0	0	Transfer to RDR	Framing error + Parity error
0	0	0	1	0	0	0	Transfer to RDR	Manchester error
0	1	0	1	0	0	0	Transfer to RDR	Framing error + Manchester error
0	0	1	1	0	0	0	Transfer to RDR	Parity error + Manchester error
0	1	1	1	0	0	0	Transfer to RDR	Framing error + Parity error + Manchester error
1	0	0	0	0	0	0	Lost	Overrun error
1	1	0	0	0	0	0	Lost	Overrun error + Framing error
1	0	1	0	0	0	0	Lost	Overrun error + Parity error
1	1	1	0	0	0	0	Lost	Overrun error + Framing error + Parity error
1	0	0	1	0	0	0	Lost	Overrun error + Manchester error
1	1	0	1	0	0	0	Lost	Overrun error + Framing error + Manchester error
1	0	1	1	0	0	0	Lost	Overrun error + Parity error + Manchester error
1	1	1	1	0	0	0	Lost	Overrun error + Framing error + Parity error + Manchester error
0	Combination of above			0	0	1	Transfer to RDR	Errors above + Receive SYNC error ^{*2}
1				0	0	1	Lost	Errors above + Receive SYNC error ^{*2}
hold	hold	hold	hold	0	1	0	Lost	Preface error ^{*3}
hold	hold	hold	hold	1	0	0	Lost	Start bit error ^{*3}
hold	hold	hold	hold	0	1	1	Lost	Preface error ^{*3} + Receive SYNC error ^{*2}
hold	hold	hold	hold	1	0	1	Lost	Start bit error ^{*3} + Receive SYNC error ^{*2}

Note 1. Start bit error and Preface error never become 1 at the same time.

Note 2. When MCR.SYEREN = 1, SCIn_ERI interrupt / event is generated by SYER factor.

Note 3. If MCR.PFEREN = 1 or MCR.SBEREN = 1, an SCIn_ERI interrupt / event is generated when the corresponding flag is set.

Table 38.35 Errors Detectable in Each Area

	Preface error (PFER)	Start Bit error (SBER)	Manchester error (MER)	Receive SYNC error (SYER)	Parity error (PER)	Framing error (FER)
Preface area	✓	—	— ^{*1}	✓ ^{*2}	—	—
Start Bit area	—	✓	—	✓ ^{*2}	—	—
Data area	—	—	✓	✓	—	—
Parity area	—	—	✓	✓	✓	—
Multi-processor area	—	—	✓	✓	—	—
Stop Bit area	—	—	—	—	—	✓

Note: ✓: Detected, —: Not detected

Note 1. When a Manchester code error occurs in the preface area, it is defined as a preface error.

Note 2. It may not be subject to Receive SYNC error detection. For details see the text [section 38.5.11. Errors in Manchester Mode](#) (7)

Table 38.36 Operation status due to presence / absence of error in previous frame and operation status list in multiprocessor mode (1 of 2)

Previous frame	Each area of the Frame					PFEREN	SBEREN	SYEREN	Received data	Error flag	Interrupt request	Event signal
	preface	start bit	data	parity	stop							
No Error	PFER	No Error	Don't Care	Don't Care	Don't Care	0	Don't Care	Don't Care	Lost	Set PFER*1	Not output	Not output
	No SYER*1					1					Output	Output
	No Error	SBER	Don't Care	Don't Care	Don't Care	Don't Care	0	Don't Care	Lost	Set SBER*1	Not output	Not output
	No SYER*1						1				Output	Output
	SYER	No Error	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	0	Transfer to RDR	Set SYER	Not output	Not output
	No PFER							1	Lost		Output	Output
	No Error	SYER	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	0	Transfer to RDR	Set SYER	Not output	Not output
	No SBER							1	Lost		Output	Output
	No Error	No Error	SYER		No Error	Don't Care	Don't Care	0	Transfer to RDR	Set SYER	Not output	Not output
								1			Output	Output
	No Error	No Error	MER		No Error	Don't Care	Don't Care	Don't Care	Transfer to RDR	Set MER	Output	Output
	No Error	No Error	Don't Care	PER	No Error	Don't Care	Don't Care	Don't Care	Transfer to RDR	Set PER	Output	Output
	No Error	No Error	Don't Care	Don't Care	FER	Don't Care	Don't Care	Don't Care	Transfer to RDR	Set FER	Output	Output
	There is some error ORER					Don't Care	Don't Care	Don't Care	Lost	Set some flags*2	Output	Output
	No Error	No Error	No Error	No Error	No Error ORER	Don't Care	Don't Care	Don't Care	Lost	Set ORER	Output	Output

Table 38.36 Operation status due to presence / absence of error in previous frame and operation status list in multiprocessor mode (2 of 2)

Previous frame	Each area of the Frame					PFERE N	SBERE N	SYERE N	Received data	Error flag	Interrupt request	Event signal
	preface	start bit	data	parity	stop							
Some error*3 *6	PFER	No Error	Don't Care	Don't Care	Don't Care	0	Don't Care	Don't Care	Lost	Set PFER*1	Output*4	Not output*5
	No SYER*1					1						
	No Error	SBER No SYER*1	Don't Care	Don't Care	Don't Care	Don't Care	0	Don't Care		Set SBER*1		
						1						
	SYER No PFER	No Error	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	0		Set SYER		
						1						
	No Error	SYER No SBER	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	0		Set SYER		
						1						
	No Error	No Error	SYER	No Error	Don't Care	Don't Care	0	Don't set any flags				
							1					
	No Error	No Error	MER	No Error	Don't Care	Don't Care	Don't Care					
							Don't Care					
No Error	No Error	Don't Care	PER	No Error	Don't Care	Don't Care	Don't Care					
No Error	No Error	Don't Care	Don't Care	FER	Don't Care	Don't Care	Don't Care					
There is some error ORER						Don't Care	Don't Care		Don't Care			
No Error	No Error	No Error	No Error	No Error ORER	Don't Care	Don't Care	Don't Care					

Note 1. If SYER is detected, the SYER flag is also set. Other operations are as shown in this table.

Note 2. Other detected error flags including ORER are also set.

Note 3. If all the error flags are cleared before the STOP bit is judged, the operation will be the same as the case where there is no error in the previous frame of this table.

Note 4. Since the SCIn_ERI interrupt request is level output, it remains active due to errors in the previous frame regardless of the presence or absence of error in the relevant frame.

Note 5. Since the error cause is continuously detected, the SCIn_ERI event is not newly output regardless of the presence or absence of errors in the relevant frame.

Note 6. For PFER, SBER, and SYER, when each enable bit is set to disable, it is treated as no error.

Table 38.37 Operation when MPIE = 1 in multi-processor mode (MPIE = 0)

MPB ^{*1}	Each area of the frame					PFERE N	SBERE N	SYERE N	Received data	Error flag	Interrupt request	Event signal
	preface	start bit	data	parity	stop							
1	No Error	No Error	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	Transfer to RDR	Set some flags	Output ^{*2}	Output ^{*2}
	No PFER	No SBER	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	0				
	SYER ^{*3}	SYER ^{*3}						1	Lost	Don't set any flags	Not output	Not output
	PFER	No Error	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care				
	No Error	SBER	Don't Care	Don't Care	Don't Care	Don't Care	Don't Care					

Note 1. If the received MPB bit is 0, it is not received the frame, and the operation is the same as lost of the reception data of this table.

Note 2. If no error is detected, SCIn_RXI interrupt request or event is output, and if it is detected, SCIn_ERI interrupt request or event is output.

Note 3. When SYER is detected in the preface area or the start bit area, the behavior of handling as an error depending on the SYEREN bit changes.

38.6 Operation in Clock Synchronous Mode

Figure 38.61 shows the data format for clock synchronous serial data communications.

In clock synchronous mode, data is transmitted or received in synchronization with clock pulses. For single-character data transfer, data consists of 8-bit. In clock synchronous mode, no parity bit can be added.

In data transmission when CPHA = 1 and CPOL = 1, the SCI outputs data from one falling edge of the synchronization clock to the next falling edge. In data reception, the SCI receives data in synchronization with the rising edge of the synchronization clock. After 8-bit data is output, the transmission line holds the last bit as output state. When the CPHA bit is 0 in slave mode, the transmission line holds the first bit output state.

Within the SCI, the transmitter and receiver are independent units, enabling full-duplex communications by using a shared communication clock of the transmitter and the receiver. Furthermore, because both the transmitter and the receiver also have a double-buffered structure, so that the next transmit data can be written during transmission or the previous receive data can be read during reception, enabling continuous data transfer.

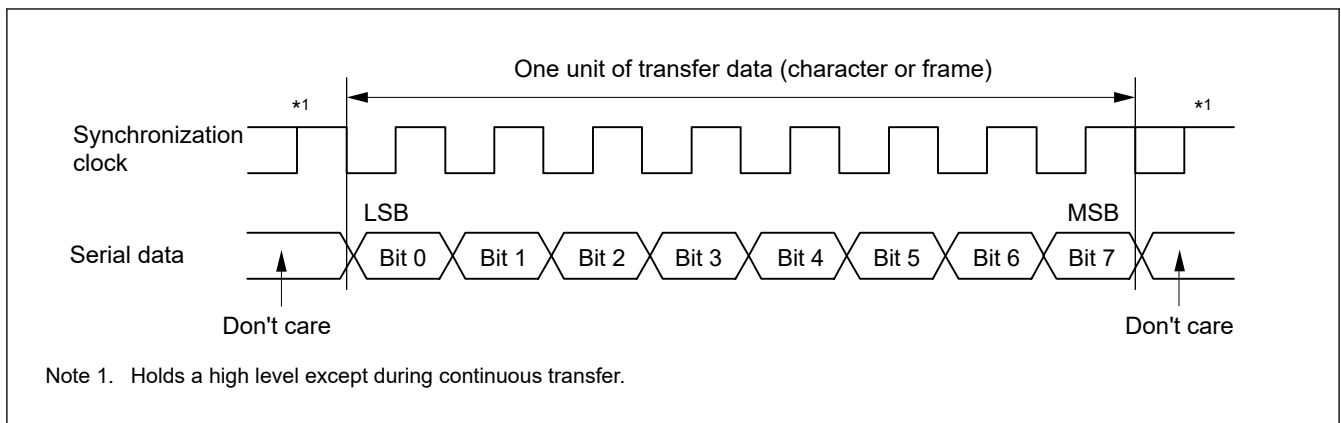


Figure 38.61 Data format in clock synchronous serial communications with LSB-first order

38.6.1 Clock

If the maximum speed of $SCK = 1/2TCLK$ is set in Clock Synchronous and Simple SPI mode, Do not make PCLK less than half the speed of TCLK. If PCLK is made slower than this, malfunction may occur.

1. When the internal clock is selected

When the SCI operates on an internal clock (CCR3.CKE[1:0] bits are set to 00b or 01b (master mode)), the synchronization clock is output from the SCKn pin. Eight synchronization clock pulses are output during single-character transmission/reception. When no data transfer is performed, the clock is held at high level.^{*1} In transmission-only or transmission/reception, the synchronization clock is not output unless transmit data is prepared. When the internal clock is selected, the clock with a delay from the SCKn signal is used for the master reception sampling clock. This ensures the data setup time and hold time for high-speed communication.

Note 1. The signal is held high while (CCR3.CPHA = 0 and CCR3.CPOL = 1) or (CCR3.CPHA = 1 and CCR3.CPOL = 1). It is held low while (CCR3.CPHA = 0 and CCR3.CPOL = 0) or (CCR3.CPHA = 1 and CCR3.CPOL = 0).

2. When the external clock is selected

When the CCR3.CKE[1:0] bits are set to 10b or 11b (slave mode), data is transmitted and received using the external clock that is input from the SCKn pin.

38.6.2 CTS and RTS Functions

In the CTS function, the CTSn_RTSn pin input controls the start of data reception or transmission when the clock source is the internal clock. Setting the CCR1.CTSE bit to 1 enables the CTS function. In clock synchronous communication, the CTS function can be used for the internal clock and the RTS function can be used for the external clock, so the CTS function and RTS function cannot be used at the same time. When the CTS function is enabled, setting the CTSn_RTSn pin low causes data reception or transmission to start.

Setting the CTSn_RTSn pin high while the data transmission or reception is in progress does not affect transmission or reception of the current frame.

In the RTS function, the CTSn_RTSn pin output is used to request the start of data reception or transmission when the clock source is an external synchronizing clock. The CTSn_RTSn output goes low when serial communication is enabled. Conditions for output of the CTSn_RTSn low and high are shown as follows:

[Conditions for low output]

Satisfaction of all the following conditions:

Non-FIFO selected when all of the following conditions are satisfied

- The value of the CCR0.RE bit or the CCR0.TE bit is 1
- Next serial communication is enabled.
 - No receive data is present before reading and not receiving. (when CCR0.RE bit = 1)
 - When the transmission data written in TDR is ready for transmission.*¹ (when CCR0.TE bit = 1)
- The CSR.ORER flag is 0

Note 1. The CTSn_RTSn pin will be High after starting transmission.

FIFO selected when all of the following conditions are satisfied

- The value of the CCR0.RE bit or the CCR0.TE bit is 1
- Next serial communication is enabled.
 - The number of receive data stored in the receive FIFO (RDR register) is less than the value set in FCR.RSTRG[4:0] (when CCR0.RE bit = 1)
 - When the transmission data written in the transmission FIFO (TDR register) is ready for transmission.*¹ (when CCR0.TE bit = 1)
- The CSR.ORER flag is 0

Note 1. The CTSn_RTSn pin will be High after the last data transmission starts.

[Condition for high output]

- The conditions for low output are not satisfied

38.6.3 SCI Initialization in Clock Synchronous Mode

Before transmitting and receiving data, start by writing the initial value 0x00 to the CCR0 register, then continue through the SCI initialization procedure given in the sections describing non-FIFO and FIFO selection in [section 38.6.2. CTS and RTS Functions](#). Anytime the operating mode or transfer format is to be changed, the CCR0.TE and CCR0.RE must write to 0 before the change can be made.

Note: Setting the CCR0.RE bit to 0 initializes neither the ORER, FER, PER, and RDRF flags in CSR nor the RDR register. When the TE bit is set to 0, the TEND flag for the selected FIFO buffer is not initialized.

Note: Switching the value of the CCR0.TE bit from 1 to 0 when the CCR0.TIE bit is 1 generates an SCIn_TXI interrupt request.

Table 38.38 Example flow of SCI initialization in clock synchronous mode with non-FIFO selected (1 of 2)

No.	Step Name	Description
1	Start initialization	—
2	Set CCR0	Set the CCR0.TEIE, TIE, RIE, TE, and RE bits to 0* ¹ . If you have not changed from the initial settings, you can skip this step.
3	Set FCR	Set the TFRST and RFRST to 1 to empty FIFO. Set the TTRG[4:0], RTRG[4:0], and RSTRG[4:0] bits.

Table 38.38 Example flow of SCI initialization in clock synchronous mode with non-FIFO selected (2 of 2)

No.	Step Name	Description
4	Set CCR3 except MOD[2:0]	Set CCR3 except of communication mode. - FIFO use/no-use - Transmission/reception format - Clock setting - Leave unused bits (CHR[1:0], STP, RXDSEL, MP, DE, ACS0, GM, BLK) at their initial values.
5	Set CCR3.MOD[2:0]	Set communication mode (MOD[2:0] = 010b) ^{*2} .
6	Set CCR2	Select clock, set bit rate ^{*3} . Leave unused bits (BCP[2:0], ABCS, ABCSE, BRME, MDDR[7:0]) at their initial values.
7	Set CCR1	Set up the loop-back function, communication pin status and the CTS/RTS function.
8	Set CCR4	Set up the adjust sampling timing function. Leave unused bits (CMPD[8:0]) at their initial values.
9	Set the I/O port functions	Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.
10	Set CFCLR, FFCLR	Write 1 to the following bits and clear the corresponding flag. CFCLR.RDRFC, FERC, PERC, MFFC, ORERC, DFERC, DPERC, DCMFC, ERSC FFCLR.BRKC, DRC
11	Set CCR0	Set the TE or RE bit to 1. ^{*1 *4} To enable interrupts, set the TE bit and TIE bit, and the RE bit and RIE bit to 1 with one instruction at the same time. Setting the TE and RE bits allows TXDn and RXDn to be used.
12	Initialization completion	—

Note 1. In simultaneous transmit and receive operations, the TE and RE bits in CCR0 should both be to 0 or set to 1 simultaneously.

Note 2. Set CPOL and CPHA before setting the communication mode.

Note 3. If you use an external clock, you do not need to set it.

Note 4. When using the internal clock (master), the setting of reception only is prohibited.

38.6.4 Serial Data Transmission in Clock Synchronous Mode

(1) Non-FIFO selected

Figure 38.62, Figure 38.63, and Figure 38.64 show examples of serial transmission in clock synchronous mode.

In serial data transmission, the SCI operates as follows:

1. The SCI transfers data from the TDR register to the TSR register when data is written to TDR in the SCIn_TXI interrupt handling routine. When starting data transmission, set the CCR0.TIE bit and the CCR0.TE bit to 1 simultaneously by a single instruction. Then a SCIn_TXI interrupt request is generated.
2. After transferring data from TDR to TSR, the SCI starts transmission. When the CCR0.TIE bit is set to 1, an SCIn_TXI interrupt request is generated. Continuous transmission is enabled by writing the next transmit data to TDR in the SCIn_TXI interrupt handling routine before transmission of the current transmit data finishes. When SCIn_TEI interrupt requests are in use, set the CCR0.TIE bit to 0 and the CCR0.TEIE bit to 1 after the last of the data to be transmitted is written to the TDR register from the handling routine for SCIn_TXI requests.
3. 8-bit data is sent from the TXDn pin in synchronization with the output clock when the clock output mode is specified and in synchronization with the input clock when the use of an external clock is specified. Output of the clock signal is suspended until the input CTS signal is low when the CCR1.CTSE bit is 1 (CTS function enabled).
4. The SCI checks for update to the TDR register on output of the last bit.
5. When the TDR register is updated, the next transmit data is transferred from TDR to TSR, and serial transmission of the next frame starts.
6. If TDR is not updated, the CSR.TEND flag is set to 1. The TXDn pin retains the output state of the last bit. If the CCR0.TEIE bit is 1, an SCIn_TEI interrupt request is generated and the SCKn pin is held high.

Figure 38.62, Figure 38.63, and Figure 38.64 show examples of serial data transmission.

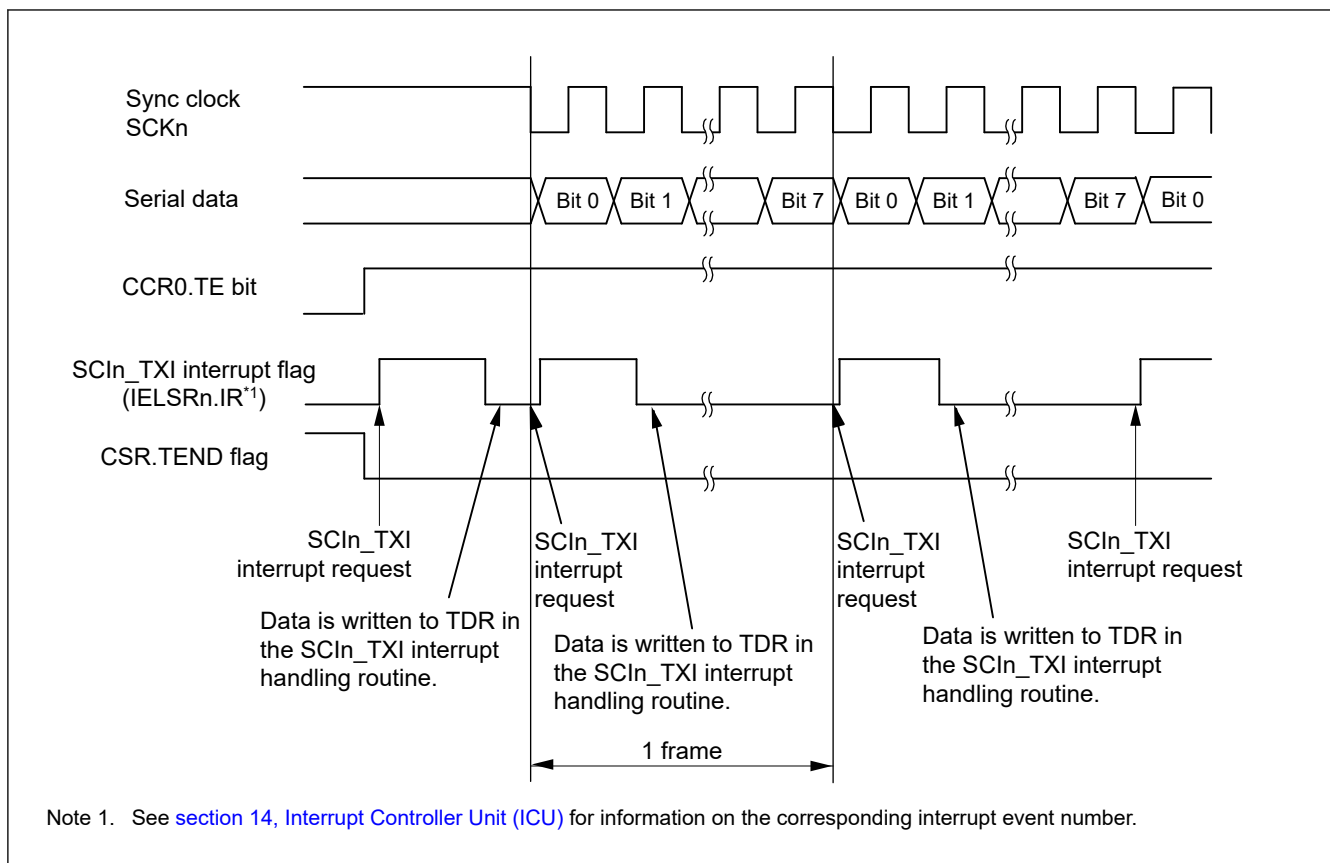


Figure 38.62 Example of serial data transmission in clock synchronous mode when the CTS function is not used at the beginning of transmission

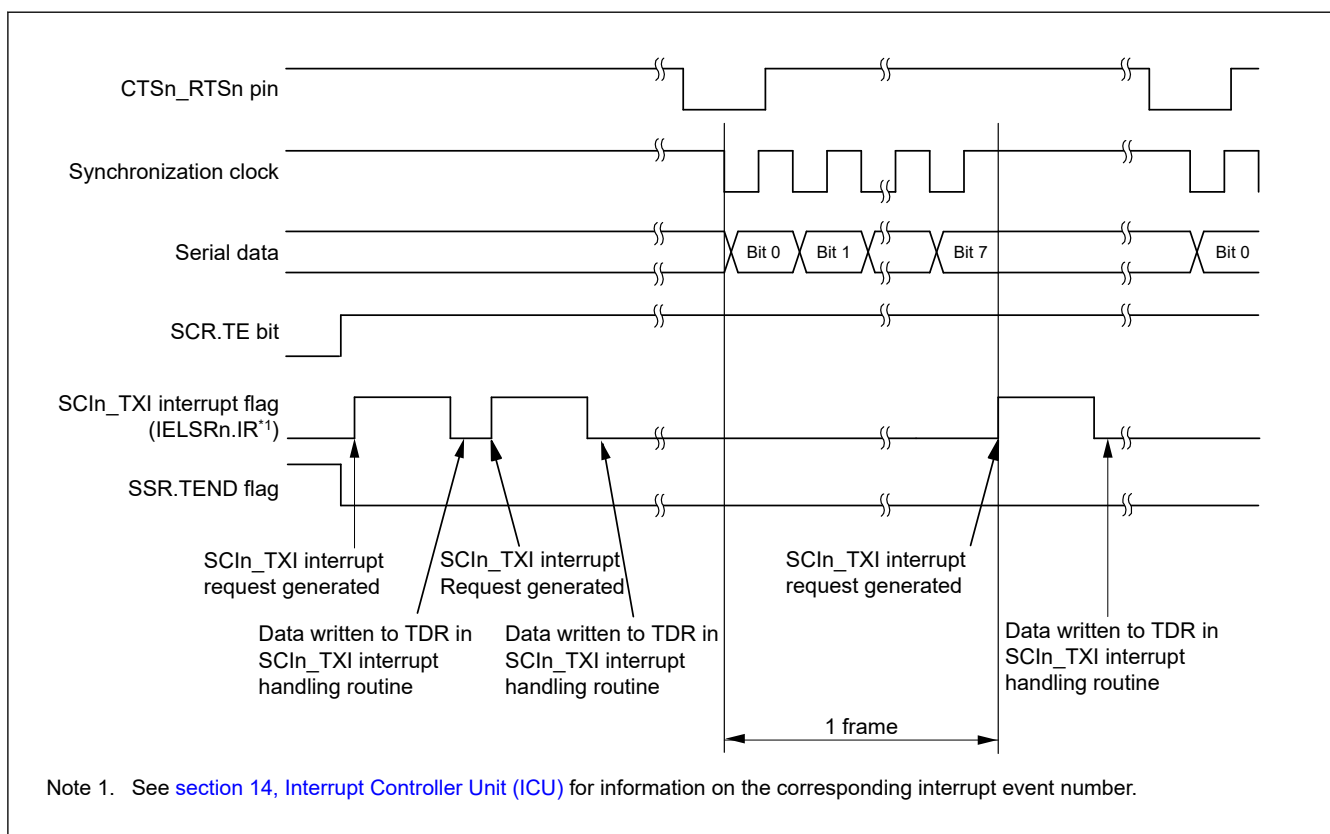


Figure 38.63 Example of serial data transmission in clock synchronous mode when the CTS function is used at the beginning of transmission

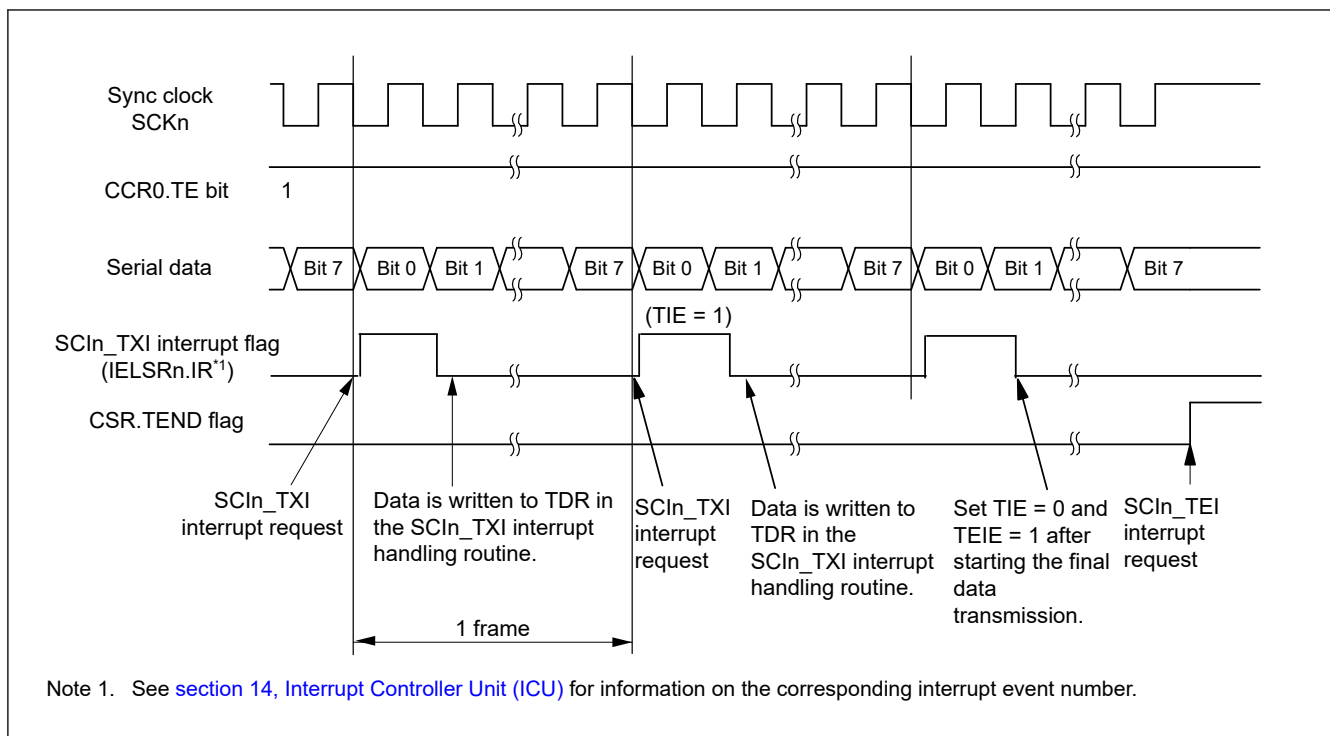


Figure 38.64 Example of serial data transmission in clock synchronous mode from the middle of transmission until transmission completion

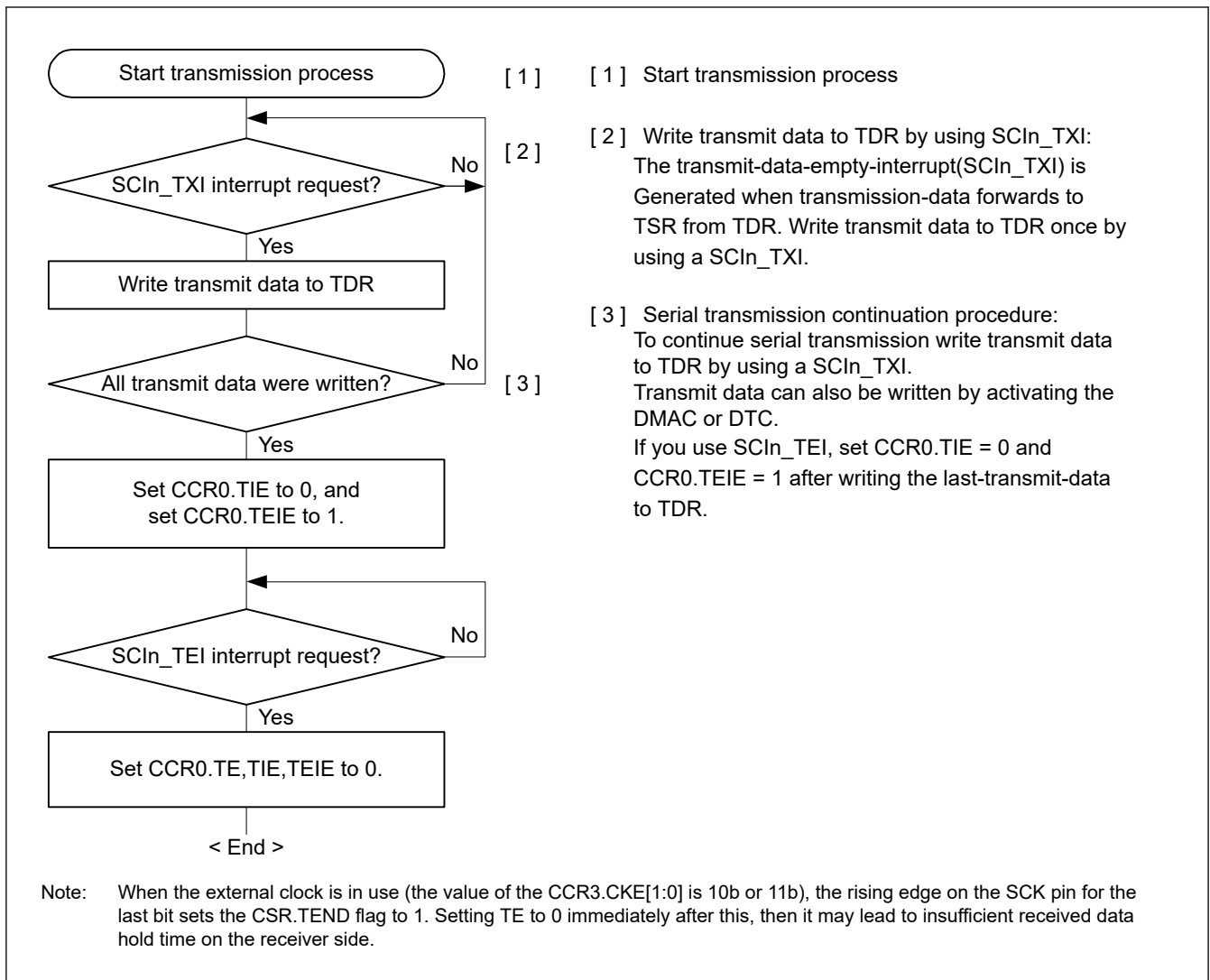


Figure 38.65 Example flow of serial transmission in clock synchronous mode with non-FIFO selected

(2) FIFO selected

Figure 38.66 shows an example of serial transmission in clock synchronous mode with FIFO selected.

In serial data transmission, the SCI operates as follows:

1. The SCI transfers data from the transmit-FIFO (TDR register) to the TSR register when data is written to transmit-FIFO (TDR register) in the SCIn_TXI interrupt handling routine. The amount of data that can be written to transmit-FIFO (TDR register) is 16 - FTSR.T[5:0] bytes. In addition, when starting data transmission, set the CCR0.TIE bit and the CCR0.TE bit to 1 simultaneously by a single instruction. Then a SCIn_TXI interrupt request is generated.
2. After transferring data from transmit-FIFO (TDR register) to TSR, the SCI starts transmission. When the amount of transmit data written in transmit-FIFO (TDR register) is equal to or less than the specified transmit triggering number, the CSR.TDRE is set to 1. When the CCR0.TIE bit is set to 1, an SCIn_TXI interrupt request is generated. Continuous transmission is enabled by writing the next transmit data to transmit-FIFO (TDR register) in the SCIn_TXI interrupt handling routine before transmission of the current transmit data has finished. When SCIn_TEI interrupt requests are in use, set the CCR0.TIE bit to 0 and the CCR0.TEIE bit to 1 after the last of the data to be transmitted is written to the transmit-FIFO (TDR register) from the handling routine for SCIn_TXI requests.
3. 8-bit data is sent from the TXDn pin in synchronization with the output clock when the clock output mode is specified and in synchronization with the input clock when the use of an external clock is specified. Output of the clock signal is suspended until the input CTS signal is low when the CCR1.CTSE bit is 1 (CTS function enabled).
4. The SCI checks whether non-transmitted data remains in transmit-FIFO (TDR register)*¹ on output of the stop bit.
5. When data is remaining in the transmit-FIFO (TDR register), the next transmit data is transferred from transmit-FIFO (TDR register) to TSR and serial transmission of the next frame starts.

6. If no data is remaining in the transmit FIFO (TDR register), the CSR.TEND flag is set to 1. The TXDn pin retains the output state of the last bit. If the CCR0.TEIE bit is 1, an SCIn_TEI interrupt request is generated and the SCKn pin is held high.

Note 1. The number of unsent transmit data stored in the TDR register (transmit FIFO) can be monitored by reading the FTSR.T[5:0] bits.

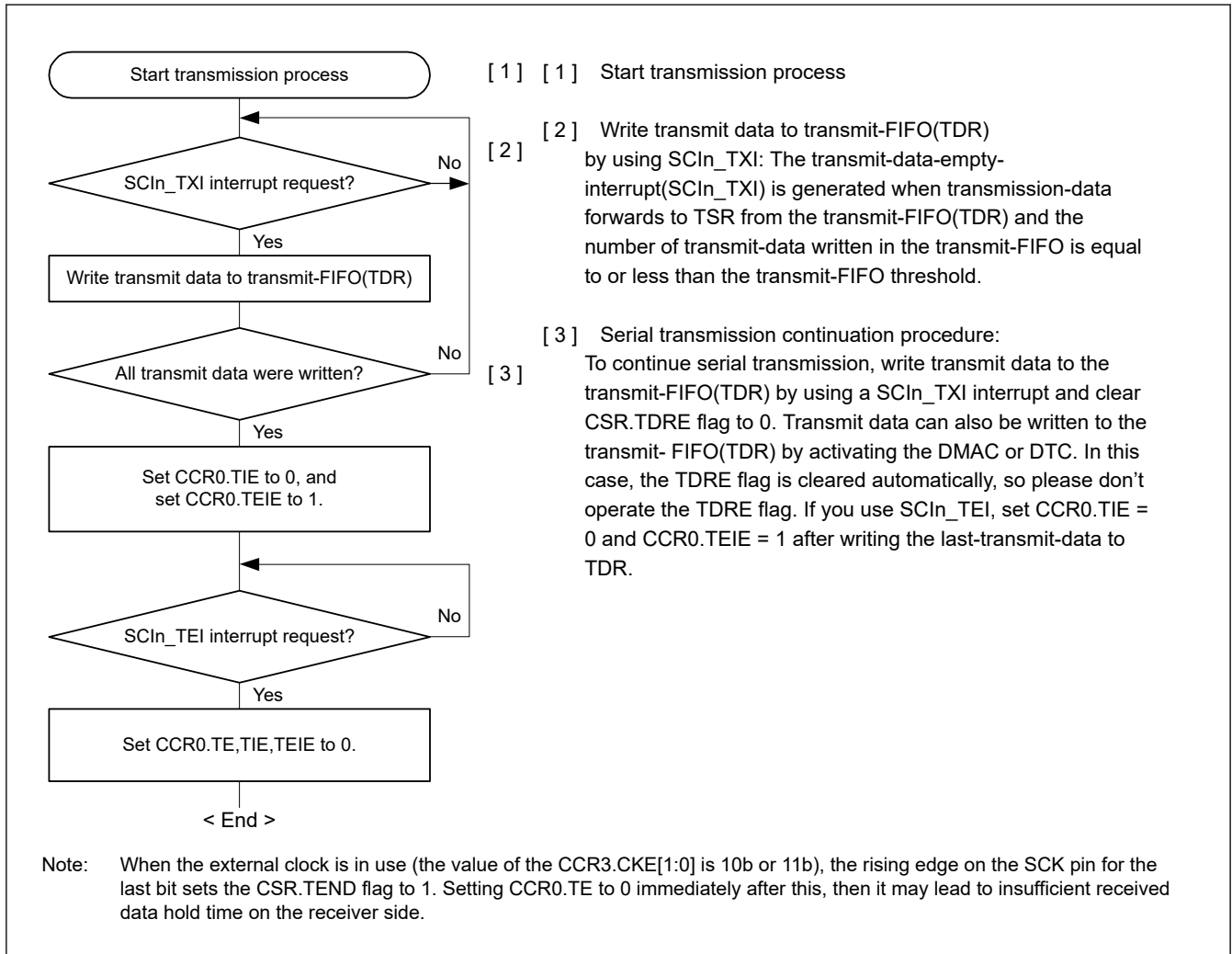


Figure 38.66 Example flow of serial transmission in clock synchronous mode with FIFO selected

38.6.5 Serial Data Reception in Clock Synchronous Mode

(1) Non-FIFO selected

Figure 38.67 and Figure 38.68 show examples of SCI operation for serial reception in clock synchronous mode.

In serial data reception, the SCI operates as follows during serial data reception. Reception-only operation is possible only in slave mode. (In master mode, reception-only operation is prohibited.)

1. When the value of the CCR0.RE bit becomes 1, the CTSn_RTSn pin goes low (when the RTS function is used).
2. The SCI performs internal initialization and starts receiving data in synchronization with a synchronization clock input or output, and stores the receive data in the RSR register.
3. If an overrun error occurs, the CSR.ORER flag is set to 1. If the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated. Receive data is not transferred to the RDR register.
4. When reception completes successfully, receive data is transferred to the RDR register. If the CCR0.RIE bit is 1, an SCIn_RXI interrupt request is generated. Continuous reception is enabled by reading the received data transferred to the

RDR register in the SCIn_RXI interrupt handling routine before reception of the next receive data completes. Reading the received data that is transferred to RDR causes the CTSn_RTSn pin to output low (when the RTS function is used).

If you want to prevent the CTSn_RTSn pin output from turning low level after the final data is received, clear the CCR0.RE bit to 0 and then read the RDR register.

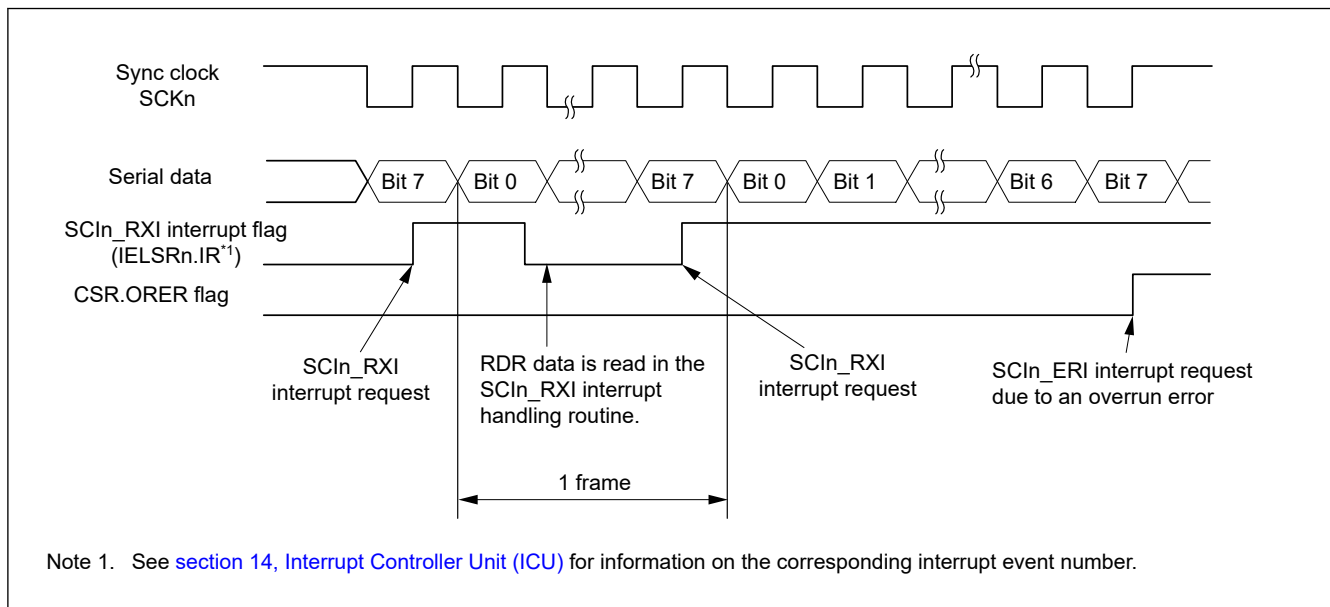


Figure 38.67 Example operation for serial reception in clock synchronous mode (1) when the RTS function is not used

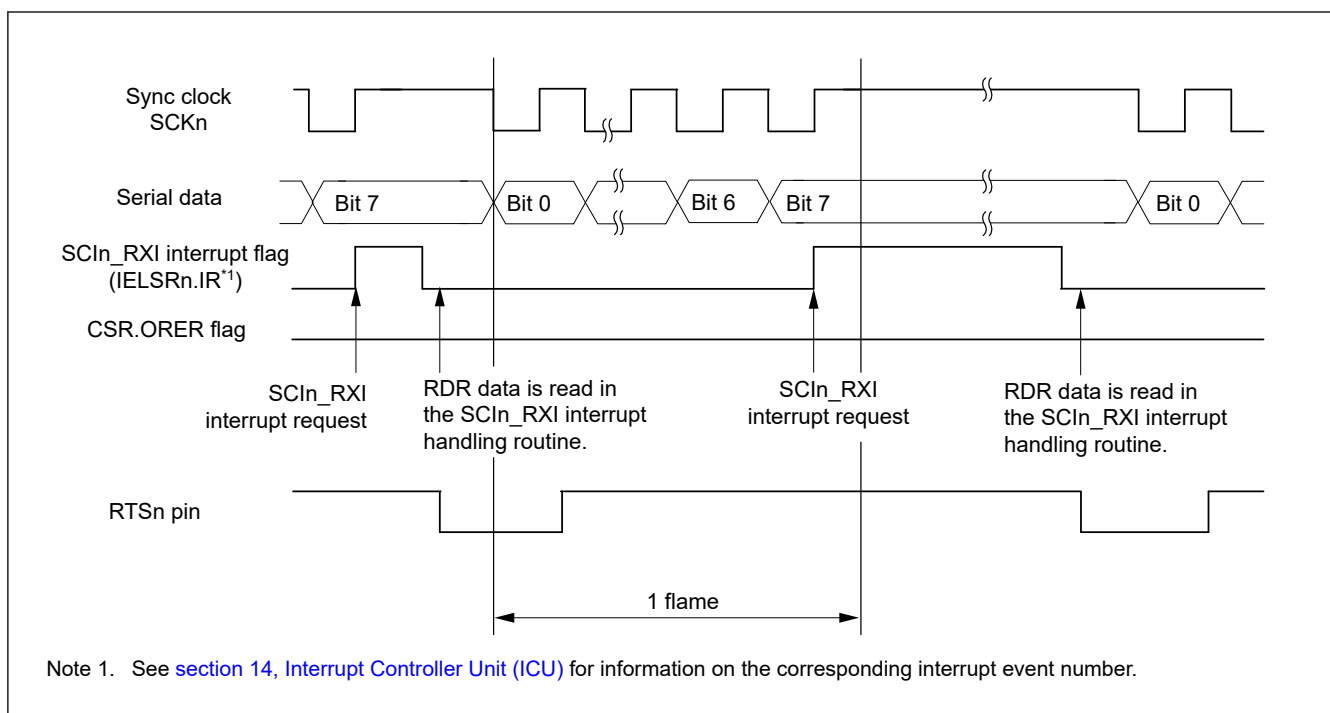


Figure 38.68 Example operation for serial reception in clock synchronous mode (2) when RTS function is used

Data transfer cannot resume while the receive error flag is 1. Therefore, clear the ORER, FER, and PER flags in the CSR register to 0 before resuming data reception. Additionally, always read the RDR register during overrun error processing. When a data reception is forced to terminate by a 0 write to the CCR0.RE bit during operation, read the RDR register because received data that is not yet read might be left in the RDR register.

[Figure 38.69](#) shows an example flow of serial data reception.

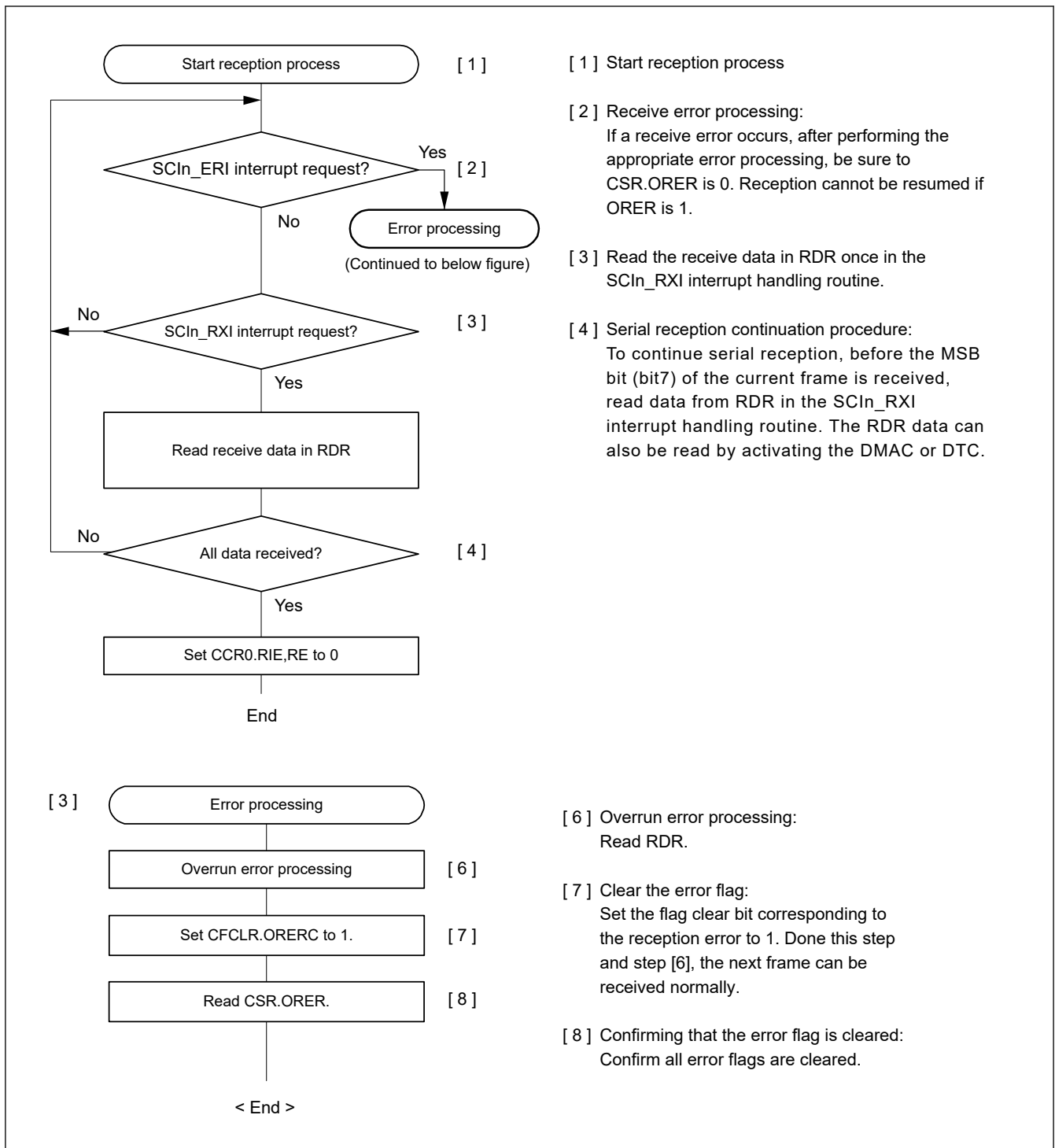


Figure 38.69 Example flow of serial reception in clock synchronous mode with non-FIFO selected

(2) FIFO selected

Figure 38.70 shows an example of serial reception in clock synchronous mode with FIFO selected.

In serial data reception, the SCI operates as follows during serial data reception. Reception-only operation is possible only in slave mode. (In master mode, reception-only operation is prohibited.)

1. When the value of the CCR0.RE bit becomes 1, the CTSn_RTSn pin goes low (when the RTS function is used).
2. The SCI performs internal initialization and starts receiving data in synchronization with a synchronization clock input or output, and transfers the received data to the receive-FIFO (RDR register).
3. If an overrun error occurs, the CSR.ORER flag is set to 1. If the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated. Received data is not transferred to the receive-FIFO (RDR register)*1.

4. When data reception completes successfully, the receive data is transferred to the receive-FIFO (RDR register)*¹. The CSR.RDRF flag is set to 1 when the amount of the receive data stored in receive-FIFO (RDR register) is equal to or greater than the specified receive triggering number. If the CCR0.RIE bit is 1, an SCIn_RXI interrupt request is generated. Continuous data reception is enabled by reading the receive data transferred to receive-FIFO (RDR register)*¹ in the SCIn_RXI interrupt handling routine before an overrun error occurs. If the amount of received data that is transferred to receive-FIFO (RDR register) is less than the specified receive triggering number, the CTSn_RTSn pin goes low (when the RTS function is used).

Note 1. In clock synchronous mode, RDR.RDAT[8] is not used.

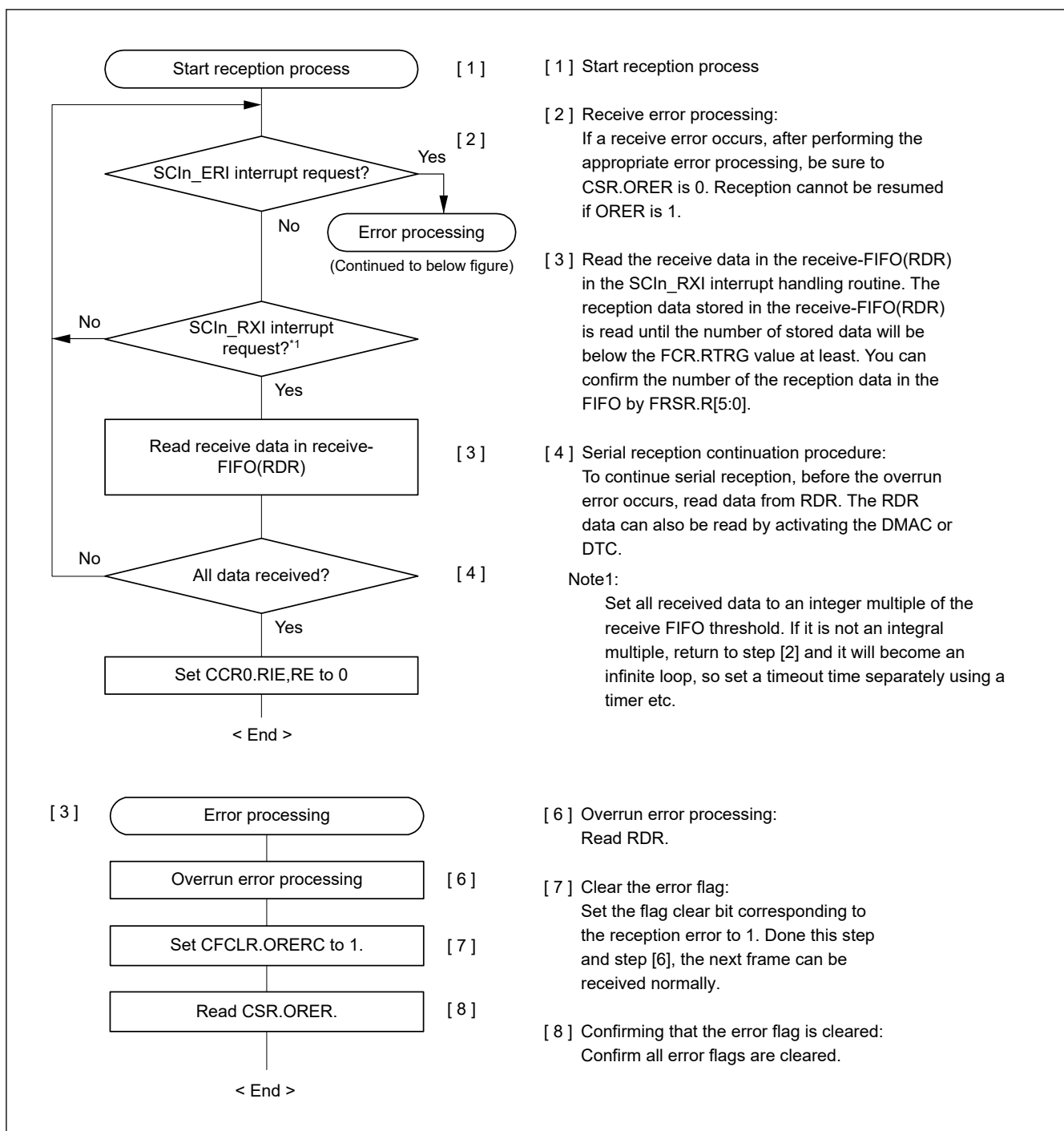


Figure 38.70 Example flow of serial reception in clock synchronous mode with FIFO selected

38.6.6 Simultaneous Serial Data Transmission and Reception in Clock Synchronous Mode

(1) Non-FIFO selected

Figure 38.71 shows an example flow of simultaneous serial transmit and receive operations in clock synchronous mode. After initializing the SCI, use the following procedure for simultaneous serial data transmit and receive operations.

To switch from transmit mode to simultaneous transmit and receive mode:

1. Check that the SCI completes the data transmission by verifying that the CSR.TEND flag is set to 1.
2. Initialize the CCR0 register, and then set the TIE, RIE, TE, and RE bits in the CCR0 register to 1 simultaneously by a single instruction.

To switch from receive mode to simultaneous transmit and receive mode:

1. Check that the SCI completes the data reception.
2. Set the CCR0.TE and RE bits to 0, and then check that the receive error flag (ORER, FER, and PER) in the CSR register is 0.
3. Set the TIE, RIE, TE, and RE bits in the CCR0 register to 1 simultaneously by a single instruction.

When the RTS function is used in the concurrent transmission/reception operation, if you want to prevent the CTSn_RTSn pin output from turning to low after the final data is received as in the reception operation, clear the RE and TE bits in CCR0 to 0 simultaneously, and then read the RDR register.

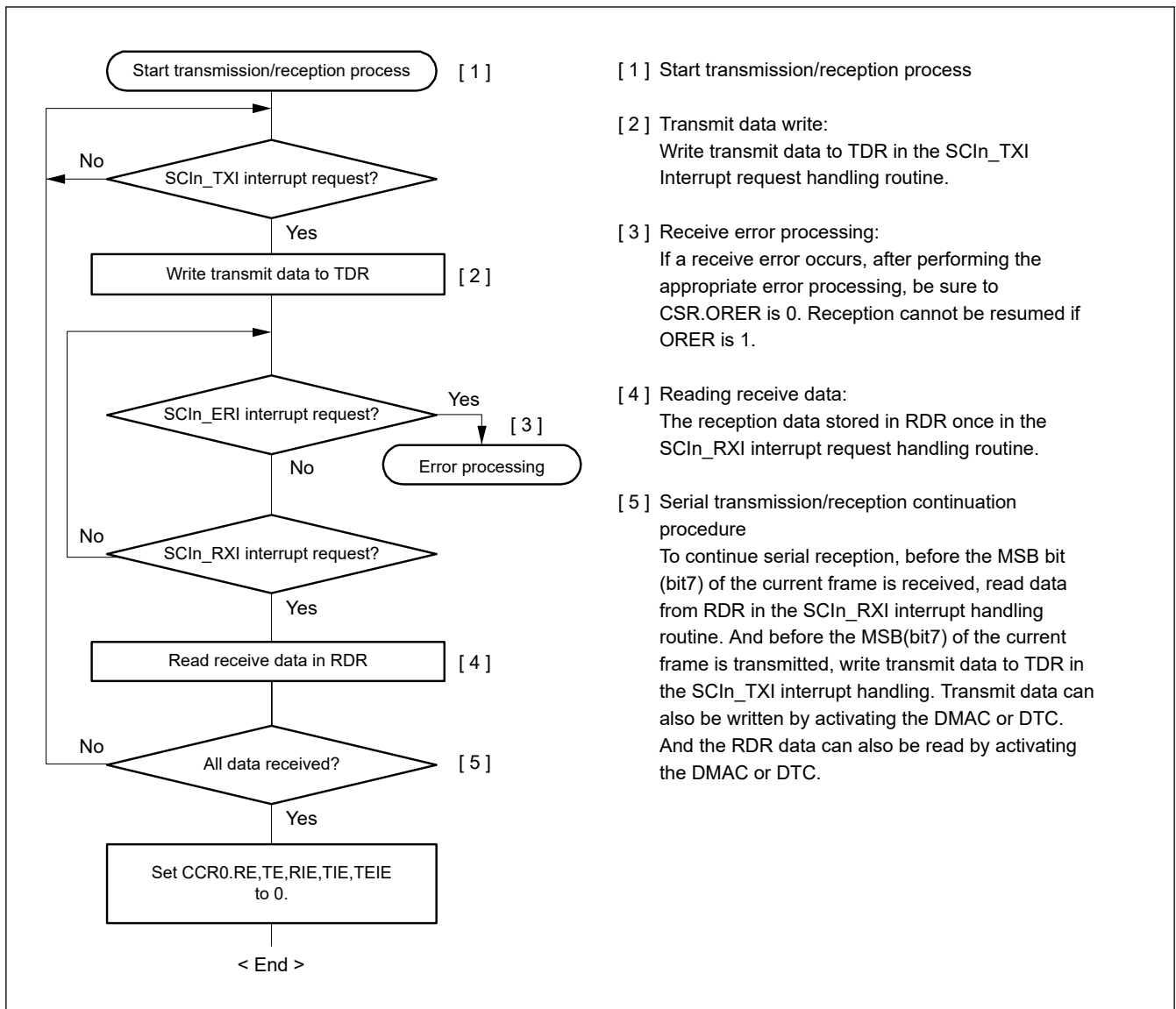


Figure 38.71 Example flow of simultaneous serial transmission and reception in clock synchronous mode with non-FIFO selected

(2) FIFO selected

Figure 38.72 shows an example flow of simultaneous serial transmit and receive operations in clock synchronous mode with FIFO selected.

After initializing the SCI, use the following procedure for simultaneous serial data transmit and receive operations.

To switch from transmit mode to simultaneous transmit and receive mode:

1. Check that the SCI completes the transmission by verifying that the CSR.TEND flag is set to 1.
2. Initialize the CCR0 register, then set the TIE, RIE, TE, and RE bits in the CCR0 register to 1 simultaneously by a single instruction.

To switch from receive mode to simultaneous transmit and receive mode:

1. Check that the SCI completes the reception.
2. Set the CCR0.TE and RE bits to 0.
3. Check that the receive error flags (ORER, FER, and PER) in the CSR register are 0, and then set the TIE, RIE, TE, and RE bits in the CCR0 register to 1 simultaneously by a single instruction.

Since clock synchronous communication performs transmission and reception at the same time, make sure that the number of data to be transmitted and received is the same.

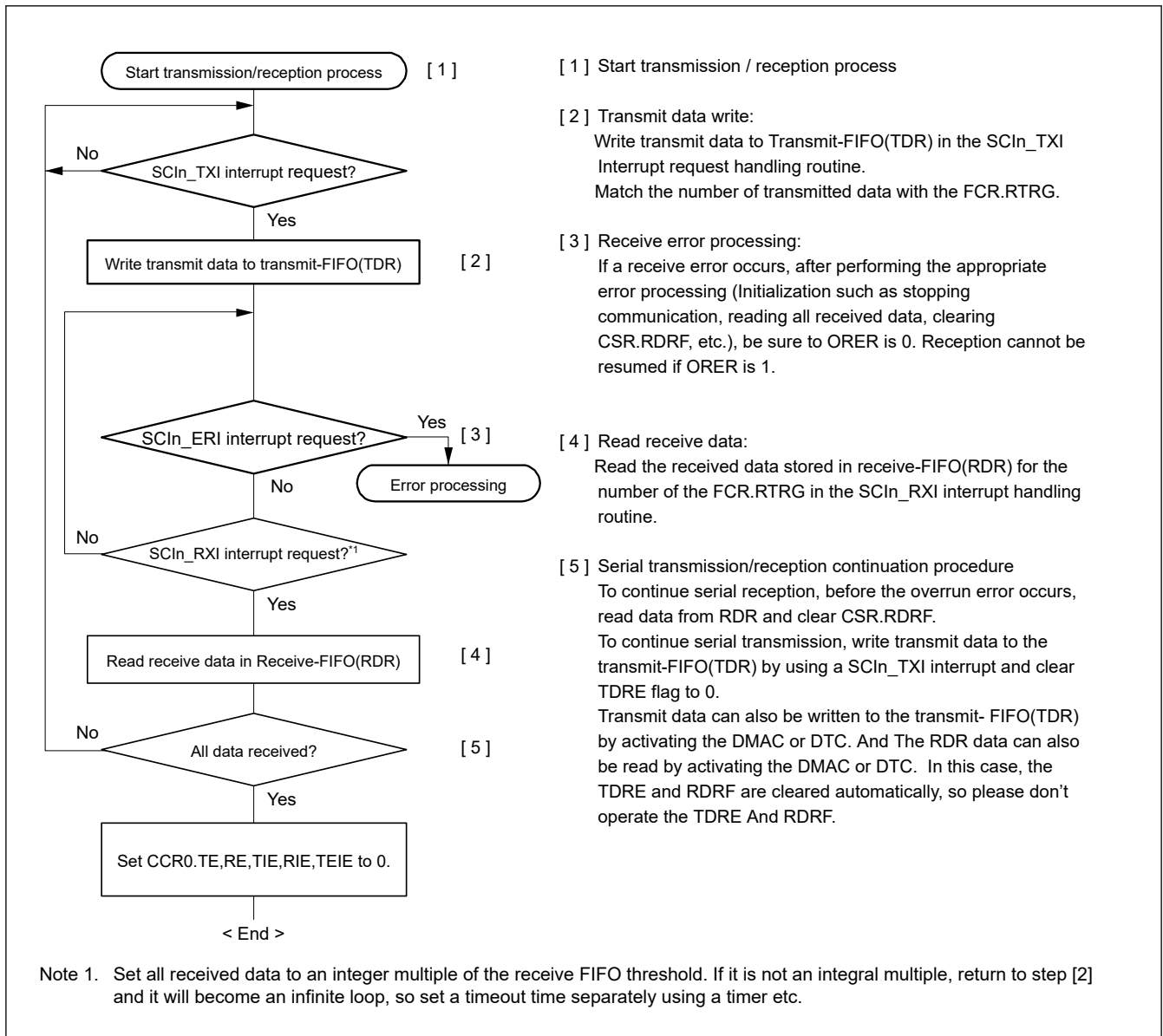


Figure 38.72 Example flow of simultaneous serial transmission and reception in clock synchronous mode with FIFO selected

38.6.7 Reception Sampling Timing Adjustment Function in Clock Synchronous Mode with Internal Clock Used

When the clock synchronous internal clock is used (master mode), MRCLK is used as a reception sampling clock.

This function adjusts the reception sampling timing by delaying MRCLK by 1 to 4 TCLK and adding a digital delay. MRCLK's analog delay cannot be adjusted by this function.

Setting the CCR4.ASEN bit to 1 enables this function. The delay value is set in CCR4.AST[1:0].

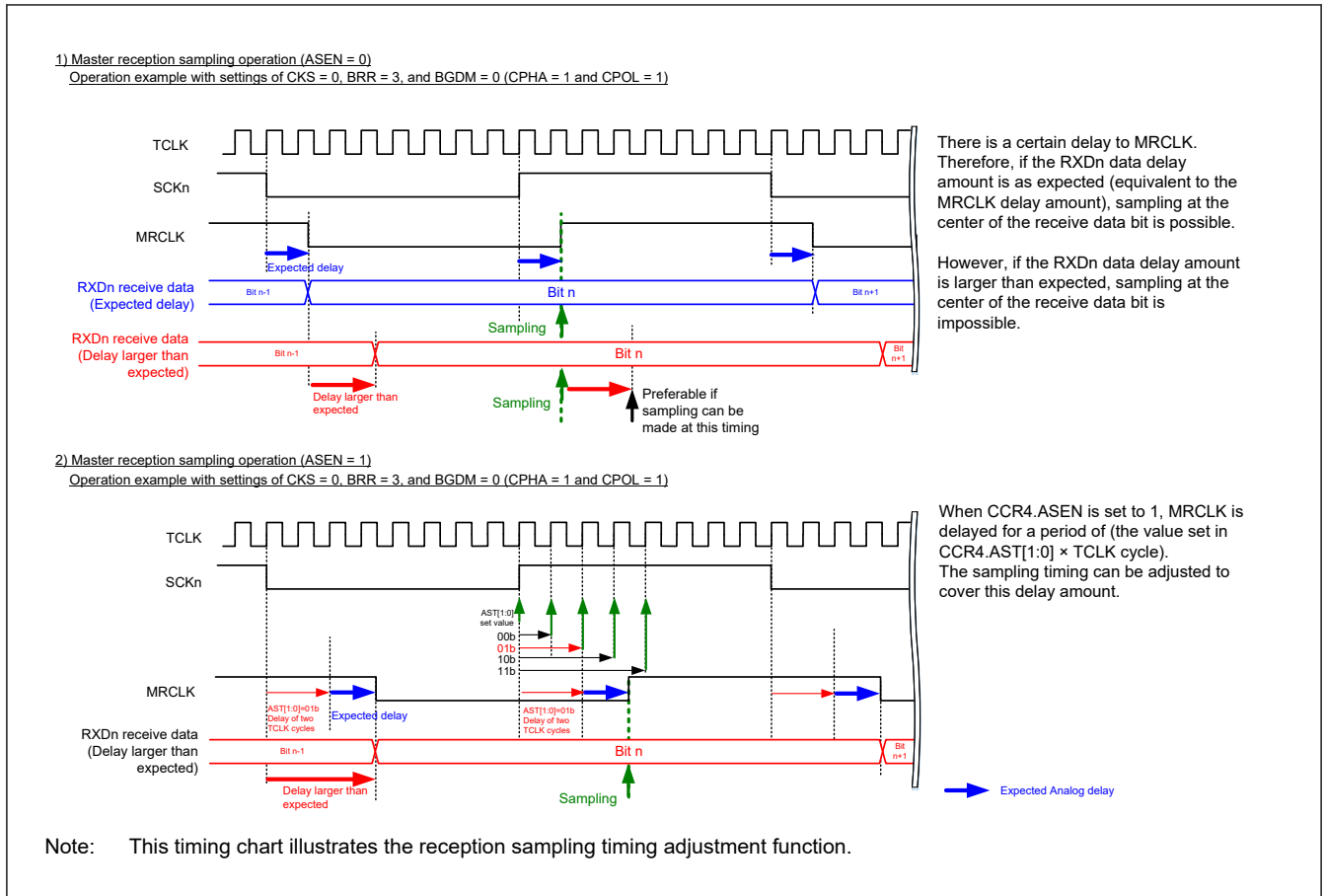


Figure 38.73 Reception Sampling Timing Adjustment Operation in Clock Synchronous Mode (Master)

38.7 Operation in Smart Card Interface Mode

The SCI supports smart card (IC card) interfaces conforming to ISO/IEC 7816-3 (standard for Identification Cards), as an extended function of the SCI.

Smart card interface mode can be selected using the appropriate register.

38.7.1 Example Connection

Figure 38.74 shows an example connection between a smart card (IC card) and the MCU. As shown in Figure 38.74, because the MCU communicates with an IC card using a single transmission line, interconnect the TXDn and RXDn pins and pull up the data transmission line to VCC using a resistor.

Setting the CCR0.TE and CCR0.RE bits to 1 with an IC card disconnected enables closed-loop transmission or reception, allowing self-diagnosis. To supply an IC card with the clock pulses generated by the SCI, input the SCKn pin output to the CLK pin of an IC card.

An output port of the MCU can be used to output a reset signal.

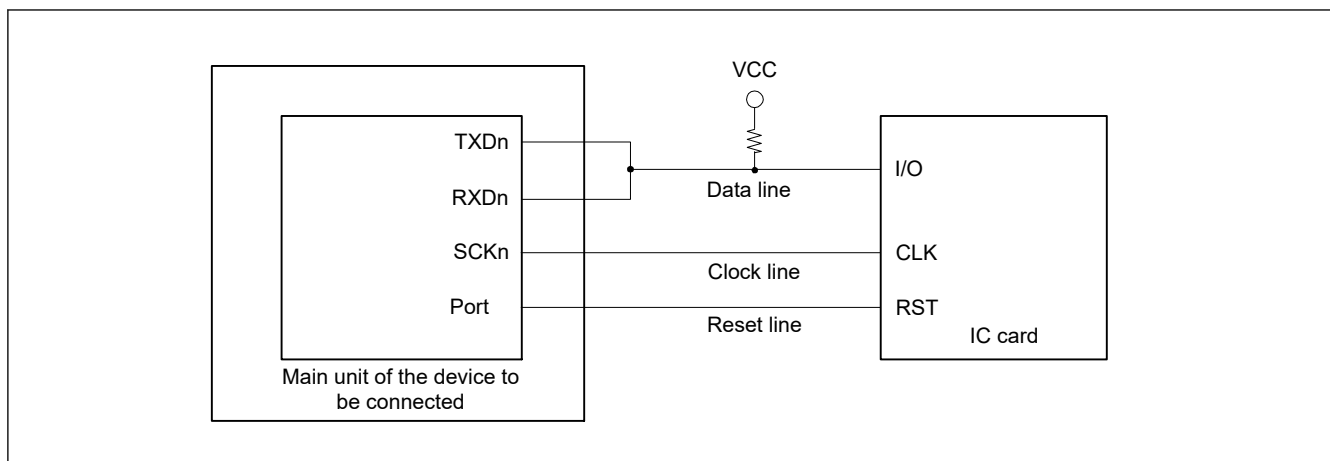


Figure 38.74 Example connection with a smart card (IC card)

38.7.2 Data Format (Except in Block Transfer Mode)

Figure 38.75 shows the data transfer formats in smart card interface mode:

- One frame consists of 8-bit data and a parity bit in Asynchronous mode.
- During transmission, at least 2 etu (elementary time unit – the time required for transferring 1 bit) is set as a guard time from the end of the parity bit until the start of the next frame.
- If a parity error is detected during reception, a low error signal is output for 1 etu after 10.5 etu elapse from the start bit.
- If an error signal is sampled during transmission, the same data is automatically retransmitted after at least 2 etu.

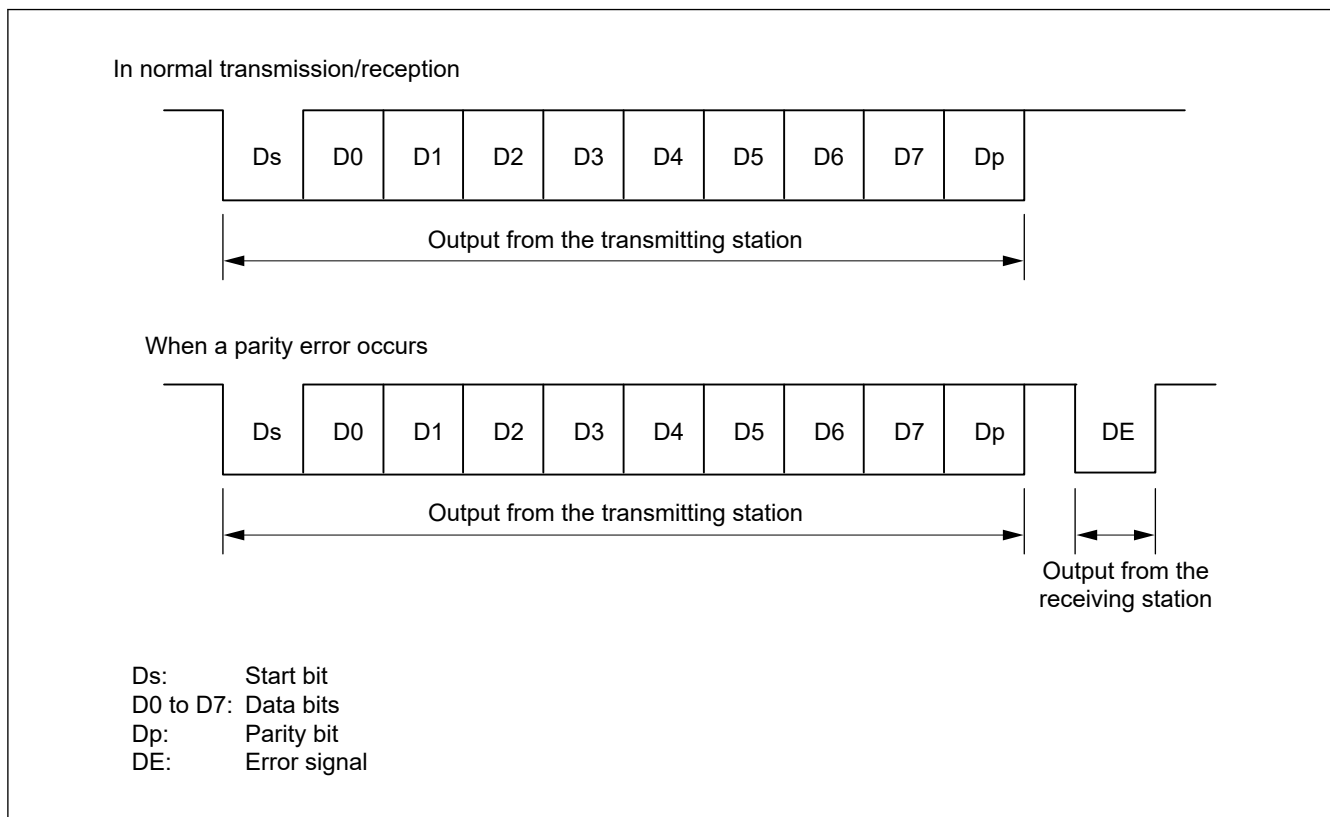


Figure 38.75 Data formats in smart card interface mode

For communications with IC cards of the direct convention type and inverse convention type, follow the procedures in this section.

(1) Direct Convention Type

For the direct convention type, logic levels 1 and 0 indicate the Z and A states, respectively, and data is transferred with LSB-first for the start character, as shown in Figure 38.76. Therefore, data in the start character in the figure is 0x3B.

When using the direct convention type, write 1 to the CCR3.LSBF and write 0 to the CCR3.SINV. Write 0 to the CCR1.PM bit to use even parity, which is prescribed by the smart card standard.

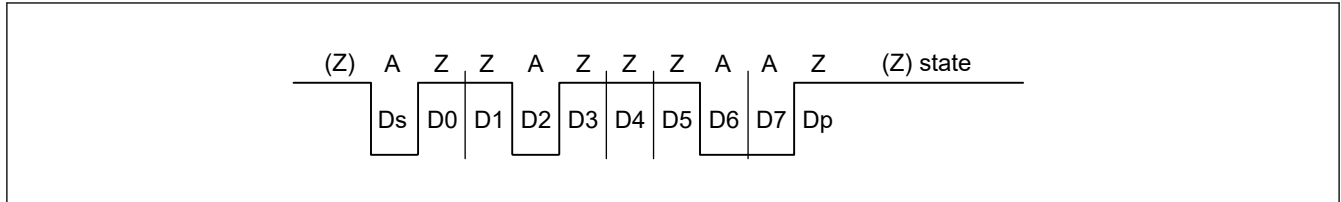


Figure 38.76 Direct convention with LSBF in CCR3 = 1, SINV in CCR3 = 0, and PM in CCR1 = 0

(2) Inverse Convention Type

For the inverse convention type, logic levels 1 and 0 indicate the A and Z states, respectively, and data is transferred with MSB-first for the start character, as shown in Figure 38.77. Therefore, data in the start character in the figure is 0x3F.

When using the inverse convention type, write 0 to the CCR3.LSBF and write 1 to the CCR3.SINV. The parity bit is logic level 0 to produce even parity, which is prescribed by the smart card standard, and corresponds to the Z state. Because the SINV bit of the MCU only inverts data bits D7 to D0, write 1 to the PM bit in CCR1 to invert the parity bit for both transmission and reception.

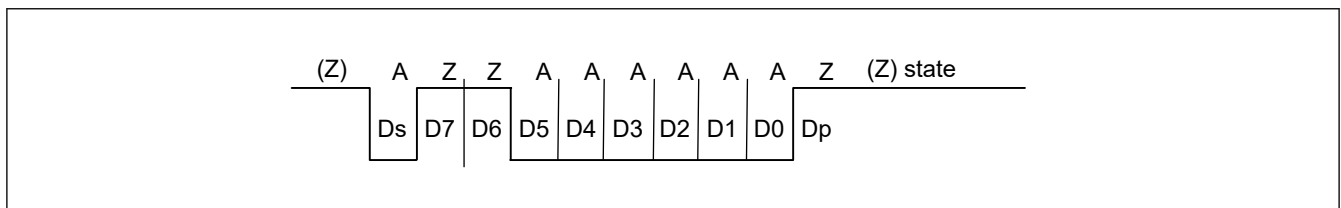


Figure 38.77 Inverse convention with LSBF in CCR3 = 0, SINV in CCR3 = 1, and PM in CCR1 = 1

38.7.3 Block Transfer Mode

Block transfer mode differs from normal smart card interface mode as follows:

- Even if a parity error is detected during reception, no error signal is output. Because the PER flag in CSR is set by error detection, clear the PER flag before receiving the parity bit of the next frame.
- During transmission, at least 1 etu is set as a guard time from the end of the parity bit until the start of the next frame
- Because the same data is not retransmitted, the TEND flag in CSR is set to 11.5 etu after transmission starts
- In block transfer mode, the ERS flag in CSR indicates the error signal status as in normal smart card interface mode, but the flag is read as 0 because no error signal is transferred

38.7.4 Receive Data Sampling Timing and Reception Margin

Only the internal clock generated by the on-chip baud rate generator can be used as a transfer clock in smart card interface mode.

In this mode, the SCI can operate on a base clock with a frequency of 32, 64, 372, 256, 93, 128, 186, or 512 times the bit rate set up in the CCR2.BCP[2:0] bits. The frequency is always 16 times the bit rate in normal Asynchronous mode.

For data reception, the falling edge of the start bit is sampled with the base clock to perform internal synchronization.

Receive data is sampled on the 16th, 32nd, 186th, 128th, 46th, 64th, 93rd, and 256th rising edges of the base clock so that it can be latched at the middle of each bit as shown in Figure 38.78. The reception margin is determined by the following formula:

$$M = \left| \left(0.5 - \frac{1}{2N} \right) - (L - 0.5)F - \frac{|D - 0.5|}{N}(1 + F) \right| \times 100 \text{ [%]}$$

M: Reception margin (%)

N: Ratio of bit rate to clock ($N = 32, 64, 372, 256$)

D: Duty cycle of clock ($D = 0$ to 1.0)

L: Frame length ($L = 10$)

F: Absolute value of clock frequency deviation

Assuming values of $F = 0$, $D = 0.5$, and $N = 372$ in the specified formula, the reception margin is determined using the following formula:

$$M = \{0.5 - 1/(2 \times 372)\} \times 100 \text{ [%]} = 49.866 \text{ %}$$

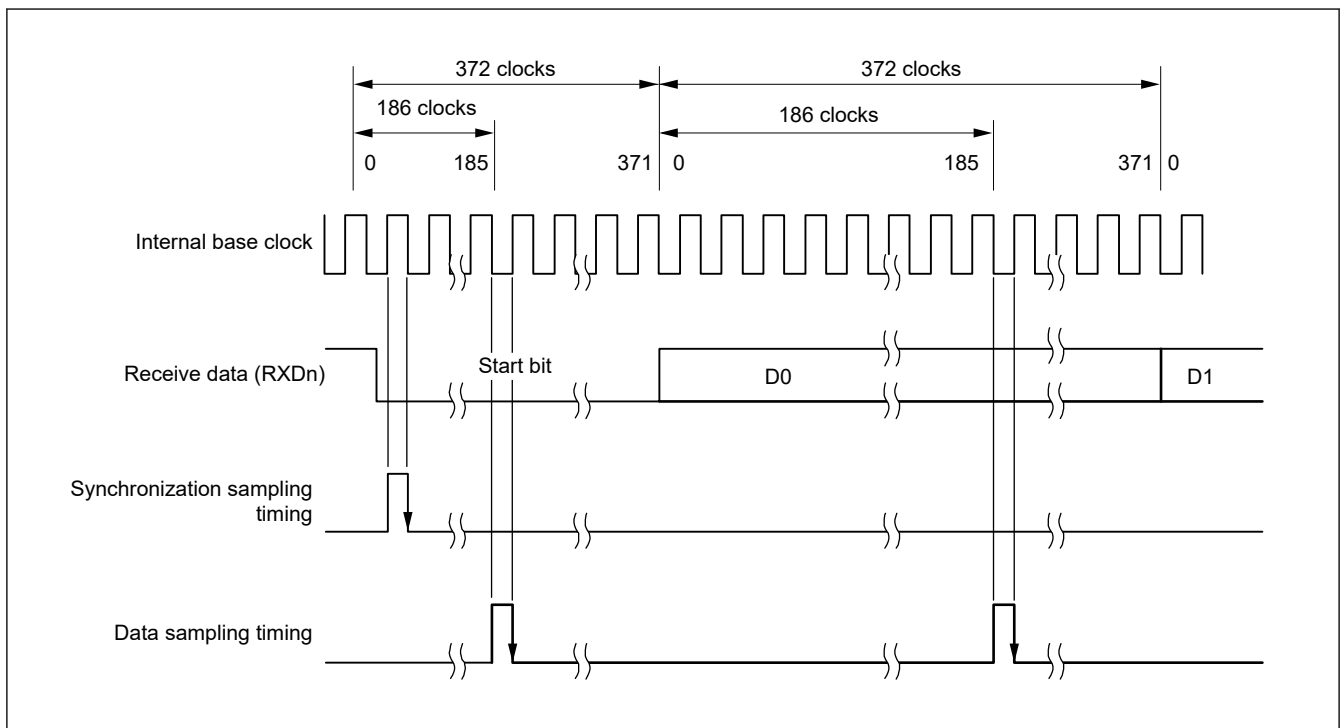


Figure 38.78 Receive data sampling timing in smart card interface mode when the clock frequency is 372 times the bit rate

38.7.5 SCI Initialization (Smart Card Interface Mode)

Before transmitting and receiving data, write the initial value 0x00 in the CCR0 register and initialize the SCI following the example flow shown in [Table 38.39](#).

Always set the initial value in the TIE, RIE, TE, RE, TEIE bits in the CCR0 register before switching from transmission to reception mode or from reception to transmission mode. When CCR0.RE is set to 0, the RDR register is not initialized.

In transmission mode, set 1 to the CCR0.TE bit and CCR0.TIE bit simultaneously, then the SCIn_TXI interrupt request is generated.

To change from reception mode to transmission mode, first check that reception has completed, then initialize the SCI. At the end of initialization, set CCR0.TE = 1 and CCR0.RE = 0. Reception completion can be verified by reading the SCIn_RXI request, ORER, or PER flag in CSR.

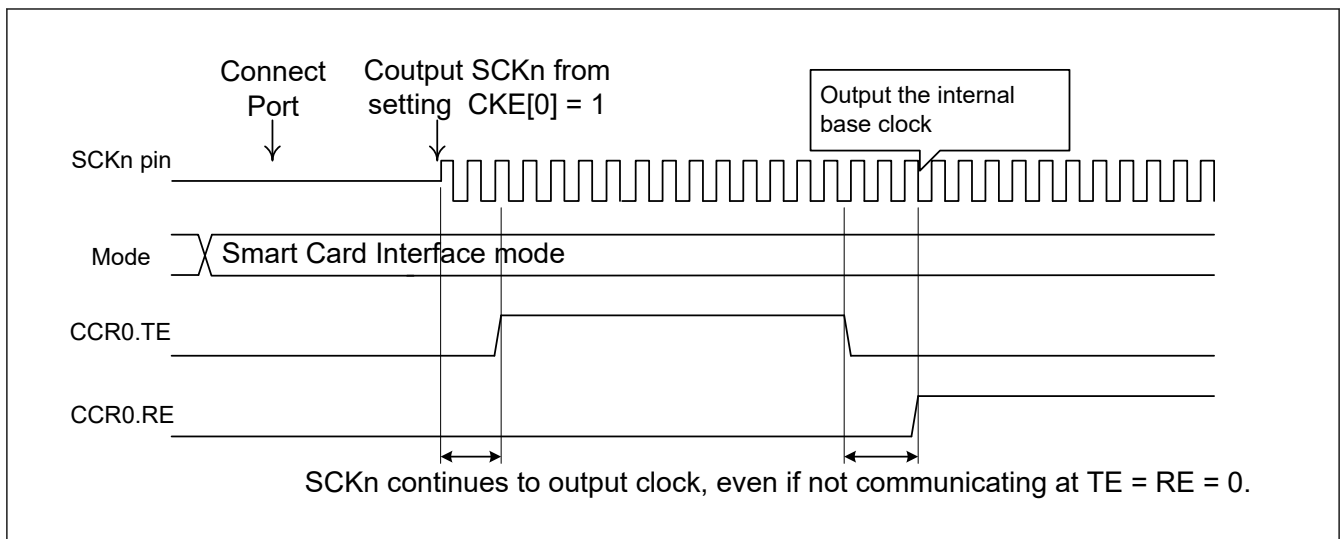
To change transmission mode to reception mode, first check that transmission has completed, then initialize the SCI. At the end of initialization, set CCR0.TE = 0 and CCR0.RE = 1. Transmission completion can be verified by reading the TEND flag in CSR.

Table 38.39 Example flow of SCI initialization in smart card interface mode

No.	Step Name	Description
1	Start initialization	—
2	Set CCR0	Set CCR0.TEIE, TIE, RIE, TE, RE to 0. If you have not changed from the initial settings, you can skip this step.
3	Set CCR3	Set communication mode (MOD[2:0] = 001b), BLK, GM, and SINV. Leave other bits at their initial values.
4	Set CCR2	Set clock-select and bit-rate. Set BRME to 0.
5	Set CCR1	Set up the Noise filter function, communication terminal status. Set NFEN, PE, CTSE to 0 and set PE to 1.
6	Set the I/O port functions	Make I/O port settings to enable input and output functions as required for TXDn, RXDn, and SCKn pins.
7	Set CCR3	Set CKE[1:0]. When the CKE[0] bit is set to "1" due on GM setting value, the clock is output from the SCKn pin.
8	Set CFCLR	Write 1 to the following bits and clear the corresponding flag. CFCLR.RDRFC, FERC, PERC, MFFC, ORERC, DFERC, DPERC, DCMFC, ERSC
9	Set CCR0	Set the TE or RE to 1. And set the TIE and RIE. Do not simultaneously set the TE and RE bits to 1 if self-diagnosis is not used.
10	Initialization completed	—

Figure 38.79 is a timing chart when data transmission is performed by making transition to the Smart Card Interface mode according to the above flow chart. The figure shows the case when CCR3.GM bit is 0. As shown in the figure, when the pin function is set to the SCKn pin, the SCKn pin is high impedance because the CCR3.CKE [0] bit is 0. When the TXDn pin is set, the TXDn pin is high impedance because the CCR0.TE bit is 0. Start clock output to the SCKn pin with the clock output setting CCR3.CKE [0] to 1, start data transmission by writing transmit data after setting CCR0.TE to 1.

In the smart card interface mode, even if not communicating at CCR0.TE = 0 and CCR0.RE = 0, the clock is continuously output if the clock output setting is used.

**Figure 38.79 Example of Timing chart of data transmission in Smart Card Interface Mode**

38.7.6 Serial Data Transmission (Except in Block Transfer Mode)

Serial data transmission in smart card interface mode (except in block transfer mode) is different from that in non-smart card interface mode, in that an error signal is sampled and data can be re-transmitted in smart card mode. Figure 38.80 shows the data re-transfer operation during transmission.

1. When an error signal from the receiver end is sampled after 1-frame data is transmitted, the CSR.ERS flag is set to 1. If the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated. Clear the ERS flag to 0 before the next parity bit is sampled.

2. For a frame in which an error signal is received, the CSR.TEND flag is not set. Data is re-transferred from TDR to TSR, allowing automatic data retransmission.
3. If no error signal is returned from the receiver, the ERS flag is not set to 1.
4. In this case, the SCI determines that transmission of 1-frame data, including the re-transfer, is complete, and the TEND flag is set. If the CCR0.TIE bit is 1, an SCIn_TXI interrupt request is generated. Write transmit data to the TDR to start transmission of the next data.

Figure 38.82 shows an example flow of serial transmission. All the processing steps are automatically performed using an SCIn_TXI interrupt request to activate the DTC or DMAC.

When the CSR.TEND flag is set to 1 in transmission and when the CCR0.TIE bit is 1, an SCIn_TXI interrupt request is generated.

The DTC or DMAC is activated by an SCIn_TXI interrupt request if the SCIn_TXI interrupt request is previously specified as a source of DTC or DMAC activation, allowing the transfer of transmit data. The TEND flag is automatically set to 0 when the DTC or DMAC transfers the data.

If an error occurs, the SCI automatically retransmits the same data. During this retransmission, the TEND flag is kept at 0 and the DTC or DMAC is not activated. Therefore, the SCI and DTC or DMAC automatically transmit the specified number of bytes, including retransmission when an error occurs. Because the ERS flag is not automatically cleared, set the RIE bit to 1 before enabling an SCIn_ERI interrupt request to be generated if an error occurs, and clear the ERS flag to 0.

When transmitting or receiving data using the DTC or DMAC, always enable the DTC or DMAC before making the SCI settings.

For DTC or DMAC settings, see [section 18, Data Transfer Controller \(DTC\)](#), [section 17, DMA Controller \(DMAC\)](#).

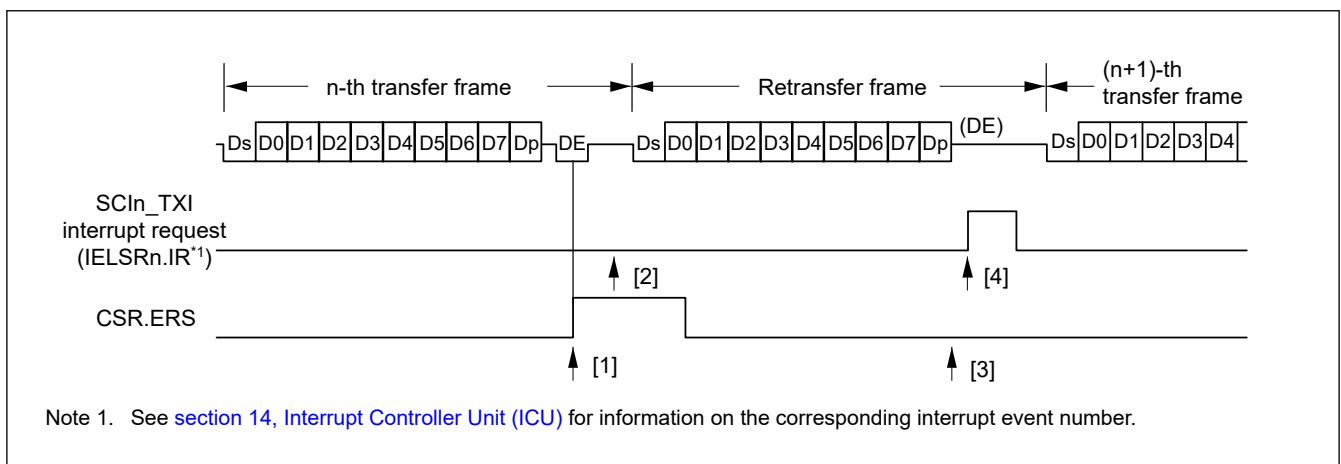


Figure 38.80 Data re-transfer operation in smart card interface transmission mode

The CSR.TEND flag is set at different timings depending on the CCR3.GM bit setting. [Figure 38.81](#) shows the TEND flag generation timing.

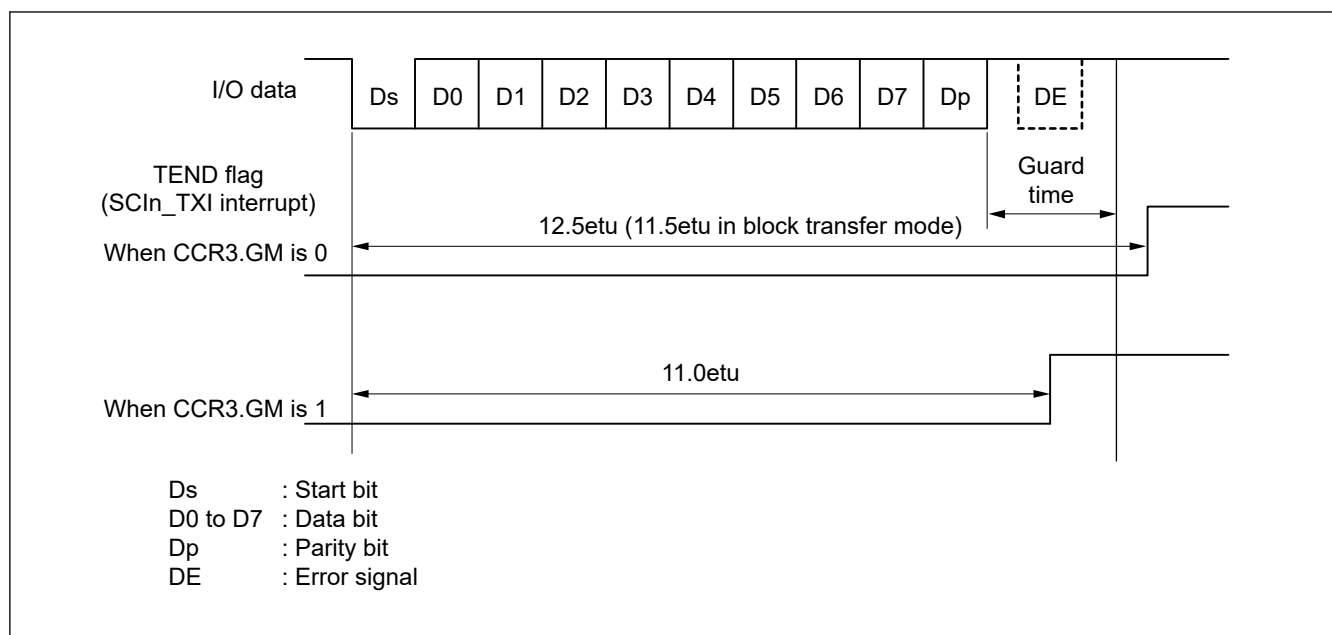


Figure 38.81 CSR.TEND flag generation timing during transmission

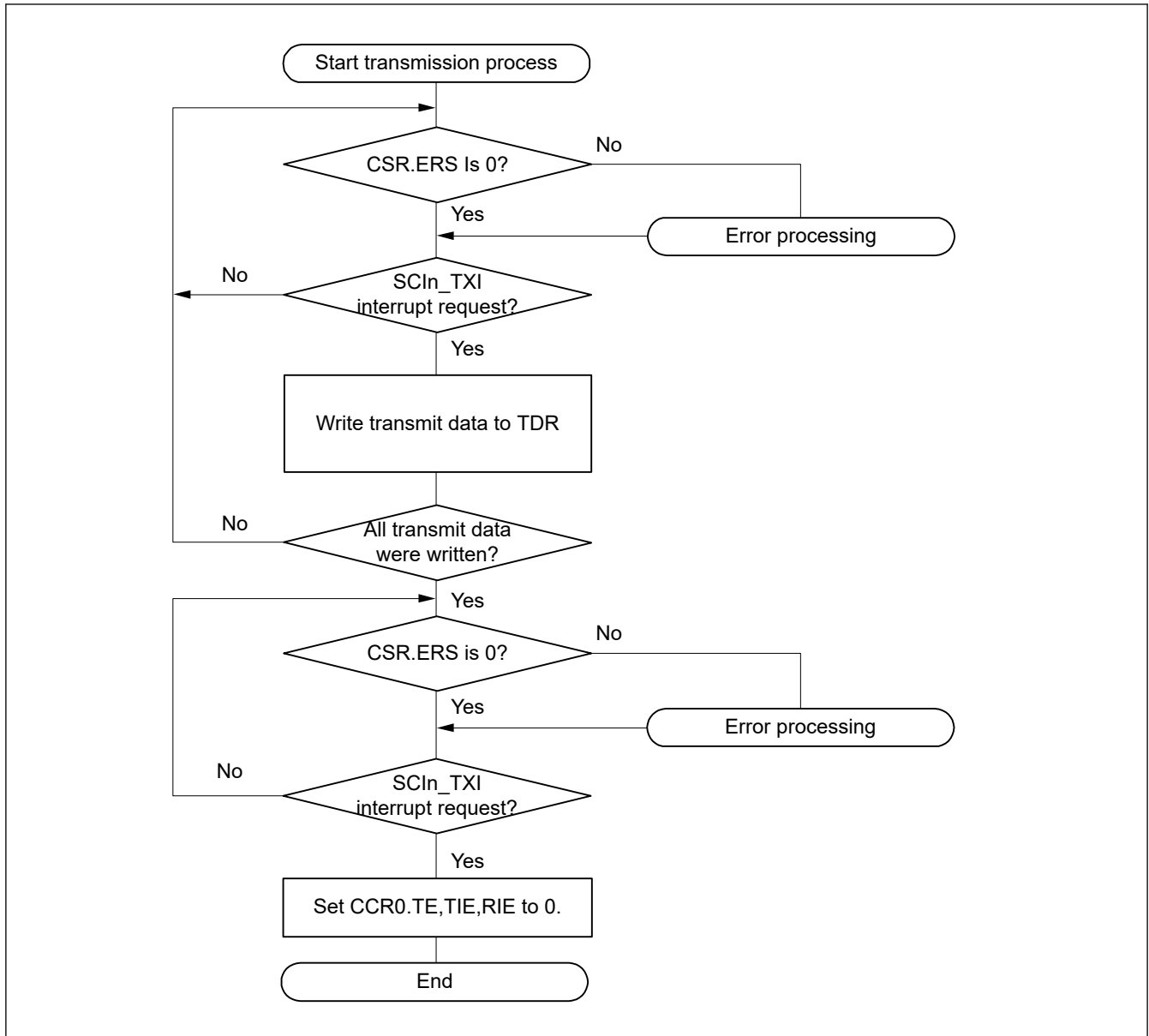


Figure 38.82 Example flow of smart card interface transmission

38.7.7 Serial Data Reception (Except in Block Transfer Mode)

Serial data reception in smart card interface mode is similar to that in non-smart card interface mode. [Figure 38.83](#) shows the data re-transfer operation in reception mode.

1. If a parity error is detected in the receive data, the CSR.PER flag is set to 1. When the CCR0.RIE bit is 1, an SCIn_ERI interrupt request is generated. Clear the PER flag to 0 before the next parity bit is sampled.
2. For a frame in which a parity error is detected, no SCIn_RXI interrupt is generated.
3. When no parity error is detected, the CSR.PER flag is not set to 1.
4. In this case, data is determined to be received successfully. When the CCR0.RIE bit is 1, an SCIn_RXI interrupt request is generated.

[Figure 38.84](#) shows an example flow of serial data reception. All the processing steps are automatically performed using an SCIn_RXI interrupt request to activate the DTC or DMAC.

In reception, setting the RIE bit to 1 allows an SCIn_RXI interrupt request to be generated. The DTC or DMAC is activated by an SCIn_RXI interrupt request if the SCIn_RXI interrupt request is previously specified as a source of DTC or DMAC activation, allowing the transfer of receive data.

If an error occurs during reception and either the ORER or PER flag in CSR is set to 1, a receive error interrupt (SCIn_ERI) request is generated. Clear the error flag after the error occurrence. If an error occurs, the DTC or DMAC is not activated and receive data is skipped. Therefore, the number of bytes of receive data specified in the DTC or DMAC is transferred.

If a parity error occurs and the PER flag is set to 1 during reception, the receive data is transferred to RDR, allowing the data to be read.

When a reception is forced to terminate by setting CCR0.RE to 0 during operation, read the RDR register because the received data that is not yet read might be left in the RDR.

Note: For operations in block transfer mode, see [section 38.3.9. Serial Data Reception in Asynchronous Mode](#).

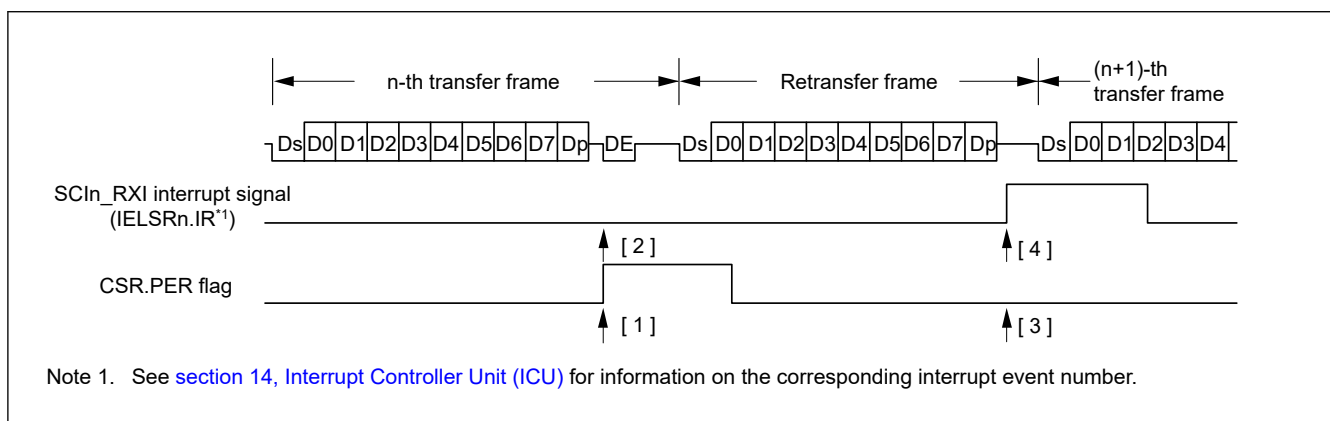


Figure 38.83 Data re-transfer operation in smart card interface reception mode

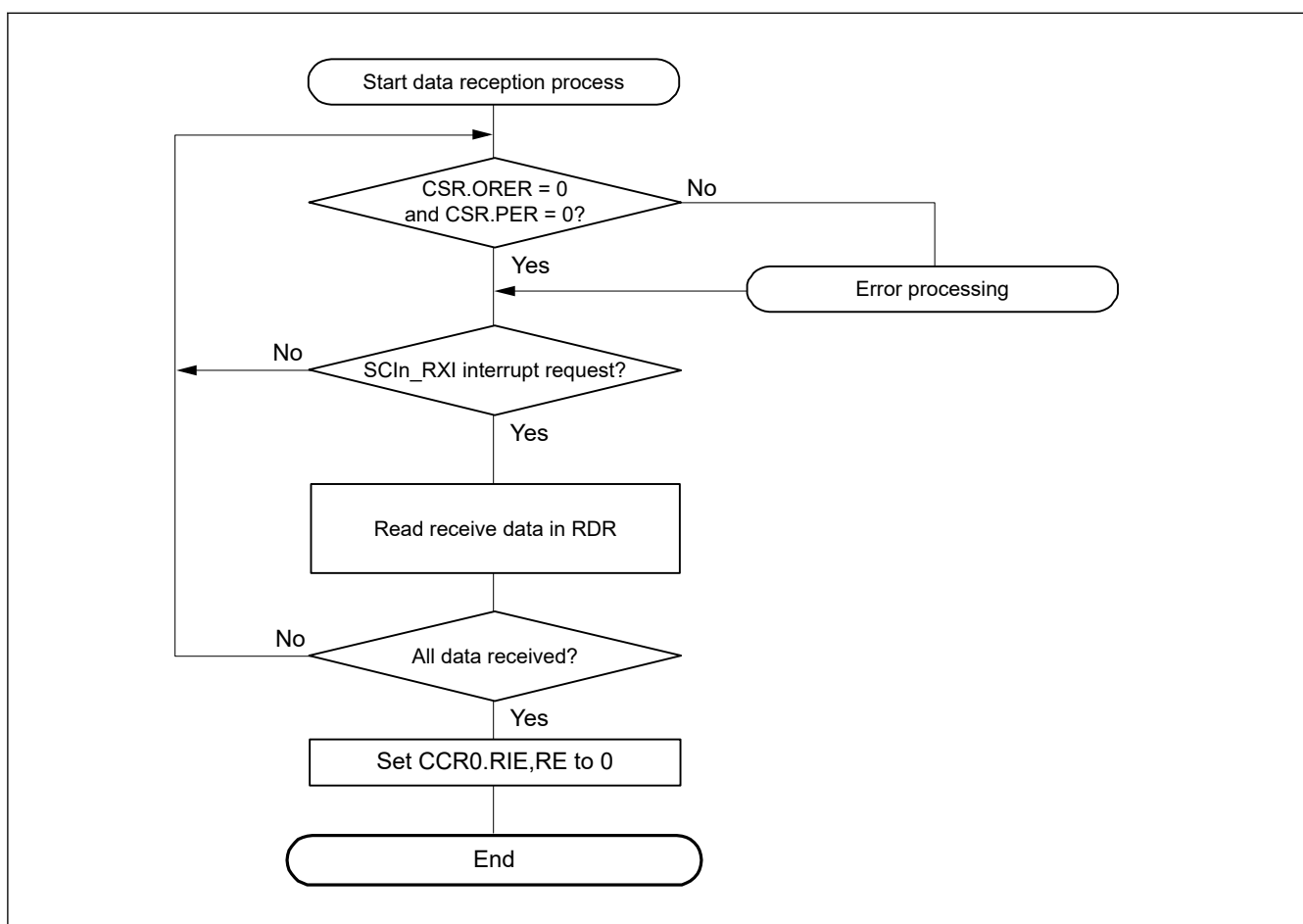


Figure 38.84 Example flow of smart card interface reception

38.7.8 Clock Output Control

When the GM bit in CCR3 is set to 1, the clock output can be controlled by the CKE[1:0] bits in CCR3. For details on the CKE[1:0] bits, see [section 38.2.8. CCR3 : Common Control Register 3](#). When setting the clock output, the base clock described in [section 38.7.4. Receive Data Sampling Timing and Reception Margin](#) the bit rate is set by CCR2.CKS, CCR2.BCP[2:0] and BRR[7:0].

[Figure 38.85](#) shows an example timing for the clock output control when the CKE[1] bit in CCR3 is set to 0 and the CKE[0] bit in CCR3 is controlled.

When the GM bit in CCR3 is 0, output control by the CKE[0] bit in CCR3 is immediately reflected on the SCKn pin, so there is a possibility that pulses with an unintended width may be output from the SCKn pin.

When the GM bit in CCR3 is 1, the output pulse control by the CCR3.CKE [0] controls the pulse width set to be based on the state of the base clock.

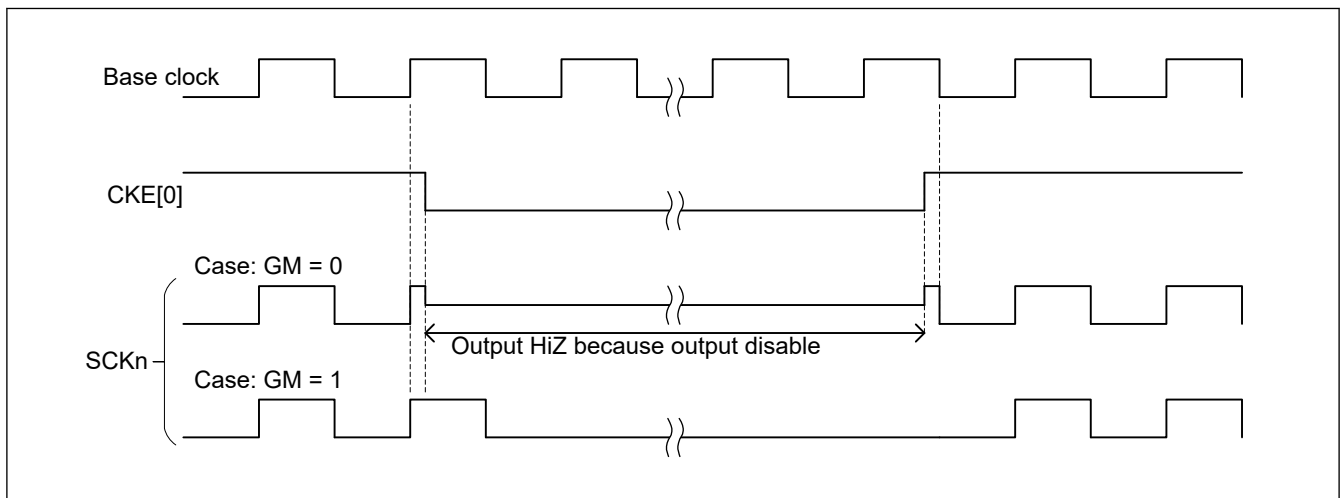
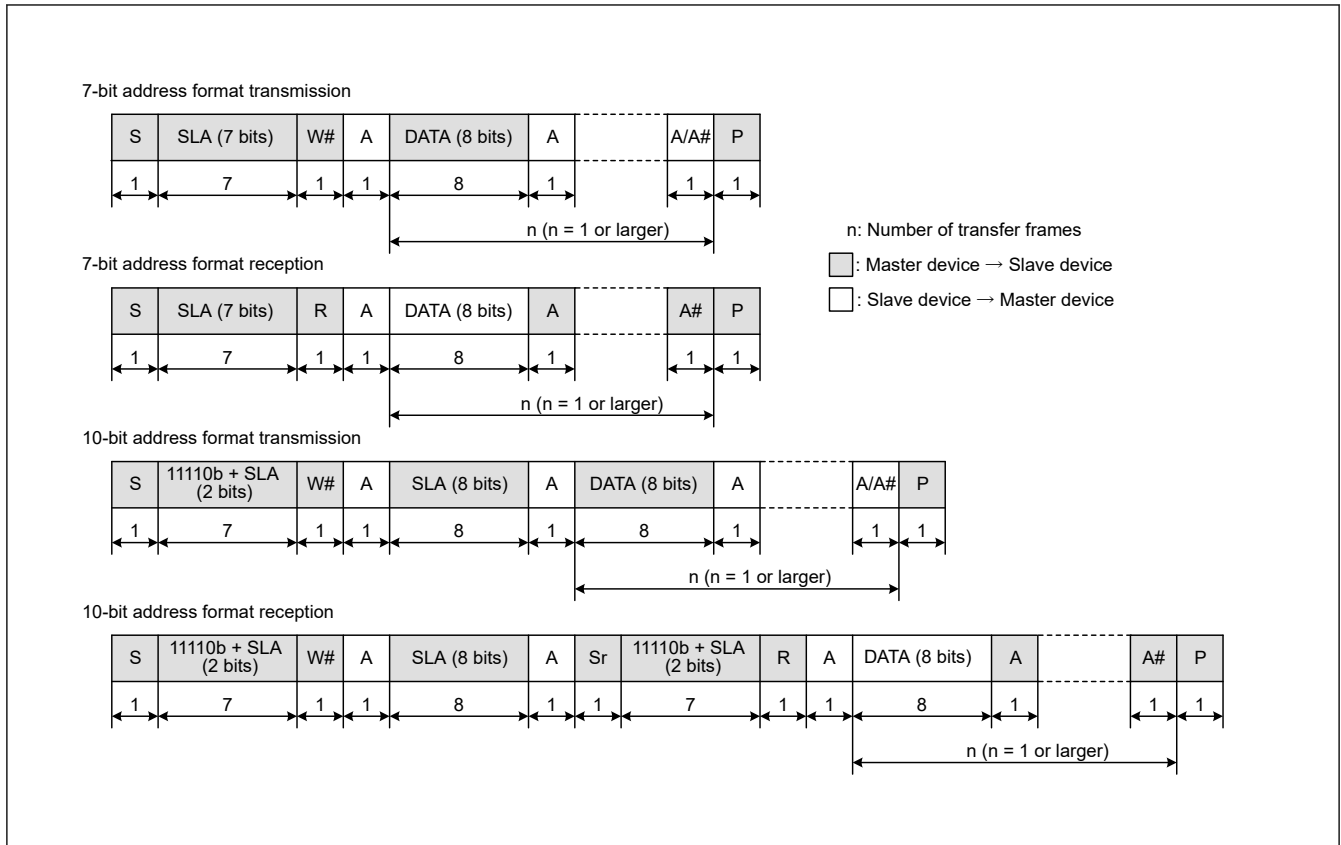
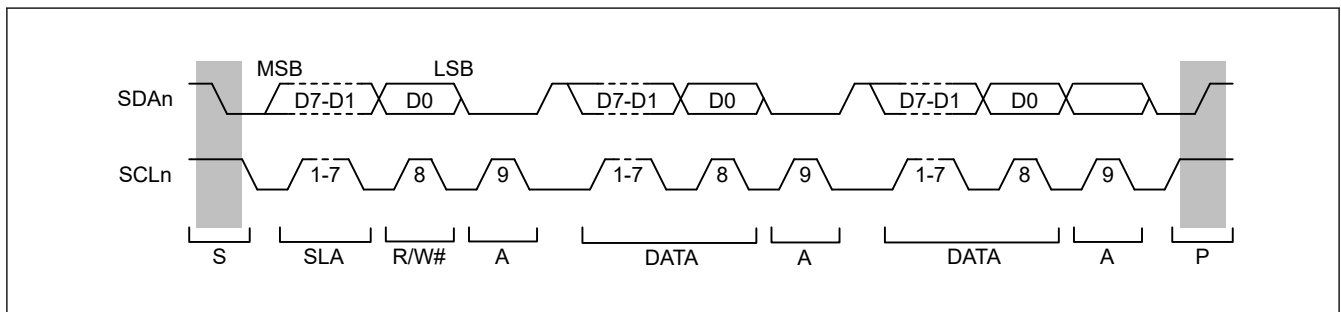


Figure 38.85 Clock Output timing

38.8 Operation in Simple IIC Mode

Simple IIC mode format is composed of 8 data bits and an acknowledge bit. By continuing into a slave-address frame after a start condition or restart condition, a master device can specify a slave device as the partner for communications. The currently specified slave device remains valid until a new slave device is specified or a stop condition is satisfied. The 8 data bits in all frames are transmitted in order from the MSB.

The I²C bus format and timing of the I²C bus are shown in [Figure 38.86](#) and [Figure 38.87](#).

Figure 38.86 I²C bus formatFigure 38.87 I²C bus timing when SLA is 7 bits

- S: Indicates a start condition, when the master device changes the level on the SDAn line from high to low while the SCLn line is high
- SLA: Indicates a slave address, by which the master device selects a slave device
- R/W#: Indicates the direction of transfer (reception or transmission). The value 1 indicates transfer from the slave device to the master device and 0 indicates transfer from the master device to the slave device.
- A/A#: Indicates an acknowledge bit. This is returned by the slave device for master transmission and by the master device for master reception. Return low indicates ACK and return high indicates NACK.
- Sr: Indicates a restart condition, when the master device changes the level on the SDAn line from high to low while the SCLn line is high and after the setup time elapses
- DATA: Indicates the data being received or transmitted
- P: Indicates a stop condition, when the master device changes the level on the SDAn line from low to high while the SCLn line is high

38.8.1 Generation of Start, Restart, and Stop Conditions

Writing 1 to the ICR.IICSTAREQ bit causes the generation of a start condition. The generation of a start condition proceeds through the following operations:

- The level on the SDA_n line falls (from the high level to the low level) and the SCL_n line is kept in the released state
- The hold time for the start condition is set as half of a bit period at the bit rate determined by the CCR2.BRR setting
- The level on the SCL_n line falls (from the high level to the low level), the IICSTAREQ bit in ICR is set to 0, and a start-condition generated interrupt is output

Writing 1 to the IICRSTAREQ bit in ICR causes the generation of a restart condition. The generation of a restart condition proceeds through the following operations:

- The SDA_n line is released and the SCL_n line is kept at the low level
- The period at low level for the SCL_n line is set as half of a bit period at the bit rate determined by the CCR2.BRR setting
- The SCL_n line is released (transition from the low to the high level)
- When a high level is detected on the SCL_n line, the setup time for the restart condition is set as half of a bit period at the bit rate determined by the CCR2.BRR setting
- The level on the SDA_n line falls (from the high level to the low level)
- The hold time for the restart condition is set as half of a bit period at the bit rate determined by the CCR2.BRR setting
- The level on the SCL_n line falls (from the high level to the low level), the ICR.IICRSTAREQ bit is set to 0, and a restart-condition generated interrupt is output

Writing 1 to the ICR.IICSTPREQ bit causes the generation of a stop condition. The generation of a stop condition proceeds through the following operations:

- The level on the SDA_n line falls (from the high level to the low level) and the SCL_n line is kept at the low level
- The period at low level for the SCL_n line is set as half of a bit period at the bit rate determined by the CCR2.BRR setting
- The SCL_n line is released (transition from the low to the high level)
- When a high level is detected on the SCL_n line, the setup time for the stop condition is set as half of a bit period at the bit rate determined by the CCR2.BRR setting
- The SDA_n line is released (transition from the low to the high level), the ICR.IICSTPREQ bit is set to 0, and a stop-condition generated interrupt is output

Figure 38.88 shows the timing of operations in the generation of start, restart, and stop conditions.

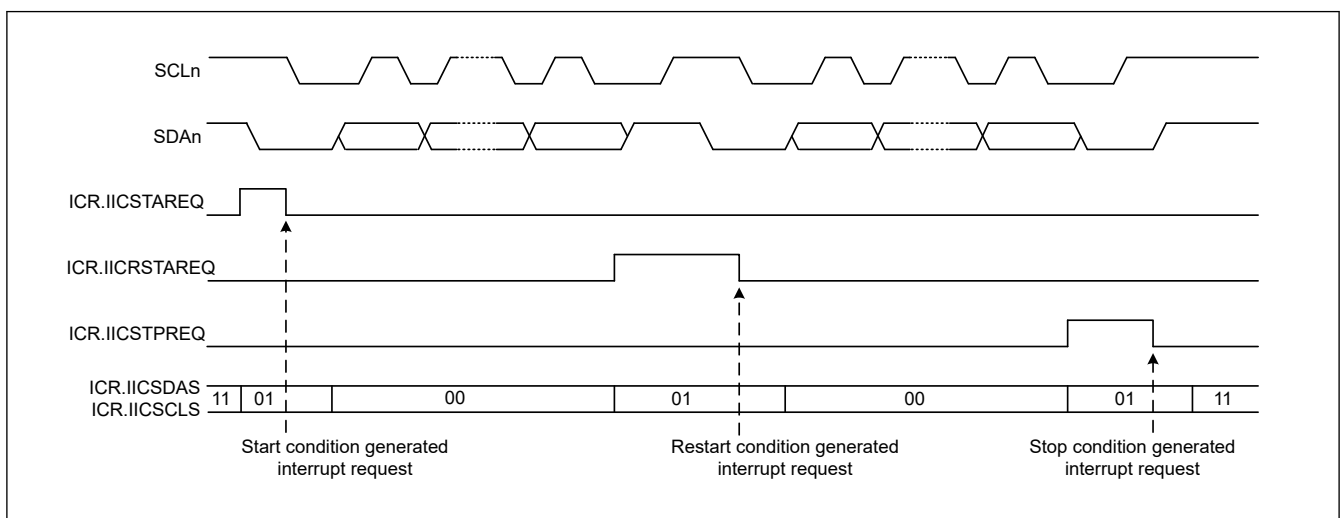


Figure 38.88 Timing of operations in generation of start, restart, and stop conditions

38.8.2 Clock Synchronization

The SCLn line can be driven low if a wait is inserted by a slave device at the other side of the transfer. Setting the ICR.IICCSC bit to 1 applies control to obtain synchronization when a difference arises between the levels of the internal SCLn clock signal and the level being input on the SCLn pin.

When the ICR.IICCSC bit is set to 1, the level of the internal SCLn clock signal changes from low to high. Counting to determine the period at a high level stops while the low level is being input on the SCLn pin. Counting to determine the period at a high level starts after the transition of the input on the SCLn pin to the high level.

The interval from this time until counting to determine the period at high level starts on the transition of the SCLn pin to the high level, is the total time which contains the SCLn input delay, delay for noise filtering of the input on the SCLn pin (2 or 3 cycles of sampling clock for the noise filter), and delay for internal processing (1 or 2 cycles of PCLK). The period at high level of the internal SCLn clock is extended even when other devices do not place the low level on the SCLn line.

If the ICR.IICCSC bit is 1, synchronization is obtained for the transmission and reception of data by taking the logical AND of the input on the SCLn pin and the internal SCLn clock. If the ICR.IICCSC bit is 0, synchronization with the internal SCLn clock is obtained for the transmission and reception of data.

If a slave device inserts a wait period into the interval until the transition of the internal SCLn clock signal from the low to the high level after a request for the generation of a start, restart, or stop condition is issued, the time until generation is prolonged by that period.

If a slave device inserts a wait period after the transition of the internal SCLn clock signal from the low to the high level, although the generation-completed interrupt is issued without stopping the waiting period, generation of the condition itself is not guaranteed.

Figure 38.89 shows an example operation for synchronizing the clocks.

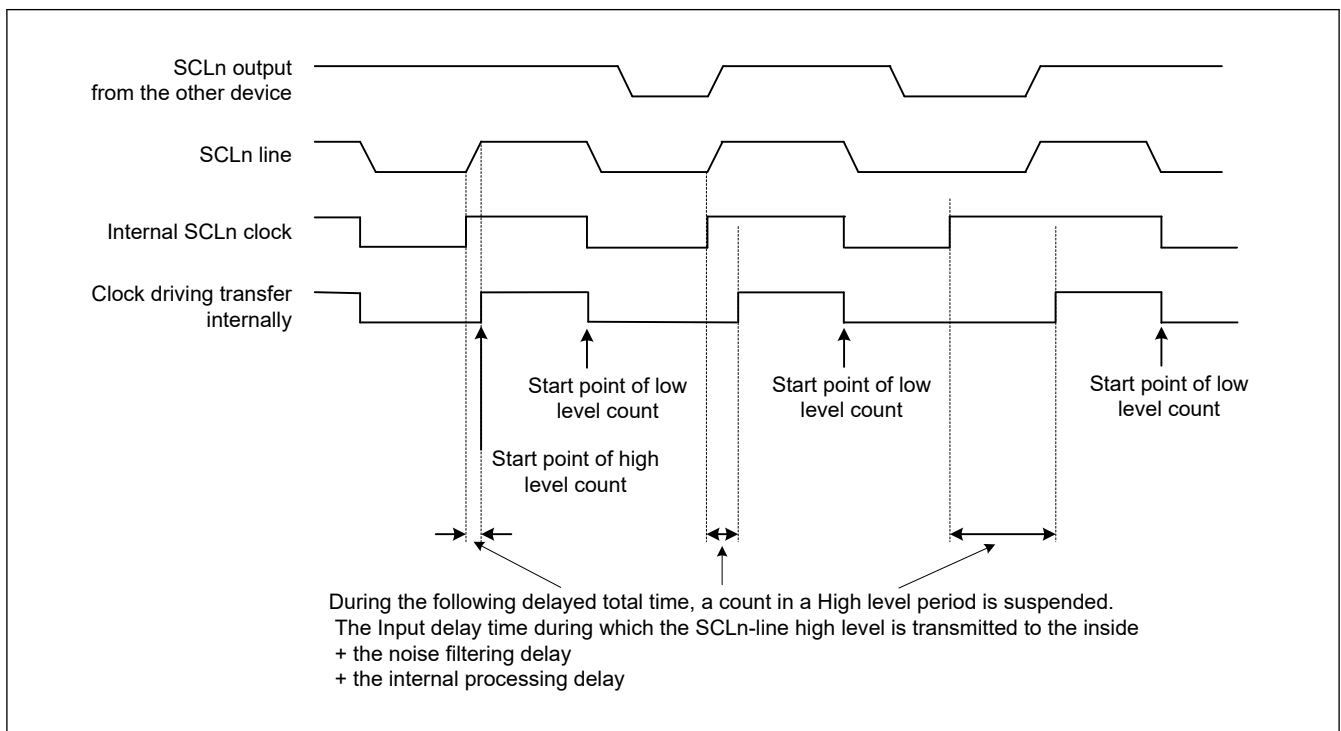


Figure 38.89 Example operations for clock synchronization

38.8.3 SDAn Output Delay

The ICR.IICDL[4:0] bits can be used to set a delay for output on the SDAn pin relative to falling edges of output on the SCLn pin. Delay settings from 0 to 31 are selectable, representing periods of the corresponding numbers of cycles of the clock signal from the on-chip baud rate generator (derived by frequency-dividing the base clock, TCLK, by the divisor selected in the CCR2.CKS[1:0] bits). A delay for output on the SDAn pin applies to the start condition/restart condition/stop condition signal, 8-bit transmit data, and acknowledge bit.

If the SDAn output delay is shorter than the time for the level on the SCLn pin to fall, the change of the output on the SDAn pin starts while the output level on the SCLn pin is falling, creating a possibility of erroneous operation for slave devices. Ensure that settings for the delay of output on the SDAn pin specify times greater than the time output on the SCLn pin takes to fall (300 ns for IIC in normal mode and fast mode).

Figure 38.90 shows the timing of delays in SDAn output.

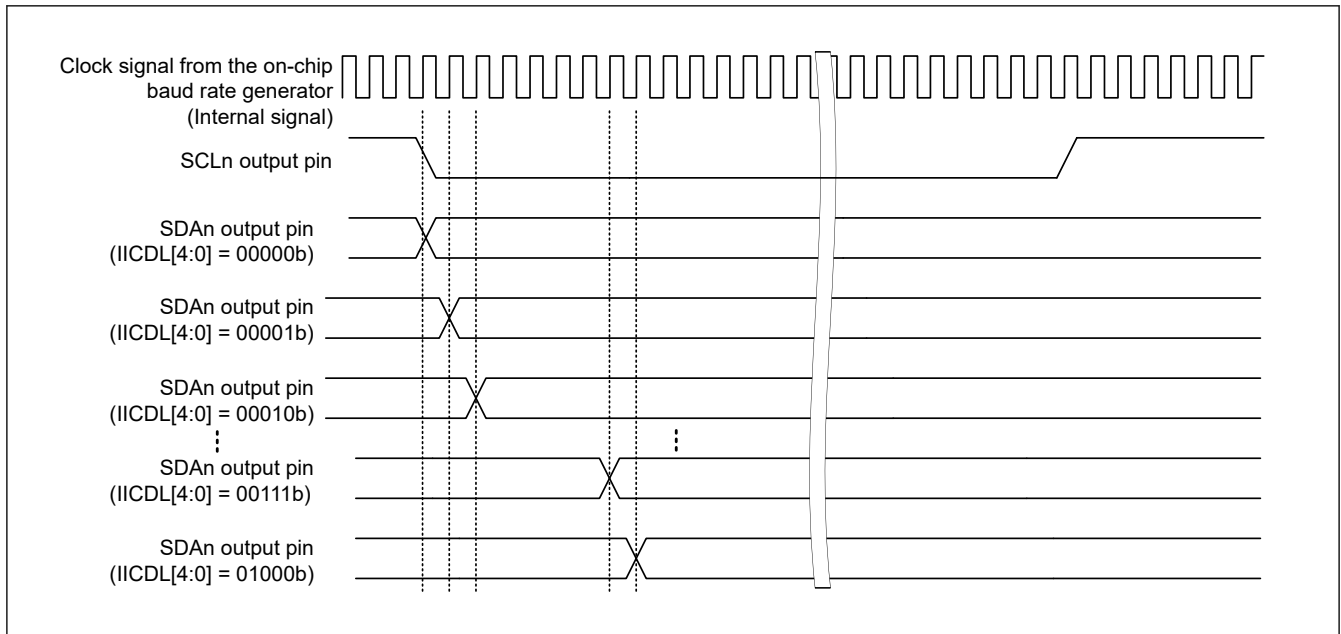


Figure 38.90 Timing of delays in SDAn output

38.8.4 SCI Initialization in Simple IIC Mode

Before transferring data, write the initial value 0x00 to CCR0 and initialize the interface following the example shown in Table 38.40.

Before making any changes to the operating mode or transfer format, be sure to set CCR0 to its initial value. In simple IIC mode, the open-drain setting for the communication ports should be made on the port side.

Table 38.40 Example flow of SCI initialization in simple IIC mode

No.	Step Name	Description
1	Start initialization	—
2	Set CCR0	Set CCR0.TEIE, TIE, RIE, TE, RE to 0. If you have not changed from the initial settings, you can skip this step.
3	Set ICR	Set the IICSDAS[1:0] and IICSCLS[1:0] to 11b. Set the IICDL[4:0] and IICINTM as required. Set the IICACKT and the IICCS bits to 1.
4	Set CCR3	Set the transmission / reception format as the communication mode (MOD [2:0] = 100b) and CKE [1:0] = 00b.
5	Set CCR2	Set the bit rate modulation function ^{*1} , the clock selection, and the bit rate.
6	Set CCR1	Set noise filter, communication pin status, parity check, and CTSn / RTSn function.
7	Set the I/O port functions	Set I/O port settings that allow use (on NMOS open-drain output pins and Hi-Z) of the SCLn and SDAn pin functions.
8	Set CFCLR, ICFLR	Write 1 to the following bits and clear the corresponding flag. CFCLR.RDRFC, FERC, PERC, MFFC, ORERC, DFERC, DPERC, DCMFC, ERSC ICFLR.IICSTIFC
9	Set CCR0 (TE, RE, TIE, RIE)	Set the TE and RE bits to 1. To enable interrupts, set the TE, TIE, RE and RIE bits to 1 with one instruction at the same time (for transmission and when the IICINTM bit is 1, clear the RIE bit). Setting the TE and RE bits to 1 makes the SCLn and SDAn pins functions available.
10	Initialization completed	—

Note: Set the CCR0.TE and RE bits to 0 or 1 at the same time.

Note 1. If you do not use the bit rate modulation function, you do not need to set it.

38.8.5 Operation in Master Transmission in Simple IIC Mode

Figure 38.91 and Figure 38.92 show examples of master transmission and Figure 38.93 shows an example flow of data transmission.

Figure 38.91 shows the operation example when ICR.IICINTM bit is 1 (use reception and transmission interrupts). In this case, you can start DMAC or DTC by SCIn_TXI interrupt. However, if you use DMAC or DTC, ACK/NACK cannot be confirmed. If you want to confirm ACK/NACK, prepare the transmit data by CPU.

In simple IIC mode, SCIn_TXI interrupt is generated when communication of one frame is completed. And it is not used SCIn_RXI interrupt in master transmission, so the CCR0.RIE set to 0.

See Table 38.45 for more information on the STI interrupt.

Figure 38.93 shows a flow chart in the case of ICR.IICINTM is 1 and address transmission by CPU and data transmission by DTC or DMAC. When 10-bit slave addresses are in use, steps [3] and [4] are repeated twice.

In simple IIC mode, the transmit data empty interrupt (SCIn_TXI) is generated when communication of one frame is complete, unlike the SCIn_TXI interrupt request generation timing during clock synchronous transmission.

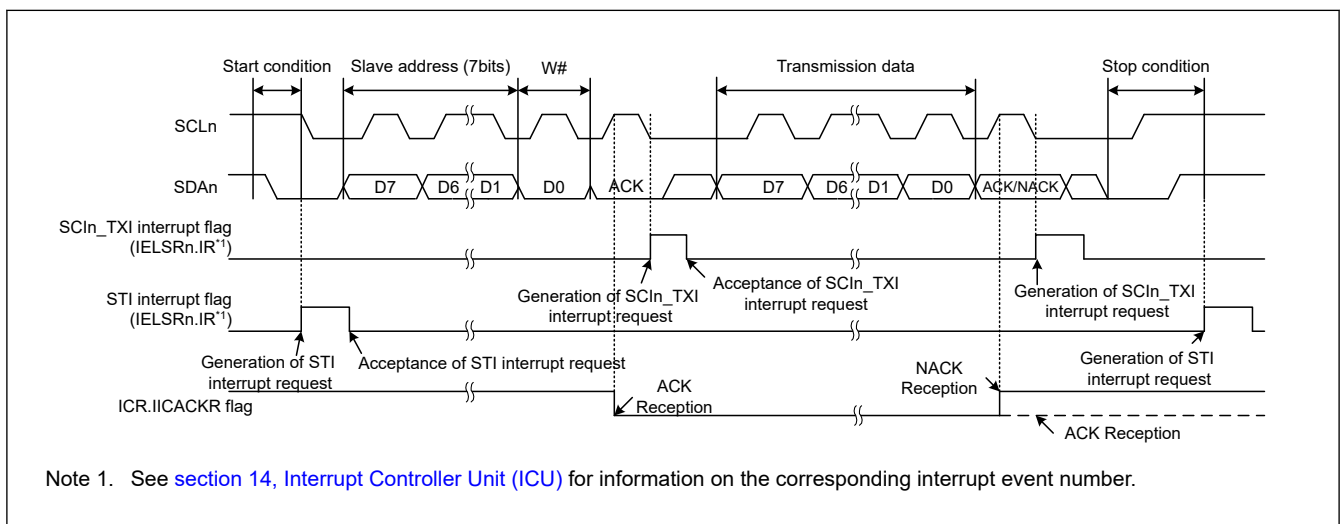


Figure 38.91 Example 1 of operations for master transmission in simple IIC mode with 7-bit slave addresses, transmission interrupts, and reception interrupts (ICR.IICINTM = 1)

When the ICR.IICINTM bit is set to 0 (use ACK/NACK interrupts) during master transmission, the DTC or DMAC is activated by the ACK interrupt as the trigger and required number of data bytes are transmitted. When the NACK is received, error processing such as transmission stop and retransmission is performed using the NACK interrupt as the trigger.

To restart communication for some reason after writing data in the TDR register, use the following procedure:

1. Set the TE and RE bits in the CCR0 register to 0 to stop communication.
2. Set ICR.IICSCLS[1:0] and ICR.IICSDAS[1:0] bits to 11b, release the I²C bus, and clear the generation of a condition.
3. If the RDRF flag in the CSR register is set to 1, the RDR register is read by dummy and the RDRF bit is set to 0.
4. Set the TE and RE bits in the CCR0 register to 1 and start the next communication.

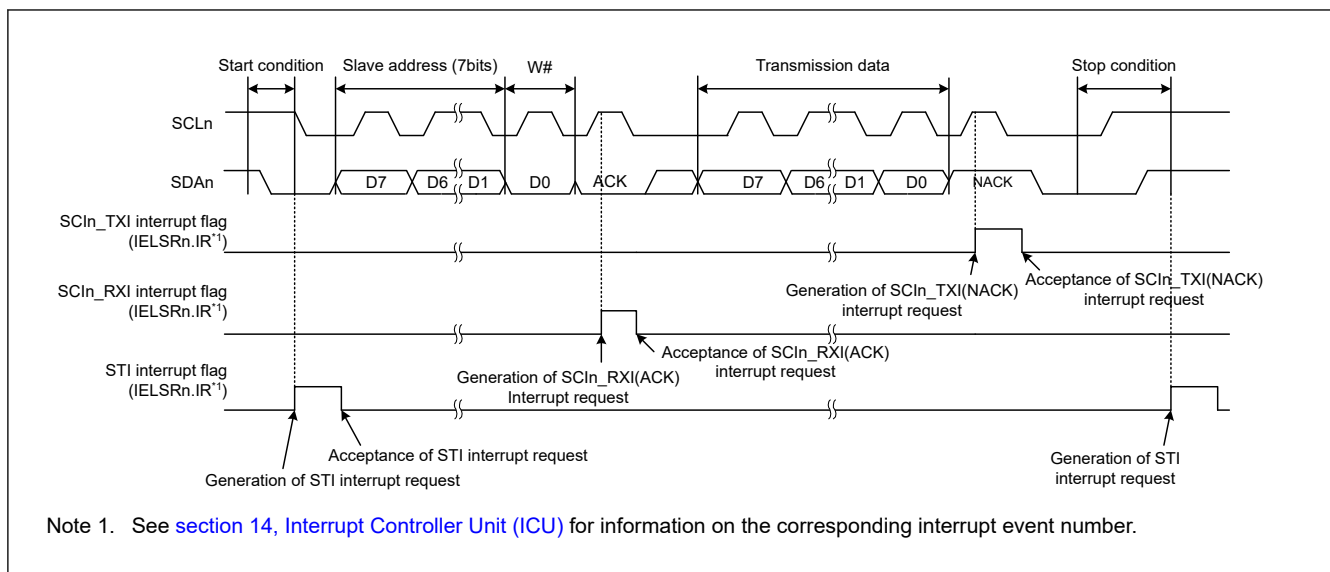


Figure 38.92 Example 2 of operations for master transmission in simple IIC mode with 7-bit slave addresses, ACK interrupts, and NACK interrupts (ICR.IICINTM = 0)

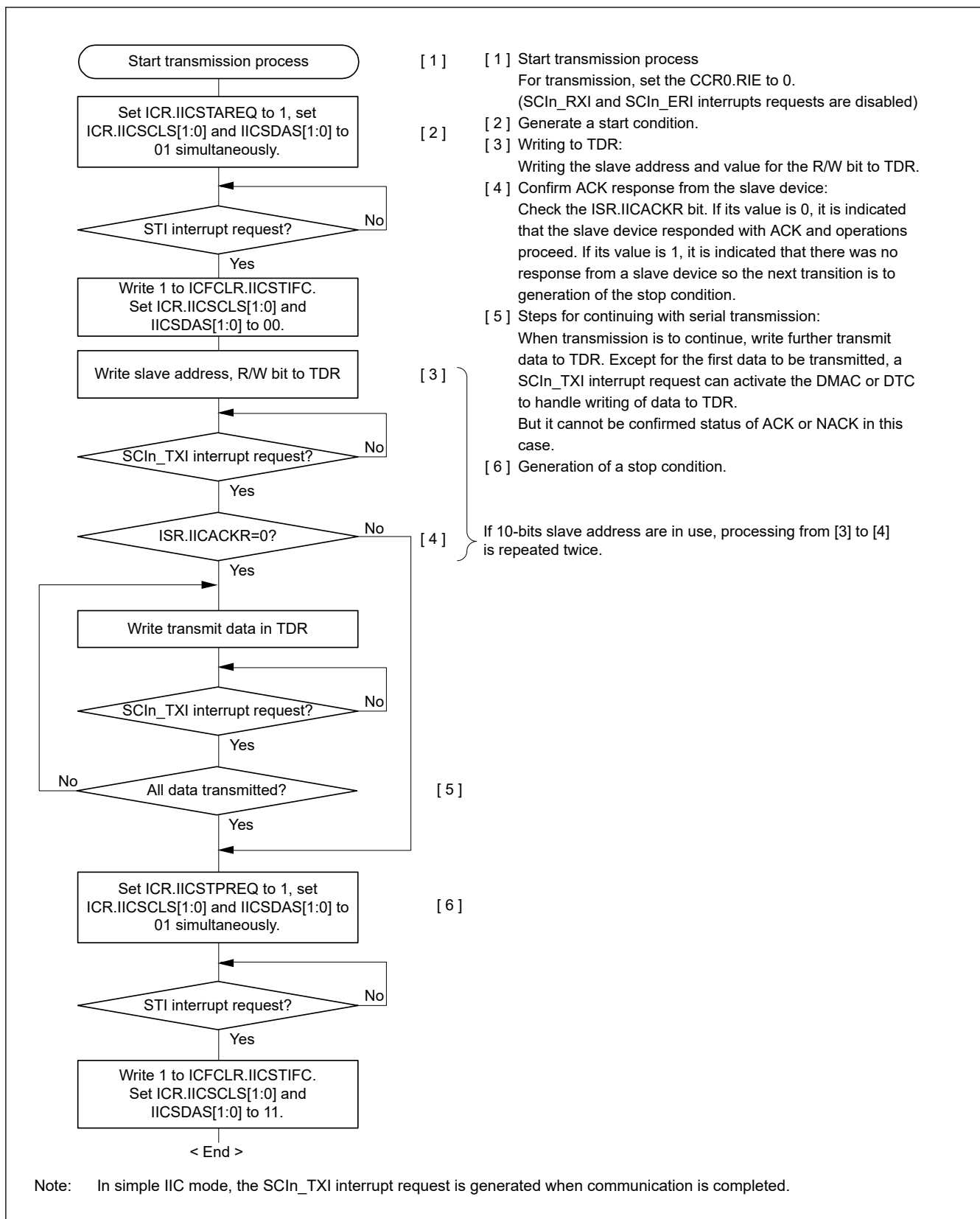


Figure 38.93 Example flow of master transmission in simple IIC mode with transmission interrupts and reception interrupts

38.8.6 Master Reception in Simple IIC Mode

Figure 38.94 shows an example operation in simple IIC mode master reception and Figure 38.96 shows an example flow of master reception.

The value of the ICR.IICINTM bit is assumed to be 1 (use reception and transmission interrupts) and 0 (use ACK and NACK interrupts).

In simple IIC mode, the transmit data empty interrupt (SCIn_TXI) is generated when communication of one frame is complete, unlike the SCIn_TXI interrupt request generation timing during clock synchronous transmission.

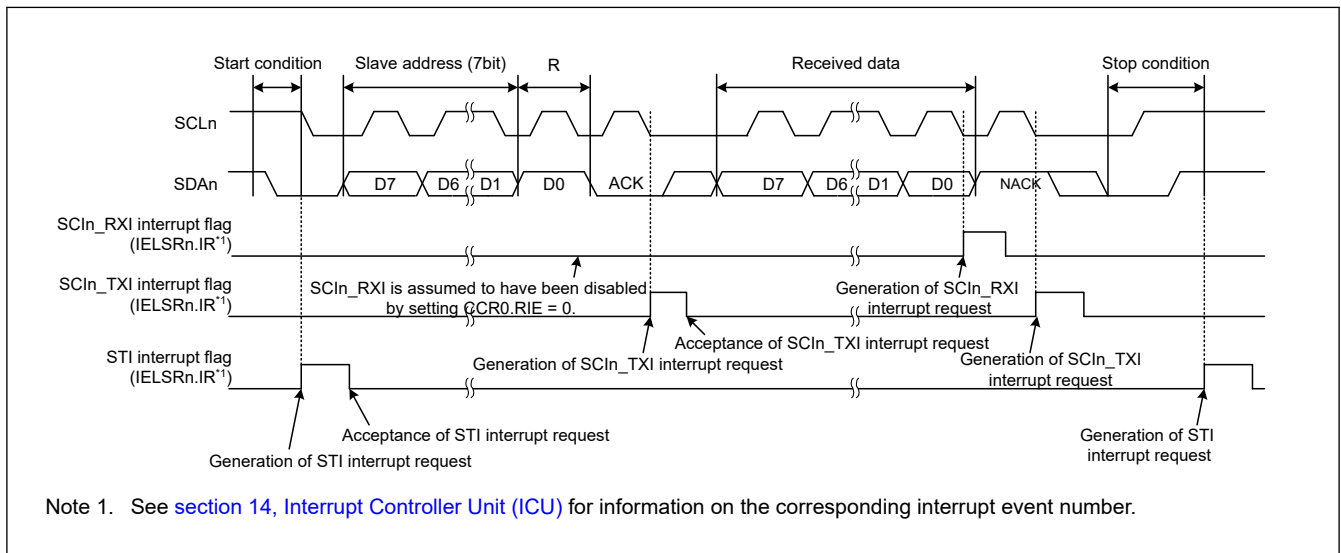


Figure 38.94 Example operations for master reception in simple IIC mode with 7-bit slave addresses, transmission interrupts, and reception interrupts (ICR.IICINTM = 1)

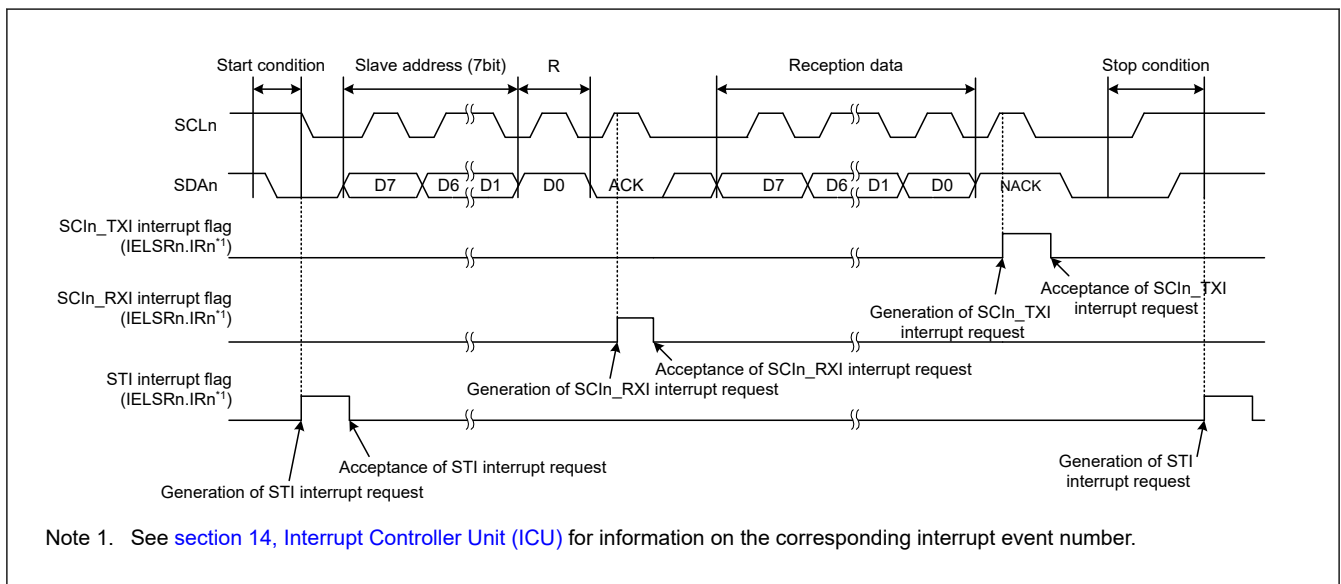
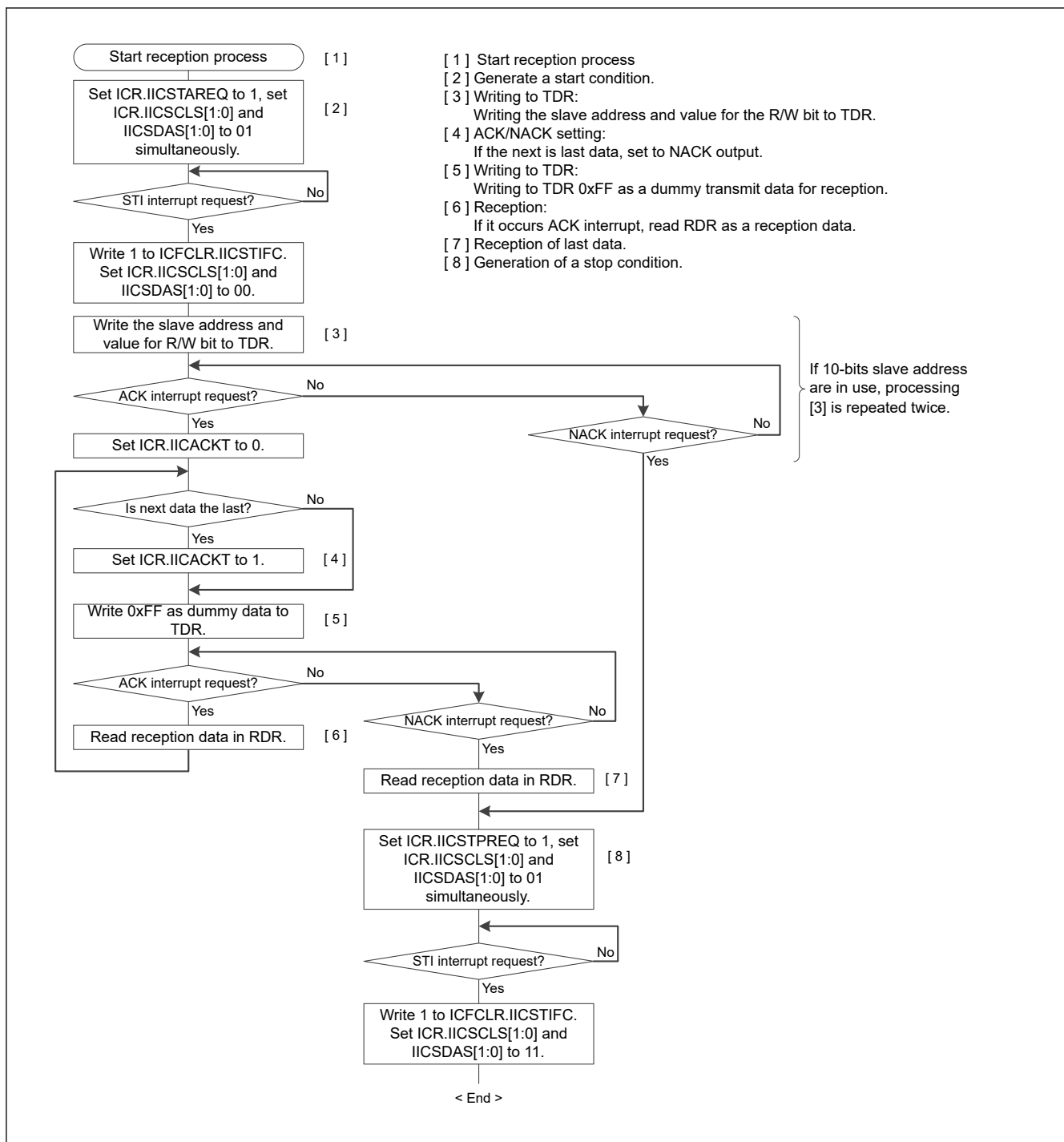


Figure 38.95 Example of Operations for Master Reception in Simple IIC Mode (7bit Slave address, ACK and NACK interrupt in use (ICR.IICINTM = 0))



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38.9 Operation in Simple SPI Mode

As an extended function, the SCI supports a simple SPI mode that handles transfer among one or multiple master devices and multiple slave devices.

Using the settings for Simple SPI mode setting (CCR3.MOD[2:0] bit = 011b) and setting the CCR0.SSE bit to 1 place the SCI in simple SPI mode. However, the SS_n pin function on the master side is not required for connection of the device used as the master in simple SPI mode when the configuration only has a single master. Therefore, set the CCR0.SSE bit to 0 in such cases.

Figure 38.98 shows an example of connections for simple SPI mode. Control a general port pin to produce the SS_n output signal from the master.

In simple SPI mode, data is transferred in synchronization with clock pulses in the same way as in clock synchronous mode. One character of data for transfer consists of 8 bits of data, and parity bits cannot be appended. The data can be inverted by setting the CCR3.SINV bit to 1.

Because the receiver and transmitter are independent of each other within the SCI module, full-duplex communications are possible, with a shared clock signal. Additionally, because both the transmitter and receiver have a buffered structure, writing the next transmit data while transmission is in progress and reading previously received data while reception is in progress are both possible. This enables continuous transfer.

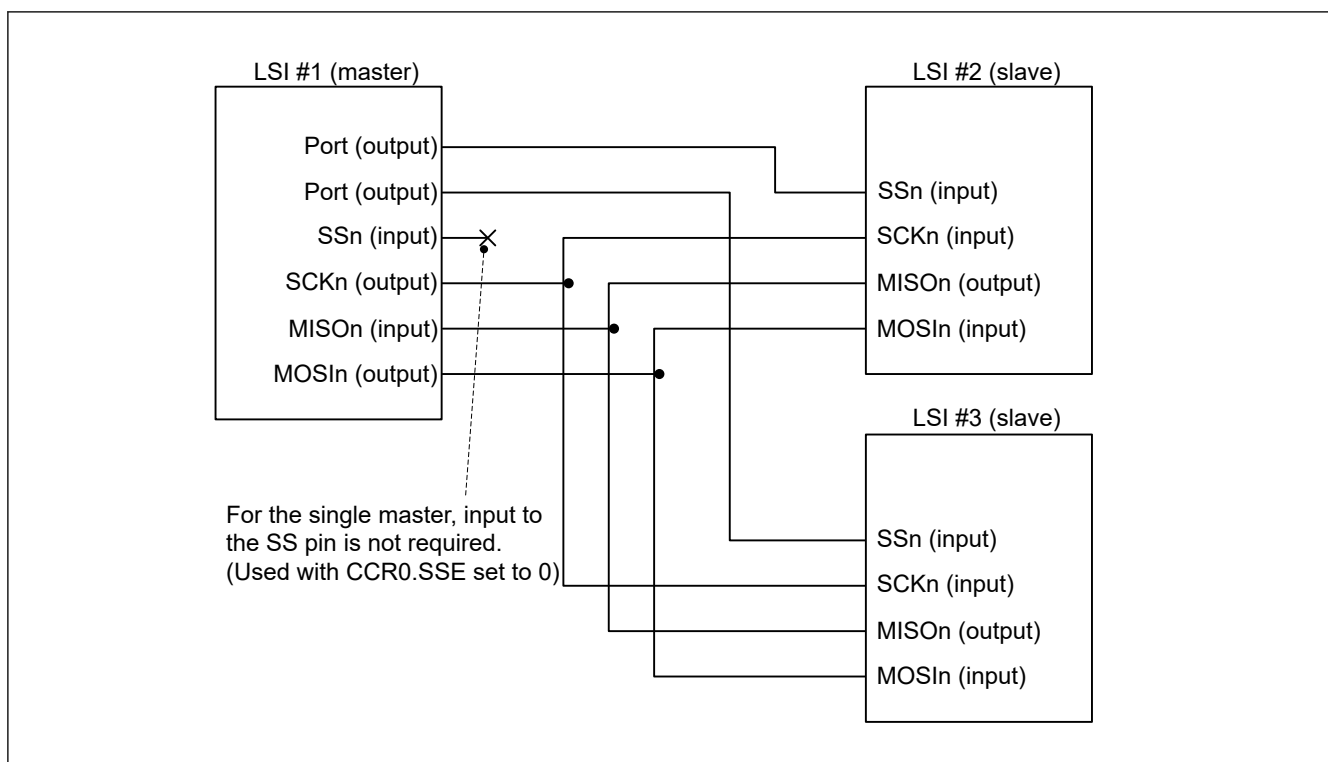


Figure 38.98 Example connections using simple SPI mode in single master mode with CCR0.SSE bit = 0

38.9.1 States of Pins in Master and Slave Modes

The direction (input or output) of pins for the simple SPI mode interface differs according to whether the device is a master (CCR3.CKE[1:0] = 00b or 01b) or slave (CCR3.CKE[1:0] = 10b or 11b).

Table 38.41 lists the relationship between the pin states, mode, and level on the SSn pin.

Table 38.41 States of pins by mode and input level on SSn pin

Mode	Input on SSn pin	State of MOSIn pin	State of MISOIn pin	State of SCKn pin
Master mode ^{*1}	High level (transfer can proceed)	Output for data transmission ^{*2}	Input for received data	Clock output ^{*3}
	Low level (transfer cannot proceed)	High-impedance	Input for received data (but disabled)	High-impedance
Slave mode	High level (transfer cannot proceed)	Input for received data (but disabled)	High-impedance	Clock input (but disabled)
	Low level (transfer can proceed)	Input for received data	Output for data transmission	Clock input

Note 1. When there is only a single master (CCR0.SSE = 0), transfer is possible regardless of the input level on the SSn pin. This is equivalent to input of a high level on the SSn pin. The SSn pin is not used and is available for other purposes.

Note 2. The MOSIn pin output is in the high-impedance state when serial transmission is disabled (CCR0.TE bit = 0).

Note 3. The SCKn pin output is in the high-impedance state when serial transmission is disabled (CCR0.TE = 0 and CCR0.RE = 0) in a multi-master configuration (CCR0.SSE = 1).

38.9.2 SS Function in Master Mode

Setting the CKE[1:0] bits in the CCR3 to 00b or 01b selects master mode operation. The SSn pin is not used in single-master configurations (CCR0.SSE = 0), so transmission or reception can proceed regardless of the value of the SSn pin.

When the level on the SSn pin is high in a multi-master configuration (CCR0.SSE = 1), a master device outputs clock signals from the SCKn pin before starting transmission or reception to indicate that there are no other masters or another master is performing reception or transmission.

When the level on the SSn pin is low in a multi-master configuration (CCR0.SSE = 1), there are other masters, and a transmission or reception is in progress. The MOSIn output and SCKn pins are placed in the high-impedance state and starting transmission or reception is not possible. In addition, the value of the CSR.MFF bit is 1, indicating a mode fault error. In a multi-master configuration, start error processing by reading CSR.MFF flag. If a mode fault error occurs while transmission or reception is in progress, transmission or reception does not stop, but the MOSIn and SCKn outputs are in the high-impedance state after completion of the transfer.

When the SSn pin input becomes high level, the SCKn pin outputs a clock signal and the MOSIn outputs data. Even if the SCKn pin and the MOSIn pin are in the high impedance state, internal transmission or reception operation continues, but it stops after transmission or reception of a single character is complete. In this case, any of SCIn_TXI, SCIn_RXI, and SCIn_TEI interrupts occurs.

Use a general port pin to produce the SS output signal from the master.

38.9.3 SS Function in Slave Mode

Setting the CCR3.CKE[1:0] bits to 10b or 11b selects slave operation. When the SSn pin is high, the MISOn output pin is in the high-impedance state and clock input through the SCKn pin is ignored. When the SSn pin is low, clock input through the SCKn pin is valid and transmission or reception can proceed.

If the input on the SSn pin changes from low to high during transmission or reception, the MISOn output pin is placed in the high-impedance state. Transmission / reception operation is immediately suspended. If the transmission is in progress, the CSR.TEND flag will not be set, a transmit end interrupt will not be output, and an abnormal stop status will occur. So, do not negate the SSn pin during slave transmission / reception. If an abnormal stop occurs, set CCR0.RE and CCR0.TE to 0 to stop transmission / reception. To resume transmission / reception, set CCR0.RE and CCR0.TE to 1 after at least $TCLK \times 3$ cycles + $PCLK \times 3$ cycles.

38.9.4 Relationship Between Clock and Transmit/Receive Data

The CPOL and CPHA bits in the CCR3 register can be used to set up the clock for use in transmission and reception in four different ways. The relation between the clock signal and the transmission and reception of data is shown in [Figure 38.99](#). The relation is the same for both master and slave operation. This is the same as when the level on the SSn pin is high.

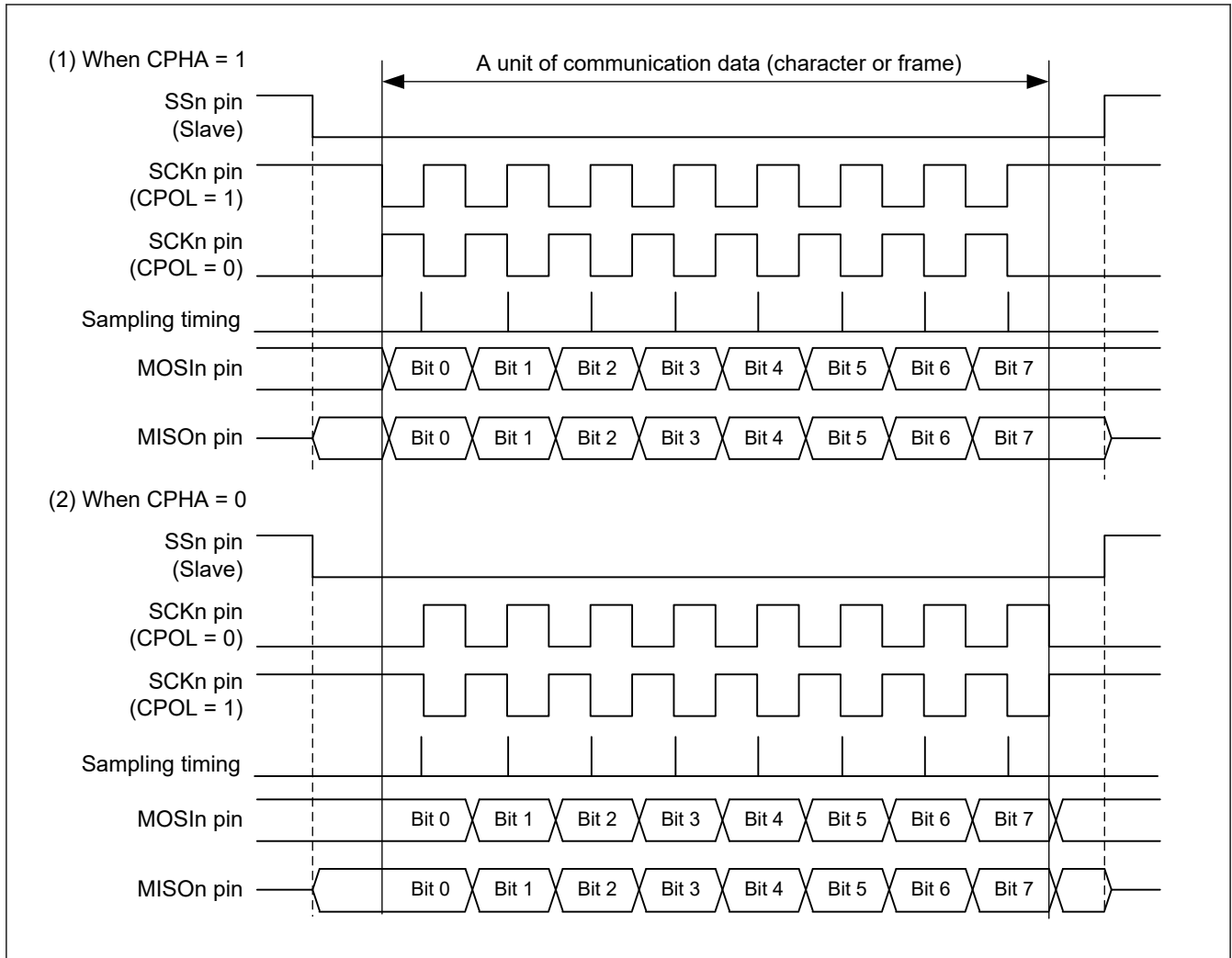


Figure 38.99 Relation between clock signal and transmit or receive data in simple SPI mode

38.9.5 SCI Initialization in Simple SPI Mode

Initialization in simple SPI mode is the same as in clock synchronous mode. See [section 38.6.3. SCI Initialization in Clock Synchronous Mode](#) for an example initialization flow. The CPOL and CPHA bits in the CCR3 register must be set to ensure that the clock signal is suitable for both master and slave devices.

Always initialize the CCR0 register before making any changes to the operating mode or transfer format.

Note: Only the RE bit is set to 0. The CSR.ORER, FER, PER, and RDR flags are not initialized.

Changing the value of the TE bit from 1 to 0 or from 0 to 1 when the TIE bit in the CCR0 register is 1 at the same time, leads to the generation of a transmit data empty interrupt (SCIn_TXI).

38.9.6 Transmission and Reception of Serial Data in Simple SPI Mode

In master operation, ensure that the SSn pin of the slave device on the other side of the transfer is at the low level before starting the transfer and at the high level on completion of the transfer. In multiple master operation with CCR0.SSE = 1 even in master mode, a mode fault error will occur if the SSn pin goes low. Therefore, make sure that no mode fault error has occurred before starting communication, and start communication, and make sure that no mode fault error has occurred even after communication ends. If a mode fault error has occurred, communication may be incomplete, so measures such as retransmission are required. Otherwise, the procedures are the same as in clock synchronous mode.

In slave mode, it operates according to the SSn pin input level. Other steps are the same as those of clock synchronous mode.

38.9.7 Reception Sampling Timing Adjustment Function in Simple SPI Mode with Internal Clock Used

The reception sampling timing adjustment function in simple SPI mode is the same as the reception sampling timing adjustment function in clock synchronous mode. For the description of operation, see [section 38.6.7. Reception Sampling Timing Adjustment Function in Clock Synchronous Mode with Internal Clock Used](#).

38.10 Bit Rate Modulation Function

Using the bit rate modulation function, the bit rate can be evenly corrected using the number specified in the MDDR register among 256 clock cycles of internal clocks which is selected by the CKS[1:0] bits in CCR2.

[Figure 38.100](#) shows an example where the PCLK is selected in the CKS[1:0] bits in CCR2, the BRR bit is set to 0, and the MDDR is set to 160 in Asynchronous mode. In this example, the cycle of the base clock is evenly corrected ($256/160$) and the bit rate is also corrected ($160/256$).

Note: Enabling an internal clock causes bias, and expansion and contraction are generated in the pulse width of the internal base clock.

Do not use this function in clock synchronous mode, simple SPI mode, Smart Card Interface mode, Manchester mode and Simple LIN mode.

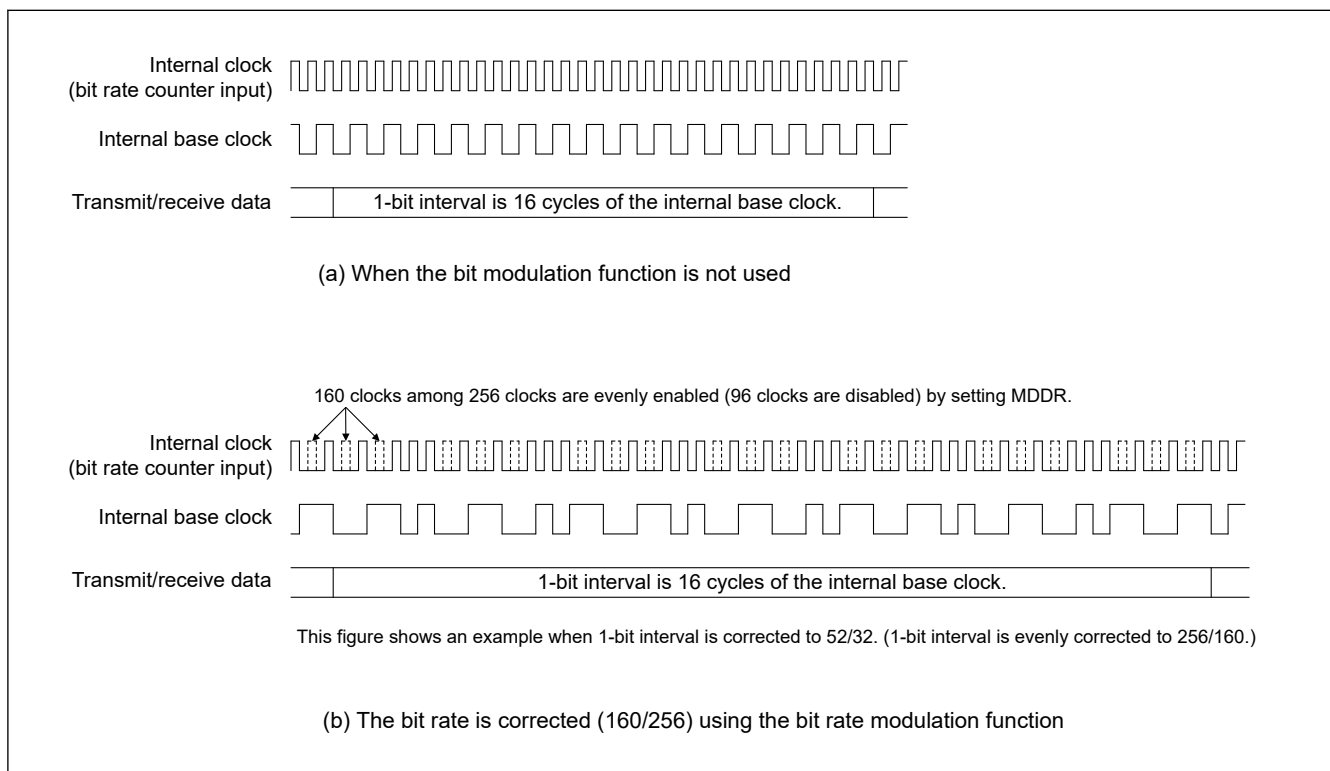


Figure 38.100 Example internal base clock when bit rate modulation function is used

38.11 Simple LIN Mode

As an extended function of the SCI, the SCI supports the serial communication protocol ([Figure 38.101](#)) consisting of a Start Frame and an Information Frame as Simple LIN. Simple LIN mode is enabled by the CCR3.MOD[2:0] = 110b. Since the Simple LIN mode uses the same circuit as the Asynchronous mode for transmission / reception control other than Break Field, the basic communication settings required for the Asynchronous mode are also required for the Simple LIN mode.

(For the setting value when using simple LIN, see the explanation in [section 38.2. Register Descriptions](#). In particular, note that CCR3.RXDESEL needs to be changed from the initial value and set to 1.)

The Start Frame consists of a Break Field, Control Field 0, and Control Field 1. The Information Frame can be configured with some Data Fields, a CRC16 Upper Field, and a CRC16 Lower Field.

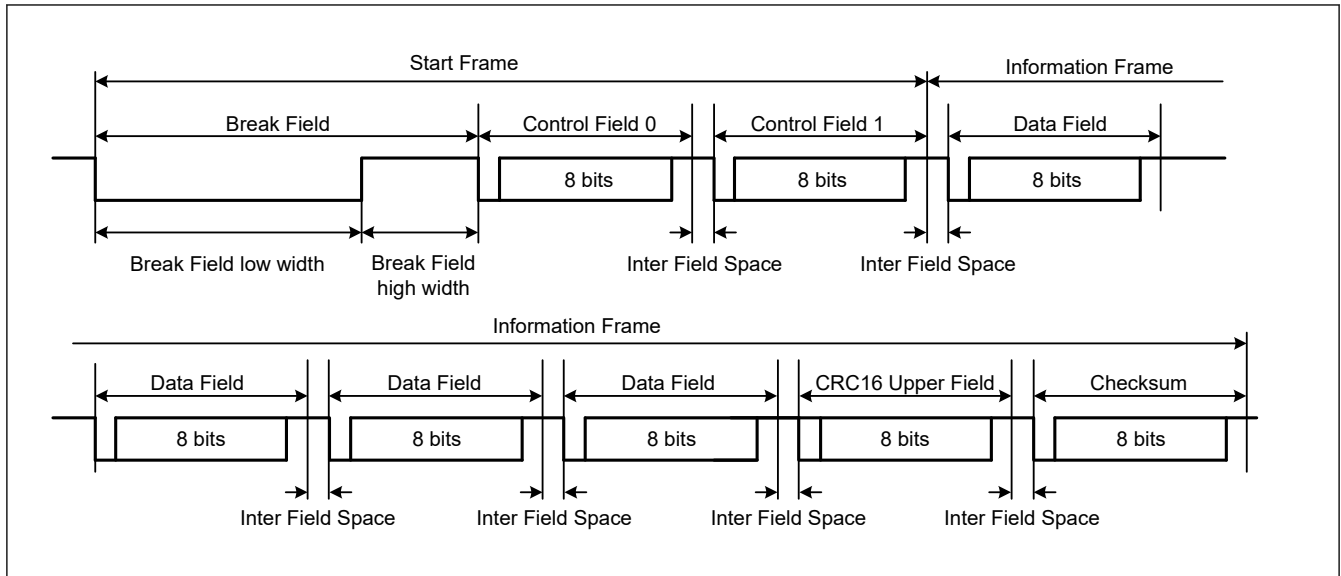


Figure 38.101 Simple LIN Protocol Example

The following describes operations when the Simple LIN is used. In this section, operations are described with the following conditions:

Communication pin (RXDn / TXDn) level inversion function: OFF (RINV = TINV = 0)

When using the simple LIN with the communication pin (RXDn / TXDn) level inversion function enabled, replace the RXDn and TXDn signal levels with their inverted levels.

38.11.1 Simple LIN Start Frame Transmission

Figure 38.102 shows an example of transmission of the Start Frame consisting of a Break Field, Control Field0, and Control Filed1. (Omit Break Field and Control Field0 according to the Start Frame configuration.)

Figure 38.103 shows a flowchart for Start Frame transmission.

The SCI operates as follows during Start Frame transmission.

1. Make the initial settings for the SCI according to the SCI initialization flow (Figure 38.68) in Asynchronous mode. In Simple LIN mode, do not set CCR0.TE and TIE to 1 at the same time to avoid SCIn_TXI output before the Break Field. Therefore, perform the following two steps sequentially to set the SCI initialization flow (Asynchronous mode) procedure [9].
 - Set the bits except CCR0.TIE. (CCR0.TIE = 0, CCR0.TE = 1, and CCR0.RE = 0)
 - Set CCR0.TIE to 1.
2. When 1 is written to TCST, the Break Field output timer starts counting and outputs a low level (Break Field) from the TXDn pin for the period set in XCR2.BFLW[15:0]. A timer count clock source can be selected by XCR0.TCSS[1:0]. Writing 0 to XCR1.TCST suspends output of the Break Field. After the suspension, set CCR0.TE = 0 and turn off the transmission.
3. When the Simple LIN module timer count value matches the set XCR2.BFLW[15:0] value, the timer stops counting and inverts the TXDn pin output level, and the XSR0.BFOF flag is set to 1^{*1}. Furthermore, if XCR0.BFOIE has been set to 1 at this time, a SCIn_TXI interrupt is generated.
4. After the SCIn_TXI interrupt and confirming XSR0.BFOF = 1, write the transmitted data then the Control Field 0 data is transmitted using the SCI^{*2}.
5. After the Control Field 0 data has been transmitted, write the Control Field 1 data to TDR. And it is transmitted.
6. After the Control Field 1 data has been transmitted, the Information Frame data is transmitted.

Note 1. After XSR0.BFOF is set to 1, if 1 is written to XCR1.TCST without clearing it, no SCIn_TXI interrupt is output at the end of Break Field transmission. Clear XSR0.BFOF before writing 1 to XCR1.TCST.

Note 2. LIN communication requires a Break Delimiter (IDLE period) of 1 bit or more from the end of Break Field output until the next data transmission starts. For this reason, the Break Delimiter length is counted upon completion of Break

Field output. If transmit data is written while the Break Delimiter length is being counted, transmission does not start until the Break Delimiter length counting is completed. When transmit data is written after the Break Delimiter length has been counted, transmission starts at the same timing as normal data transmission.

Break Delimiter length count time after Break Field output:

1-bit to 2-bit length (CCR3.STP = 0)

2-bit to 3-bit length (CCR3.STP = 1)

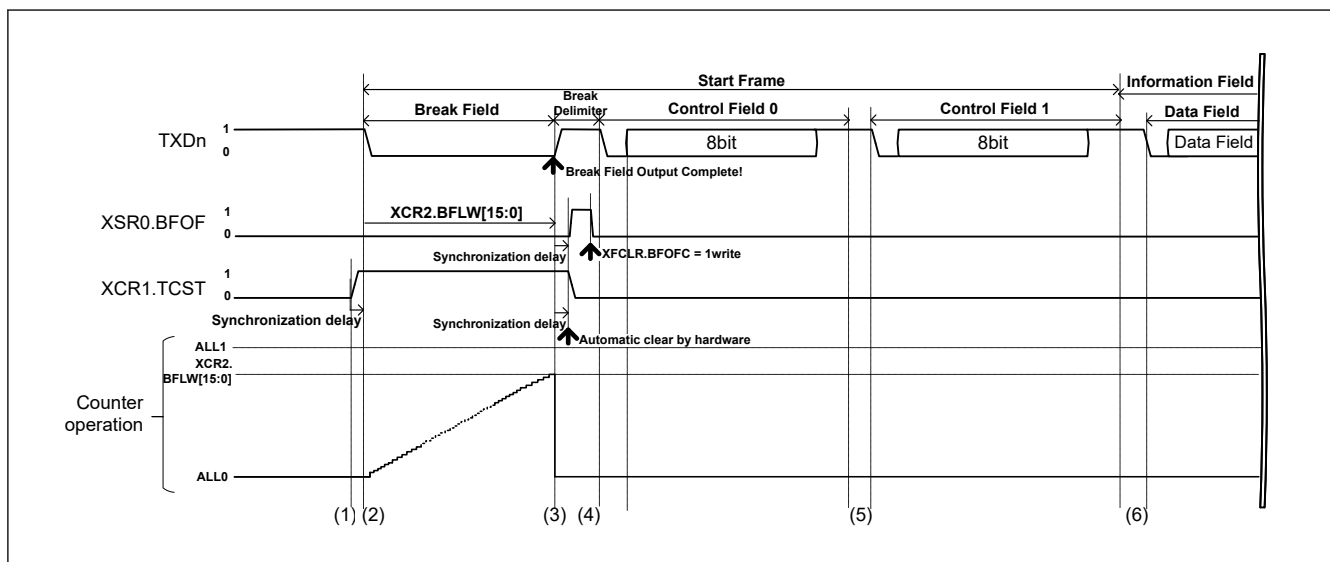


Figure 38.102 Start Frame Transmission Example

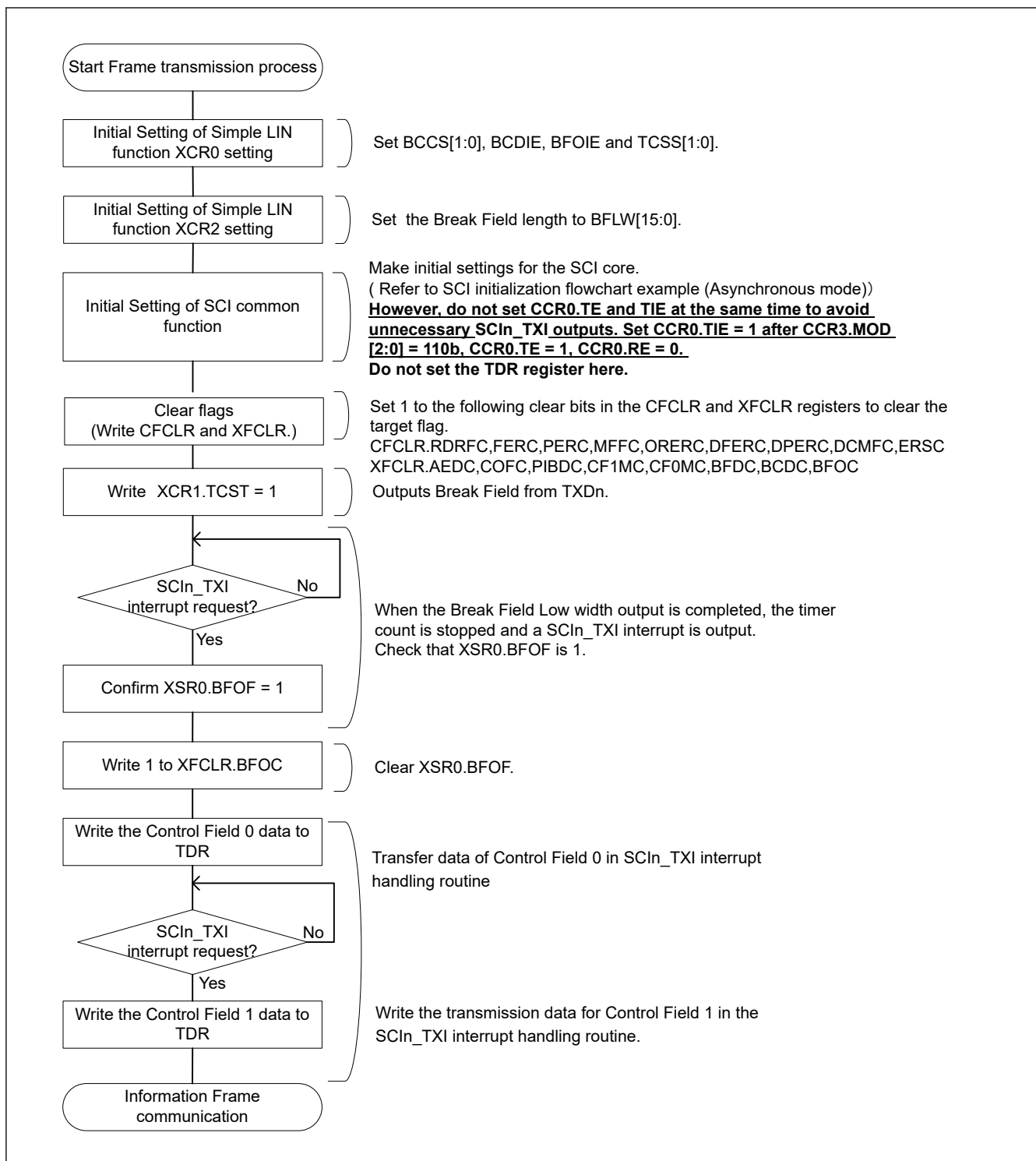


Figure 38.103 Start Frame Transmission Flowchart Example

38.11.2 Simple LIN Start Frame Reception

The SCI can detect Start Frames configured as shown in [Figure 38.104](#).

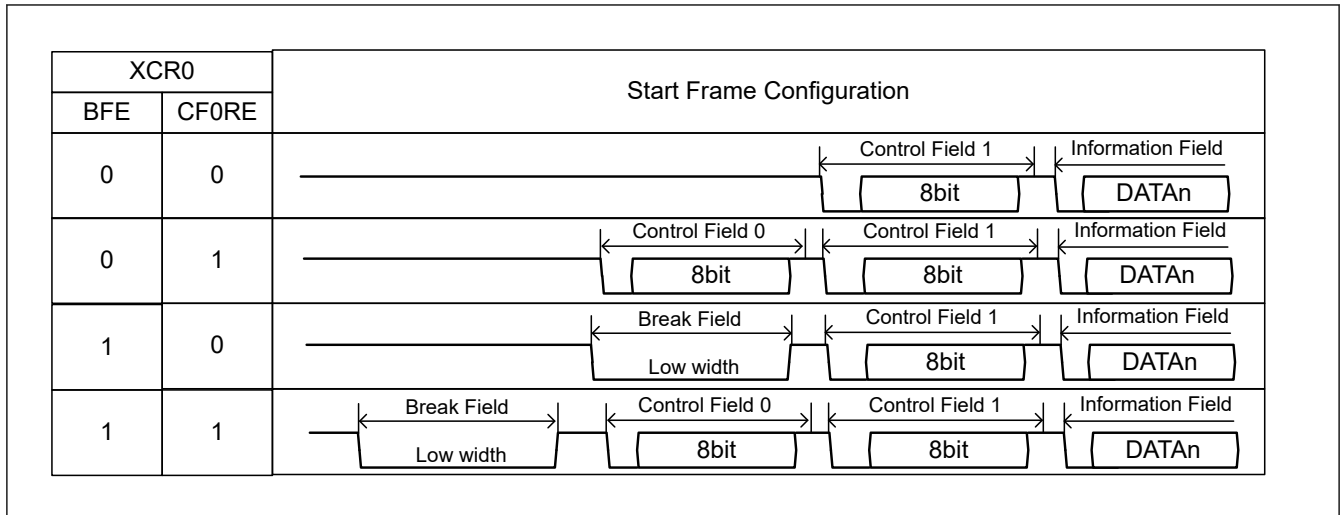
**Figure 38.104 Start Frame Configuration****(1) Simple LIN normal reception of Start Frame (PIB not used)**

Figure 38.105 shows an example of normal reception of the Start Frame consisting of a Break Field, Control Field0, and Control Field1. Figure 38.106 shows an example of reception to detect the Break Field during Control Field 1. Figure 38.107 shows a flowchart to receive the Start Frame, and Figure 38.108 shows a state transition diagram.

When receiving the Start Frame, the SCI operates as follows. Omit the processing of Break Field and Control Field0 according to the Start Frame configuration.

- Writing 1 to XCR1.SDST makes it possible to detect the Start Frame. When XCR0.BFE = 1, RXDn input to the SCI core is disabled until the Break Field is detected (because XSR0.RXDSF is set to 1). Once the Break Field is detected, RXDn input can be received to the SCI core (XSR0.RXDSF = 0).
- When a low level is input from the RXDn pin, the Break Field detection count starts. A timer count clock source can be selected by XCR0.TCSS[1:0].
- When a low level that is equal to or longer than the period set in XCR2.BFLW[15:0] is input from the RXDn pin, it is determined as Break Field. At this time, XSR0.BFDF is set to 1. If XCR0.BFDIE has been set to 1 at this time, a SCIn_BFD interrupt is generated.
The timer continues counting until the RXDn rising edge or counter overflow.
- After the Break Field is detected, when the input level from the RXDn pin becomes high, the count value is captured to XSR1.TCNT[15:0] when BMEN = 0. At this time, XSR0.RXDSF is cleared to 0 and the SCI core starts receiving the RXDn input.
- The SCI core starts receiving Control Field 0. Because the simple LIN continuously counts the edge interval, it determines a low level that is equal to or longer than the period set in XCR2.BFLW[15:0] as detection of the Break Field. When the Break Field is detected in the Control Field 0 phase, the SCI core waits for reception of Control Field 0 again (Figure 38.106).
- When Control Field 0 has been received, an SCIn_RXI interrupt is generated and the Control Field 0 data is stored in XSR0.CF0RD[7:0]. When the received data matches the set XCR2.CF0D[7:0] value, XSR0.CF0MF is set to 1. If the received data differs from the set XCR2.CF0D[7:0] value, the SCI transitions to the state before the Break Field is detected.
- The SCI core starts receiving Control Field 1. When BFE = 1, the Break Field detection function is continuously enabled while SDST = 1 as in the case of Control Field 0. When the Break Field is detected in the Control Field 1 phase, the SCI core waits for reception of Control Field 0 again.
- When Control Field 1 has been received, an SCIn_RXI interrupt is generated and the Control Field 1 data is stored in XSR0.CF1RD[7:0]. When the received data matches the set XCR1.PCF1D[7:0] value or the set XCR1.SCF1D[7:0] value, XSR0.CF1MF is set to 1. If the received Control Field 1 data matches neither the set XCR1.PCF1D[7:0] value nor the set XCR1.SCF1D[7:0] value, the SCI transitions to the state before the Break Field is detected.
- The SCI core performs Information Frame communication.
- When communication is completed, write 0 to XCR1.SDST and 0 to CCR0.RE to stop reception.

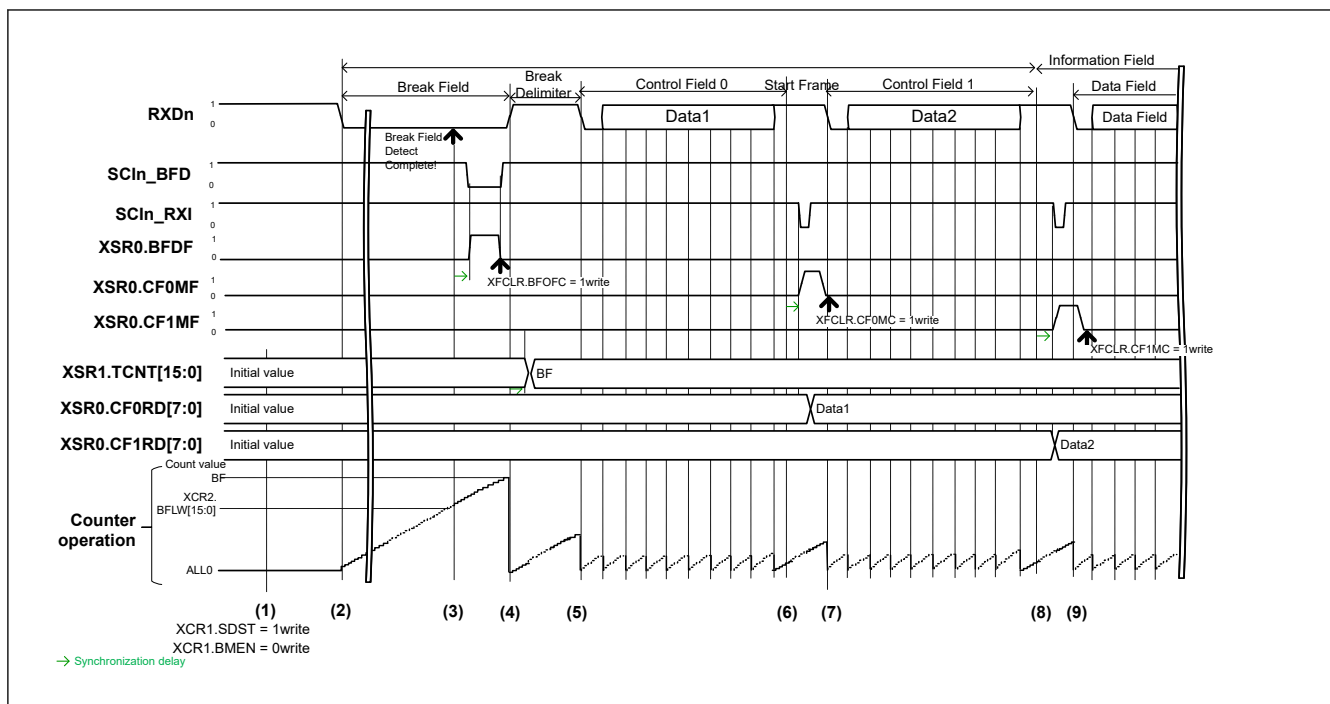


Figure 38.105 Normal Reception Example of Start Frame (PIB Not Used)

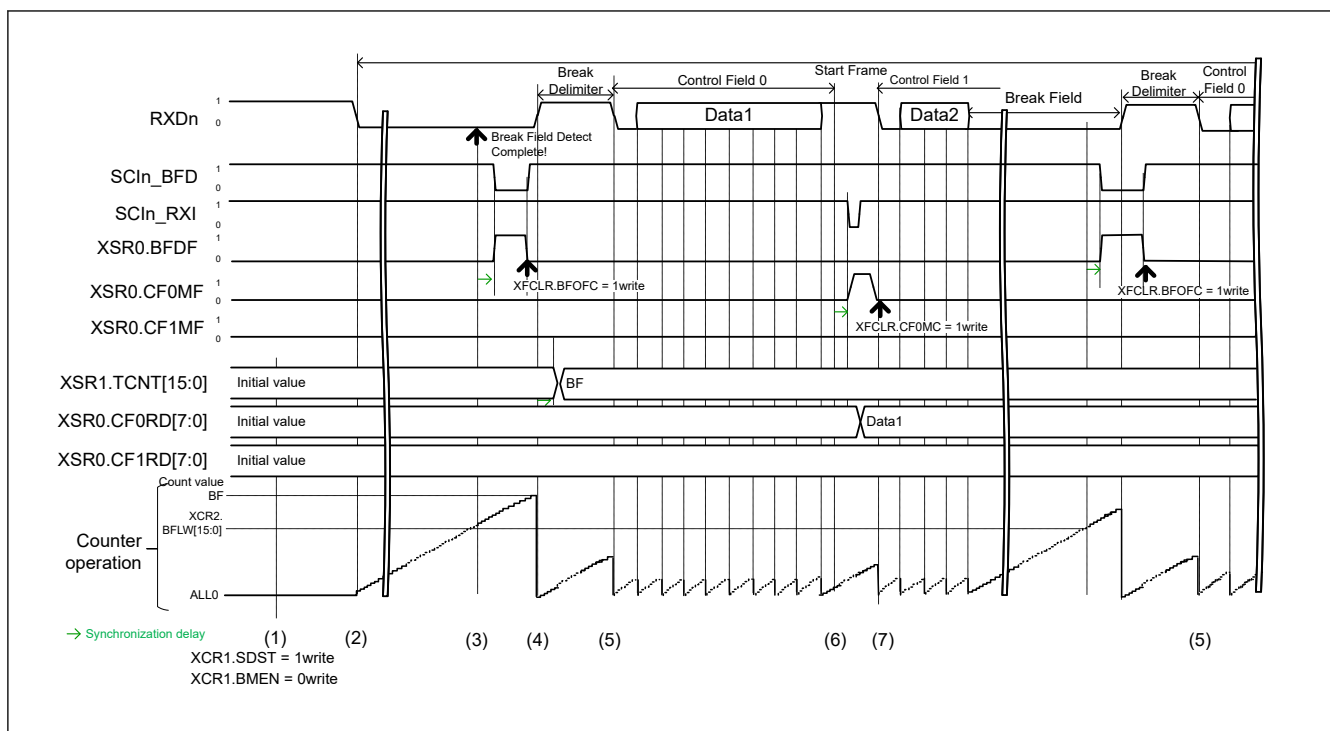


Figure 38.106 Start Frame Reception Example (PIB Not Used) Break Field Detected during Control Field 1

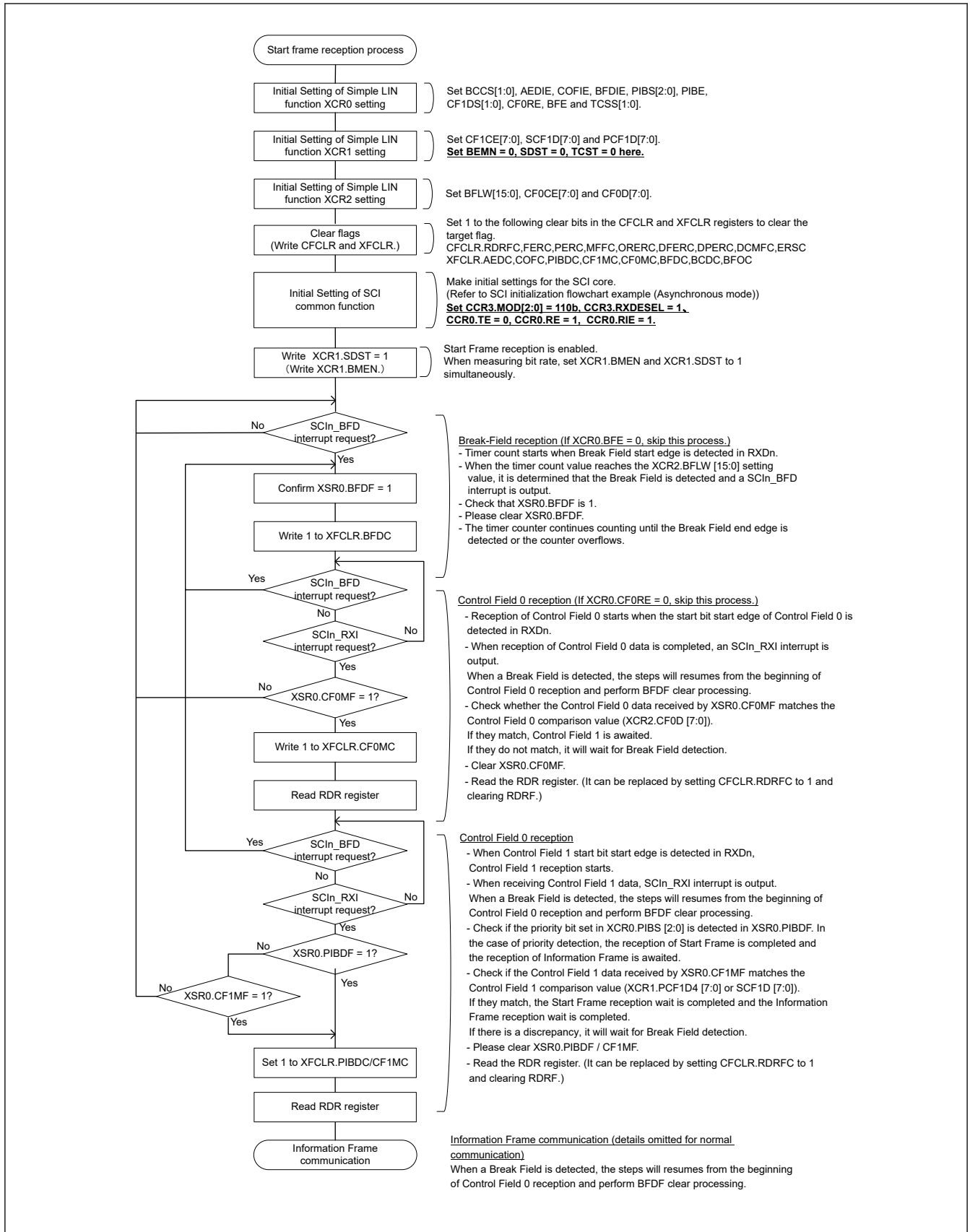


Figure 38.107 Example of Start Frame Reception Flowchart

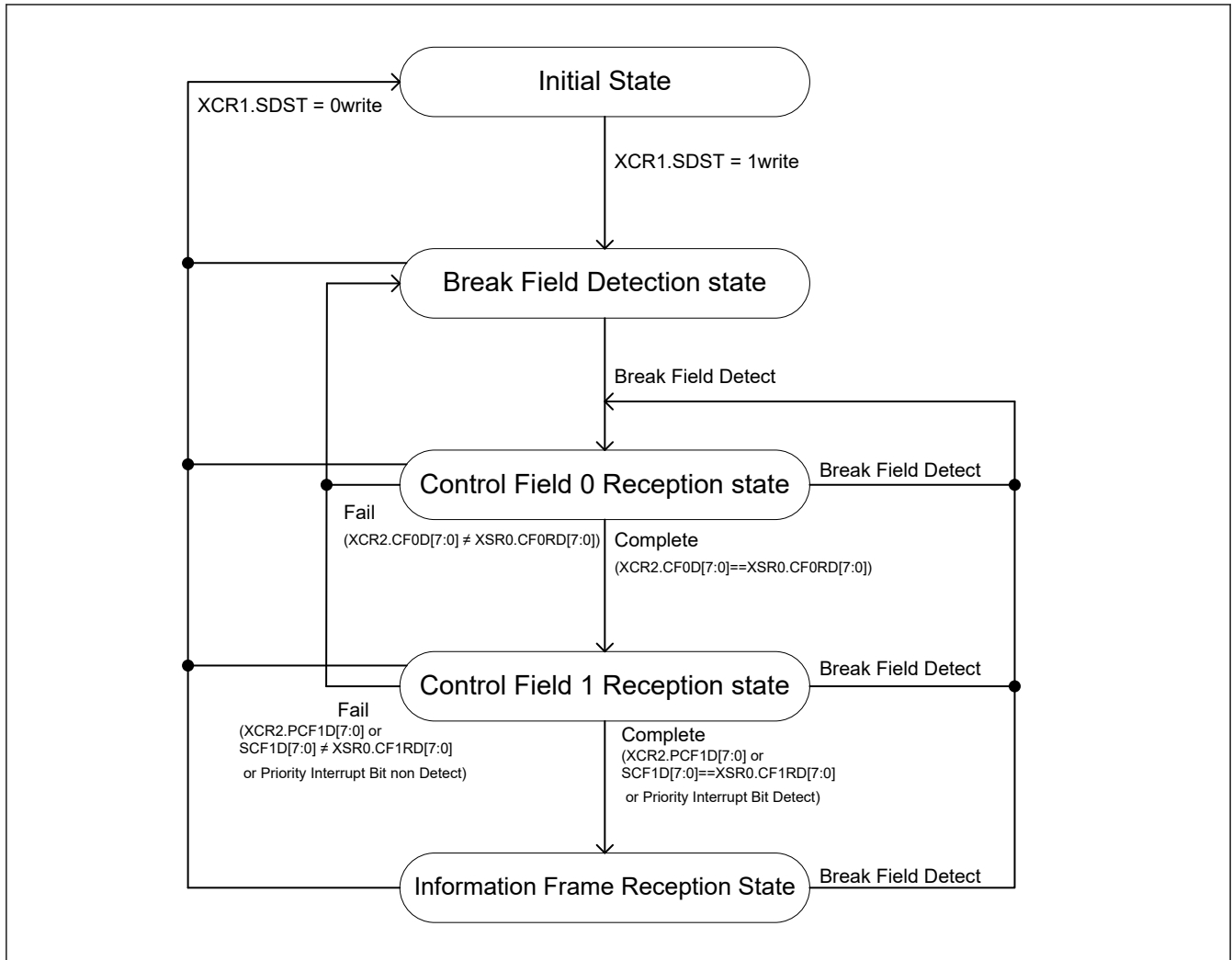


Figure 38.108 State Transition Diagram of Start Frame Reception

(2) Simple LIN Start Frame reception (using the priority interrupt bit)

Figure 38.109 shows an example of Start Frame reception using the priority interrupt bit. The priority interrupt bit is enabled by setting **XCR0.PIBE** to 1.

The SCI operates as follows during Start Frame reception using the priority interrupt bit.

Steps (1) to (7) are the same as steps (1) to (7) in the Start Frame reception example in Figure 38.105.

(8) When the value specified in the **XCR0.PIBS[2:0]** bits matches the set **XCR1.PCF1D[7:0]** value, **XSR0.PIBDF** is set to 1 and the SCI core performs communication of the Information Frame. If the data received in Control Field 1 matches neither the set **XCR1.PCF1D[7:0]** value nor the set **XCR1.SCF1D[7:0]** value and the priority interrupt bit is not detected, the SCI transitions to the state before the Break Field is detected.

(9) Communicate information frame at the SCI core.

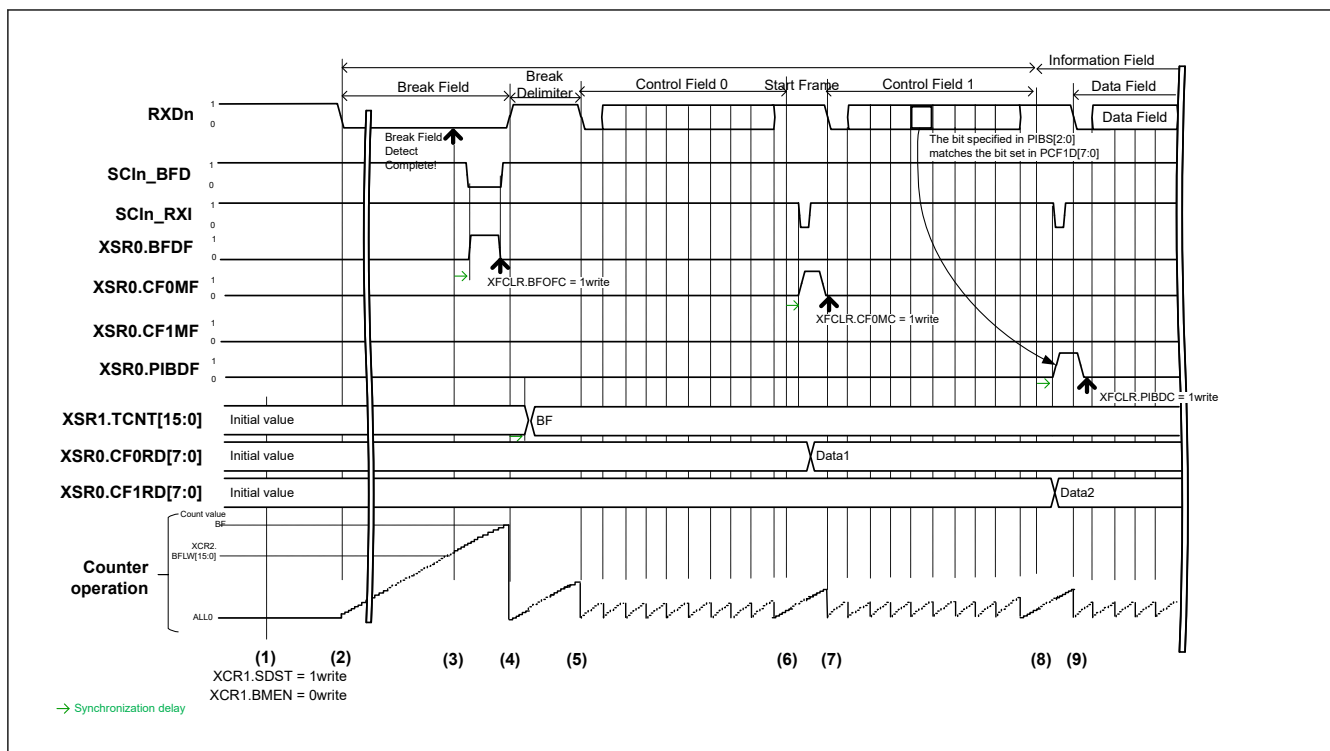


Figure 38.109 Start Frame Reception Example (Priority Interrupt Bit Used)

38.11.3 Simple LIN Bus Conflict Detection Function

In Simple LIN mode (CCR3MOD[2:0] = 110) when TE = 1, the bus conflict detection function works during Break Field output and during data transmission.

Figure 38.110 shows an operation example of the bus conflict detection function. The TXDn pin output and the RXDn pin input are sampled by the bus conflict detection clock set in XCR0.BCCS[1:0]. When a mismatch occurs three times in a row, XSR0.BCDF is set to 1, and if XCR0.BCDIE has been set to 1 at this time, an SCIIn_ERI interrupt is generated.

When an SCIIn_ERI interrupt is generated, stop transmission according to Figure 38.111. Check the bus state to decide whether to resume transmission.

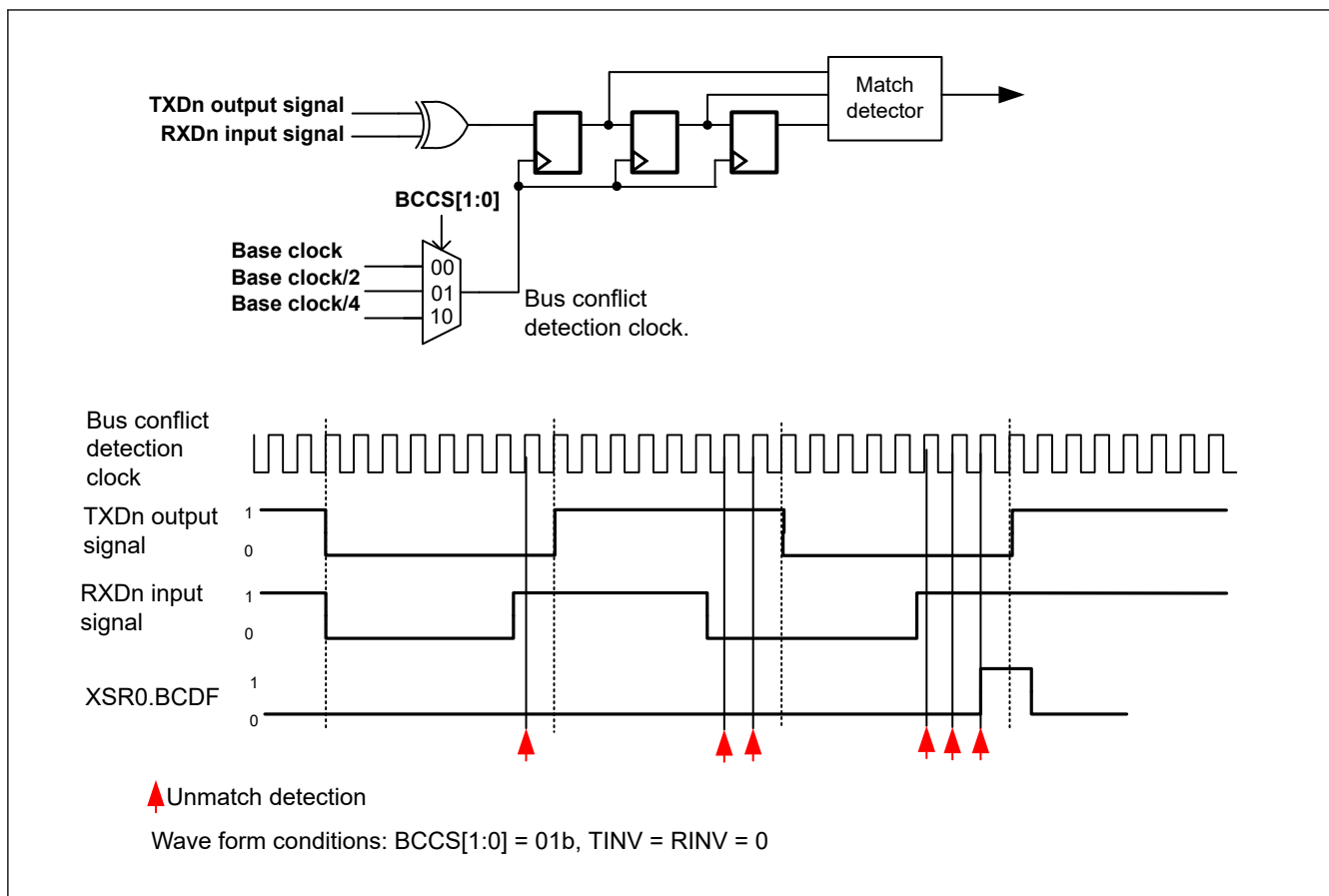


Figure 38.110 Operation Example of the Bus Conflict Detection Function

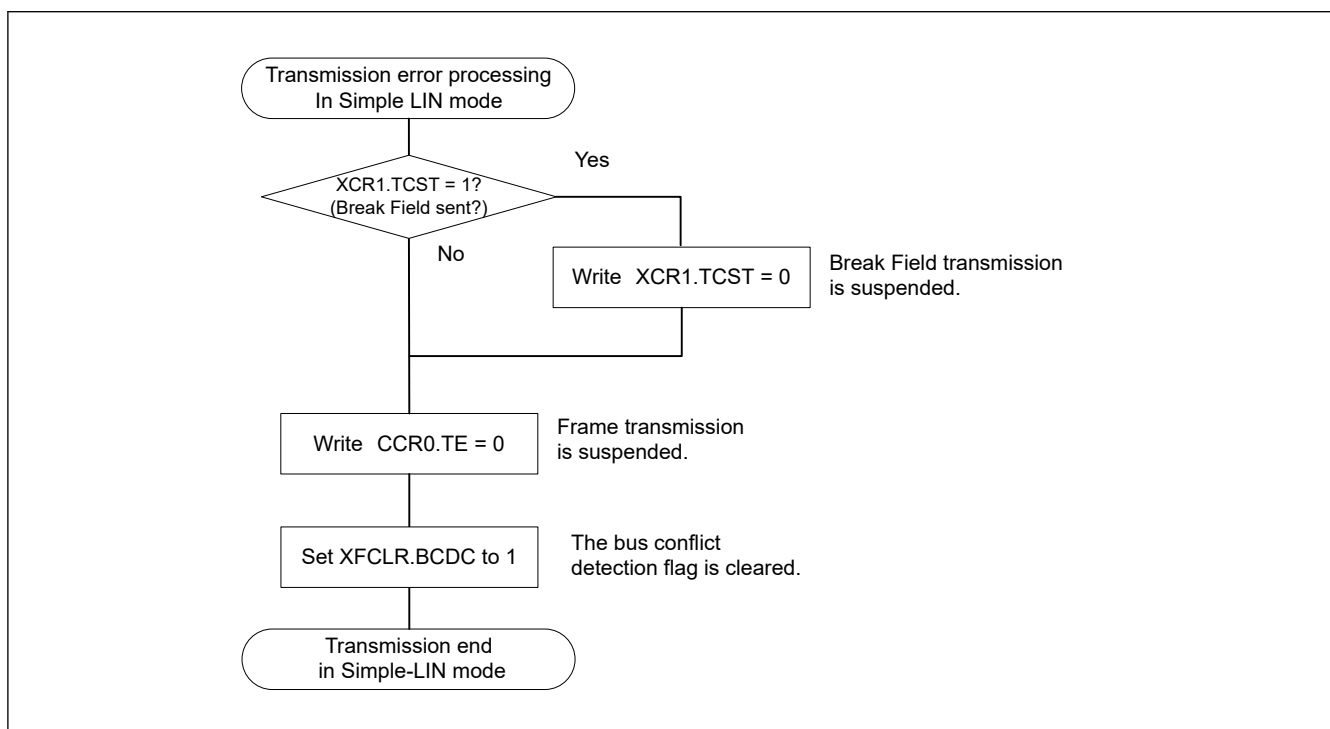


Figure 38.111 SCIn_ERI Interrupt handling flow at transmission in Simple LIN mode

38.11.4 Simple LIN Bit Rate Measurement Function

This function measures a bit rate between the effective edges of the input signal from the RXDn pin. Figure 38.112 shows an operation example of the bit rate measurement function.

1. Writing 1 to XCR1.SDST and XCR1.BMEN enables bit rate measurement. When this bit is set to 1, the valid edge interval of Control Field 0 and Control Field 1 data is measured. However, bit rate is not measured between the Break Field and the Break Delimiter. Set XCR1.BMEN and XCR1.SDST to 1 simultaneously, only when measuring bit rate.
2. Because bit rate is not measured in the Break Field, the effective edge detection flag is not set to 1 at the rising edge at the end of the Break Field, and the counter capture value is not stored in XSR1.TCNT[15:0].
3. The counter starts counting from the falling edge of the start bit in Control Field 0. The Break Delimiter count value is not captured in XSR1.TCNT[15:0].
4. The rising edge of the start bit is detected as an effective edge, and then the XSR0.AEDF flag is set to 1. If XCR0.AEDIE has been set to 1 at this time, an SCIn_AED interrupt is output. The start bit count value is stored in XSR1.TCNT[15:0]. The XSR1.TCNT[15:0] value is retained until the effective capture value is read.
5. Even if an effective edge is input from the RXDn input pin, the count value of this effective edge timing is not captured because the XSR1.TCNT[15:0] value has not been read and retention has not been released. In this case, an SCIn_AED interrupt is not output.
6. The XSR1.TCNT[15:0] value is read. Then the retention of XSR1.TCNT[15:0] is released and the XSR0.AEDF flag is cleared by hardware.
7. Because the retention of XSR1.TCNT[15:0] has been released, the count value is captured at the effective edge and is retained. At the same time, the XSR0.AEDF flag is set to 1, and if XCR0.AEDIE has been set to 1, an SCIn_AED interrupt is output. The bit rate can be adjusted by calculating it from the count value between effective edges by software and by changing the SCI settings.
8. To disable bit rate measurement, write 0 to XCR1.BMEN.
9. The XSR0.AEDF value and the XSR1.TCNT[15:0] value remain unchanged at the effective edge timing because the bit rate measurement function is disabled.

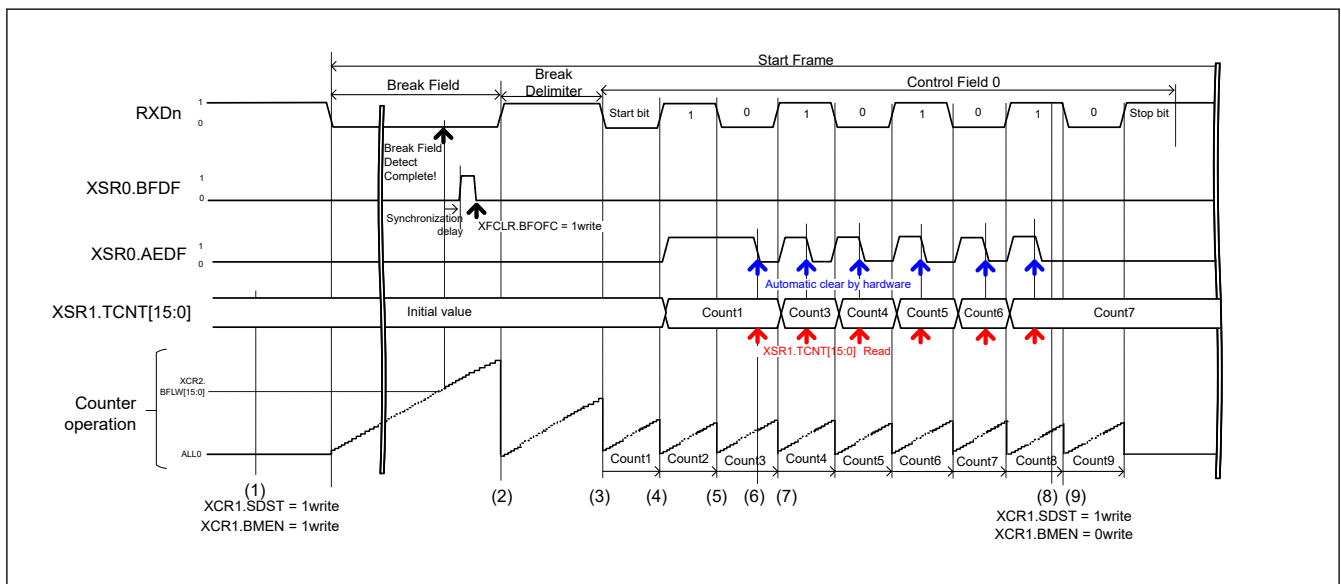


Figure 38.112 Operation Example of the Bit Rate Measurement Function

38.12 Interrupt Sources

38.12.1 Buffer Operation for SCIn_TXI and SCIn_RXI Interrupts

If the conditions for an SCIn_TXI and SCIn_RXI interrupt are satisfied while the interrupt status flag in the ICU is 1, the ICU does not output the interrupt request but saves it internally with a capacity for retention of one request per source.

38.12.2 Interrupts in Asynchronous, Manchester, Clock Synchronous, and Simple SPI Modes

(1) Non-FIFO selected

Table 38.42 lists interrupt sources in Asynchronous mode, Manchester mode, clock synchronous mode, and simple SPI mode.

A different interrupt vector can be assigned to each interrupt source. Individual interrupt sources can be enabled or disabled with the enable bits in the CCR0 register.

If the CCR0.TIE bit is 1, an SCIn_TXI interrupt request is generated when transmit data is transferred from the TDR register to the TSR register. An SCIn_TXI interrupt request can also be generated by using a single instruction to set the CCR0.TE and CCR0.TIE bits to 1 at the same time. An SCIn_TXI interrupt request can activate the DTC or DMAC to handle data transfer.

An SCIn_TXI interrupt request is not generated by setting the CCR0.TE bit to 1 when CCR0.TIE is 0 or by setting the CCR0.TIE bit to 1 when the CCR0.TE is 1.*¹

When new data is not written by the time of transmission of the last bit of the current transmit data and CCR0.TEIE is 1, the CSR.TEND flag is set to 1 and an SCIn_TEI interrupt request is generated. Additionally, when CCR0.TE is 1, the CSR.TEND flag retains the value 1 until more transmit data is written to the TDR register, and setting CCR0.TEIE to 1 leads to the generation of an SCIn_TEI interrupt request.

Writing data to the TDR register leads to clearing of the CSR.TEND flag and, after a certain time, discarding of the SCIn_TEI interrupt request.

If the CCR0.RIE bit is 1, an SCIn_RXI interrupt request is generated when received data is stored in the RDR register. An SCIn_RXI interrupt request can activate the DTC or DMAC to handle data transfer.

Setting any of the CSR.ORER, FER, PER or MSR.MER*², SYER*², PFER*², and SBER*² flags to 1 when the CCR0.RIE bit is 1 leads to the generation of an SCIn_ERI interrupt request.

An SCIn_RXI interrupt request is not generated in this case. Clearing all these flags (ORER, FER, PER, MER*², SYER*², PFER*² and SBER*²) leads to discarding of the SCIn_ERI interrupt request.

Note 1. To temporarily prohibit SCIn_TXI interrupts on transmission of the last of the data when a new round of transmission is to be started, after handling the transmission-completed interrupt, control activation of the interrupt by using the interrupt request enable bit in the ICU rather than using the CCR0.TIE bit. This approach can prevent the suppression of SCIn_TXI interrupt requests in the transfer of new data.

Note 2. MER, SYER, PFER, and SBER work as a factor of SCIn_ERI interrupt only in Manchester mode. SYER, PFER, and SBER also only work if its enable bits (SYEREN, PFEREN, SBEREN in MECR) are set to "1".

(2) FIFO selected

Table 38.43 lists interrupt sources in FIFO selected mode.

If the CCR0.TIE bit is 1, an SCIn_TXI interrupt request is generated when the stored amount of data in the transmit-FIFO (TDR) register becomes the threshold value indicated in FCR.TTRG or below. An SCIn_TXI interrupt request can also be generated by using a single instruction to set the CCR0.TIE and CCR0.TE bits to 1 simultaneously or by setting CCR0.TIE to 1 when CCR0.TE is 1.

An SCIn_TXI interrupt request is not generated by setting CCR0.TE to 1 when CCR0.TIE is 0 or by setting the CCR0.TIE bit to 1 while the setting of the CCR0.TE bit is 1.

If CCR0.TEIE is 1 and if the next data is not written to the transmit-FIFO (TDR) register by the time the last bit of the transmit data is sent, the CSR.TEND flag is set to 1 and the SCIn_TEI interrupt request is generated.

If CCR0.RIE is 1, the SCIn_RXI interrupt request is generated when the stored amount of data in the transmit-FIFO (TDR) register is equal to or greater than the threshold value indicated in FCR.RTRG. When RTRG is 0, an SCIn_RXI interrupt does not occur even when the amount of data in the receive FIFO is equal to 0.

If the CCR0.RIE bit is 1, when the CSR.ORER flag is set to 1 or data with a framing error or a parity error is stored in the transmit-FIFO (TDR) register, the SCIn_ERI interrupt request is generated. When the amount of data stored in the transmit-FIFO (TDR) register is at the threshold value or above, the SCIn_RXI interrupt request is also generated. The SCIn_ERI interrupt request can be canceled, in which case CSR.ORER, FER, and PER flags are all cleared.

Table 38.42 SCI interrupt sources with non-FIFO selected

Name	Interrupt source	Interrupt flag	Interrupt enable	DTC or DMAC activation
SCIn_ERI (n = 0 to 9)	Receive error	CSR.ORER, CSR.FER, CSR.PER, CSR.DFER, CSR.DPER, (MSR.MER, MSR.SYER, MSR.PFER, MSR.SBER)*1	CCR0.RIE	Not possible
SCIn_RXI (n = 0 to 9)	Receive data full	CSR.RDRF	CCR0.RIE	Possible
	Address match	CSR.DCMF	CCR0.RIE	Possible
SCIn_TXI (n = 0 to 9)	Transmit data empty	CSR.TDRE	CCR0.TIE	Possible
	TE = 0->1 detection			
SCIn_TEI (n = 0 to 9)	Transmit end	CSR.TEND	CCR0.TEIE	Not possible

Note: Only SCI0 supports Manchester mode.

Note 1. MER, SYER, PFER, and SBER work as a factor of SCIn_ERI interrupt only in Manchester mode. SYER, PFER, and SBER also only work if its enable bits (SYEREN, PFEREN, SBEREN in MCR) are set to 1.

Table 38.43 SCI interrupt sources with FIFO selected

Name	Interrupt source	Interrupt flag	Interrupt enable	DTC or DMAC activation
SCIn_ERI (n = 0 to 9)	Receive error	CSR.ORER, CSR.FER, CSR.PER, CSR.DFER, CSR.DPER	CCR0.RIE	Not possible
		FRSR.DR (when FCR.DRES = 1)	CCR0.RIE	Not possible
SCIn_RXI (n = 0 to 9)	Receive data full	CSR.RDRF	CCR0.RIE	Possible
	Receive data ready	FRSR.DR (when FCR.DRES = 0)	CCR0.RIE	Possible
	Address match	CSR.DCMF	CCR0.RIE	Possible
SCIn_TXI (n = 0 to 9)	Transmit data empty	CSR.TDFE	CCR0.TIE	Possible
	TE=0->1 detection			
SCIn_TEI (n = 0 to 9)	Transmit end	CSR.TEND	CCR0.TEIE	Not possible

Note: Only SCI0 supports Manchester mode.

38.12.3 Interrupts in Smart Card Interface Mode

Table 38.44 lists interrupt sources in smart card interface mode. A transmit end interrupt (SCIn_TEI) request and an address match (SCIn_AM) request cannot be used in this mode.

Table 38.44 SCI Interrupt sources in Smart Card Interface Mode

Name	Interrupt source	Interrupt flag	Interrupt enable	DTC or DMAC activation
SCIn_ERI (n = 0 to 9)	Receive error or error signal detection	CSR.ORER, CSR.PER, CSR.ERS	CCR0.RIE	Not possible
SCIn_RXI (n = 0 to 9)	Receive data full	CSR.RDRF	CCR0.RIE	Possible
SCIn_TXI (n = 0 to 9)	Transmit data empty	CSR.TEND	CCR0.TIE	Possible
	When set TE = 0->1			

Data transmission or reception using the DTC or DMAC is also possible in smart card interface mode, similar to normal SCI mode. In transmission, when the CCR0.TEND flag is set to 1, an SCIn_TXI interrupt request is generated. This SCIn_TXI interrupt request activates the DTC or DMAC, allowing transfer of transmit data if the SCIn_TXI request is previously specified as a source of DTC or DMAC activation. The TEND flag is automatically set to 0 when the DTC or DMAC transfers the data.

If an error occurs, the SCI automatically retransmits the same data. During the retransmission, the TEND flag is kept at 0 and the DTC or DMAC is not activated. Therefore, the SCI and DTC or DMAC automatically transmit the specified

number of bytes, including retransmission after an error occurrence. However, the CSR.ERS flag is not automatically set to 0 at error occurrence. Therefore, the ERS flag must be cleared by previously setting the CCR0.RIE bit to 1 to enable an SCIn_ERI interrupt request to be generated at error occurrence.

When transmitting or receiving data using the DTC or DMAC, always enable the DTC or DMAC before making the SCI settings. For DTC or DMAC settings, see [section 18, Data Transfer Controller \(DTC\)](#), [section 17, DMA Controller \(DMAC\)](#).

In reception, an SCIn_RXI interrupt request is generated when receive data is set to the RDR register. This SCIn_RXI interrupt request activates the DTC or DMAC, allowing transfer of the receive data if the SCIn_RXI request is previously specified as a source of DTC or DMAC activation. If an error occurs, the error flag is set. Therefore, the DTC or DMAC is not activated and an SCIn_ERI interrupt request is issued to the CPU instead. The error flag must be cleared.

38.12.4 Interrupts in Simple IIC Mode

[Table 38.45](#) lists the interrupt sources in simple IIC mode. The STI interrupt is allocated to the transmit end interrupt (SCIn_TEI) request. The receive error interrupt (SCIn_ERI) and the address match (SCIn_AM) request cannot be used.

The DTC or DMAC can also be used to handle transfer in simple IIC mode.

When the ICR.IICINTM bit is 1:

- An SCIn_RXI request is generated on the falling edge of the SCLn signal for the 8th bit. If SCIn_RXI is previously set up as an activation source for the DTC or DMAC, the SCIn_RXI request activates the DTC or DMAC to handle transfer of the received data.
- An SCIn_TXI request is generated on the falling edge of the SCLn signal for the 9th bit (acknowledge bit). If SCIn_TXI is previously set up as an activation source for the DTC or DMAC, the SCIn_TXI request activates the DTC or DMAC to handle transfer of the transmit data.

When the ICR.IICINTM bit is 0:

- An SCIn_RXI request (ACK detection) is generated if the input on the SDAn pin is low on the rising edge of the SCLn signal for the 9th bit (acknowledge bit)
- An SCIn_TXI request (NACK detection) is generated if the input on the SDAn pin is high on the rising edge of the SCLn signal for the 9th bit (acknowledge bit)
- If SCIn_RXI is previously set up as an activation source for the DTC or DMAC, the SCIn_RXI request activates the DTC or DMAC to handle transfer of the received data.

If the DTC or DMAC is used for data transfer in reception or transmission, always set up and enable the DTC or DMAC before setting up the SCI.

When the IICSTAREQ, IICRSTAREQ, and IICSTPREQ bits in ICR are used to generate a start condition, restart condition, or stop condition, the STI request is issued when generation is complete.

Table 38.45 SCI interrupt sources in Simple IIC Mode

Name	Interrupt source		Interrupt flag	Interrupt enable	DTC or DMAC activation
	ICR.IICINTM = 1	ICR.IICINTM = 0			
SCIn_RXI (n = 0 to 9)	Reception end	—	—	CCR0.RIE	Possible* ¹
	—	ACK detection	—		Possible
SCIn_TXI (n = 0 to 9)	Transmission end	—	—	CCR0.TIE	Possible* ¹
	—	NACK detection	—		Possible
SCIn_TEI(STIn) (n = 0 to 9)	Completion of generation of a start, restart, or stop condition		ICR.IICSTIF	CCR0.TEIE	Not possible

Note 1. If the DMAC or DTC are being used, you cannot confirm whether ACK or NACK.

38.12.5 Interrupts in Simple LIN Mode

[Table 38.46](#) lists interrupt sources in Simple LIN mode.

Table 38.46 SCI interrupt sources in Simple LIN mode

Name	Interrupt Sources	Interrupt Flag	Flag the needs to be confirmed	Interrupt Enable	DTC/DMAC Activation
SCIn_ERI	Receive error	CSR. ORER, CSR.FER, CSR.PER	—	CCR0.RIE	Not Possible
		XSR0.BCDF		XCR0.BCDIE	
		XSR0.COF		CCR0.RIE, XCR0.COFIE	
SCIn_RXI	Receive data full flag	CSR.RDRF	XSR0.CF0MF XSR0.CF1MF XSR0.PIBDF	CCR0.RIE	XSR0.SFSF = 0: Possible XSR0.SFSF = 1: Not Possible
SCIn_AED	Active edge detection	XSR0.AEDF	—	XCR0.AEDIE	Possible
SCIn_TXI	Transmit data empty interrupt	CSR.TDRE	—	CCR0.TIE	Possible
	When set TE = 0->1				
	Break Field output completion	XSR0.BFOF		CCR0.TIE, XCR0.BFOIE	
SCIn_TEI	Transmit end	CSR.TEND	—	CCR0.TEIE	Not Possible
SCIn_BFD	Break Field Detection	XSR0.BFDF	—	XCR0.BFDIE	Not Possible (Unnecessary)

In Simple LIN mode, in addition to reception errors (ORER, FER, PER), an SCIn_ERI interrupt request is output when a bus conflict is detected during transmission, or when a counter overflow of the Simple LIN module occurs. At this time, a SCIn_RXI interrupt request is not output. The SCIn_ERI interrupt request can be canceled by clearing all the flags.

When transmitting Start Frame, if CCR0.TIE = 1 and XCR0.BFOIE = 1, a SCIn_TXI interrupt request is output when Break Field transmission is completed. When Control Field 0 data is written to the TDR register, data transmission starts. Therefore, transmission using DTC or DMAC is possible.

Set CCR0.TEIE = 1 after writing the last transmit data to the TDR register and transmission starts.

During Start Frame reception (XSR0.SFSF = 1), reception using DTC or DMAC by SCIn_RXI interrupt is not possible. Check the CSR register and XSR0 register, check the reception status (See [Figure 38.83](#)), and then clear the flag. Also read the RDR register (if you do not need to check the received data value, clear the RDRF flag without reading the RDR register). When reception of Control Field 1 is completed (XSR0.CF1MF = 1), Start Frame detection is disabled (XSR0.SFSF = 0) and reception using DTC or DMAC is possible. Be sure to read the RDR register.

When Start Frame / Break Field detection is enabled (XCR1.SDST = 1), if a Break Field longer than the period set in XCR2.BFLW [15:0] is received, the BFDF flag is set and a SCIn_BFD interrupt request is output. Then SCI becomes the Start Frame reception state. Clear the BFDF flag.

When Start Frame / Break Field detection is enabled (XCR1.SDST = 1) and the bit rate measurement function is enabled (XCR1.BMEN = 1), an SCIn_AED interrupt factor is output when an active edge is detected. Read the timer count capture value (XSR1.TCNT [15:0]).

38.13 Event Linking

By using interrupt request signals as event signals, the SCIn can provide linked operation through the ELC for modules selected in advance.

Event signals can be output regardless of the values of the associated interrupt request enable bits.

(1) Error event output (receive error or error signal detected) (SCIn_ERI, n = 0 to 9)

- Indicates abnormal termination because of a parity error during reception in Asynchronous mode
- Indicates abnormal termination because of a framing error during reception in Asynchronous mode
- Indicates abnormal termination because of an overrun error during reception
- Indicates abnormal termination due to a Manchester error during reception (Only in Manchester mode).

- Indicates that a preface error occurred upon reception and abnormal termination occurred (only in Manchester mode and MCR.PFEREN = 1).
- Indicates that a start bit error occurred during reception and abnormal termination occurred (only in Manchester mode and MCR.SBEREN = 1).
- Indicates that a reception sync error occurred during reception and abnormal termination occurred (only in Manchester mode and only when MCR.SYEREN = 1).
- Indicates detection of the error signal during transmission in smart card interface mode
- The CSR.FER and PER flags are 0, and receive data less than the receive FIFO data trigger number is set in a reception FIFO buffer, and it indicates that 15 etu elapse when FIFO is selected and the FCR.DRES bit is 1
- In Simple LIN mode, indicates that the 16-bit counter in the Simple LIN module has overflowed.
- In Simple LIN mode, a bus collision is detected during transmission (CCR0.TE = 1).

(2) Receive data full event output (SCIn_RXI, n = 0 to 9)

- Indicates that ACK is detected if the ICR.IICINTM bit is 0 in simple IIC mode
- Indicates that the 8th-bit SCLn falling edge is detected if the ICR.IICINTM bit is 1 in simple IIC mode
- When the ICR.IICINTM bit is 1 during master transmission in simple IIC mode, set the ELC so that receive data full events are not used

Non-FIFO selected

- Indicates that received data is set in the Receive Data Register (RDR).

FIFO selected

- Using this event output is prohibited.

(3) Transmit data empty event output (SCIn_TXI, n = 0 to 9)

- Indicates that the CCR0.TE bit is changed from 0 to 1
- Indicates that transmission is complete in smart card interface mode
- Indicates that NACK is detected if the ICR.IICINTM bit is 0 in simple IIC mode
- Indicates that the 9th-bit SCLn falling edge is detected if the ICR.IICINTM bit is 1 in simple IIC mode
- In Simple LIN mode, indicates that Break Field output is complete.

Non-FIFO selected

- Indicates that transmit data is transferred from the Transmit Data Register (TDR) to the Transmit Shift Register (TSR).

FIFO selected

- Using this event output is prohibited.

(4) Transmit end event output (SCIn_TEI, n = 0 to 9)

- Indicates the completion of transmission
- Indicates that the starting condition, resumption condition, or termination condition is generated in simple IIC mode
- In Smart Card mode, the transmit end event is not output.

Note: When FIFO is selected, using this event output is prohibited

(5) Address match event output (SCIn_AM, n = 0 to 9)

- Indicates a match of the comparison data (CCR4.CMPD) with one frame of receive data when CCR0.DCME is set to 1 in Asynchronous mode, including multi-processor mode.

(6) Active edge detection event output (SCIn AED, n= 0, 1)

- In Simple LIN mode, when CCR1.BMEN is 1, it indicates that a valid edge has been detected in the RXD input signal.

38.14 Noise Cancellation Function

Figure 38.113 shows the configuration of the noise filter used for noise cancellation. The noise filter consists of a 2-stage flip-flop circuit and a match detection circuit. When the input signals of the noise filter and the output signals of the 2-stage flip-flop circuits completely match, the matched level is conveyed as an internal signal. Unless otherwise matched, the previous value is retained. When the same level is retained for 3 cycles or longer on the sampling clock of the noise filter, it is considered as a valid receive signal. A change in pulse for 3 cycles or shorter is considered as noise, not as a receive signal. (When CCR1.NFM = 0)

When CCR2.ABCSE2 = 1 in asynchronous mode, the above noise cancellation cannot be used because one bit has only 4 base clocks. Therefore, only in this case, set the noise cancellation mode to transmit the majority value internally by a majority vote of 3 sampling values instead of matching 3 consecutive sampling values. (CCR1.NFM = 1)

In Asynchronous mode, Manchester and Simple LIN modes, the noise cancellation function can be applied to the receive signal input to the RXDn pin. The sampling period of the noise filter can be selected from the base clock period and the divided clock of the baud rate generator clock source by CCR1.NFCS[2:0].

- When CCR1.NFCS[2:0] = 000b, CCR2.ABCS = 0 and CCR2.ABCSE = 0, the cycle is 1/16 of a 1-bit period.
- When CCR1.NFCS[2:0] = 000b, CCR2.ABCS = 1 and CCR2.ABCSE = 0, the cycle is 1/8 of a 1-bit period.
- When CCR1.NFCS[2:0] = 000b, CCR2.ABCSE = 1, the cycle is 1/6 of a 1-bit period.
- When CCR1.NFCS[2:0] = 000b, CCR2.ABCSE2 = 1, the cycle is 1/4 of a period 1 transfer bit.
- For the CCR1.NFCS value, select “000b” or “001b” if ABCSE = 1 or ABCSE2 = 1. At least, the sampling period of Noise Filter should not be greater than the base clock due to the CCR2.BRR setting value.

In simple IIC mode, this function can be used for each input on SDAn and SCLn. The sampling clock is selected from divided clock of baud rate generator settings by CCR1.NFCS[2:0].

If the base clock is stopped once with the noise filter enabled and then the base clock input is restarted again, the noise filter operation resumes from the state where the clock was stopped. When CCR0.TE and CCR0.RE are set to 0 during base clock input, all of the noise filter flip-flop values are initialized to 1. Accordingly, if the input data is 1 when reception operation resumes, the function determines that a level match is detected and the result is conveyed as an internal signal. If the input data is 0, the Noise Filter output holds its initial value until all three sampled pin levels match (CCR1.NFM = 0) or two or more match (CCR1.NFM = 1).

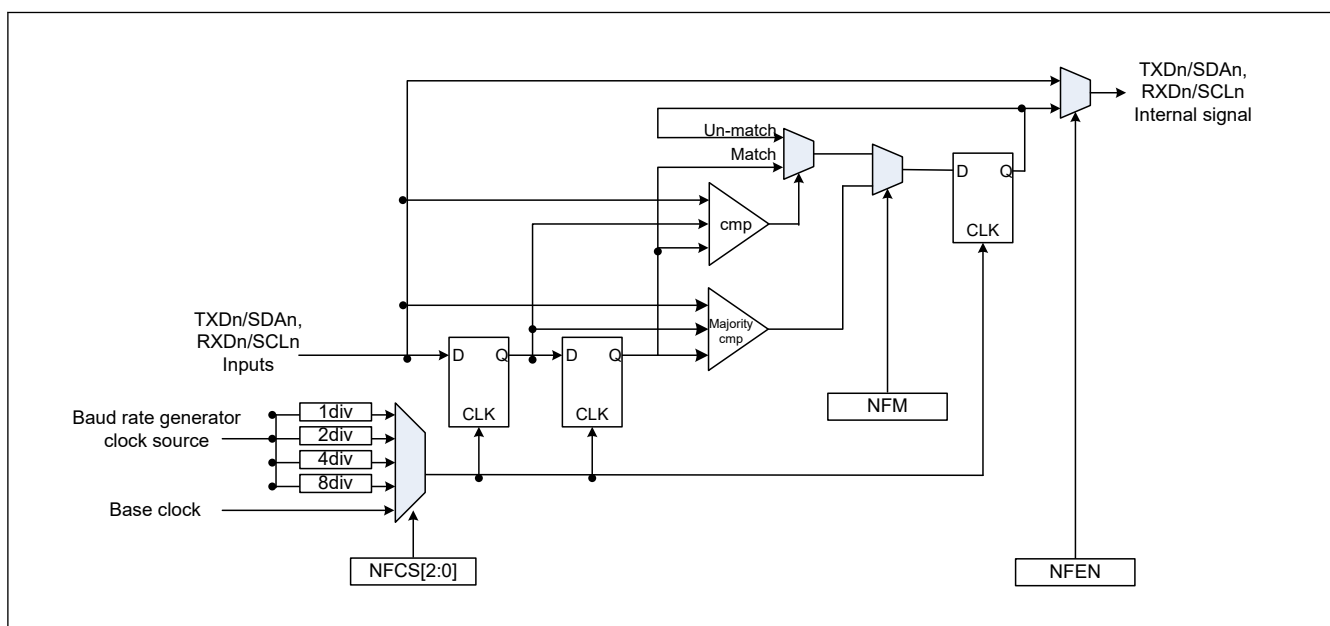


Figure 38.113 Digital noise filter circuit block diagram

38.15 RS-485 Driver Control Function

Setting the DEN bit in the SCI common control register3 (CCR3) to 1 enables the RS-485 driver control function and generates a DEN (Driver Enable) signal that enables the external transceiver transmission mode. The DEN signal outputs a valid level for the period with driver assertion time and driver negate time added before and after data transmission. The DEN signal valid level is set by the DEPOL bit in the driver control register (DCR).

The driver assertion time is the time from when the DEN signal is valid until the start bit starts. Set by DEAST [4:0] of driver control register (DCR).

The driver negate time is the time from the end of the last stop bit of the transmitted message to the invalidation of the DEN signal. Set with DENG [4:0] of the driver control register (DCR).

DEAST and DENG are expressed in base clock period (1/4, 1/6, 1/8, or 1/16 bit period, see [Table 38.6](#)). For details, see [section 38.2.13. DCR : Driver Control Register](#).

When this function is used (CCR3.DEN = 1), the CSR.TEND set timing and SCIn_TEI interrupt output timing are at the end of the driver negation time.

When transmission is completed and the next transmission data is not written before the DEN signal is negated, the DEN signal is negated once. If the timing for writing the next transmit data is not in time, assert the DEN signal after negating it again, insert the driver assertion time, and transmit the next data. If you want to perform the next transmission with the DEN signal asserted, write the next transmission data to the TDR quickly enough in consideration of the synchronization delay time of the register.

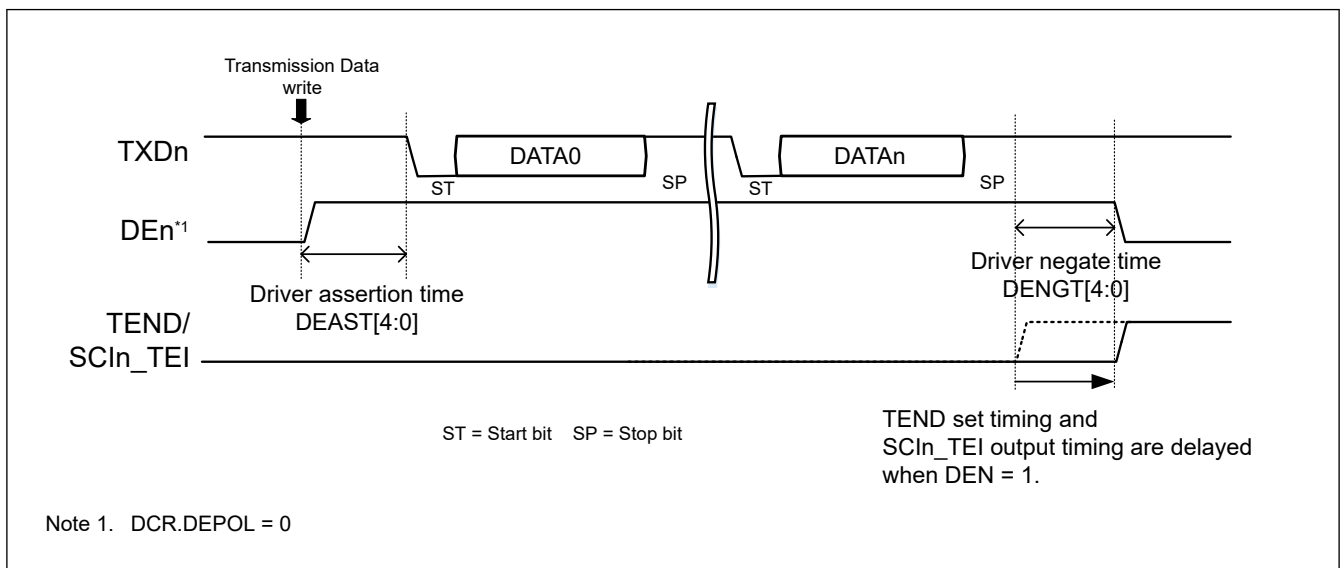


Figure 38.114 The image waveform for RS-485 driver control DE signal output

38.16 Loopback Function

The loopback function can be used in Asynchronous mode with the internal clock, and Manchester mode with the internal clock, and Clock synchronous mode with the internal clock.

When 1 is written to the SPLP bit in the CCR1 register, SCI blocks the external input (RXDn) path and connects the output path of the transmit data register and the input path of the receive data register.

When this function is used with TINV bit = 1, inversion of transmission data becomes reception data. However, this function can be used with TINV = 1 only when operating in clock synchronous mode internal clock.

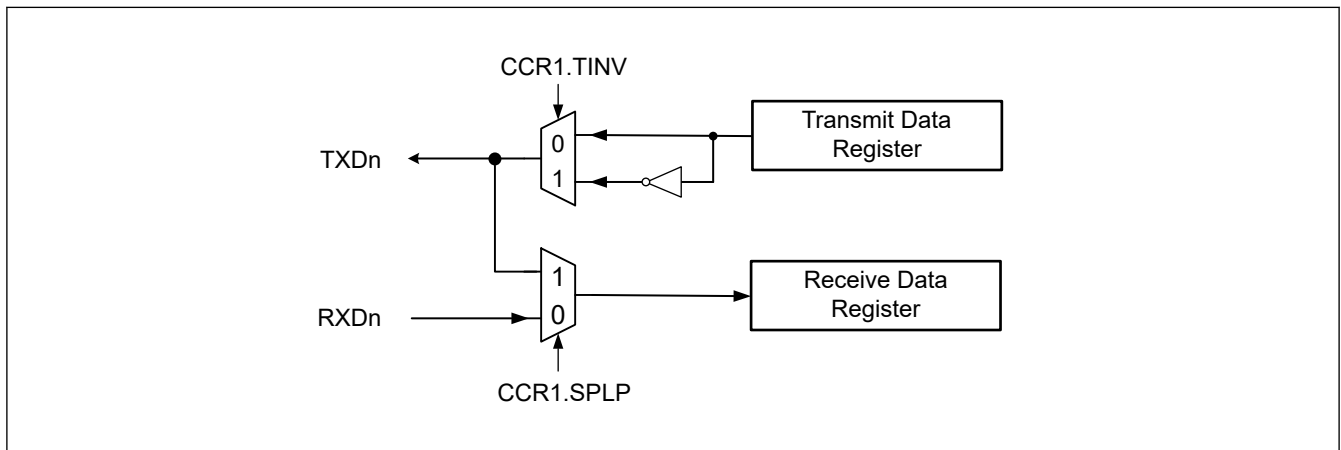
[Table 38.47](#) shows the relationship between the TINV and SPLP bit settings and the received data.

Table 38.47 TINV and SPLP bit settings and received data

CCR1.TINV	CCR1.SPLP	Receive Data	Communication mode		
			Asynchronous	Manchester	Clock synchronous
			internal clock	internal clock	internal clock
—	0	Receive Data from RXDn pin	Possible	Possible	Possible
0	1	Transmit Data	Possible	Possible	Possible
1	1	Inverted transmit data	Impossible	Impossible	Possible

Note: —: don't care

Figure 38.115 shows the configuration of the shift register input / output path in loopback mode.

**Figure 38.115** Shift register input output configuration image in loopback mode

38.17 Half-Duplex Communication Function

Do not use the half-duplex communication function in Simple IIC, Simple SPI and Smart Card Interface modes.

In other communication modes, if the CCR1.SHARPS bit is set to 1, half-duplex communication using the TXDn pin is possible. When half-duplex communication is used, transmission and reception must be performed exclusively. Transmission and reception settings (CCR0.TE = 1 and CCR0.RE = 1) is prohibited.

However, if half-duplex communication is performed as the master reception in clock synchronous mode, perform transmission / reception settings (CCR0.TE = 1 and CCR0.RE = 1) and perform dummy transmission. By dummy transmission (arbitrary transmission data is written to TDR), SCKn is output and reception is enabled. The dummy transmission data is discarded inside the IP and is not actually transmitted.

During half-duplex communication, only the TXDn pin is used as the communication pin. Output when CCR0.TE = 1, input when CCR0.TE = 0.

38.18 Synchronizer Bypass Function

The SCI has a bus clock (PCLK) and the operation clock (TCLK). And these have each operating circuit. Therefore, there is a synchronization circuit for signal transfer between different clocks, and synchronization delay time is required for signal propagation between different clocks.

However, the synchronization circuit can be bypassed by the CCR3.BPEN bit only when the same clock is input to the bus clock and the operation clock. In this case, eliminates synchronization delay time and improves responsiveness. Figure 38.116 shows the image waveform of the bypass function.

The SCI also has a synchronization circuit between the communication clock (SCKn) and the operation clock (TCLK), but this synchronization circuit cannot be bypassed.

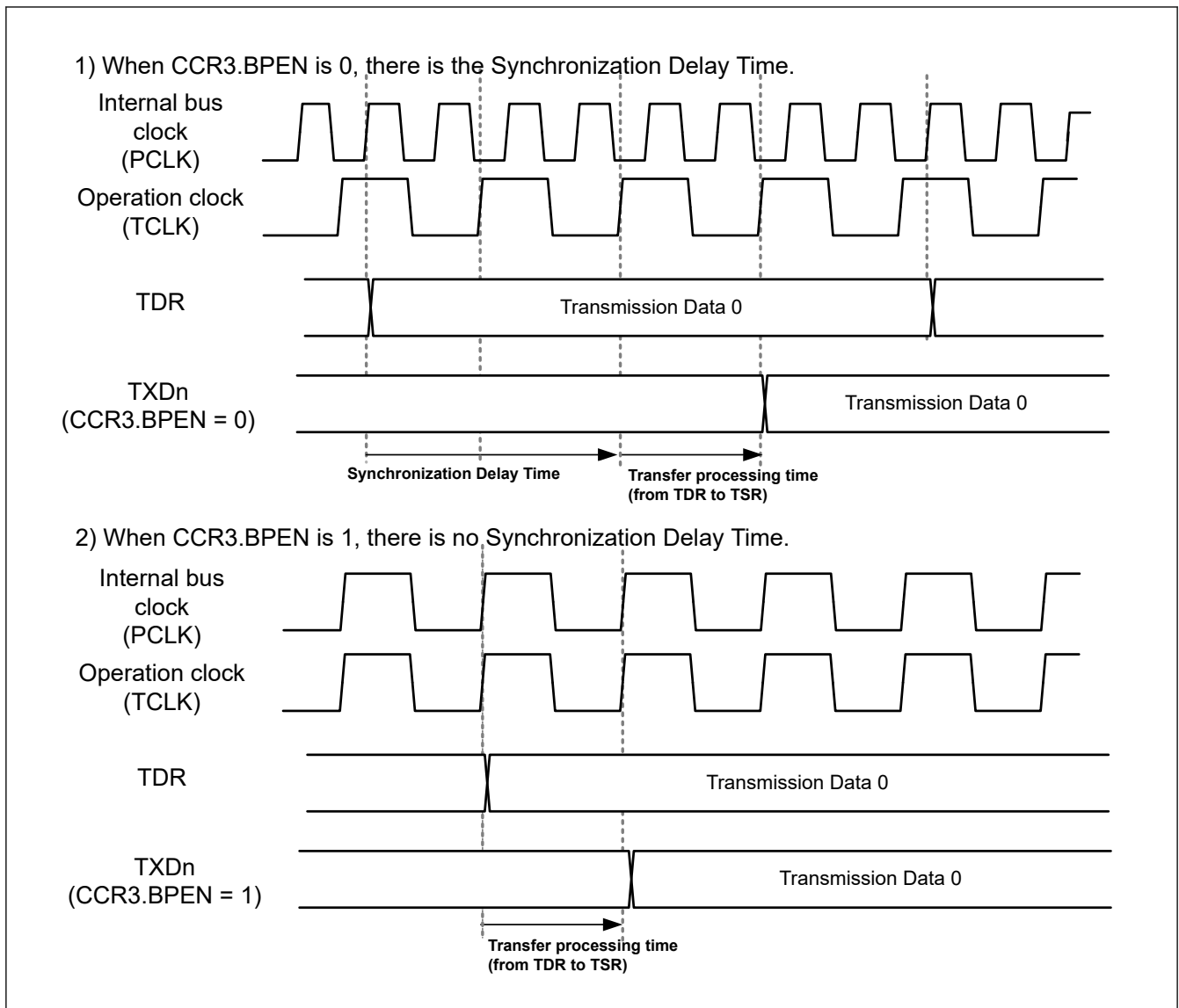


Figure 38.116 Image waveform of Synchronizer bypass function

38.19 Usage Notes

38.19.1 Settings for the Module-Stop Function

The Module Stop Control Register B (MSTPCRB) can enable or disable SCI operation. The SCI is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

38.19.2 Software Standby Mode

(1) Transmission

Before using the power consumption reduction function to reduce SCI's power consumption, do the following to confirm transmission end (CSR.TEND = 1):

- Set the output pin state after transmission operation is stopped by CCR1.SPB2DT, SPB2IO.
- Stop the transmission (CCR0.TIE = 0, TE = 0, TEIE = 0)

When transitions to these states are made during transmission, the data being transmitted become indeterminate.

To transmit data in the same operation mode after cancellation of the low power consumption state, set the TE bit to 1, read CSR, and write data to TDR sequentially to start data transmission. To transmit data with a different operation mode, initialize the SCI first.

To start transmission using the DMAC or DTC after cancellation from software standby mode, set the CCR0.TE and CCR0.TIE bit to at the same time. Then SCI_{In}_TXI interrupt flag is generated, which causes the DMAC or DTC to write the transmit data, which starts transmission.

Figure 38.117 shows a sample flowchart for transition to software standby mode during transmission. Figure 38.118 and Figure 38.119 show the port pin states during transition to software standby mode.

(2) Reception

When Address match function is non used as condition of resumption (wake-up)

Before specifying the module stop state or making a transition to software standby mode, stop the receive operations (CCR0.RE = 0). If transition is made during data reception, the data being received will be invalid.

Figure 38.120 shows a sample flowchart for reception to software standby mode during reception.

When Address match function is used as condition of resumption (wake-up)

When using the power consumption reduction function to reduce SCI's power consumption, do the following:

- Set the operation mode to Asynchronous mode
- Set the compare data to CCR4.CMPD and set 1 to CCR0.DCME
- Set the receive operation (CCR0.RE = 1)

Set CCR3.RXDESEL = 0. Because there is a possibility that a start bit (fall edge of RXD_n pin) cannot be detected at the time of low power consumption mode release.

Figure 38.121 shows a sample flowchart for reception to software standby mode during reception.

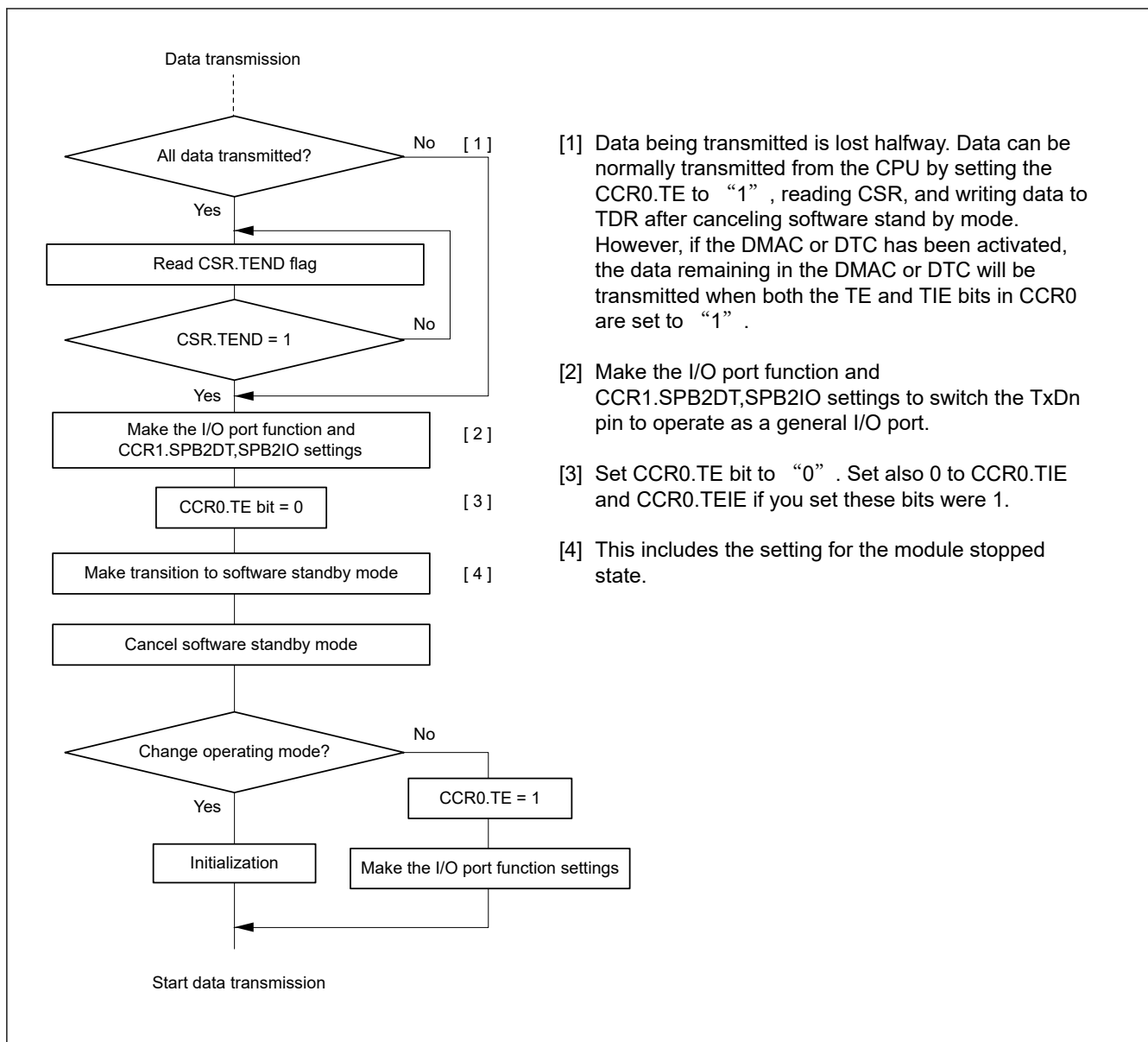


Figure 38.117 Example of Flowchart for Transition to Software Standby Mode during Transmission

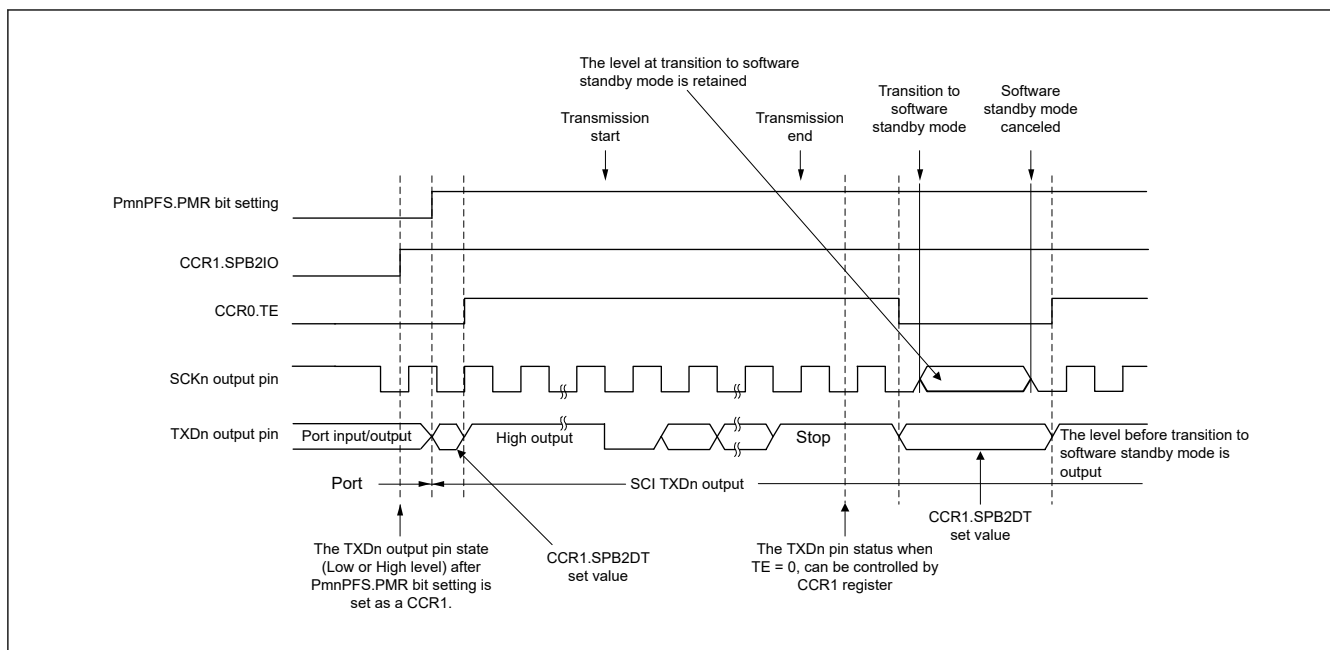


Figure 38.118 Port Pin States during Transition to Software Standby Mode (Internal Clock, Asynchronous Transmission)

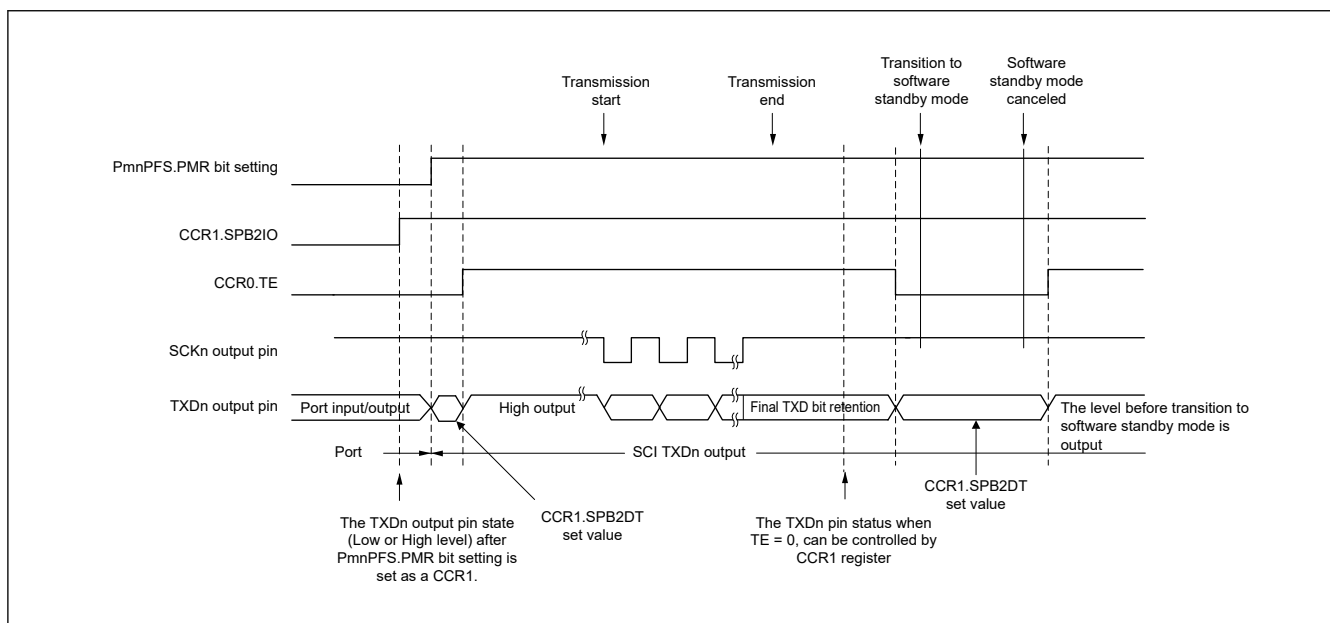


Figure 38.119 Port Pin States during Transition to Software Standby Mode (Internal Clock, Clock Synchronous Transmission)

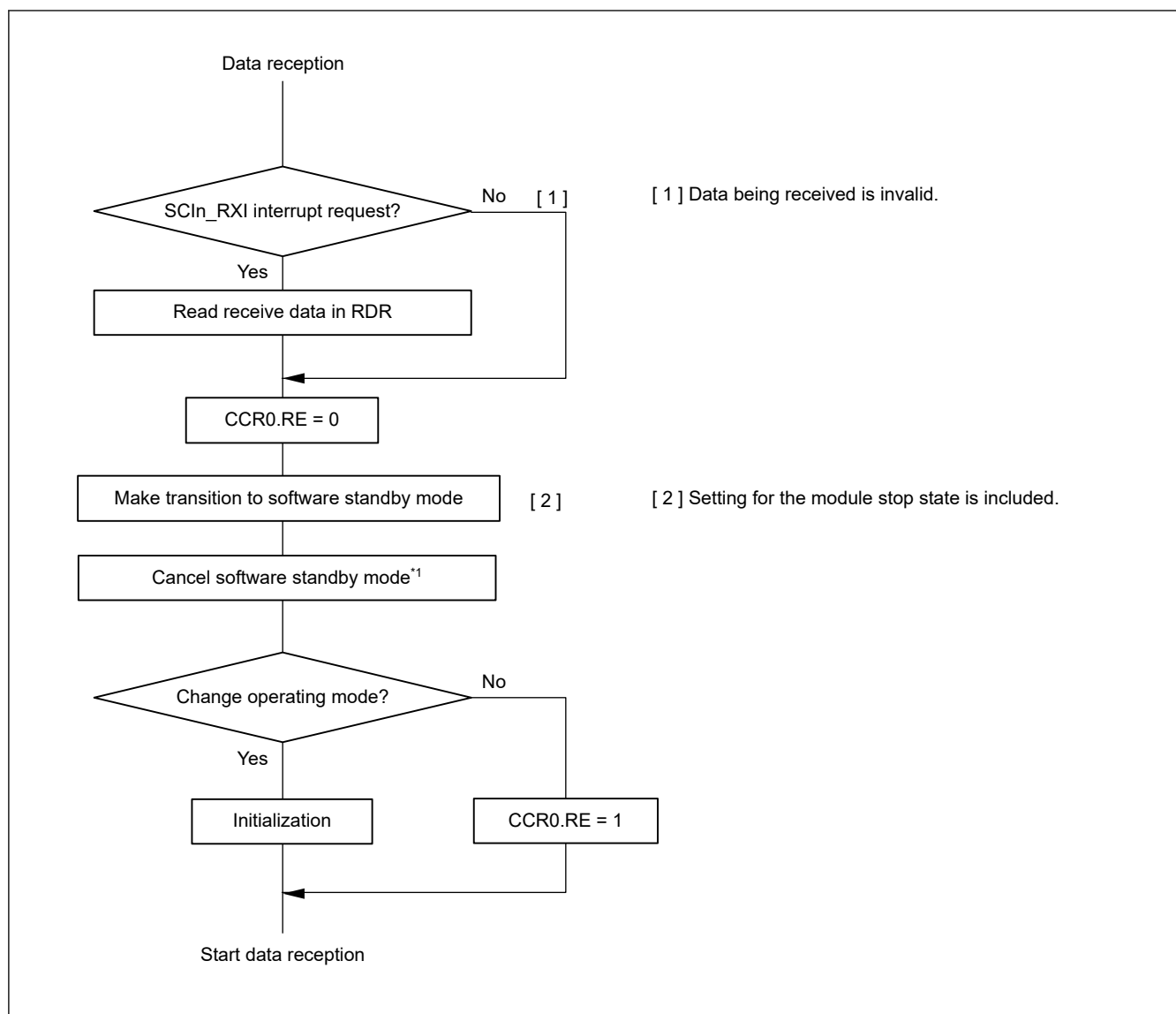


Figure 38.120 Example of Flowchart for Reception to Software Standby Mode during Reception

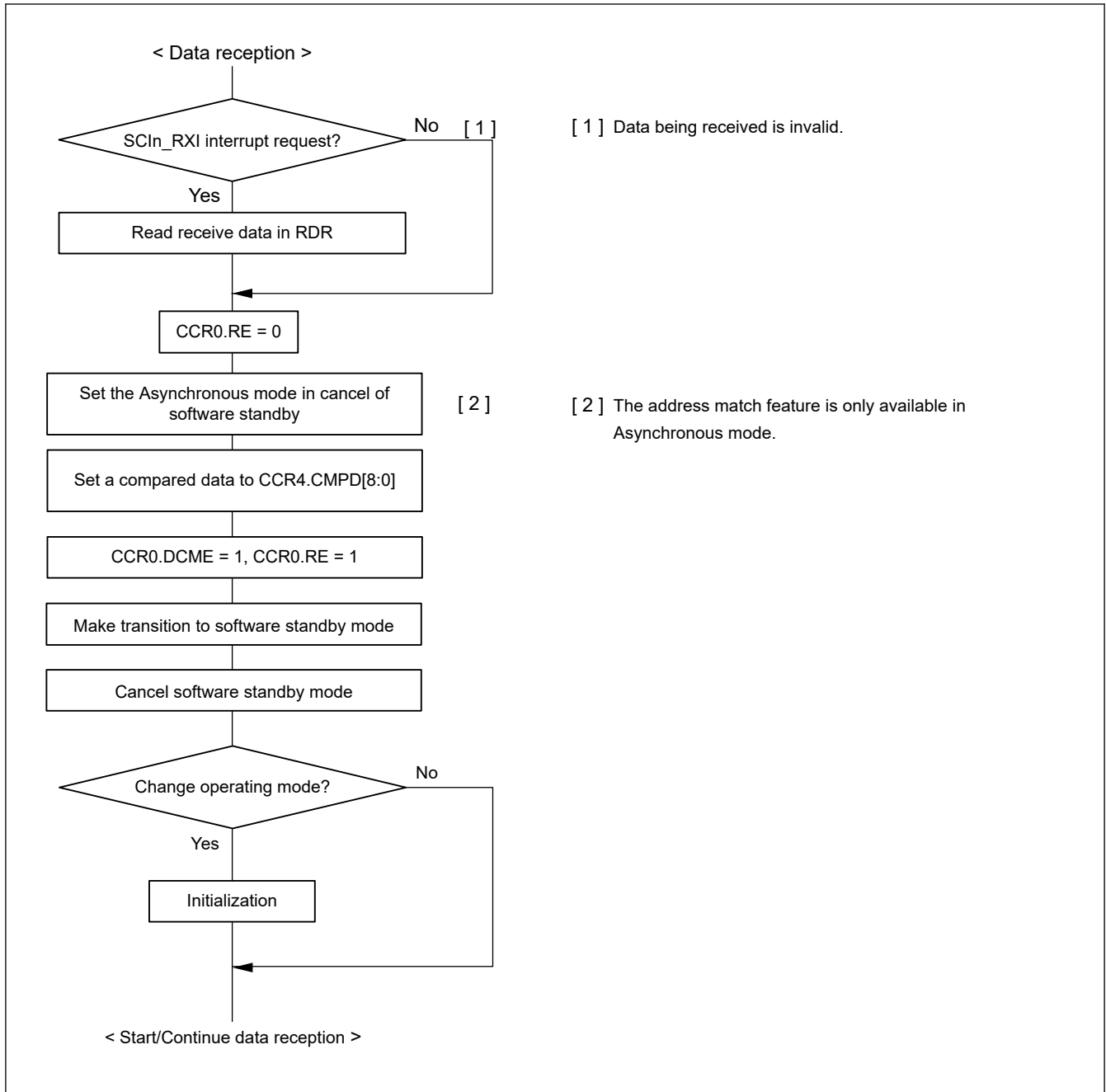


Figure 38.121 Example of Flowchart for Reception to Software Standby Mode during Reception with Address match

38.19.3 Break Detection and Processing

(1) Non-FIFO selected

When a framing error is detected, a break can be detected by reading CSR.RXDMON bit value. In a break, the input from the RXDn pin becomes all 0s, and the CSR.FER flag is set to 1 to indicate a framing error, and the CSR.PER flag might also be set to 1 to indicate a parity error. The SCI continues the receive operation even after a break is received. Therefore, even if the FER flag is 0, indicating that no framing error occurred, it is set to 1 again. When the CCR3.RXDESEL bit is 1, the SCI sets the CSR.FER flag to 1 and stops receiving operations until a start bit of the next data frame is detected. If the CSR.FER flag is set to 0, the CSR.FER flag retains 0 during the break.

When the RXDn pin is set to 1 and the break ends, detecting the beginning of the start bit on the first falling edge of the RXDn pin allows the SCI to start the receiving operation.

(2) FIFO selected

After a framing error is detected and when the SCI detects that continuous receive data is 0 for 1 frame, reception stops. When a framing error is detected, a break can be detected by reading the CSR.RXDMON flag value. After the RXDn signal is in high and the break is finished, data reception to the receive-FIFO (RDR) register resumes.

38.19.4 Mark State and Production of Breaks

When the CCR0.TE bit is 0, disabling serial transmission, the state of the TXDn pin can be set using the CCR1.SPB2IO and CCR1.SPB2DT bits. With this approach, a TXDn pin can be placed in the mark state to transmit a break.

Before setting the CCR0.TE bit to 1, enabling serial transmission, set the SPB2IO and SPB2DT bits to put the communication line in the mark state (the state of 1), and change the TXDn pin using I/O port function. To output a break on data transmission, after setting the TXDn pin to output 0 by setting the SPB2IO and SPB2DT bits, change the TXDn pin using the I/O port function and set the CCR0.TE bit to 0. When the CCR0.TE bit is set to 0, the transmitter is initialized regardless of the current state of transmission.

38.19.5 Receive Error Flags and Transmit Operations (Clock Synchronous Mode and Simple SPI Mode)

Transmission can be start by writing transmit-data to TDR even if CSR.ORER is 1. However, reception cannot be started. Note also that the receive error flags cannot be set to 0 even if the CCR0.RE is set to 0 (serial reception is disabled).

38.19.6 Writing Data to TDR

(1) Non-FIFO selected

Data can be written to TDR anytime when CCR0.TE is 1. However, if new data is written to TDR when transmit data is remaining in TDR, the previous data in TDR is lost because it has not been transferred to TSR yet. If you use DTC or DMAC, be sure to write transmit data to TDR in the SCIn_TXI interrupt request handling routine.

(2) FIFO selected

Data can be written to transmit-FIFO(TDR) when CCR0.TE is 1. Check the number of writable data with the FTSR.T [5:0] bit.

38.19.7 Restrictions on Clock Synchronous Transmission (Clock Synchronous Mode and Simple SPI Mode)

When the external clock source is used as a synchronization clock, the following restrictions apply.

(1) Start of transmission

Update TDR by the CPU, DMAC, or DTC and wait at least the following time until the start of the external clock input: (See [Figure 38.122](#))

Take the following time into account: the output AC spec of the MISO pin of this product and the input AC spec of the master reception + 1 PCLK cycle + synchronization delay.

(2) Continuous transmission

Write the next transmit data to TDR before the falling edge^{*1} of the transmit clock for bit 7. Write the transmit data to TDR in consideration of synchronization delay. If the transmit data cannot be written in time, the previous frame data is resent. (See [Figure 38.122](#))

Note 1. When CCR3.CPOL = 1 and CCR3.CPHA = 0, or CCR3.CPOL = 0 and CCR3.CPHA = 1. In the case of CCR3.CPOL = 0 and CCR3.CPHA = 0, or CCR3.CPOL = 1 and CCR3.CPHA = 1, it's the rising edge.

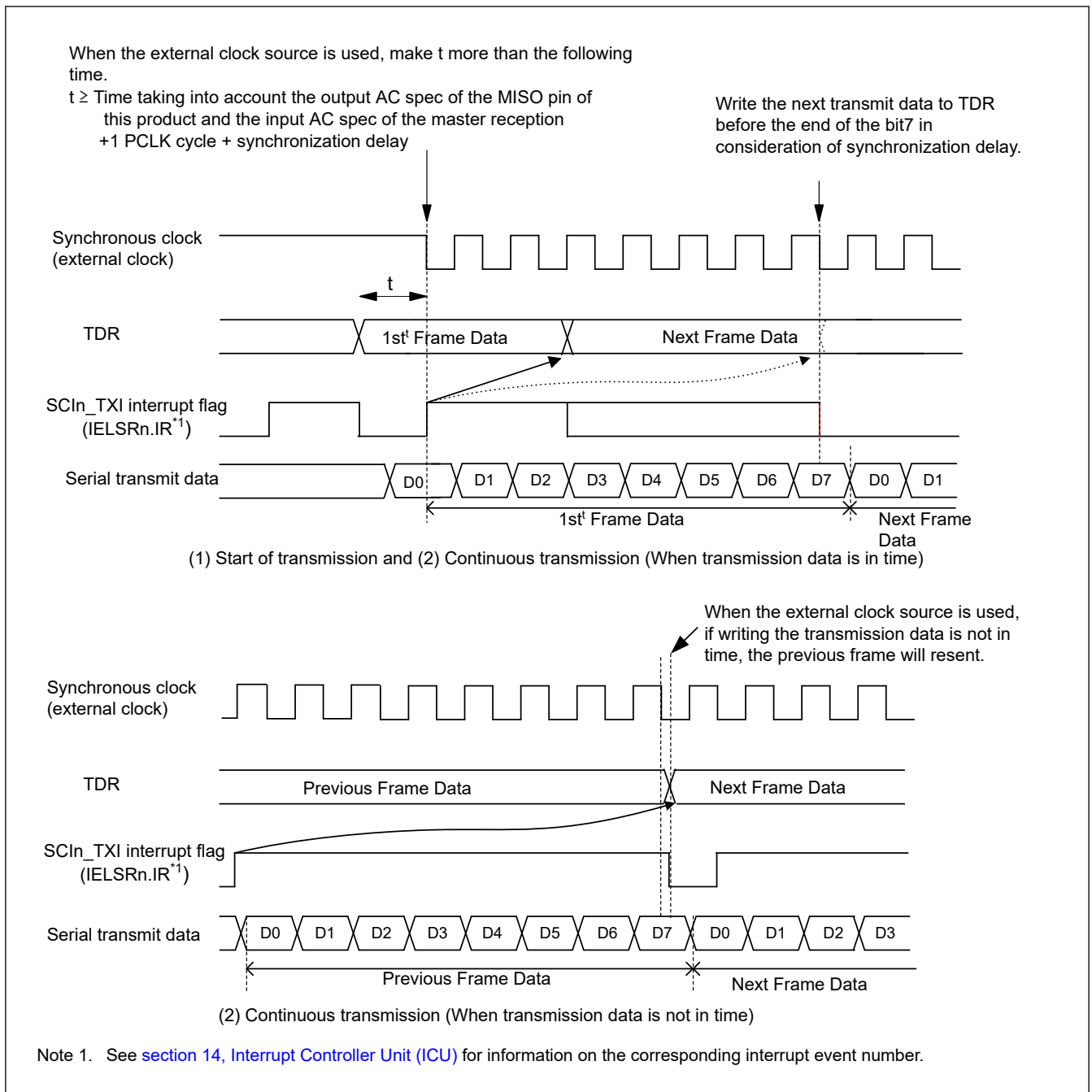


Figure 38.122 Restrictions on Use of External Clock in Clock Synchronous Transmission

38.19.8 Restrictions on Using DMAC or DTC

When using the DMAC or DTC to read RDR, be sure to set the receive data full interrupt (SCIn_RXI) as the activation source of the relevant SCI.

During the operation in transmission / reception using the DMAC or DTC, it should not set transfer information of DMAC or DTC.

38.19.9 Notes on Starting Transfer

At the point where transfer starts when the interrupt status flag (IELSRn.IR flag) in the ICU is 1, follow the procedure in this section to clear interrupt requests before permitting operations (by setting the CCR0.TE or CCR0.RE bit to 1). For details on the interrupt status flag, see [section 14, Interrupt Controller Unit \(ICU\)](#).

1. Confirm that transfer has stopped (the CCR0.TE or CCR0.RE bit is 0)

2. Set the associated interrupt enable bit (CCR0.TIE or CCR0.RIE bit) to 0
3. Read the associated interrupt enable bit (CCR0.TIE or CCR0.RIE bit) to check that it actually becomes 0
4. Set the interrupt status flag, IELSRn.IR, in the ICU to 0

38.19.10 Limitations on Simple SPI Mode

(1) Master mode

- Use a resistor to pull up or pull down the clock line matching the initial settings for the transfer clock set in the CCR3.CPHA and CPOL bits when the CCR0.SSE bit is 1.

This prevents the clock line from being placed in the high-impedance state when the CCR0.TE bit is set to 0 or unexpected edges from being generated on the clock line when the CCR0.TE bit changes from 0 to 1. When the CCR0.SSE bit is 0 in single master mode, pulling up or pulling down the clock line is not required because the clock line is not placed in the high-impedance state even when the SCR.TE bit is set to 0.

- For the clock delay setting (CCR3.CPHA bit is 0), the receive data full interrupt (SCIn_RXI) is generated before the final clock edge on the SCKn pin as indicated in Figure 38.123. If the TE and RE bits in the CCR0 register become 0 before the final edge of the clock signal on the SCKn pin, the SCKn pin is placed in the high-impedance state, so the width of the last clock pulse of the transfer clock is shortened. Additionally, an SCIn_RXI interrupt might lead to the input signal on the SSn pin of a connected slave going to the high level before the final edge of the clock signal on the SCKn pin, leading to incorrect operation of the slave.
- In a multi-master configuration, the SCKn pin output goes to high-impedance while the input on the SSn pin is at the low level if a mode fault error occurs while a character is being transferred, stopping supply of the clock signal to the connected slave. Reset the connected slave to avoid misaligned bits when transfer is restarted.

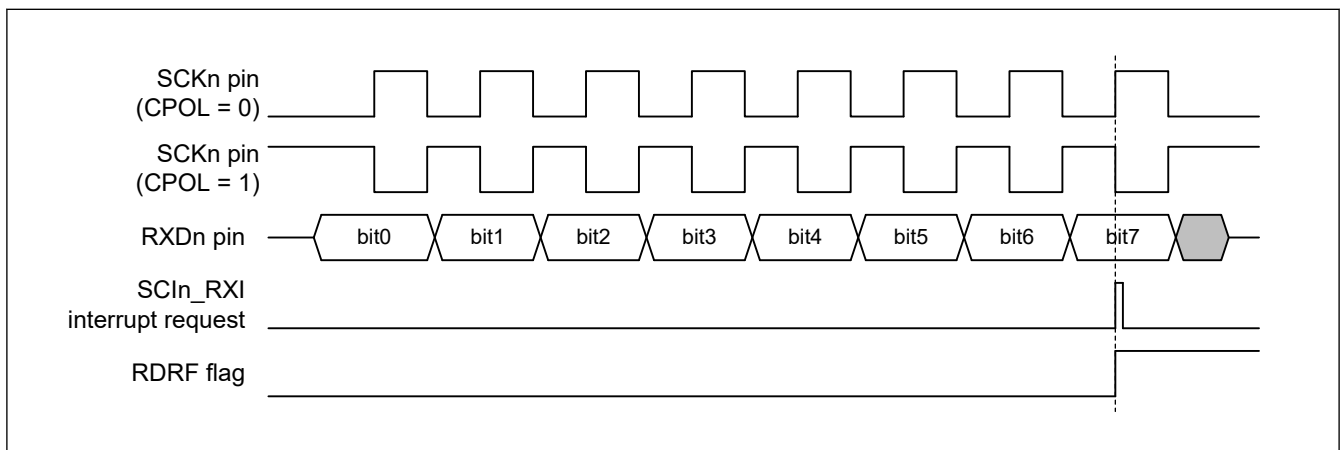


Figure 38.123 Timing of SCIn_RXI interrupt in simple SPI mode with clock delay

(2) Slave mode

- It takes 1PCLK + synchronization delay time + data output delay time (AC spec) from writing the transmit data to the TDR register until the data is output to the RXDn pin. Take these into account when starting external clock input.
- Provide an external clock signal to the master the same as the data length for transfer
- Secure the SSn input setup time (AC spec) from the SSn low-level input to the start of external clock input.
- Control the input on the SSn pin before the start and after the end of data transfer
- When the input level on the SSn pin is to be changed from low to high while a character is being transferred, set the TE and RE bits in the CCR0 register to 0 and, after restoring the settings, restart transfer of the first byte

38.19.11 Notes on Transmit Enable bit (CCR0.TE)

In initial register value, when CCR0.TE = 0, the state of the TXDn pin is high impedance. The TXDn line should not be high impedance by the following one of ways.

1. The pull-up or pull-down resistance is connected to the TXDn line.

2. Set CCR1 and decided level of TXDn terminal during TE is 0.

38.19.12 Notes on Simple LIN Mode

In Simple LIN mode (CCR3.MOD[2:0] = 110), the following functions cannot be used.

- Multi-processor communication function
- Bit Rate Modulation function
- Loopback function
- FIFO buffer

38.19.13 Notes on RS-485 Driver Control Function

RS-485 Driver control function is valid only in Asynchronous mode.

When RS-485 Driver control function is active (CCR3.DEN = 1), the CSR.TEND set timing / SCIn_TEI output timing changes as follows. Wait for the SCIn_TEI interrupt and set the CCR0.TE bit in SCI to 0.

When RS-485 Driver control function is inactive: When STOP bit output is completed.

When RS-485 Driver control function is active: At the end of DEN negation time.

38.19.14 Notes on Loopback Function

The Loopback function is valid in Asynchronous mode with internal clock, in Manchester mode with internal clock and Clock synchronous mode with internal clock.

38.19.15 Notes Regarding Register Access When Operation Clock (TCLK) Is Slower than Bus Clock (PCLK)

If the operating clock (TCLK) is slower than the bus clock (PCLK), the time until this information is transmitted internally after writing to the CCR0.TE and CCR0.RE registers is slower than the bus access time. In particular, when trying to change the setting register after writing 0 and interrupting communication, do not change the setting register before the signal inside the IP is in the communication stopped state. To prevent this, after setting CCR0.TE and CCR0.RE to 0, check the CESR.TIST and CESR.RIST bits until they are 0 before setting the next register.

38.19.16 Notes on Interrupting Operation

If 0 is written to CCR0.RE during data reception and the reception operation is interrupted, there is a possibility of an invalid state, do not use the received data (RDR register stored value) and the flag value of each status register. To interrupt the reception operation, stop the interrupt or event link reception side and then write 0 to the CCR0.RE bit.

38.19.17 Notes on CCR3.BPEN Bit Setting

Set the BPEN bit only once when setting the CCR3 register in the SCI initialization flow.

This bit cannot be changed after the initialization.

When this bit setting is changed, start from the SCI initialization flow again.